



NAVFAC
Naval Facilities Engineering Systems Command

NAVFAC MIDLANT, Public Works Department, Maine

ATTACHMENT 2

**PNSY
DIVISION 1 & 2
SPECIFICATIONS**

November 2023

PREPARED BY:

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Building 59
Portsmouth Naval Shipyard
Kittery, Maine 03801-2032

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SECTION 01 14 00.00 22

WORK RESTRICTIONS (PWD ME)

10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard only.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES (for Design Build Projects):

SD-01 Preconstruction Submittals

List of Contact Personnel; G

1.3 SPECIAL SCHEDULING REQUIREMENTS (PNSY)

- a. Work requiring interruption (outage) of any Portsmouth Naval Shipyard roads, crane rail, railroads, and/or utility services must be coordinated/aligned with stakeholders prior to submission of the Outage package for final 15 calendar day approval. See Paragraph 1.8 for Outage timing and instructions. .
- b. Permission to interrupt any Portsmouth Naval Shipyard roads, crane rail, railroads, and/or utilities or services must be submitted to the Contracting Officer in writing a minimum of 30 calendar days (20 working days) prior to the desired date of interruption. See Paragraph 1.8 for FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS.
- c. When the project is located in and adjacent to restricted area . The Contractor is not allowed in these areas without continuous Government escorts. See the Scope of Work to determine if the project limits include any security Island in or adjacent to the work area.

- d. Coordinate the work with the sequencing/phasing requirements outlined in the Scope of Work Memo and as outline in the Construction Drawings.

1.4 ACCESS AND USE OF PREMISES (PNSY)

1.4.1 Portsmouth Naval Shipyard Regulations

Notify the Contracting Officer 30 calendar days in advance for any wide loads. Contact the Contracting Officer to determine if there are other access limitations at the Portsmouth Naval Shipyard.

Ensure that personnel employed on the Portsmouth Naval Shipyard become familiar with and obey Portsmouth Naval Shipyard regulations including safety, fire, traffic, and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE). Do not enter restricted areas unless required to do so and until cleared for such entry.

All equipment must be conspicuously marked with company signage for identification.

1.4.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel to include Subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.4.1.2 Identification Badges and Installation Access

- a. Application for and use of badges will be as directed. Obtain access to the Portsmouth Naval Shipyard by submitting the SECNAV Form 5512-NSNPT Jul 2019 (Attachment C) a minimum of 15 working days in advance for vetting and approval, or by obtaining passes each day from the Portsmouth Naval Shipyard's Pass and Identification/Security Office. One-day passes, issued through the Portsmouth Naval Shipyard's Pass and Identification Office, will be furnished without charge. Report any instances of lost or stolen badges to the Contracting Officer immediately.

Prime contractor is responsible for recovering any badges that are no longer required, including from workers who are no longer employed by prime or subs, and immediately turning badges into the Pass/ID Office. Failure to maintain accountability of issued badges could cause delays in future badge requests. Contractors are warned that badges are checked for currency and any badge not used within 30 days will be disabled.

- b. All Contractors who possess a Commercial Access Control (CAC) System Card are also required to submit an access request (5512) to support a valid purpose to be on the Portsmouth Naval Shipyard. A second form of valid ID may also be required, if requested, upon arrival at the Portsmouth Naval Shipyard's Entrance Gate. All personnel without CAC cards will need two (2) forms of approved identification for access to the Portsmouth Naval Shipyard.
- c. One-Day Passes: Participation in the vetting process is mandatory, unless a valid emergency exists. Then it is possible to utilize an escort from the department effected. If the Contractor chooses to not participate in the vetting process, they will not be allowed on the

Portsmouth Naval Shipyard. The Government will not be responsible for any cost or lost time associated with obtaining daily passes or added vehicle inspections.

- d. PNS has an online appointment system. Appointments may be made using <https://kiosk.na4.qless.com/kiosk/app/home/191>.

1.5 PORTSMOUTH NAVAL SHIPYARD (PNS) REGULATIONS

1.5.1 Radiological

1.5.1.1 Radiological Indoctrination (PNSY)

All personnel working at the Portsmouth Naval Shipyard are required to view a 15 minute video briefing on radiological postings and controls in use at the Portsmouth Naval Shipyard. The briefing will be given at the Pass Office prior to issue of security badges and vehicle passes.

Any employee who disregards, alters, moves, or otherwise tampers with a radiological posting, or who disobeys a radiological instruction, may be removed from the Portsmouth Naval Shipyard and denied future access.

1.5.1.2 Yellow Materials (PNSY)

Do not use yellow or orange-yellow colored materials at the Portsmouth Naval Shipyard for the following purposes: Protective clothing, hoods, sheeting, tarpaulins, polyethylene bottles or other containers, tapes, bags, banding, identification marks on tools, boundary markers, ribbons, vent ducts, temporary erosion control devices, survey ribbon, etc. Contact the Contracting Officer for a list of yellow items that have been approved for use on the Portsmouth Naval Shipyard. Dispose of generated yellow colored waste off of the Portsmouth Naval Shipyard. Portsmouth Naval Shipyard refuse containers must not be used for disposal of yellow colored waste materials. Yellow colored items such as described above are of special significance within the Portsmouth Naval Shipyard and are subject to strict controls. Yellow colored Contract generated debris must be bagged in non-translucent containers, and promptly removed from the Portsmouth Naval Shipyard.

1.5.1.3 Smoke Detectors (PNSY)

Ionization type smoke detectors and duct smoke detectors contain radioactive material and are prohibited from use on the Portsmouth Naval Shipyard. Photoelectric smoke detectors are the only type authorized for use on the Portsmouth Naval Shipyard.

1.5.1.4 Radioactive Sources (PNSY)

All Contracts involving radiation generating devices must conform to the requirements listed in Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) and U.S. Army Corps of Engineers Safety Manual EM 385-1-1. All requirements are to be submitted to the Contracting Officer at least 14 days prior to commencement of operations involving radiation generating devices. A requirements checklist will be provided by NAVFAC (COTs) Contractor Oversight Technician.

1.5.2 Laser Control

Comply with laser safety requirements under 21 CFR 1040 and ANSI

2136.1-1986 for any work under this Contract utilizing lasers.

1.5.3 Energy Conservation

In cooperation with Government representatives, participate in an active program directed toward the efficient use of energy.

1.5.4 Fire Prevention (PNSY)

Require employees to become familiar with fire prevention regulations within the Portsmouth Naval Shipyard to include the proper method of turning in a fire alarm, storage of flammable and combustible materials, and control of combustible waste and trash. Any HOT WORK (welding, burning, grinding, cutting, etc.) requires a HOT WORK PERMIT prior to commencing such work. This permit is obtained from the Portsmouth Naval Shipyard's Fire Department via the Contracting Officer and must be submitted at least three(3) working days prior to commencing any Hot Work.

1.5.5 Identification and Control of Seamed (Welded) Pipe and Tubing (PNSY)

Submarine Safety regulations prohibit the use of seamed (welded) pipe or tubing within the Portsmouth Naval Shipyard, unless such pipe or tubing is identified and controlled so as to prevent its inadvertent substitution for seamless pipe or tubing. The following requirements apply and will be strictly enforced:

Any seamed (welded) copper-nickel, carbon steel, carbon-moly steel, stainless steel, nickel-chromium-iron alloy, or nickel-copper pipe or tubing intended for use on the Portsmouth Naval Shipyard must be identified in the following manner PRIOR TO DELIVERY TO THE PORTSMOUTH NAVAL SHIPYARD:

Use a lead-free white paint, to mark a 24-inch long stripe and the word "welded" alternately along the entire length of the pipe or tubing. Apply a one-half inch wide stripe unless the size of the pipe or tubing requires use of a narrower stripe.

Maintain positive control over seamed pipe or tubing until worked into place or removed from the Portsmouth Naval Shipyard.

Seamless pipe or tubing may be substituted for any seamed (welded) pipe or tubing specified in the technical specifications.

The above requirements do not apply to square or rectangular tubing, copper or brass pipe or tubing, or to piping or tubing which has been incorporated into equipment or fixtures prior to delivery to the Portsmouth Naval Shipyard.

1.5.6 Pesticide and Herbicide Control

Do not apply pesticides nor herbicides unless specifically required by this Contract. Where application of pesticides or herbicides is required, provide the submittals required by the specification and obtain written approval prior to any application. The Contracting Officer will be required to review and approve pesticides or herbicides submittals.

1.5.7 Smoking Policy

In accordance with NAVFAC policy, smoking is prohibited inside all buildings and other facilities except those areas specifically identified

as smoking areas (e.g., smoking shelters). Smoking is not permitted within 20 feet of air intakes, doorways, or windows.

1.5.8 Portal Crane Clearance Zone (PNSY)

Ensure there is no construction debris or materials within the Crane Clearance Zone (i.e., between the painted yellow lines on each side of the rail) unless a rail outage/securement has been approved.

1.6 WORKING HOURS (PNSY)

Regular working hours must consist of a period established by the Contracting Officer between 7 AM and 3:30 PM, Monday through Friday, excluding Government holidays. The regular working hours must be confirmed with the Contracting Officer.

1.6.1 Work Outside Regular Hours (PNSY)

Work outside regular working hours requires Contracting Officer approval. Provide written requests fifteen (15) Calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress and to allow scheduling of full time escorts in the building(s) if required. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer.

As part of the requests, provide the specific dates, hours, location(s), type(s) of work to be performed, Contract number, and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours.

Working on Weekends/Holidays: Any request to work on a weekend must be submitted to the Contracting Officer no later than two (2) working days prior to the requested work weekend/Holiday.

1.7 WORK IN OCCUPIED AND EXISTING BUILDING(S)

Work under this Contract may be located in an occupied building. Move unfixed furniture away from working area(s) as required to perform the work; protect; and replace in original locations upon completion of the work. Leave fixed equipment in place and protect against damage or temporarily disconnect, relocate, protect, and reinstall at completion of work. If determined necessary by the Contracting Officer, the Government will remove and relocate other Government property in the areas of the buildings scheduled to receive work. After providing written notification, allow 15 calendar days for the Government to relocate Government property.

1.8 FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS

1.8.1 FACILITY/UTILITY/SERVICE INTERRUPTIONS

Make utility/service cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays as approved by the Contracting Officer. Conform to procedures required in the paragraph entitled WORK OUTSIDE REGULAR HOURS (PNSY) herein. Anticipated costs must be included in the bid.

Ensure that new utilities/services are complete, except for the connection, before interrupting existing services.

Interruption to Water, Sanitary Sewer, Storm Sewer, Telecommunication Services(Telephone, internet, radio), Electric Service, Air Conditioning, Heating, Fire Alarm, Compressed Air, and other services must be considered service interruptions pursuant to the paragraph entitled WORK OUTSIDE REGULAR HOURS (PNSY) herein. This time limit includes time for deactivation and reactivation. Also, see paragraph 1.8.2 of this Section for Outage Requirements.

Operation of Portsmouth Naval Shipyard Utility/Service: Do not operate nor disturb the setting of control devices in the Portsmouth Naval Shipyard's utilities/service systems, including water, sewer, electrical, air and steam services. The Government will operate the control devices as required for normal conduct of the work.

1.8.2 TEMPORARY OUTAGE PROCESS

a. Outage Requirements

The Outage process includes development of a plan and outage package, vetting and aligning the plan and associated impacts with stakeholders, and final outage submittal and Government approval.

Temporary outages are required when normal system operation is or could be impacted for utility, facility, fire lane, road, and ground level track. This situation can occur when inspecting, repairing, or upgrading, utility systems and equipment, buildings and structures (facilities), and areas on and within 10' adjacent to ground level trackage (that extends below pavement depth). These temporary outages may affect one or several buildings, facilities, production processes, and geographical locations. Although most outage requirements are addressed in this section, the entire specification must be reviewed for additional outage requirements.

Outages involving Critical Shipyard Infrastructure require a specific Risk Analysis and Risk Mitigation Plan as outlined in Paragraph 1.19. The risk mitigation analysis and risk mitigation plan must be submitted to the Contractor Officer's Representative for review and approval prior to the submission of outage request.

Significant outages required to execute the work may be outlined on project Drawings. The Drawings do not include all the outages that will be required to support the execution of all work. If not already provided in the project drawing package, the contractor will coordinate with the PWD ME Design Manager (DM)/Appropriate Technical code, via the ET as liaison, to develop an outage submittal package and a target outage date. The Contractor is responsible for the final outage submittal package.

All Outages packages must be submitted to the Back Shift Rep via the Contracting Officer's Representative for final stakeholder concurrence only after coordination/alignment with stakeholders on scope, impact and target outage date, and at least 15 calendar days in advance of the target outage date. The Final Government Outage concurrence period is 15 calendar days. Any Outage requests denied due to incomplete stakeholder alignment or information required to support the requested outages, will not be justification for a delay claim. The Contractor must review any planned outage with the NAVFAC Construction Management Team/stakeholders prior to submitting the outage request.

The Contracting Officer's Representative will provide the Outage Request Form and the local Outage Instruction.

The Outage Coordination Meeting described in Paragraph 1.8.c is intended to provide a forum to review the outage plan and information required to support any outage reviews.

At a minimum, the outage submittal package must address:

- 1) Specific location/system requiring the outage.
- 2) Duration of outage.
- 3) Scope of work required for the outage.
- 4) Outage work plan, including organization responsible for each step to accomplish the outage.
- 5) Scope of outage impact, identifying specific affected systems and stakeholders.
- 6) Hazardous energy control.
- 7) For Critical Shipyard Infrastructure (see Section 01 14 00.00 22, Paragraph 1.19) a Critical Infrastructure Risk Analysis and Risk Mitigation plan is required.
- 8) Adequate sketches that clearly depict location, situation, and relevant information/interfaces.
- 9) Include LOTO boundaries and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy. These activities include, but are not limited to, a review of HECF and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the outage process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is not feasible.

b. Additional Outage Requirements:

The Three Week Look Ahead schedule must include upcoming outages planned for that three week period. Outages planning must start early in the construction process to allow adequate coordination and alignment with affected stakeholders and to integrate timing with critical shipyard operations.

The Work Sequencing and Preparation Plan must address how outages will be accomplished with an absolute minimum of disruption to Government operations.

Provide an Outage Schedule that indicates all the required Outages that will be required to support the Construction. See Section 01 32 17.00 20.

c. Outage Coordination Meeting:

Outage Coordination Meetings are to be held as necessary, prior to submittal of an outage request package. The meetings are intended to develop a mutual understanding of the administrative and technical scope and requirements needed to support development of the outage request package, and execution of the outage.

This meeting should include applicable personnel from the Prime Contractor, the Prime and Subcontractors to perform the work, the Contracting Officer, PWD-ME DM, Designer of Record, the NAVFAC PWD ME UEM representatives, and applicable stakeholders.

All parties must fully coordinate and be aligned with the activities and dates required to support the outages.

d. Outage Safety Review Meeting

During the Outage Safety Review meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

1.9 CRANE AND RAILROAD TRACKAGE INTERRUPTIONS (PNSY)

Crane and railroad trackage are facilities that are subject to strict scheduling and outage approvals. When Contract work is planned, within 10' adjacent to ground level trackage (that extends below pavement depth) or on ground level track, submit written requests for Outages & Securements a minimum of 15 calendar days prior to the desired date of interruption to the Contracting Officer: Coordination/alignment on the desired date of interruption should be accomplished with the crane department (Code 722) before submitting the written request. (See Contracting Officer Representative for Track Outage & Securement Request Forms.)

A. Track Outage is required for:

Any excavation, that is within 10 feet of a rail that will extend below depth of the pavement.

Any work performed that will penetrate a track foundation.

Any work involving trackage system replacement or repair.

Work must not be conducted in affected areas until a written approval to the request for outage in accordance with the requirements specified in Paragraph 1.8 is received.

B. Track Securement:

Track Securement differs from Track Outage in that it does not affect track certification. Track Securement requests must be prepared and submitted to the Contracting Officer to obtain scheduling approval from Code 740.2 before work that would curtail the use of tracks (e.g., placing equipment or hoses within the crane envelope or within the defined Crane Clearance Zones).

No work must be conducted in affected areas until a written approval to the request for securement is received.

1.10 WORK ADJACENT TO CIA SECURITY FENCING (PNSY)

Work adjacent to Portsmouth Naval Shipyard Controlled Industrial Area (CIA) fencing is strictly controlled to ensure security is maintained at all times.

Work which will breach CIA fencing is prohibited unless approval has been obtained from Head of Security Operations (Code 1122) and a Portsmouth Naval Shipyard Police representative is at the worksite during the period that the fence has been breached.

A minimum of 30 calendar days prior to performing work which requires breaching the CIA security fence, arrange through the Contracting Officer to obtain Head of Security Operations approval and scheduling of the Portsmouth Naval Shipyard Police representative. "Breaching the fence" is any repair, alteration, or other work which would allow access into the CIA either over, under, or through an opening in a CIA fence.

Conditions which breach the fence must be eliminated during all non-work periods to the satisfaction of the Portsmouth Naval Shipyard Police representative. Do not leave the worksite until such conditions are eliminated. All materials used to close openings in fencing and method of installation must be the same type and construction as adjacent, undisturbed CIA fencing.

Except for temporary off-loading of materials, the clear zone (10 ft on the inside & outside of the security fence) adjacent to CIA fencing must remain clear of vehicles, materials, equipment or tools. Contractor personnel must be at the location throughout the entire time of any off-loading or work. Under no circumstances shall vehicles, materials, equipment, or tools be left unattended in the clear zone. Any work within the clear zones requires approval from Portsmouth Naval Shipyard Security. Any requests to work in the clear zone must be provided to the Contracting Officer at least five (5) working days in advance.

If the Contractor violates the security requirements related to the CIA Fencing, Security will issue a Security Deficiency Log (SDL). Upon receiving a SDL, the Contractor must take immediate corrective actions as directed in the SDL including responding to the SDL with documentation of

the corrective actions taken and what actions are being taken to prevent recurrence of any further violations. (Depending the security violation, the Contractor may be required to have a Security Stand Down in which all staff including subcontractor will need to be present for full review of the PNSY Security Requirements.)

If a second offense occurs, the Contractor will have a Security Stand Down in which all staff including subcontractor will need to be present for full review of the PNSY Security Requirements.

If a third offense occurs, the Project Superintendent is subject to removal by the Contracting Officer for non-compliance with security requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to Stand Downs, stop work orders, or removal of the Project Superintendent are acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor. Work adjacent to Portsmouth Naval Shipyard Controlled Industrial Area (CIA) fencing is strictly controlled to ensure security is maintained at all times.

"Breaching the fence" is any repair, alteration, or other work which would allow access into the CIA either over, under, or through an opening in a CIA fence. Work which will breach CIA fencing is prohibited unless approval has been obtained through the outage process from Head of Security Operations (Code 1122), and a Portsmouth Naval Shipyard Police representative is at the worksite during the period that the fence has been breached. The contractor shall follow the outage process for any planned breach in the fence-line, or for the need for Naval Security Force (NSF) to man a CIA perimeter crossing due to contractor work.

Arrange through the Contracting Officer with the Head of Security Operations for scheduling of the CIA fence breach and the presence of a Portsmouth Naval Shipyard Police representative a minimum of 30 calendar days prior to performing work which requires breaching the CIA security fence. Final approval from Head of Security Operations will be through the outage process. Refer to paragraph 1.8 of this section for the outage process.

Conditions which breach the fence must be eliminated during all non-work periods to the satisfaction of the Portsmouth Naval Shipyard Police representative. Do not leave the worksite until such conditions are eliminated. All materials used to close openings in fencing and method of installation must be the same type and construction as adjacent, undisturbed CIA fencing.

Any work (including surveys and inspections) within the 10 ft CIA fence clear zones requires prior approval from Portsmouth Naval Shipyard Security via the outage process. Refer to paragraph 1.8 of this section for the outage process.

The only outage exception is for short term (less than 4 hours) temporary off-loading of materials in the CIA clear zone, which must have prior Shipyard Security approval (code 1122 verbal or email). During temporary off-loading, Contractor personnel must be at the location throughout the entire time. Under no circumstances shall vehicles, materials, equipment, or tools be left unattended in the clear zone.

If the Contractor violates the security requirements related to the CIA Fencing, Security will issue a Security Deficiency Log (SDL). Upon

receiving a SDL, the Contractor must take immediate corrective actions as directed in the SDL including responding to the SDL with documentation of the corrective actions taken and what actions are being taken to prevent recurrence of any further violations. (Depending the security violation, the Contractor may be required to have a Security Stand Down in which all staff including subcontractor will need to be present for full review of the PNSY Security Requirements.)

If a second offense occurs, the Contractor will have a Security Stand Down in which all staff including subcontractor will need to be present for full review of the PNSY Security Requirements.

If a third offense occurs, the Project Superintendent is subject to removal by the Contracting Officer for non-compliance with security requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to Stand Downs, stop work orders, or removal of the Project Superintendent are acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.10.1 CIA Lighting

CIA lighting outages must be approved through Code 1122, and temporary lighting must be provided at the contractor's expense. The lighting required to be provided shall be equal to or greater than the current foot-candle rating of the area. Refer to paragraph 1.8 of this section for the outage process.

1.11 WORK ADJACENT TO AN OVERHEAD CRANE

Provide a minimum vertical clearance of three (3) inches between the highest point of the crane and the lowest overhead obstruction. For buildings where truss sag becomes a factor, increase the clearance as necessary to maintain the minimum required clearance.

The horizontal clearance between the end of the crane and the building columns, knee braces or any other obstructions must not be less than two (2) inches with the crane centered on the runway rails. Pipes, conduits, etc. must not reduce this clearance.

The vertical clearance beneath a bridge crane is to be at least three (3) inches. This clearance is not applicable to the hook block unless it is in its up most position.

For work involving installation, adjustment, or replacement of rail fasteners (e.g. clips/bolts), verify the existing crane envelope and clearance measurements around the rail head prior to beginning work. Notify the Government if the work will reduce the clearance between the bridge crane and rail fasteners and ensure newly installed items will not obstruct bridge crane travel.

Notify the Government to verify that crane clearance has been maintained when the work performed may have changed any physical dimensions of objects or structures adjacent to the crane (e.g., changing or servicing lighting fixtures/pendant assemblies, removal and reinstallation of pipes, conduits, junction boxes, etc.). If the crane is not available (e.g., undergoing maintenance, inspection, etc.), verify crane clearance by taking measurements using reference points (e.g., vertical and horizontal distance

from the top of crane rail with respect to the crane envelope, vertical distance from the floor with respect to the crane envelope, etc.).

Where bridge cranes within buildings exist, the contractor must ensure that all installations, modifications or other work included in this contract does not create interferences that fall within the bridge crane clearance zone. The bridge crane clearance zone is 2" on any side of the crane, and 3" above or below the crane. It is the responsibility of the contractor to verify the accuracy of the building and crane dimensions in the field prior to conducting work.

1.12 FIRE PROTECTION

1.12.1 Compliance (PNSY)

Comply with COE EM 385-1-1, NFPA 241, and Portsmouth Naval Shipyard fire regulations. Obtain approval no later than two (2) working days from the Portsmouth Naval Shipyard Fire Chief via the Contracting Officer prior to commencement of hot work operations.

1.12.2 Fire Lanes

1.12.2.1 Fire Lane Width

Fire lanes shall be a minimum of Twenty (20) feet in width.

1.12.2.2 Vertical Clearance for Fire Lane

Fire lanes shall have a minimum of Thirteen (13) feet Six (6) inches nominal vertical clearance, which shall be provided and maintained over the full width of the fire lane.

1.12.2.3 Angle of Approach/Departure

Minimum inside turn radius shall be Twenty-Five (25) feet and minimum outside turn radius shall be Fifty (50) feet.

1.12.2.4 Waiver Request for Fire Lane size and dimensions

Due to existing infrastructure, some requirements of this section may not be achievable. The Contractor shall submit a Waiver Request to the Fire Department for review. Additional provisions may be required for the approval of the waiver.

1.12.2.5 Parking in or blocking fire lanes

Parking in or blocking fire lanes is expressly prohibited for any reason without prior Fire Department approval. Vehicles left unattended in fire lanes are subject to towing. Contractors will be issued a Non-Compliance Notice for vehicles left unattended in fire lanes.

- a. First Offense: The Contractor will remove the vehicle from the Project site immediately and the vehicle will not be allowed to return to the Project site for Seven (7) Days. The contractor will perform a safety stand down within three (3) days of the incident and the Fire Department shall be invited to attend the safety stand down.
- b. Second Offense: Contractor and Sub-Contractors vehicles will be permanently barred from the Project site and a Non-Compliance Notice

will be issued. Individuals performing deliveries in support of the Project will be allowed but are expected to be on site for the shortest duration required to perform the delivery, and the driver/operator shall remain in the immediate vicinity of the vehicle at all times.

1.12.2.6 Approval to Temporarily Block a Roadway/Fire Lane

The Contractor may request approval to temporarily block a roadway/fire lane as part of a Road Outage to complete work associated with the Project. Approval may only be given by the Fire Chief or his/her assigned representatives. Arrange through the Contracting Officer with the Fire Chief for scheduling the temporary blockage prior to submitting the outage package. Final approval will be through the outage process. Refer to paragraph 1.8 of this section for the outage process. Add the following to the outage package:

- a. Alternative analysis that discusses whether a practical alternative to the outage exists. The analysis must address the activity purpose, need, and why the activity cannot be completed by:
 - (1) Utilizing, managing, or expanding the work area to avoid the fire lane.
 - (2) Reducing the size, scope, configuration or density of the activity as proposed (phasing).
- c. Illustration showing the location of contractor equipment during the proposed outage.
- d. Provisions for emergency access (i.e. a steel plate shall be kept with 10 feet of the excavation).
- e. Additional provisions may be required by the Fire Department on a site by site basis.

1.13 SECURITY REQUIREMENTS

1.13.1 General

Employees and representatives performing work under this Contract are required to be United States citizens. If naturalized, the individual must present his/her naturalization papers to the Security Officer for inspection. Foreign born personnel must present evidence of citizenship regardless of citizenship of parents, as required by immigration laws.

1.13.2 Access to the Portsmouth Naval Shipyard (PNSY)

Contract Clause "FAR 52.204-2, Security Requirements and Alternate II" and the following apply:

Access to areas designated as restricted NAVSEA spaces will require an escort by a Government Representative. Notify the Contracting Officer at least 14 Calendar Days in advance of the date access is required.

Obtain security badges and vehicle passes to enter the Portsmouth Naval Shipyard at the Portsmouth Naval Shipyard's Pass/Security Office. Furnish proof that employees are U.S. citizens to obtain badges to enter the Portsmouth Naval Shipyard.

Complete Department of Homeland Security Form I-9; Employment Eligibility

Verification for each employee and furnish proof that employees are U.S. citizens to obtain badges to enter Portsmouth Naval Shipyard.

1.13.3 Application and Issue of Security Badges

"Temporary" Security Badges will be issued to Contractor personnel requiring access to the shipyard for less than 90 days. Badge(s) will be issued upon satisfactory proof of U.S. citizenship, in the form of an original or certified birth certificate, passport, or naturalization papers. A picture ID is required in addition to satisfactory proof of citizenship. See Attachment B.

"Permanent" (photo) Standard Access Control Badges will be issued to Contractor personnel requiring access to the shipyard for 90 or more days. Contractor personnel will be required to complete an authorization application form for local record check, and a personal information sheet. The forms will be furnished to the Contractor following award of any contract resulting from this solicitation, at time of pre-performance or pre-construction conference."

In the event access is required to Contract work areas not permitted by the level of security badge issued, such need must be demonstrated and an escort obtained.

STANDARD ACCESS CONTROL BADGES MUST BE ATTACHED TO THE OUTER GARMENT AND DISPLAYED ABOVE THE WAISTLINE AT ALL TIMES WHILE ON THE PORTSMOUTH NAVAL SHIPYARD.

PERSONNEL MUST NOT ENTER AREAS FOR WHICH THEY HAVE NOT BEEN CLEARED. WHERE A NEED HAS BEEN DEMONSTRATED TO ENTER SUCH AREAS, PERSONNEL WILL BE UNDER CONSTANT ESCORT BY PERSONNEL WHO HAVE BEEN CLEARED. FAILURE TO ADHERE TO POSTED SECURITY REQUIREMENTS MAY RESULT IN REMOVAL OF THE EMPLOYEE FROM THE PORTSMOUTH NAVAL SHIPYARD WITH FUTURE ACCESS DENIED.

1.13.4 Application and Issue of Vehicle Passes (PNSY)

Vehicle passes will be issued upon satisfactory proof of a valid Operator's License, Vehicle Insurance, and State Vehicle Registration. Temporary passes will be issued for short term or single trip requirements on a case by case basis. All vehicles permitted to enter or park on the Portsmouth Naval Shipyard must comply with the Portsmouth Naval Shipyard's traffic and parking regulations and must only park in assigned areas, which may or may not be in the vicinity of the site of the Contract work. Do not park a vehicle in such a manner that crane tracks, railroad tracks, and vehicle access routes are blocked(including associate clearance zones). Vehicles left unattended which are blocking such access routes are subject to towing and loss of vehicle passes. Parking on the Portsmouth Naval Shipyard may be in excess of one-half mile from the worksite.

1.13.5 Application and Issue of Vehicle Passes for Entry into Portsmouth Naval Shipyard's Controlled Industrial Areas (CIA)

Vehicular access to the CIA must be minimized and all vehicles must comply with the following requirements:

Vehicles must visibly display a CIA vehicle entry pass and inspection pass from the Commercial Vehicle Inspection Station (CVIS), Building 386. CIA passes will only be issued to company owned or leased vehicles, rental vehicles rented in the company name, or privately owned vehicles the

company has certified in writing, to be necessary in the performance of Contracted work. CIA passes will be issued on weekends and holidays at Building 29, from the Watch Supervisor. Personnel not possessing the level security badge required for CIA access must be accompanied by a properly badged escort to obtain the CIA vehicle pass.

Vehicles will only be allowed in the CIA for the transportation of tools, parts, and materials to and from the worksite. An exception to this policy, employees may be transported to and from the worksite after full vehicle inspection at CVIS if a specific security plan has been developed and approved by the Portsmouth Naval Shipyard Security Officer.

Parking of privately owned or company owned vehicles not utilized in the execution of construction activities (e.g. not a working truck or specialty vehicle) within the CIA or in any project laydown space is prohibited.

For Vehicles remaining in the CIA for more than 24 hours: Obtain a CIA Overnight Parking Permit from the PWD ME Construction ET. Vehicles without this pass are subject to being towed. Passes will only be issued to those demonstrating a valid reason as determined by Portsmouth Naval Shipyard Security.

1.13.6 Application and Issue of Crane Passes (PNSY)

Comply with EM 385-1-1. For Crane Passes at the Portsmouth Naval Shipyard to be valid, the Certificate of Compliance (FORM 16-1) must be completed and accepted.

1.13.7 Return of Badges and Vehicle Passes

Ensure all vehicle access permits and personnel badges are returned to the Portsmouth Naval Shipyard Security Officer when the need has ended. Account in writing for each missing pass or badge prior to final payment being made on the Contract. Any vehicle passes issued must be returned to Portsmouth Naval Shipyard Security daily.

1.13.8 Security Responsibilities (PNSY)

Employees must not transport, drink, or have in their possession any alcoholic beverages. Possession of any controlled substances without a physician's prescription is also prohibited. Any employee appearing to be under the influence of intoxicating liquor or narcotics will be apprehended by Portsmouth Naval Shipyard Police, escorted off of the Portsmouth Naval Shipyard, and turned over to the local Police Department.

Any vehicle found to contain controlled substances, including usable residue, may be seized and impounded. Within 24 hours of the work day following any vehicle seizure, the Portsmouth Naval Shipyard Police will have determined whether forfeiture of the vehicle is required. If not, the vehicle will be returned to the owner or authorized agent. If the vehicle is determined to be appropriate for forfeiture, the Portsmouth Naval Shipyard's Legal Officer will notify the Drug Enforcement Administration of such seizure and impoundment, for initiation of forfeiture proceedings pursuant to Title 21, U.S. Code, Section 881. Such actions may be taken regardless of whether the owner/operator of the vehicle had knowledge of the presence of drugs in the vehicle. The Government may pursue criminal or other disciplinary actions pursuant to Title 18, U.S. Code, Section 1382.

Possession of firearms, ammunition, and/or explosives is prohibited. In

the event explosives are required for construction work, specific handling requirements and approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Cameras, video equipment, or similar photographic equipment must not be introduced into nor removed from the Portsmouth Naval Shipyard. In the event such equipment is required for performance of Contract work, approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Weapons (firearms, personal knives with blades 2-1/2 inch long or greater, Mace, Pepper Spray etc.) are not permitted aboard the Portsmouth Naval Shipyard.

The following are the penalties for any personnel that are found to be in possession of a camera or cell phone equipped with a camera taking photographs inside & outside NAVSEA spaces:

Camera/Camera Equipped Cell Phone Outside a NAVSEA Space:

The following are the penalties for any Contractor personnel or Subcontractor personnel that are found to be in possession of a camera or cell phone equipped with a camera taking pictures outside of NAVSEA spaces:

1st offense with pictures: Full investigation with security forces and NCIS, camera or cell phone confiscation, and depending on results of investigation 30 days restriction.

2nd Offense with pictures: The offender will be permanently barred from access to NAVSEA spaces.

Camera/Camera Equipped Cell Phone Inside a NAVSEA Space:

Cameras, or cell phones equipped with cameras, are not allowed in any NAVSEA space such as the CIA, NWA, CNIA or Security Island. If a Contractor is found with a camera equipped cell phone in a NAVSEA space, the Standard Access Control Badge (SACB) and Defense Biometric Identification System card (DBIDS) will be confiscated. Access to the installation will be denied and a Security Deficiency Log (SDL) will be issued.

The following are the penalties for any Contractor personnel or Subcontractor personnel that are found to be in possession of a camera or cell phone with & without photos of NAVSEA Spaces:

Camera/Cell phones with no pictures of NAVSEA spaces:

1st Offense: SDL will be written to document with a 10-day deadline for completion. Credentials will be reissued to the offender once the SDL response is received from the contracting company and reviewed/accepted by the Security office.

2nd Offense: The offender will be permanently barred from access to NAVSEA spaces.

Camera/Cell phones with pictures of NAVSEA spaces:

1st offense: The offender will be permanently barred from access to NAVSEA

spaces.

At a minimum, all Superintendents, QC Manager and SSHO (and anyone acting in this capacity) must have a CIA compliant mobile phone on their persons any time while work is occurring at the Contract jobsite(s). The phone number(s) associated with these phones must be provided during the pre-con meeting and work will not be allowed to proceed without one.

Laptops and any other computer, regardless of form factor, must not be introduced nor removed from the Portsmouth Naval Shipyard without authorization. Only business use computers are allowed. Personally owned computers are not authorized on the Portsmouth Naval Shipyard. If a computer is required to perform work on the Portsmouth Naval Shipyard, a Visitors Portable Computer Equipment Registration form must be completed and submitted for approval. These forms can be obtained and completed at the pass office upon entrance to the Portsmouth Naval Shipyard or through the Portsmouth Naval Shipyard POC.

Any computing devices that also have or had an installed camera, must follow the same requirements as cell phones. If the computing device is visiting a NAVSEA space, it cannot have a camera. If the device has a camera, the only acceptable remediation is that the camera be professionally removed or drilled out and the resulting void must be filled with an opaque epoxy.

In the event that a camera is required for the performance of the Contracted work, approval must be obtained from the C1120 Security Office. In the event of any deviation from the requirement to remove and epoxy the camera, authorization must be sought from C1120. All requests for camera use and deviations must come to C1120 through the Contracting Officer's Representative or Portsmouth Naval Shipyard Point of Contact (POC). Authorizations and approved computer registrations must remain with the devices and be presentable upon request. Devices without a documented approval or observed being used in a manner that violates the terms of the Visitor Portable Computer Equipment Registration agreement are subject to search and seizure.

Basic guidance for use of Portable Electronic Devices (PED) at Portsmouth Naval Shipyard: Obtain Contracting Officer approval of their intent to use, of any portable electronic device which has the capability to record, copy, store, export and transmit data, photography, digital images, video or audible information, inside the CIA or any NAVSEA controlled spaces. Provide the manufacturer's data describing the electronic capabilities of the device(s).

Use of wireless networking and communication on the Portsmouth Naval Shipyard requires prior authorization. Communication devices such as cellular hotspots and wireless routers must also be registered using the same Visitors Portable Computer Equipment Registration. This does not include air-cards and straight cellular data. It does include any device that transmits or broadcasts Wi-Fi, including cellular phones with the personal hot-spot feature enabled.

Broadband internet connections must be authorized prior to use on the Portsmouth Naval Shipyard. All requests for physical cable service from an internet service provider (ISP) must be authorized and coordinated through C109. Work through the COR to request broadband access.

Driver use of a hand-held cellular phone in a moving vehicle on the

Portsmouth Naval Shipyard is prohibited. This prohibition does not include hands-free cellular phone devices. Hands-free devices include console/dash-mounted or otherwise secured cellular phones with integrated features such as voice-activation, speed dial, speakerphone or other similar technology for sending and receiving calls.

Driver use of any portable, personal listening device worn inside the aural canal, around or covering the driver's ear while operating a motor vehicle, is prohibited. Listening devices include wired or wireless earphones and headphones (including blue tooth or similar technology), and do not include hearing aids or devices designed and required for hearing protection.

The use of radar or laser detection devices to indicate the presence of speed recording instruments or to transmit simulated erroneous speeds is prohibited in accordance with OPNAVINST 5100.2H.

Indoctrinate personnel on access limitations to ensure security control is maintained as an integral part of their work pattern and habit.

Unescorted personnel found in security areas requiring a higher level clearance than the level represented by the badge displayed will be removed from the area with possible confiscation of security badges and vehicle passes.

1.13.9 Access to Unclassified Information

Access to U.S. Navy technical information manuals, documents, drawings, plans, specifications, and other information (e.g., photos, presentations, renderings, papers, etc.) is Government property and restricted to an official need-to-know basis. Handle, control, and safeguard to prevent oral, visual, and documentary disclosure to the public, to foreign sources, and to personnel not having an official need-to-know.

1.13.10 Disclosure of Information

- a. Do not release to anyone outside the Contractor's organization any unclassified information, regardless of medium (e.g., hard copies, computer files, film, tape, document), pertaining to any part of this Contract or any program related to this Contract, unless the Contracting Officer has given prior written approval.
- b. Requests for approval under paragraph (a) must identify the specific information to be released, the medium to be used, and the purpose for the release. Submit the request to the Contracting Officer at least 10 business days before the proposed date for release.
- c. The Contractor agrees to include a similar requirement, including this paragraph (c), in each subcontract under this Contract. Subcontractors must submit requests for authorization to release through the Prime Contractor to the Contracting Officer.

1.13.11 Portsmouth Naval Shipyard Operations Security Statement

During the course of this Contract, in addition to those restrictions, instructions, and guidelines delineated in the Contract Statement of Work, Contract Data Requirements List (CDRL), and/or other references provided, adhere to the following minimum requirements in support of the Portsmouth Naval Shipyard (PNS) OPSEC Program:

Introduction of personal electronic devices, laptops, tablet PCs, cellular phones, cameras, recording devices, and data recording/storage devices into Government spaces is STRICTLY controlled and forbidden in most cases. Company issued equipment required for the performance of work must be approved by the PNS Code 1120 Security Office. Photography and recording is not allowed except for official use and by permit only. Photographs will be reviewed by Code 1123.2 to ensure sensitive information is not revealed.

Do not discuss Government operations in public or over unprotected or unencrypted communications. Do not post to company websites, publications, newsletters or other media, any images, data, or information that reveal sensitive Government operations, personnel, equipment, and/or classified or controlled unclassified information. When in doubt, company press releases related to this Contract must be coordinated through the Contracting Officer's Representative (COR) or Technical Point of Contact, as applicable, and in conjunction with PNS Public Affairs Office (PAO).

Due to observation of events, operations, physical changes, etc. which may reveal National Security Information or Naval Nuclear Propulsion Information (NNPI), specific restrictions are needed to preclude unintentional release of this information to unauthorized parties. (Unauthorized disclosure and transfer of National Security Information is punishable under 18 USC § 793.) Therefore, do not disclose to unauthorized third parties or post to unofficial sites (including Social Networking sites) any images, data or information, or observed events that reveal sensitive Government operations, personnel, equipment, including, but not limited to:

Tactics, techniques and procedures, production or work schedules, any visible or concealed modifications, upgrades, additions to vessels, weapons or equipment; increases, changes, or decreases in work/deployment frequency or Government personnel, vehicle, vessel movements; specialized equipment orders, deliveries, shipments, etc. Unauthorized disclosures and attempts to solicit this type of information by unauthorized third parties or others not affiliated with this Contract must be reported to PNS Code 1120 Security and/or PNS Police Department. Non-Disclosure requirements remain in effect during the duration of this Contract and indefinitely thereafter.

Government issued badges and identification must be removed and/or concealed from plain sight when off PNS and must not be left in vehicles or unprotected. Badges and passes may not be duplicated, copied, or loaned to others. Lost or stolen identification badges, vehicle passes, etc., must be immediately reported to PNS Code 1120 Security Office and/or PNS Police Department.

It is strongly recommended the Contractor mark and protect related internal production schedules, deliverables, inventories and shortages, and identified vulnerabilities related to production of Government material. Internal company markings (e.g., Business Sensitive, etc.) are appropriate for identifying the aforementioned as sensitive information. All Government information must be destroyed at Contract termination or returned to the Government at the Government's discretion.

1.14 MARINE ACTIVITIES

- a. Coordinate all marine vessel movements with the Contracting Officer's Representative and the Portsmouth Naval Shipyard's Port Operations Department. Submit a weekly updated schedule showing proposed docking

locations and vessel movements to the Contracting Officer's Representative. Meet with the Contracting Officer's Representative and Portsmouth Naval Shipyard Port Operations Representative weekly to review the vessel schedule.

- b. Any waterborne craft or vessel movements which will be adjacent to any naval vessels must be made under the direction of the Portsmouth Naval Shipyard's Pilot. Notify the Contracting Officer's Representative at least 15 calendar days in advance of any movements that will require the Portsmouth Naval Shipyard Pilot.
- c. All Contract activities must maintain a minimum standoff from submarines of 100 feet for in-water work and 50 feet for landside work. All waterborne craft must, at all times, maintain a minimum of twenty-five (25) feet clearance to any Government Barge in the vicinity of the work. This applies to Subcontractors and materials suppliers as well as to the Prime Contractor. This minimum clearance must also take into account any materials or equipment present on the craft that could reduce this effective clearance distance. This restriction is in effect at all times 24/7 including overnight hours and weekends.
- d. All marine activities must be completed to a manner that ensures the stability of caissons, piers, berths, bulkheads, fender systems, mooring hardware, and other structures adjacent to the work site. Repair any damage caused by operations or vessels required to complete the work.
- e. When not in use, vessels must be tied up at a location approved by the Contracting Officer's Representative and the Portsmouth Naval Shipyard's Port Operations Department.
- f. Any waterborne craft which is deemed to be unsafe by the Contracting Officer's Representative will be prohibited from working at the Portsmouth Naval Shipyard. Copies of all inspections and certificates must be submitted to the Contracting Officer's Representative for approval prior to bringing any vessel to the Portsmouth Naval Shipyard.

1.15 WORK WITHIN PUMPWELLS AND DRY DOCKS (PNSY)

All work completed within the dry docks and pumpwells must have a Flooding Mitigation Strategy Plan. The process for development of this plan must start with a meeting to determine any locations of possible flooding. The plan must cover all penetrations through structures which may be sources of flooding of the pumpwell, the dry dock, or an associated structure. Flooding sources considered include the Piscataqua River as well as surface rain water. The plan must identify the source of flooding, provide a strategy for mitigating flooding from that source and details on when that mitigation will be in place (e.g. pipe that is currently open to the river will have double valve protection when not actively being worked). Provide product data sheets with the plan. The plan must also include an inspection program to verify proper installation and maintenance of all mitigation strategies. The plan must be submitted and approved by the Government prior to commencing the work within the pumpwells and dry docks.

1.16 CONSTRUCTION VEHICLES

Do not utilize any vehicle that will exceed an HS20 wheel load. The use of "off road", "Utility", "ATV", or any vehicles which cannot be legally

operated on State roadways or highways is prohibited.

1.17 EMPLOYEE PARKING

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. Contractor's employees & Subcontractor's employees must park privately owned vehicles in the Jamaica Island Parking Lot only. Parking is not allowed in any other parking lot or area on the Portsmouth Naval Shipyard. Employee parking must not interfere with existing and established parking requirements of the Portsmouth Naval Shipyard. The Contractor & subcontractors must not violate any Town of Kittery Parking regulations. If a privately owned vehicle is found parked outside the Jamaica Island parking lot or found violating any Shipyard or Town of Kittery Parking Regulations, the vehicle will be subject to towing, a non-compliance notice will be issued and may reflect poorly on the Prime Contractor's performance rating (CPARS) and noted in their final performance evaluation. It should be noted that there may not be sufficient on-shipyard parking available for all contractor & subcontractor employees. The Contractor may need to make arrangement for off shipyard parking and shuttling employees on to the Shipyard. Limited employee parking will not be a basis for a claim for additional time or costs. The Contractor must provide an employee parking plan that must include the estimated amount employee parking required. See Section 01 50 00.00 22. There is a Maine DOT Park & Ride located in Kittery (<https://www.kitteryme.gov/home/news/maine-department-transportation-launches-new-park-and-ride-kitte>)

1.18 PORTSMOUTH NAVAL SHIPYARD 6010 TRAINING

Personnel working inside of or within 10 feet of the coping of the dry dock and its associated structures must have Portsmouth Naval Shipyard 6010 training. Associated structures include pumpwells, service tunnels, altar pits, etc. This training will be provided, at the Portsmouth Naval Shipyard, at no cost and is approximately 4 hours in duration. Coordinate training schedule with the Contracting Officer.

Personnel that complete 6010 training can serve as an escort for 6010 purposes. The escort must be within 10 feet or audio range, whichever is closer, and is responsible for the safety of the personnel they are escorting in an emergency situation.

1.19 CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS

All construction activities (including excavation adjacent to Critical Infrastructure) and outages directly impacting or within three feet of shipyard critical infrastructure are required to have a Critical Infrastructure Risk Analysis and Risk Mitigation plan (CI Risk Mitigation Plan). The Contract Documents identify the Critical Infrastructure that could be impacted by the work associated with this project.

The CI Risk Mitigation Plan (See Example below) is the Contractor's responsibility. General information related to Project Specific Outage Phasing & Sequencing may be provided in the Contract Documents. See Paragraph 1.8 for Outage Requirements.

The Contractor's proposed means and methods related to any work that may impact shipyard critical infrastructure are subject to Government approval. The anticipated phasing & sequencing for completing this work may be outlined in the Drawings. Any additional work and costs associated with any changes to the phasing & sequencing based on the Contractor's means &

methods are the responsibility of the Contractor and will not be a basis for a claim for additional cost and time.

The Contracting Officer will require approval of each CI Risk Mitigation Plan prior to work, and when associated with an outage, it must be submitted along with the Outage Request.

DEFINITION:

- a. "Critical Infrastructure" are shipyard facilities (utilities and structures) that directly affect personnel safety, radiological controls, nuclear and reactor safety, and ship safety and require unhindered reliable operation. This definition is intended to be used in a conservative manner when determining if work impacts Critical Infrastructure, and must be reviewed by the Contracting Officer representatives through appropriate Shipyard personnel in those cases when there is any question as to the applicability.
 - i. Critical structures (and their foundation systems) include: dry dock structures, pump well structures, caissons as well as caisson seat and apron, submarine piers, reactor servicing complexes, crane and rail road rails, and bridges 1 and 2. Also, Radiological Material Storage facilities and Emergency Control Center. All work within three feet of a dry dock pump well and dry dock (when a submarine is docked) must follow critical shipyard infrastructure requirements.
 - ii. Critical utilities/services support dry docks and piers with submarines, reactor servicing complexes, Radiological Material Storage facilities, and Emergency Control Center including: dry dock pumping systems (drainage, dewatering, and ASW- both temporary and permanent systems) high voltage electrical systems and low voltage electrical systems, communication systems, high pressure air systems, low pressure air (breathing air), gas lines to the power plant, fresh water, firefighting water supply, and high pressure steam.

CRITICAL INFRASTRUCTURE RISK ANALYSIS (RA) AND RISK MITIGATION PLAN (RMP)
(Example additional information may be required to address the RA & RMP requirements)

1. Work: (describe in enough detail to allow someone not involved to have a good general understanding of scope):
2. Risks to Critical Infrastructure:
 - A. Describe what could go wrong with the work and the Impact to shipyard operations (Air, Water, Electrical Power, Telecom/IT, Rail or Structures).
 - i. Risk Assessment: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact. See RISK DETERMINATION TABLE - in the Outage Form.)
 - ii. Mitigated Risk Assessment: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below.
 - B. What could go wrong #2:

- i. Risk Assessment #2: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact)
- ii. Mitigated Risk Assessment #2: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below. State why or how the risk is reduced.

3. Work Control Plan:

- A. Describe specific steps and actions (how, when, and by who) to mitigate the risks to critical infrastructure that were identified above. (examples only shown below)

- i. Specific utilities/services in the area will be secured prior to work, or, potholing/GPR or other utility/service locating has been performed to identify specific utility/service location and depth, which is included in Preparatory Phase meeting agenda.
- ii. Protective timber matting installed over crane and train rail to prevent damage to rails supporting train and crane movements.
- iii. Noise/Dust/Vibration. Managed per contract requirements, and to be briefed during Initial Phase meeting.
- iv. Detailed work plan.
- v. Specific selection of equipment/tools for work.
- vi. Foreman/safety pre-job reviews.
- vii. Foreman pre-job work brief with crew.

4. Oversight: Describe specific oversight work leading up to and for the execution of work (how, when, and by who) to mitigate the risks to critical infrastructure that were identified above. (examples only shown below)

- A. Daily Contractor Production Reports.
- B. Daily Contractor Quality Control (QC) Reports.
- C. QC Manager field visits (include periodicity).
- D. Periodic project Coordination Meeting (include parties involved).

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

ARE YOU A CONTRACTOR, VENDOR, OR SUPPLIER WHO NEEDS BASE ACCESS?

IF YES, please read below:

Effective 14 August 2017, NCACS credentials will no longer be accepted for base access. Please read below:

- If you applied for an NCACS credential **PRIOR to 17 April 2017**, your NCACS credential will remain in effect for base access through 14 August 2017, when NCACS credentials will no longer be accepted.
- If you applied for an NCACS credential **BETWEEN 17 April and 31 May 2017** you will also be required to obtain a DBIDS credential in order to obtain base access.
- **AFTER 31 May** no new NCACS applications will be accepted. All new contractors, vendors and/or suppliers requesting base access will be required to obtain a DBIDS credential.
- **AFTER 14 August 2017 only DBIDS credentials will be accepted for base access.**
- There is no cost to obtaining a DBIDS credential.

FOR MORE INFORMATION VISIT:

<https://www.cnic.navy.mil/om/dbids.html>

DBIDS ACCESS REQUEST FORM (NAVSHIPYD PTSMH 5500)

SUBMISSION INFORMATION

1. Access Requested For(Full Name):

2. Title:

3. Phone Number :

4. COMPANY NAME :

DBIDS INFORMATION

Is the company requesting access to the installation the primary Contractor Yes ☐ No ☐

If no, who do they subcontract for:

(IF REQUEST IS FOR MORE THAN ONE PERSON PLEASE ATTACH EMPLOYEE LIST)

COMPANY REQUESTING ACCESS INFORMATION

Company Name (Corporate Legal name):

Company Address :

Company Phone Number(Corporate Office):

Company Local POC:

Email and Phone:

Are Any Personnel Foreign Nationals?

Yes ☐ No ☐ (If Yes provide separate list of Names)

For access to: **PORTSMOUTH NAVAL SHIPYARD**

Timeframe of Access Required: ☐ 0600-1800 ☐ 1400-2300 ☐ 24 Hours

☐ All Week ☐ M-F only ☐ Weekends Only

SPONSOR INFORMATION (MUST BE SHIPYARD PERSONNEL)

Sponsor Name:

Sponsor Organization: **NAVFAC**

Sponsor Title:

Sponsor Phone Number:

Sponsor Email Address:

Contract or project number:

Contract or Project ACCESS EXPIRATION Date:

Only ONE request for Company access should be submitted to; PNsDBIDS@navy.mil (all requests for employees are on the 5512 and are carried by the employee to pass and ID once the company is approved)

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installation.

(2) Persons requesting access who are not in possession of an approved, government issued credential listed in paragraph 1204.b. shall provide an unexpired document for identity proofing purposes listed in Table 12-1. Any fraudulent information passed during the process may lead to prosecution under appropriate legal authorities.

(3) Authorized government representatives shall, prior to acceptance, visually and tactilely (by touch or feel) inspect documents for evidence of tampering, alteration, or other indications of falsified/fraudulent documents. Authorized government representatives will not accept documents that appear to be fraudulent, forged, or counterfeit and follow the CO's directed standards for response actions to include detention of persons attempting to provide fraudulent documents. Indications of tampered documents include:

(a) Strange text, fonts, slightly altered text, incomplete letters, misaligned words, strange spacing and errors in punctuation or spelling.

(b) Texture or physical indication the photograph has been glued over the original.

Table 12-1. List of Identity Proofing Documents

Documents that Establish Identity
1. U.S. Passport or Passport Card
2. Permanent Resident Card or Alien Registration Receipt Card (Form I-551)
3. Foreign passport that contains a temporary I-551 stamp or temporary I-551 printed notation on a machine-readable immigrant visa (MRIV)
4. Employment Authorization Document (Card) that contains a photograph (Form I-766)
5. In the case of a nonimmigrant alien authorized to work for a specific employer incident to status: (a) Foreign passport; and (b) Form I-94 or Form I-94A has the following: (1) Bearing the same name as the passport; and (2) An endorsement of the alien's nonimmigrant status, as long as the period of endorsement has not yet expired and the proposed employment is not in conflict with any restrictions or limitations identified on the form.
6. Driver's license or ID card issued by a RealID Act compliant state or outlying possession of the U.S., provided it contains

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a photograph and biographic information such as name, date of birth, gender, height, eye color, and address. Licenses or IDs possessing "NOT APPLICABLE FOR FEDERAL PURPOSES" will not be accepted.
7. State-issued Enhanced Driver's licenses
8. Driver's license issued by the U.S. Department of State
9. Border Crossing Card (From DSP-150)
10. Identification card issued by Federal, State, or local government agencies, provided it contains a photograph and biographic information such as name, date of birth, gender, height, eye color, and address.
11. Veteran Health Identification Card (VHIC) issued by the Department of Veterans Affairs
12. Department of Homeland Security "Trusted Traveler" Cards (Global Entry, NEXUS, SENTRI, FAST)
13. U.S. Certificate of Naturalization or Certificate of Citizenship (Form N-550)
14. School identification card with a photograph
15. Persons under the age of 18 who are unable to present a document listed above may present one of the below documents. (a) School record or report card (b) Day care or nursery school record (c) Birth certificate (original or certified copy)
16. Native American Tribal Photo ID cards
17. U.S. Coast Guard Merchant Mariner Credential (MMC) or Merchant Mariner's Documents (MMD)
18. Other documents that may be provide for identity proofing purposes, but must be accompanied by a second form of ID with photograph and biographical information. a. Social Security Number card b. Original or certified copy of a birth certificate issued by a state, county, municipal authority, or outlying possession of the U.S. bearing an official seal. c. Certification of birth Abroad issued by the U.S. Department of State (Form FS-545) d. Certification of Report of Birth issued by the U.S. Department of State (Form DS-1350) e. Voter's Registration Card

d. Outside the Continental United States (OCONUS)

(1) NSF will utilize appropriate identity proofing credentials such as a passport or nationally issued identity card, or other COCOM approved ID.

(2) ISO will codify approved non-U.S. identity proofing

DEPARTMENT OF THE NAVY LOCAL POPULATION ID CARD/BASE ACCESS PASS REGISTRATION

PRIVACY ACT STATEMENT:

AUTHORITY: 10 U.S.C. 5013, Secretary of the Navy; 10 U.S.C. 5041, Headquarters, Marine Corps; OPNAVINST 5530.14E, Navy Physical Security; Marine Corps Order 5530.14A, Marine Corps Physical Security Program Manual; and E. O. 9397 (SSN), as amended, SORN **NM05512-2**.

PURPOSE(S): To control physical access to Department of Defense (DoD), Department of the Navy (DON) or U.S. Marine Corps Installations/Units controlled information, installations, facilities, or areas over which DoD, DON, or U.S. Marine Corps has security responsibilities by identifying or verifying an individual through the use of biometric databases and associated data processing/information services for designated populations for purposes of protecting U.S./Coalition/allied government/national security areas of responsibility and information; to issue badges, replace lost badges, and retrieve passes upon separation; to maintain visitor statistics; collect information to adjudicate access to facility; and track the entry/exit times of personnel.

ROUTINE USE(S): To designated contractors, Federal agencies, and foreign governments for the purpose of granting Navy officials access to their facility.

DISCLOSURE: Providing registration information is voluntary. Failure to provide requested information may result in denial of access to benefits, privileges, and DoD installations, facilities and buildings.

IDENTITY PROOFING AND APPLICANT INFORMATION

1. LAST NAME:		2. FIRST NAME:		3. MIDDLE NAME:		4. NAME SUFFIX: ☐ Jr. ☐ Sr. ☐ I ☐ II ☐ III ☐ IV	
5. HISPANIC OR LATINO (Check one): ☐ YES ☐ NO		6. RACE (Check one or more): ☐ WHITE ☐ AFRICAN AMERICAN OR BLACK ☐ ASIAN ☐ AMERICAN INDIAN OR ALASKAN NATIVE ☐ NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER					
7. GENDER (Check one): ☐ MALE ☐ FEMALE		8. DATE OF BIRTH:		9. CITY OF BIRTH:		10. STATE OF BIRTH:	
11. BIRTH COUNTRY:		12. US CITIZEN (Check): ☐ YES ☐ NO					
13. DUAL CITIZENSHIP: ☐ YES ☐ NO CITIZENSHIP IF OTHER THAN US (Country):							

U.S. Citizen Minimum Documentation Required:

By Birth - Social Security No and/or State ID/Drivers License.

Naturalized - Certification Number, Petition Number, Date, Place and Court, United States passport number, Social Security No and/or State ID/Drivers License.

Derived - Parent's certification number, Social Security No and/or State ID/Drivers License.

Alien Minimum Documentation Required:

Registration Number, Expiration date, Date of entry, Port of entry.

14. IDENTITY SOURCE DOCUMENTS PRESENTED:	15. DOCUMENT NUMBER:	16. ISSUED BY STATE/COURT:	17. ISSUED BY COUNTRY:	18. ISSUED:	19. EXPIRES:
☐ Social Security No.			United States		
☐ State ID/Drivers License			United States		
☐ Passport No.					
☐ Certification Number and Petition Number					
☐ Derived - Parent's Certification Number:			United States		
☐ Alien Registration No.			United States		
		Date of Entry:		Port of Entry:	

OTHER APPROVED IDENTITY SOURCE DOCUMENTS:

☐					
☐					

20. WEIGHT (Pounds):	21. HEIGHT (Inches):	22. HAIR COLOR (Check one): ☐ Blond ☐ Brown ☐ Black ☐ Gray ☐ Red ☐ White ☐ Silver ☐ Auburn ☐ Bald	23. EYE COLOR (Check one): ☐ Brown ☐ Green ☐ Blue ☐ Hazel ☐ Black ☐ Gray ☐ Violet ☐ Unknown
24. HOME ADDRESS (Include city, state, zip code):			HOME PHONE (Include Area Code):
25. BASE SPONSOR'S NAME:			SPONSOR PHONE (Include Area Code):

EMPLOYMENT ACTIVITY INFORMATION

26. EMPLOYER NAME AND ADDRESS (Include city/state/zip code):	EMPLOYER PHONE (Include Area Code):
27. SUPERVISOR NAME AND ADDRESS (Include city/state/zip code):	SUPERVISOR PHONE (Include Area Code):

28. Check the applicable box for WORK HOURS box or check the OTHER box and enter the work hours, then check the applicable for WORK DAYS:

WORK HOURS: ☐ 0600-1800 ☐ 0800-1700 ☐ OTHER _____ WORK DAYS: ☐ SN ☐ M ☐ T ☐ W ☐ TH ☐ F ☐ ST

PRIOR FELONY CONVICTIONS

29. Have you ever been convicted of a Felony? ☐ YES ☐ NO _____ Initial

REQUIREMENT TO RETURN LOCAL POPULATION ID CARD

30. I understand that I am required to return my Local Population Identification Card to the Base Pass Office when it expires or if my employment is terminated for any reason. _____ (initial)

AUTHORIZATION AND RELEASE AND CERTIFICATION

31. I hereby authorize the DOD/DON and other authorized Federal agencies to obtain any information required from the Federal government and/or state agencies, including but not limited to, the Federal Bureau of Investigation (FBI), the Defense Security Service (DSS), the U.S. Department of Homeland Security (DHS).

I have been notified of DON right to perform minimal vetting and fitness determination as a condition of access to DON installation/facilities. I understand that I may request a record identifier; the source of the record and that I may obtain records from the State Law Enforcement Office as may be available to me under the law. I also understand that this information will be treated as privileged and confidential information.

I release any individual, including records custodians, any component of the U.S. Government or the individual State Criminal History Repository supplying information, from all liability for damages that may result on account of compliance, or any attempts to comply with this authorization. This release is binding, now and in the future, on my heirs, assigns, associates, and personal representative(s) of any nature. Copies of this authorization that show my signature are as valid as the original release signed by me.

FALSE STATEMENTS ARE PUNISHABLE BY LAW AND COULD RESULT IN FINES AND/OR IMPRISONMENT UP TO FIVE YEARS.

BEFORE SIGNING THIS FORM, REVIEW IT CAREFULLY TO MAKE SURE YOU HAVE ANSWERED ALL QUESTIONS FULLY AND CORRECTLY.

I DECLARE UNDER PENALTY OF PERJURY THAT THE STATEMENTS MADE BY ME ON THIS FORM ARE TRUE, COMPLETE AND CORRECT

DATE _____ SIGNATURE _____

FINAL DETERMINATION ON YOUR ACCESS: The Base Commanding Officer has final authority for determination on granting physical access to DON controlled installations/facilities under his/her jurisdiction.

BELOW COMPLETED BY BASE REGISTRAR PERSON CONDUCTING IDENTITY PROOFING and NCIC CHECK

32. INFORMATION VERIFIED BY: _____ 33. ENTERED IN C/S SYSTEM BY: _____ 34. PASS ISSUE DATE: _____ 35. PASS EXPIRATION DATE: _____

36. NCIC CHECK PERFORMED BY: _____	37. RESULTS OF NCIC CHECK:	38. RESULTS OF LOCAL RECORDS CHECK:
	☐ NO RECORDS ☐ RECORD IDENTIFIER RECORD NUMBER: _____	☐ NO RECORDS ☐ RECORD IDENTIFIER RECORD NUMBER: _____

Office of Under Secretary of Defense Directive-Type Memorandum (DTM) 09-012, "Interim Policy Guidance for DoD Physical Access Control," December 8, 2009. DTM 09-012 requires that DoD installation government representatives query the National Crime Information Center (NCIC) and Terrorist Screening Database to vet the claimed identity and to determine the fitness of non-federal government and non-DoD-issued card holders (i.e. visitors) who are requesting unescorted access to a DoD installation. The minimum criteria to determine the fitness of a visitor is: 1) not on a terrorist watch list; 2) not on an DoD installation debarment list; and 3) not on a FBI National Criminal Information Center (NCIC) felony wants and warrants list. Additionally, SECNAV Memo, Policy for Sex Offender Tracking and Assignment and Access Restrictions within the Department of the Navy, of 7 Oct 08 and OPNAVINST 1752.3 established the Navy's policy on sex offenders, requiring Region Commanders (REGCOMs) and Installation Commanding Officers (COs) to prohibit sex offender access to DoN facilities and Navy owned, leased or PPV housing. This form describes the authority and purpose to collect and share the required information; and identifies the applicant/visitor and sponsor; and authorizes the DoD to perform the minimum vetting and fitness determination criteria. A favorable response on the vetting and fitness determination is required to receive access to DOD-controlled installation/facilities.

Instruction for completing the Local Population Access Registration Form

INSTRUCTIONS: Please complete all information in black ink (printed) or by typing. By voluntarily providing your Personal Information, you agree to the following terms and restrictions:

RESTRICTIONS: Local Population Identification Card/Base Access Pass may only be used by person to whom they are issued and for the specific business/purpose issued. Applicants are reminded that soliciting (i.e., door-to-door sales) is prohibited on the base, and that such activity is grounds for cancellation of the Pass. Additionally, such action may result in debarment from the base and legal action. The Base Commanding Officer has discretion over specifying the period of validity for any Local Population ID Cards/Base Access Passes that are issued under his/her jurisdiction. Review the Privacy At Statement that is printed at the top of the form

Block 1: Enter the Last Name. Block 2: Enter the First Name. Block 3: Enter the Middle Name. Block 4: If applicable, check the box for Name Suffix. Block 5: Check the applicable box for Hispanic or Latino. Block 6: Check the applicable box for Race. Block 7: Check the applicable box for Gender. Block 8: Enter Date of Birth. Block 9: Enter City of Birth. Block 10: Enter State of Birth. Block 11: Enter Country of Birth. Block 12: Check the applicable box for US Citizenship. Block 13: If not a US Citizen, enter the name of the Country of Citizenship. Block 14: Two forms of identity source documents from the list of acceptable documents listed below must be presented to the base registrar with this completed form. Check the box for the type of Documents that will be presented for identity proofing. If the document type is not listed, use the two rows under Other Approved Identity Source Documents to enter the type of document(s) that you will present. Block 15: Enter the Document Number located on the Identity Proofing Source document that was checked in Block 14. Block 16: Enter the State that issued the Identity Source Document. Block 17: Enter the Country that issued the Identity Source Document.	Block 18: Enter the Date that the Identity Source Document was issued. Block 19: Enter the Date that the Identity Source Document will expire. Block 20: Enter Weight in pounds. Block 21: Enter Height in inches. Block 22: Check the applicable box for Hair Color. Block 23: Check the applicable box for Eye Color. Block 24: Enter Home Address Including City, State, Zip Code, and Home Telephone Number. Block 25: Enter Name of Registrant's Base Sponsor and Base Sponsor's Telephone Number. Block 26: Enter Employer Name and address including City, State, Zip Code, and Employer's Telephone Number. Block 27: Enter Supervisor's Name including City, State, Zip Code, and Supervisor's Telephone Number. Block 28: Check the applicable box for Work Hours box or check the OTHER box and enter the work hours, then check applicable boxes for Work Days. Block 28: Check the applicable answer if you have been convicted of Felony and enter initials. Block 29: Check the applicable box for felony conviction. Block 30: Enter initials to accept terms for returning Local Population Identification Card. Block 31: Sign and date the form to attest that the foregoing information is true and complete to best of your knowledge.
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LIST OF ACCEPTABLE DOCUMENTS - All documents must not be expired.

Must present one selection from List A or a combination of one selection from List B and one selection from List C.

List A - Documents that Establish Identity and Employment Authorization	OR	List B - Documents that Establish Identity	AND	List C - Documents that Establish Employment Authorization
- U.S. Passport or U.S. Passport Card. - Permanent Resident Card or Alien Registration Receipt Card (Form I-551). - Foreign passport that contains a temporary I-551 stamp or temporary I-551 printed notation on a machine-readable immigrant visa. - Employment Authorization Document that contains a photograph (Form I-766). - For a nonimmigrant alien authorized to work for a specific employer because of his or her status: - Foreign Passport; and - Form I-94 or Form I-94A that has the following: - The same name as the passport; and - An endorsement of the alien's nonimmigrant status as long as that period of endorsement has not yet expired and the proposed employment is not in conflict with and restrictions or limitations identified on form. - Passport from the Federal States of Micronesia (FSM) or the Republic of the Marshall Islands (RM) with Form I-94 or Form I-94A indicating nonimmigrant admission under the Compact of Free Association Between the United States and FSM or RM.		- Driver's license or ID card issued by a State or outlying possession of the United States provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address. - ID card issued by federal, state or local government agencies or entities, provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address. - School ID card with a photograph - Voter's registration card. - U.S. Military card or draft record. - Military dependent's ID card. - U.S. Coast Guard Merchant Mariner Card. - Native American tribal document. - Driver's license issued by a Canadian government authority. For persons under age 18 who are unable to present a document listed above: - School record or report card. - Clinic, doctor, or hospital record. - Day-care or nursery school record.		- A Social Security Account Number card, unless the card includes one of the following restrictions: - NOT VALID FOR EMPLOYMENT - VALID FOR WORK ONLY WITH INS AUTHORIZATION. - VALID FOR WORK ONLY WITH DHS AUTHORIZATION. - Certification of Birth Abroad issued by the Department of State (Form FS-545). - Certification of Birth issued by the Department of State (Form DS-1360). - Original or certified copy of birth certificate issued by a State, county, municipal authority or territory of the United States bearing an official seal. - Native American tribal document. - U.S. Citizen ID Card (Form I-197). - Identification Card for Use of Resident Citizen in the United States (Form I-179). - Employment authorization document issued by the Department of Homeland Security.

The remainder of the form will be completed by the Base Registrar Person conducting Identify Proofing process and NCIC check.

AGENCY DISCLOSURE STATEMENT:

The public reporting burden for this collection of information is estimated to average 10 minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing the burden, to the Department of Defense, Washington Headquarters Services, Executive Services Directorate, Information Management Division, 4800 Mark Center Drive, East Tower, Suite 02G09, Alexandria, VA 22350-3100 OMB 0703-0061. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.

PLEASE DO NOT RETURN COMPLETED FORM TO THE ABOVE ADDRESS.

Completed form should be submitted to the Base Registrar.

SECTION 01 20 00.00 22

PRICE AND PAYMENT PROCEDURES (PWD ME)

05/22

PART 1 GENERAL

[This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.]

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP-1110-1-8

(2016) Construction Equipment Ownership
and Operating Expense Schedule

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Schedule of Prices; G

1.3 SCHEDULE OF PRICES

1.3.1 Data Required

Within 15 calendar days of notice of award, prepare and deliver to the Contracting Officer a Schedule of Prices (construction Contract) as directed by the Contracting Officer. Provide a detailed breakdown of the Contract price, giving quantities for each of the various kinds of work, unit prices, and extended prices. Provide labor, material, equipment for each line item. Contractor overhead and profit including salaries for field office personnel, if applicable, must be proportionately spread over all pay items and not included as individual pay items. Summarize costs and for each construction category.

1.3.2 Payment Schedule Instructions

Payments will not be made until the Schedule of Prices has been submitted to and accepted by the Contracting Officer. For design phase progress payment(s), the Schedule of Prices must include detailed design activities and general (summarized) approach for the construction phase(s) of the project. The Schedule of Prices must be fully developed with detailed construction line items as design progresses. The complete design and construction Schedule of Prices must be submitted and accepted prior to starting construction work.

For Fast-Tracked or Critical Path Submittals of construction projects, the

Schedule of Prices must include detailed design and construction line items for each fast-tracked/critical path phase(s), submitted to and accepted by the Contracting Officer during the Post Award Kickoff Meetings and confirmed prior to starting construction work in that phase.

Additionally, the Schedule of Prices must be separated as follows:

a. Primary Facilities Cost Breakdown:

Defined as work on the primary facilities out to the 5 foot line. Work out to the 5 foot line includes construction encompassed within a theoretical line 5 foot from the face of exterior walls and includes attendant construction, such as pad mounted HVAC cooling equipment, cooling towers, and transformers placed beyond the 5 foot line.

b. Supporting Facilities Cost Breakdown:

Defined as site work, including incidental work, outside the 5 foot line.

If as-built drawings, eOMSI, or operations and maintenance manuals are required, payments for either will not be made until complete and final submissions are approved by the Contracting Officer. The following minimum amounts must be included as Schedule of Prices line items:

Minimum Schedule of Prices values for complete final approved as-built drawings (based on awarded Contract amount: \$0-\$150K 4%; \$150k-\$700k 3%; \$700k - \$2.49M 2%; \$2.5M + 2% (max \$200k))

Minimum Schedule of Prices value for complete final approved Operation and Maintenance Manuals (based on awarded Contract amount: \$0-\$150K 2%; \$150k-\$700k 2%; \$700k - \$2.49M 1%; \$2.5M + 1% (max \$200k))

Minimum Schedule of Prices value for complete final approved eOMSI Workbook (based on awarded Contract amount: \$0-\$150K 2%; \$150k-\$700k 2%; \$700k - \$2.49M 1%; \$2.5M + 1% (max \$200k))

1.3.3 Real Property Assets

The Government will provide the Draft DD Form 1354, Transfer and Acceptance of Military Real Property filled in with the appropriate Real Property Unique Identifiers (RPUID) and related construction Category Codes to summarize the designed real property assets that apply to this Contract. Meet with the Contracting Officer and the Real Property Accounting Officer during the Pre-Construction Meeting and the Project Closeout Meetings to modify and include any necessary changes to the DD Form 1354. Provide the Interim DD Form 1354 that uses the appropriate division of the RPUIDs/Category Codes to represent the final constructed facility and include all associated cost. Coordinate the Price and Payment structure with the structure of the RPUIDs/Category Codes.

Divide detailed asset breakdown into the RPUIDs and related construction Category Codes and populate associated costs which represent all aspects of the work. Where assets diverge into multiple RPUID/Category Codes, divide the asset and provide the proportion of the assets in each RPUID/Category Code. Assets and related RPUID/Category Codes may be modified by the Contracting Officer as necessary during course of the work. Coordinate identification and proportion of these assets with the Government Real Property Accounting Officer.

Cost data accumulated under this Section are required in the preparation of DD Form 1354.

1.3.4 Schedule Requirements for HVAC TAB

The field work of HVAC TABS must be broken down in the Schedule of Prices and in the Construction Progress Documentation by separate line items which reflect measurable deliverables. Specific payment percentages for each line item must be determined on a case by case basis for each Contract. The line items must be as follows:

- a. Approval of Design Review Report: The TABS Agency is required to conduct a review of the project plans and specifications to identify any feature, or the lack thereof, that would preclude successful testing and balancing of the project HVAC systems. The resulting findings must be submitted to the Government to allow correction of the design. The progress payment must be issued after review and approval of the report.
- b. Approval of the pre-field engineering report: The TABS Agency submits a report which outlines the scope of field work. The report must contain details of what systems will be tested, procedures to be used, sample report forms for reporting test results and a quality control checklist of work items that must be completed before TABS field work commences.
- c. Season I field work: Incremental payments are issued as the TABS field work progresses. The TABS Agency mobilizes to the project site and executes the field work as outlined in the pre-field engineering report. The HVAC water and air systems are balanced and operational data must be collected for one seasonal condition (either summer or winter depending on project timing).
- d. Approval of Season I report: On completion of the Season I field work; the data is compiled into a report and submitted to the Government. The report is reviewed, and approved, after ensuring compliance with the pre-field engineering report scope of work.
- e. Completion of Season I field QA check: Contract QC and Government representatives meet the TABS Agency at the jobsite to retest portions of the systems reported in the Season I report. The purpose of these tests is to validate the accuracy and completeness of the previously submitted Season I report.
- f. Approval of Season II report: The TABS Agency completes all Season II field work, which is normally comprised mainly of taking heat transfer temperature readings, in the season opposite of that under which Season I performance data was compiled. This data must be compiled into a report and submitted to the Government. On completion of submittal review to ensure compliance with the pre-field engineering report scope, progress payment is issued. Progress payment is less than that issued for the Season I report since most of the water and air balancing work effort is completed under Season I.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause "DFARS 252.236-7000, Modification Proposals-Price Breakdown," and where actual ownership and operating costs

of construction equipment cannot be determined from accounting records, equipment use rates must be based upon the applicable provisions of the EP-1110-1-8.

1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not completed in accordance with Contract requirements will be returned to the Contractor for correction of the deficiencies. The requests for payment must include the documents listed below.

- a. The Contractor's invoice, on NAVFAC Form 4330 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 4330 must include certification by the Quality Control (QC) Manager as required by the Contract.
- b. The Estimate for Voucher/ Contract Performance Statement on NAVFAC Form 4330 furnished by the Government. Use NAVFAC Form 4330, unless otherwise directed by the Contracting Officer, on NAVFAC Contracts when a Monthly Estimate for Voucher is required.
- c. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.
- d. Monthly Work-hour report.
- e. Updated Construction Progress Schedule and tabular reports required by the Contract.
- f. Contractor Safety & QC Self Evaluation Checklist.
- g. Updated submittal register.
- h. Solid Waste Disposal Report.
- i. Certified payrolls.
- j. Updated testing logs.
- k. Monthly Progress As-Builts (Provide one (1) set of working as-built drawings (CADD) in the specified software and format, hard copy and electronic).
- l. Other supporting documents as requested.

1.5.2 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area Work Flow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow

(WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.5.3 Final Invoice

- a. A final invoice must be accompanied by the Final Release. If incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.
- b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form has been confirmed by the Contracting Officer, submit the final invoice and attach a copy of the Final Release Form in WAWF.
- c. Final invoices not accompanied by the Final Release will be considered incomplete and will be returned.

1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests which comply with the requirements of this Section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and/or suspensions permitted under the FAR and agency regulations including the following in accordance with FAR 32.103 Progress Payments Under Construction Contracts:

- a. Reasonable deductions due to defects in material or workmanship;
- b. Claims which the Government may have against the Contractor under or in connection with this Contract;
- c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and
- d. Failure to provide up to date record drawings not current as stated in Contract Clause FAR 52.236.21.

1.6.2 Payment for Onsite and Offsite Materials

Progress payments may be made for materials delivered on the site, for materials stored off construction sites, or materials that are in transit

to the construction sites under the following conditions:

- a. FAR 52.232-5(b) Payments Under Fixed Price Construction Contracts.
- b. Materials delivered on the site but not installed, including completed preparatory work, and off-site materials to be considered for progress payment must be major high cost, long lead, special order, or specialty items, not susceptible to deterioration or physical damage in storage or in transit to the construction site. Examples of materials acceptable for payment consideration include, but are not limited to, structural steel, non-magnetic steel, non-magnetic aggregate, equipment, machinery, large pipe and fittings, precast/pre-stressed concrete products, plastic lumber (e.g., fender piles/curbs), and high-voltage electrical cable. Materials not acceptable for payment include consumable materials such as nails, fasteners, conduits; gypsum board, glass, insulation, and wall coverings.
- c. Materials to be considered for progress payment prior to installation must be specifically and separately identified in the estimates of work submitted for the Contracting Officer's approval in accordance with Schedule of Prices requirement of this Contract. Requests for progress payment consideration for such items must be supported by documents establishing their value and that the title requirements of the clause at FAR 52.232-5 "Payments Under Fixed-Price Construction Contracts" have been met.
- d. Materials are adequately insured and protected from theft and exposure.
- e. Provide a written consent from the surety company with each payment request for offsite materials.
- f. Materials to be considered for progress payments prior to installation must be stored within 50 miles of the Installation (Portsmouth Naval Shipyard). Other locations are subject to written approval by the Contracting Officer.
- g. Materials in transit to the job site or storage site are not acceptable for payment.
- h. Provide written consent allowing the Govt. access to the remote storage location for inspection of the stored materials.
- i. Materials shall be separated from other stored materials and be clearly identifying the contract number where the materials will be used.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00.00 22

ADMINISTRATIVE REQUIREMENTS (PWD ME)

10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES (for Design Build Projects):

SD-01 Preconstruction Submittals

Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM; G

NAVFAC PWD ME Internal Service Requirements List; G

Archaeological Coordination Meeting; G

1.3 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 LIABILITY, during the entire period of performance under this Contract. Provide other insurance coverage as required by State law.

1.4 SUPERVISION

1.4.1 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the Project Site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills. Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents.

1.4.2 Superintendent Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

The Contractor may propose individuals that do not meet the qualifications specified above by submitting a RFI to the Contracting Officer Representative (COR) along with the individual's qualifications. The COR will review the qualifications of the individuals being proposed and will make a determination to waive the qualifications specified based on the information submitted. If it is determined the proposed individual fails to perform the duties required herein, the Contractor must replace the individual with an individual that meets the qualifications specified. If the Superintendent is dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out. Any schedule delays or additional costs incurred are the Contractor's responsibility.

The Project Superintendent must be on site during working hours. The Superintendent **cannot** be the Quality Control Manager nor the Site Safety and Health Officer (SSHO).

1.4.2.1 Duties

The project superintendent is primarily responsible for managing Subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Pre-Construction Meeting, Environmental Brief, Outage Coordination Meetings, Outage Safety Review Meeting, NAVFAC Red Zone meetings, partnering meetings, Preparatory meetings and quality control meetings. The superintendent or qualified alternate must be on-site at all times during the performance of this Contract until the work is completed and accepted.

1.4.3 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.5 PRECONSTRUCTION MEETING

Notify the Construction Manager assigned to this project to arrange and hold a preconstruction meeting with all interested parties prior to start of work. **The Pre-Construction meeting must be held no later than 30 Calendar days from receiving the Signed Contract or Task Order from the Contracting Officer, but prior to commencement of any work at the site.** The purpose of this meeting is to discuss and develop a mutual

understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, Base-access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices , shop drawings, submittals, cybersecurity, prosecution of the work, Government acceptance, final inspections, and Contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing. For Design-Build projects, the Preconstruction Meeting is to be held upon completion of the projects design and design acceptance by the Government.

1.5.1 Attendees

Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager, and major Subcontractor.

The following Preconstruction submittals must be submitted to the Contracting Officer fifteen (15) calendar days prior to the pre-construction meeting:

<u>Specification</u>	<u>Description</u>
01 32 17.00 20	Baseline Network Analysis Schedule (NAS)
01 35 26.00 22	Accident Prevention Plan (APP)
01 45 00.00 22	Construction Quality Control (QC) Plan

The following Pre-construction submittals must be submitted to the Contracting Officer at the pre-construction meeting:

<u>Specification</u>	<u>Description</u>
01 14 00.00 22	List of Contact Personnel
01 20 00.00 22	Schedule of Prices
01 30 00.00 22	NAVFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM
01 30 00.00 22	NAVFAC PWD ME Internal Service Requirements List
01 31 23.13 20	List of Personnel (eCMS)
01 32 17.00 20	Three-Week Look Ahead Schedules
01 32 17.00 20	Outages Schedule
01 33 00	Submittal Register
01 33 10.05 20	Consolidated RFP Documents
01 33 00.05 20	Submittal Register (DB)
01 50 00.00 22	Construction site plan

The following Pre-construction submittals must be submitted to the Contracting Officer prior to the start of construction:

<u>Specification</u>	<u>Description</u>
01 44 00.00 22	Archeological Monitoring Plan
01 50 00.00 22	Traffic control plan
01 50 00.00 22	Employee Parking Plan
01 57 19.00 22	Preconstruction Survey
01 57 19.00 22	Solid Waste Management Plan
01 57 19.00 22	Regulatory Notifications
01 57 19.00 22	Environmental Management Plan (EMP)
01 57 19.00 22	Dirt and Dust Control Plan
01 57 19.00 22	Contractor Hazardous Material Inventory Log

01 57 19.00 22 Storm Water Management/Erosion and Sedimentation
Control Plan
01 57 19.00 22 Spill Prevention, Control, and Countermeasures (SPCC) Plan
01 74 19 Waste Management Plan

Confirm, in writing, the construction start date with the Contracting Officer Representative at least two (2) working days prior to start date.

Prepare and distribute minutes of all meetings to the attendees within three (3) working days of the preconstruction meeting.

1.6 OUTAGE COORDINATION MEETING

The Outage Coordination Meeting is to be held prior submittal of the outage request package and prior to beginning work on the utility system requiring shut-down (outage), attend a pre-outage coordination meeting. The meeting shall include a discussion to develop a mutual understanding of the administrative requirements including but not limited to outage requests.

This meeting must include the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, PWD ME DM, Designer of Record and the NAVFAC PWD ME UEM representative.

All parties must fully coordinate all the activities required to support the outages.

1.7 Outage Safety Review Meeting

During the Outage Safety Review meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECF training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

1.8 ENVIRONMENTAL BRIEF

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. The Brief will be conducted by the Contracting Officer's Representative (COR) with PWD ME & Shipyard C106 Environmental Representatives. The Brief will be part of the Pre-Construction Meeting agenda or held as a separate meeting depending on the size and scope of work and potential environmental impact, as determined by the Government.

The meeting will discuss details for all the submittals that are required in Section 01 57 19.00 22 including review of the submittal requirements for the Environmental Management Plan, important notes regarding

Installation Restoration (IR), Historic, and Cultural Resources, Wetlands, and Waste Generation, Concrete Wash Water/Wastewater, Hazard Materials Management including sampling, testing and disposal requirements and other applicable topics.

At the Brief, Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the Portsmouth Naval Shipyard; types and quantities of wastes/wastewater that may be generated during the Contract; types and quantities of oil that will be brought onto the activity; and pollution control measures for spill prevention and control, and any bulk oil storage container information including quantity and type of product stored. Discuss the results of the Preconstruction Survey at this time.

Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural resources, historic properties, required reports, required permits, specific permit requirements, and other measures to be taken. Identify additional environmental concerns specific to the site (e.g., historic, archaeological and natural resources, IR, erosion and sediment control, spill prevention and control, soil management, testing and disposal requirements, etc.) (See Section 01 57 19.00 22, Paragraph 1.5.3.)

1.9 ARCHAEOLOGICAL COORDINATION MEETING

Prior to the start of any ground disturbing/excavation work that is located in an Archaeological Site identified on the Drawings, the Archaeological Coordinator/archaeologist, Contractors QC Manager, PWD ME CM/ET and PWD-ME Cultural Resources Manager must meet to review the proposed approach to the work, establish a protocol for collection of artifacts, and review the proposed and review relevant documentation.

For projects with a duration exceeding one year, re-initial coordination meeting must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor; Archaeological Coordinator; QC Manager, Site Superintendent, Project Manager, etc.)

1.10 MEP COORDINATION MEETING

The Contractor shall schedule in person coordination meetings with the mechanical, electrical, piping, and fire protection trades on a monthly basis at a minimum. The intent of the meetings is to review the progress of the coordination drawings and coordinate work being installed.

Meeting minutes must be provided within 24 hours of completion of the MEP coordination meeting.

The QC organization will be responsible for leading the MEP Coordination Meetings with other trades encouraged to participate.

1.11 FACILITY TURNOVER PLANNING MEETINGS (NAVFAC Red Zone - RZ)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start the turnover process at the Pre-Construction Meeting with a discussion of the Red Zone (RZ) process and convene at regularly scheduled RZ Meetings beginning at approximately 75 percent of construction completion. Include

the following in the facility Turnover effort:

1.11.1 RZ Checklist

See Appendix A of this Section for an example of the Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM. Contracting Officer's Representative (COR) will provide the Contractor a copy of the Red Zone Checklist template in advance of the RZ turnover meeting.

At the initial Red Zone Facility Turnover meeting, NAVFAC, the Client and Contractor will modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. This becomes the Red Zone POAM which will be utilized through to the contract completion.

Items listed on the checklists are required to remain on the checklists if they are part of the project/Contract or required by construction convention. Items not listed on the checklists, but required in the Contract or by construction convention, must be added to the checklists by the Contractor, Client and NAVFAC. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary. Checklists are applicable to all Contracts no matter what Category of Work. The Point of Contact and due date must initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, Client and Contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) must indicate date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this must be noted on the checklist and a new due date established and indicated.

1.11.2 Meetings

- a. Upon Government acceptance of the RZ Checklist-POAM, the COR will send out the accepted RZ Checklist-POAM to all attendees. The Project Superintendent is required to lead regular Red Zone Meetings beginning at approximately 75 percent project completion, or three (3) to six (6) months prior to Beneficial Occupancy Date (BOD), whichever comes first.
- b. The Contracting Officer will determine the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.
- c. Using RZ Checklist-POAM as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.
- d. Discuss and coordinate with the CM/ET for upcoming activities that require Government involvement.
- e. All parties will maintain their copy of the RZ Checklist-POAM documenting the actual completion dates as work is completed and update the RZ Checklist-POAM with revised planned completion dates as necessary to match progress. The CM will maintain the master RZ Checklist-POAM, periodically distributing a scanned copy of the current RZ Checklist-POAM to attendees via email after significant progress is made.

1.11.3 NAVFAC PWD ME Internal Service Requirements List

An initial, pre-edited draft of the NAVFAC PWD ME Internal Service Requirements List is included in Appendix B of this Section.

Include all information usually listed on manufacturer's name plate. The Internal Service Requirements List must include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), manufacturer, model number, serial number, capacity, floor coverings, wall and ceiling surfaces; types and square footage of coverage, lighting fixtures, bathroom fixtures, windows, and HVAC filters.

Submit a preliminary Internal Service List to the COR at the initial Facility Turnover Meeting. Provide the final completed Internal Service List with all required facility system/equipment information to the COR at least ninety (90) calendar days prior to the project BOD.

1.12 PARTNERING

To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget, and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user (who will occupy the facility), NAVFAC, PWD Maine Design and Construction team and Subject Matter Experts (SME's), FEAD, Project Manager (PM), Design Manager (DM), Construction Manager (CM), Engineering Technician (ET), Contractor, key Subcontractors, and the Designer of Record are required to participate in the Partnering process.

The Contracting Officer will provide Information on the Partnering Process and a list of key and optional personnel who are to attend the Partnering meeting.

1.13 NOT USED

1.14 ELECTRONIC MAIL (E-MAIL) ADDRESS

Establish and maintain electronic mail (e-mail) capability along with the capability to open various electronic attachments as text files, pdf files, and other similar formats. Within 10 days after Contract award, provide the Contracting Officer a single (only one) e-mail address for electronic communications from the Contracting Officer related to this Contract including, but not limited to, Contract documents, invoice information, request for proposals, and other correspondence. The Contracting Officer may also use email to notify the Contractor of Base access conditions when emergency conditions warrant, such as hurricanes or terrorist threats. Multiple email addresses are not allowed.

It is the Contractor's responsibility to make timely distribution of all Contracting Officer initiated e-mail with its own organization including field office(s). Promptly notify the Contracting Officer, in writing, of any changes to this email address.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

01 30 00 APPENDIX A**NAVFAC Red Zone
Facility Turnover Planning Meeting****AGENDA****I. Introduction and Overview – Purpose****CM**

The purpose of the Facility Turnover Planning Meeting is to address elements within the project team's purview – schedule management, assure completed facility complies with contract requirements, and other contractual issues. Each member of the project delivery team (Client, NAVFAC, and the contractor) has critical responsibilities to ensure timely completion and turnover of the new facility and each member should execute the NRZ process to achieve this end. The NRZ process provides a final re-focusing of attention to details of those key elements critical for a successful construction contract completion. In implementing NRZ processes, the NAVFAC/Contractor/Client team take a collective "snapshot" of contract status, identifying remaining actions to be accomplished, and confirm required resources needed for successful contract completion and turnover to the Client.

The Facility Turnover Planning Meeting is a collaborative effort between the Client, NAVFAC, and the contractor and results in a completed "NRZ Checklist/POAM Items" list that identifies the major items (and their due dates) that must be completed by the Contractor, the Client and the NAVFAC team to ensure timely completion of the contract.

II. Attendees

NAVFAC Echelon IV (PM); NAVFAC FEAD/ROICC Team (AROICC, CM, ET/QA, Contracting Officer); Client Team (Project Manager, Program Coordinator, User/Tenant); Contractor Team (Project Manager, Project Superintendent, CQC Manager)

III. Schedule to Completion (POAM)**Contractor****IV. Schedule of Final Outfitting and Occupancy (POAM)****Client****V. Critical feature(s) of project (POAM)****CM****VI. Transfer of Maintenance Responsibility****CM****VII. Systems training & O&M Manuals (POAM)****CQC Manager****VIII. Other Items to include on NRZ checklists****All****IX. Summary of Required Actions and Responsibility****CM**

Guidelines for conducting Facility Turnover Planning Meeting are as follows:

- a. Meeting is held at approximately 75% construction contract completion or three to six months prior to BOD. NAVFAC representatives will include the Project Manager, Construction Manager/AROICC (CM) and Design Manager (DM), as appropriate. The contractor representatives include applicable prime contractor staff and decision-makers from major subcontractors. Design-Build contractors will have A-E representatives attending. The Client should include representatives from Public Works Officer (PWO) staff, a Client scope and financial decision maker, a user tenant representative, a facility start-up person, and others such as SPAWAR, NMCI, telephone, and furniture contractor, etc.
- b. The purpose of the meeting is to plan the remaining work, identify critical project features that still need to be completed (such as “soft” construction contract requirements as shown on the NRZ Checklist/POAM Items), and to complete the filling out of the “NRZ Checklist/POAM Items”.
- c. The contractor, client and NAVFAC provide a POC and due date for each item on their checklist. The team fills in the checklists by selecting items applicable to the project, selects due dates on each item, and appoints a person who has responsibility to ensure the item gets completed by the due date. The CM will be responsible to monitor the milestones.

NRZ Checklist/POAM Items

The table below provides typical NRZ checklist items for contractor, Client, and NAVFAC actions. Items listed on the checklists are required to remain on the checklists if they are part of the project/contract or required by construction convention. Items not listed on the checklists, but required in the contract or by construction convention, must be added to the checklists by the contractor, Client and NAVFAC. Checklists are applicable to all contracts no matter what Category of Work.

The Point of Contact and due date shall initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, client and contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) shall indicate date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this should be noted on the checklist and new due date established and indicated. The completed NRZ Checklist/POAM shall be placed in the contract file.

	A	B	C	D	E	F	G	H	I
1			Edit the document to remove Facility Delivery items that are not in the contract. To delete the rows highlight the row(s) and use Control - (minus sign). To insert additional rows use Control + (plus sign).						
2	Revision - 4/24/23		* INFO / DIRECTIONS - This NRZ Checklist/POAM is a tool to track the status of critical activities required for BOD, to help ensure their timely completion and prevent delays with facility acceptance and turnover. - The critical activities are organized by section according to requirements (e.g. NMCI & BCO acceptance and connections) - Critical items missing from this list should be added as necessary to ensure the list is comprehensive. Likewise, unnecessary items should be deleted. - Any critical items left off the NRZ Checklist that are later identified after initial NRZ meeting is conducted should be added immediately so their progress can be tracked. - A copy of the latest project customized NRZ Checklist/POAM shall be maintained in the Project contract file.						
3			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
4				Required for Facility Delivery (BOD)					
5			KTR	Electrical Systems Testing					
6			NAVFAC	Transformer Performance Verification					
7			KTR	Generator Testing					
8			NAVFAC	Generator Performance Verification					
9			KTR	Final Electrical Connections					
10			NAVFAC	Coordinate Final Electrical Connections					Coordinate w/PW Shops
11			NAVFAC	Acceptance Testing of Critical Electrical Systems					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
12			KTR	Final Gas Connections					
13			NAVFAC	Coordinate Final Gas Connections					Coordinate w/PW Shops
14			KTR	Final Water Connections					
15			KTR	Super chlorination of Potable Water Systems					
16			KTR	Plumbing/Backflow Testing					
17			NAVFAC	Coordinate Final Water Connections					Coordinate w/PW Shops
18			KTR	Critical System Start-up: (Edit & add detail accordingly)					
19									
20			KTR	Duct Air Leakage Testing (DALT)					
21			NAVFAC	DALT Field Acceptance					
22			KTR	Air Barrier Testing					
23			NAVFAC	All HVAC Commissioning/Testing Complete					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
24									
25			KTR	Elevator Testing					
26			NAVFAC	Elevator Certification					Notify ML elev inspector 30 days prior to required site visit.
27									
28			KTR	Boiler/UPV Testing					
29			NAVFAC	Boiler/UPV Certification					Notify ML boiler inspector 30 days prior to required site visit.
30									
31			KTR	Crane Testing					
32			C700	Crane Certification					Notify C700 30 days prior to required testing.
33									
34			KTR	Fire Alarm/Sprinkler Testing					
35			NAVFAC	FA & FP test results to ML & DOR					days prior to acceptance testing by FPE

NAVFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM

	A	B	C	D	E	F	G	H	I
3			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
36			NAVFAC	Fire Alarm/Sprinkler Acceptance Test (exceptions must be raised to Chief Engineer).					Notify ML FPE 30 days prior to required testing. Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
37			NAVFAC	All Life Safety Inspections/Testing complete					Exceptions must be raise up through the DCBL chain so that the Command can take action under our Technical Authority.
38									
39			KTR	Keying Plan Meeting					
40			Client	Keying Plan Meeting					
41			NAVFAC	Keying plan to NAVFAC Locksmith					
42			KTR	Deliver Lockset Cores to CM/ET					
43			NAVFAC	Lockset cores & keys and installed					Locksmith
44									
45			KTR	NMCI Connections/IT Systems Testing					
46			Client	NMCI Install or other networks (C109)					
47			Client	Secure Network Installations					
48			KTR	IDS & SCIF Testing					
49			Client	IDS & SCIF Security Insp & Acceptance					
50			NAVFAC	IDS & SCIF Insp & Test					
51			KTR	Telecommunications Connections & Test					
52			Client	Telecommunication install (BCO phones)					
53			Client	Mod service contracts for Phone/Utilities/Custodial/Grounds					
54									
55			KTR	Commissioning Functional Performance Testing (FPT)					
56			KTR	TAB (Air and Water Balancing)					
57			KTR	TAB Proportional Balancing Report					
58			NAVFAC	Proportional Balanc'g TAB Field Accept					
59			KTR	TAB Field Acceptance Testing					
60			KTR	TAB Season 1 Report*					
61			KTR	TAB Field Acceptance Testing					
62			NAVFAC	1st Season TAB Field Acceptance					
63			KTR	DDC Trend Logs / Endurance Testing					
64			KTR	ACATS (controls) Testing					
65			NAVFAC	ACATS Performance Verification					
66			KTR	Performance Verification Testing (PVT) Controls					
67									
68			NAVFAC	Roofing final acceptance					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
69									
70			NAVFAC	Underwater Structure Acceptance					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.

NAVAFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM

	A	B	C	D	E	F	G	H	I
			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
3									
71									
72			NAVFAC	Performance Verification Testing (PVT) Controls NAVFAC Acceptance					
73									
74			KTR	Facility-Related Control System Cybersecurity Commissioning					
75			NAVFAC	Completed Cyber Hygiene Checklist					
76			NAVFAC	Completed RMF Step 4 Validation					
77									
78			Client	GFE status/delivery schedule (GFCI, GFGI)					
79			Client	Client provided equipment KTR installed					
80			Client	Client provided equipment SELF installed					
81									
82			NAVFAC	Training Coordinated/Scheduled with FMS					
83			KTR	System Training of Navy Personnel					
84			Client	Attend Training					
85									
86			KTR	Provide As-Built Dwgs & floor plans in CAD					
87			NAVFAC	CM to Submit floor plan As-Built to CAD Dept.					RQ'd for Fire Bill - RQ'd before BOD(A)
88			KTR	Submit sample as-built drawings for CAD review					Submit 3 drawings from ea. discipline for review
89			NAVFAC	CM to submit to DM who must review and submit Sample As-Built documents to CAD Dept.					
90									
91			KTR	Pre-Warranty Conference					
92			KTR	Contractor's Pre Final Punch List Complete					
93			NAVFAC	Obtain a copy and ensure QA Inspections are completed.					Include a completed copy for BOD Mangament sign off.
94			NAVFAC	Obtain a copy and ensure Special Inspections are completed.					Include a completed copy for BOD Mangament sign off.
95			KTR	Pre Final Inspection					
96			NAVFAC	Client walk-thru Inspection					
97			KTR	Pre-Final Inspection					
98			NAVFAC	Pre-Final Inspection					
99			KTR	Punch List					
100			KTR	Final Inspection					
101			NAVFAC	Final Inspection					
102									
103			NAVFAC	BOD/Use and Possession					
104			Client	Ribbon-cutting ceremony					
105			Client	Planned User Move-in					
106									
107			KTR	Delivery of O&M Manuals					Must be provided before BOD
108			NAVFAC	CM to submit the O&M's to DM who must review and submit to NAVFAC PWD ME Requirements FMS					
109									
110			KTR	Delivery of Product Warranties					Must be provided before BOD

NAVAFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM

	A	B	C	D	E	F	G	H	I
3			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
111			NAVFAC	CM to submit Warranties to NAVFAC PWD ME Requirements FMS					
112									
113			KTR	As-Built Drawings					Must be provided 30 cal days before BOD
114			NAVFAC	CM to submit the As-Built to DM who must review and submit documents to CAD Dept.					
115			KTR	Submit eOMSI Facility Data Workbook					Must be provided before BOD
116			NAVFAC	CM to submit the eOMSI Workbook to the DM who must review and submit to the NAVFAC PWD ME FOS.					
117			KTR	Submit the Internal Service Requirement List (ISRL).					Must be provided before BOD
118			NAVFAC	CM to submit the ISRL to the DM who must review and submit to the NAVFAC PWD ME FSC. Confirm ISRL submitted to FSC.					
119									
120				Required for Contract Close-out:					
121			NAVFAC	Warranty documentation to FMS					Must be provided before CCD
122			KTR	Site Restoration					
123			KTR	Final Landscaping					
124			KTR	Spare Parts, Extra Stock, Special Tools, etc					
125									
126			KTR	Final Demobilization and Clean-up					
127			KTR	Temp Construction Fence Removed					
128			KTR	Project Close-out Meeting					
129			NAVFAC	As-Built to PWD ME PDC CADD Branch (Preston Gowen)					
130			KTR	Complete DD1354					
131			NAVFAC	DD 1354 Signed & turned over to RPAO					
132			KTR	Closeout & submit permits and inspection reports					Notice of Termination (NOT); Storm water insp rpts; Stormwater BMP Install rpts; Archaeological Monitor'g rpts and artifacts, salvaged mat. etc.
133			NAVFAC	Closeout & submit permits and inspection reports					NPDES, Notice of Termination (NOT); Storm water insp rpts; Stormwater BMP Install rpts; Archaeological Monitor'g rpts and artifacts, salvaged mat. etc.
134			KTR	Seasonal/Deferred Commissioning FPT					
135			KTR	2 nd Season TAB Report					
136			NAVFAC	2 nd Season TAB Field Accept Testing					
137			KTR	Annual Elevator Maintenance Complete					
138			NAVFAC	Annual Elevator Maintenance Complete					
139			NAVFAC	Authority to Operate delivered to client					
140			KTR	Commissioning 10-Month/Warranty Visit					
141			NAVFAC	Inform PM of BOD					

NAVAFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM

	A	B	C	D	E	F	G	H	I
3			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
142			NAVFAC	BOD (A) entered into eContracts					
143			NAVFAC	BOD Letter to Contractor					Sample ltr can be found here
144			NAVFAC	Acceptance Letter to Client					Sample ltr can be found here
145			NAVFAC	Process recycled/recovered materials report					
146			NAVFAC	Contractor Evaluations (CPARS) Complete					assist PM
147			NAVFAC	Finalize Outstanding Contract Mods					

SECTION 01 31 23.13 20

ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM
05/17, CHG 6: 02/21

PART 1 GENERAL

1.1 CONTRACT ADMINISTRATION

Utilize the Naval Facilities Engineering Command's (NAVFAC's) Electronic Construction and Facility Support Contract Management System (eCMS) for the transfer, sharing and management of electronic technical submittals and documents. The web-based eCMS is the designated means of transferring technical documents between the Contractor and the Government. Paper media or e-mail submission, including originals or copies, of the documents identified in Table 1 are not permitted, except where eCMS is unavailable, non-functional or specifically requested in addition to electronic submission. When specifically requested to provide documents outside of eCMS, upload all final project documentation (e.g. documents that are signed and/or adjudicated by the Government) mentioned in Table 1 into the subject eCMS document management folders that are associated with that document type. Include the identification number of the document, type of document; the name/subject or title; and for daily reports the date (day of work) with format YYYY/MM/DD in the filename. For example for RFI's 0011_RFI_Roof_Leaking.doc; For submittals 0032a_Submittals_Light_Fixture.pdf; For Daily Reports 0132_Daily_Report_20190504.xls. Contact the Contracting Officer's Representative (COR) regarding availability of eCMS training and reference materials.

1.2 USER PRIVILEGES

The Contractor will be provided access to eCMS. All technical submittals and documents must be transmitted to the Government via the COR. Project roles and system roles will be established to control each user's menu, application, and software privileges, including the ability to create, edit, or delete objects.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contractor's Personnel; G

1.4 SYSTEM REQUIREMENTS AND CONNECTIVITY

1.4.1 General

The eCMS requires a web-browser (platform-neutral) and Internet connection. Obtain from an approved vendor an External Certification Authority (ECA), Primary Key Infrastructure (PKI) certificate, or other

similar digital identification to support two-factor authentication and access to eCMS. Provide and maintain computer hardware and software for the eCMS access throughout the duration of the contract for all Contractor-designated users. Provide connectivity, speed, bandwidth, and access to the Internet to ensure adequate functionality. Neither upgrading of the Contractor's computer system nor delays associated from the usage of the eCMS will be justification or grounds for a time extension or cost adjustment to the Contract.

1.4.2 Contractor Personnel List

Within 20 calendar days of contract award, provide to the Contracting Officer a list of Contractor's personnel who will have the responsibility for the transfer, sharing and management of electronic technical submittals and documents and will require access to the eCMS. Project personnel roles to be filled in the eCMS include the Contractor's Project Manager, Superintendent, Quality Control (QC) Manager, and Site Safety and Health Officer (SSHO). Personnel must be capable of electronic document management. Notify the COR immediately of any personnel changes to the project. The Contracting Officer reserves the right to perform a security check on all potential users. Provide the following information:

- First Name
- Last Name
- E-mail Address
- Office Address
- Project Role (e.g. Project Manager, QC Manager, Superintendent)

1.4.3 NOT USED

1.5 SECURITY CLASSIFICATION

In accordance with Department of Navy guidance, all military construction contract data are unclassified, unless specified otherwise by a properly designated Original Classification Authority (OCA) and in accordance with an established Security Classification Guide (SCG). Refer to the project's OCA when questions arise about the proper classification of information.

The eCMS must only be used for the transaction of unclassified information associated with construction projects. In conformance with the Freedom of Information Act (FOIA), DoD INSTRUCTION 5200.48 CONTROLLED UNCLASSIFIED INFORMATION (CUI), and DoD requirements, any unclassified project documentation uploaded into the eCMS must be designated either "U - UNCLASSIFIED" (U) or "CUI - CONTROLLED UNCLASSIFIED INFORMATION" (CUI). Project Photos must not be uploaded to eCMS. All photos must be reviewed by PNS Security prior to any public release.

1.6 ECMS UTILIZATION

Establish, maintain, and update data and documentation in the eCMS throughout the duration of the contract.

Personally Identifiable Information (PII) transmittal is not permitted in the eCMS.

1.6.1 Information Security Classification/Identification

The eCMS must be used for the transmittal of the following documents. This requirement supersedes conflicting requirements in other sections, however,

submittal review times in Section 01 33 00 SUBMITTAL PROCEDURES remain applicable. Table 1 - Project Documentation Types provides the appropriate U and CUI designations for various types of project documents. Construction documents requiring CUI status must be marked accordingly. Apply the appropriate markings before any document is uploaded into eCMS. Markings are not required on U documents.

Table 1 also identifies which eCMS application is to be used in the transmittal of data (these are subject to change based on the latest software configuration). If a designated application is not functional within 4 hours of initial attempt, defer to the Submittal application and submit the required data as an uploaded portable document (e.g. PDF), word processor, spreadsheet, drawing, or other appropriate format. Hard copy or e-mail submission of these items is acceptable only if eCMS is documented to be not available or not functional or specifically requested in addition to electronic submission. After uploading documents to the Submittal application, transmit the submittals and attachments to the COR via the Transmittal application. For Submittals, select the following:

Preparation by = Contractor personnel assigned to prepare the submittal
 Approval by = Contracting Officer Representative (COR)
 Returned by = Design Lead/Manager
 Forwarded to = Contractor project manager

Table 1 - Project Documentation Types

SUBJECT/NAME	DESIG	REMARKS	ECMS APPLICATION
As-Built Drawings	U	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals and Transmittals
Building Information Modeling (BIM)	U	1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager 2. Design reviews will be performed in existing "Dr Checks"	Submittals and Transmittals
Construction Permits	U	Refer to rules of the issuing activity, state or jurisdiction	Submittals and Transmittals
Construction Schedules (Activities and Milestones)	U	After the schedule submittal is approved by the COR, import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline	Submittals, Transmittals and Scheduling App

SUBJECT/NAME	DESIG	REMARKS	ECMS APPLICATION
Construction Schedules (Cost-Loaded)	U	After the schedule submittal is approved by the COR, import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline	Submittals, Transmittals and Scheduling App
Construction Schedules (3-Week Lookahead)	U	Import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline	Scheduling App
DD 1354 Transfer of Real Property	U		Submittals and Transmittals
Daily Production Reports	U	Provide weather conditions, crew size, man-hours, equipment, and materials information	Daily Report
Daily Quality Control (QC) Reports	U	Provide QC Phase, Definable Features of Work Identify visitors	Daily Report
Designs and Specifications	U	1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager 2. Design reviews will be performed in existing "Dr Checks"	Submittals and Transmittals
Environmental Notice of Violation (NOV), Corrective Action Plan	U	Refer to rules of the issuing activity, state or jurisdiction	Submittals and Transmittals
Environmental Protection Plan (EPP)	U		Submittals and Transmittals
Invoice (Supporting Documentation)	U	Applies to supporting documentation only. Invoices are submitted in Wide-Area Workflow (WAWF)	Submittals and Transmittals
Jobsite Documentation, Bulletin Board, Labor Laws, SDS	U		Submittals and Transmittals

SUBJECT/NAME	DESIG	REMARKS	ECMS APPLICATION
Meeting Minutes	U		Meeting Minutes
Modification Documents	U	Provide final modification documents for the project. Upload into "Modifications -	Document Management
Operations & Maintenance Support Information (OMSI/eOMSI), Facility Data Worksheet	U	1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager 2. Design reviews will be performed in existing "Dr Checks"	Submittals and Transmittals
Photographs		MUST NOT BE SAVED TO ECMS. ALL PHOTOS MUST BE REVIEWED BY SECURITY	Submittals and Transmittals
QCM Initial Phase Checklists	U		Checklists (Site Management)
QCM Preparatory Phase Checklists	U		Checklists (Site Management)
Quality Control Plans	U		Submittals and Transmittals
QC Certifications	U		Submittals and Transmittals
QC Punch List	U		Punch Lists (Testing Logs)
Red-Zone Checklist	U		Checklists (Site Management)
Rework Items List	U		Punch Lists (Testing Logs)
Request for Information (RFI) Post-Award	U		RFIs
Safety Plan	U		Daily Report
Safety - Activity Hazard Analyses (AHA)	U		Daily Report

SUBJECT/NAME	DESIG	REMARKS	ECMS APPLICATION
Safety - Mishap Reports	U		Daily Report
SCIF/SAPF Accreditation Support Documents	CUI	Note: Some Construction Security plans may be classified as Secret. Classified information must not be uploaded into eCMS. Refer to the Site Security Manager, as applicable.	Submittals and Transmittals
Shop Drawings	U	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals and Transmittals
Storm Water Pollution Prevention (Notice of Intent - Notice of Termination)	U	Refer to rules of the issuing activity, state or jurisdiction	Submittals and Transmittals
Submittals and Submittal Log	U		Submittals and Transmittals
Testing Plans, Logs, and Reports	U		Submittals and Transmittals
Training/Reference Materials	U		Submittals and Transmittals
Training Records (Personnel)	U		Submittals and Transmittals
Utility Outage/Tie-In Request/Approval	U		Submittals and Transmittals
Warranties/BOD Letter	U		Submittals and Transmittals
Quality Assurance Reports	U		Checklists (Government initiated)
Non-Compliance Notices	U		Non-Compliance Notices (Government initiated)

SUBJECT/NAME	DESIG	REMARKS	ECMS APPLICATION
Other Government-prepared documents	U		GOV ONLY
All Other Documents	U	Refer to FOIA guidelines and contact the FOIA official to determine whether exemptions exist	As applicable

1.6.2 Markings on CUI documents

- a. Only CUI documents being electronically uploaded into the eCMS (.docx, .xlsx, .pptx, and others as appropriate), and associated paper documents described in the paragraph CONTRACT ADMINISTRATION require CUI markings as indicated in the subparagraphs below.
- b. CUI documents that are originally created within the eCMS application using the web-based forms (RFIs, Daily Reports, and others as appropriate) will be automatically watermarked by the eCMS software, and these do not require additional markings.
- c. CUI documents must be marked "CONTROLLED UNCLASSIFIED INFORMATION" at the bottom of the outside of the front cover (if there is one), the title page, the first page, and the outside of the back cover (if there is one).
- d. CUI documents must be marked on the internal pages of the document as "CONTROLLED UNCLASSIFIED INFORMATION" at top and bottom.
- e. Where Installations require digital photographs to be designated CUI, place the markings on the face of the photograph.
- f. For visual documentation, other than photographs and audio documentation, mark with either visual or audio statements as appropriate at both the beginning and end of the file.

1.7 QUALITY ASSURANCE

Requested Government response dates on Transmittals and Submittals must be in accordance with the terms and conditions of the Contract. Requesting response dates earlier than the required review and response time, without concurrence by the Government COR, may be cause for rejection.

Incomplete submittals will be rejected without further review and must be resubmitted. Required Government response dates for resubmittals must reflect the date of resubmittal, not the original submittal date.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 32 16.00 20

SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES

08/18, CHG 1: 08/20

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard and PWD ME AOR Facilities that are less than \$1,000,000 or as approved by the Contracting Officer through the PWD Maine Design Manager.

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES]:

SD-01 Preconstruction Submittals

Construction Schedule; G

Outage Schedule; G

SD-07 Certificates

Monthly Updates

1.2 ACCEPTANCE

Prior to the start of work, prepare and submit to the Contracting Officer for acceptance a construction schedule in the form of a Bar Chart Schedule in accordance with the terms in Contract Clause "FAR 52.236-15, Schedules for Construction Contracts," except as modified in this Contract.

The acceptance of a Baseline Construction Schedule is a condition precedent to:

- a. The Contractor starting work on the demolition or construction stage(s) of the Contract.
- b. Processing Contractor's invoice(s) for construction activities/items of work.
- c. Review of any schedule updates.

Submittal of the Baseline Schedule, and subsequent schedule updates, is understood to be the Contractor's certification that the submitted schedule meets all of the requirements of the Contract Documents, represents the Contractor's plan on how the work will be accomplished, and accurately reflects the work that has been accomplished and how it was sequenced (as-built logic).

1.3 SCHEDULE FORMAT

1.3.1 Bar Chart Schedule

The Bar Chart must, as a minimum, show work activities, submittals, Government review periods, material/equipment delivery, outages, on-site construction, inspection, testing, and closeout activities. The Bar Chart must be time scaled and generated using an electronic spreadsheet program.

1.3.2 Schedule Submittals and Procedures

Submit Bar Chart Schedules and updates in hard copy and on electronic media that is acceptable to the Contracting Officer. Submit an electronic back-up of the project schedule in an import format compatible with the Government's scheduling program.

1.4 SCHEDULE MONTHLY UPDATES

Update the Construction Schedule at monthly intervals or when the schedule has been revised. The updated schedule must be kept current, reflecting actual activity progress and plan for completing the remaining work. Submit copies of purchase orders and confirmation of delivery dates as directed by the Contracting Officer.

a. Narrative Report: Provide with schedule updates. Identify and justify;

- (1) Progress made in each area of the project;
- (2) Critical Path;
- (3) Date/time constraint(s), other than those required by the Contract;
- (4) Changes in the following; added or deleted activities, original and remaining durations for activities that have not started, logic, milestones, planned sequence of operations, and critical path;
- (5) Pending items and status thereof, including permits, change orders, and time extensions;
- (6) Status of Contract Completion Date and interim milestones;
- (7) Current and anticipated delays (describe cause of delay and corrective actions(s) and mitigation measures to minimize); and
- (8) Description of current and future schedule problem areas.

For each entry in the narrative report, cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

1.5 CONTRACT MODIFICATION

Submit a Time Impact Analysis (TIA) with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, and extends the Projected Finish beyond the Contract Completion Date.

- a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact, successor and predecessor activity to impact period, responsible party, describe how it originated, and how it impacts the schedule. The schedule submission must consist of three (3) native files:
 - (1) Fragnet used to define the scope of the changed condition
 - (2) Most recent accepted schedule update as of the time of the proposal or claim submission that has been updated to show all activity progress as of the time of the impact start date.
 - (3) The impacted schedule that has the fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.
- b. For claimed as-built project delay, the inserted fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used.
- c. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.
 - (1) Identify types of delays as follows:
 - (a) Force majeure delay (e.g. weather delay): Any delay event caused by something or someone other than the Government (including its agents) or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the critical path, in absence of other types of concurrent delays, the Contractor is granted an extension of Contract time, classified as a non-compensable event.
 - (b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the Contractor-delay is on the critical path, in absence of other types of concurrent delays, Contractor is not granted extension of Contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, provide a Corrective Action Plan, identifying plan to mitigate delay, to the Contracting Officer.
 - (c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.
 - (2) Use functional theory to analyze concurrent delays, where: Separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under

review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:

(a) Government-delay concurrent with Contractor-delay: Excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.

(3) A pacing delay, reacting to another delay (parent delay) equally or more critical than paced activity, must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with Contractor-pacing: Excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing: Inexcusable time extension, classified non-compensable event.

- d. Submit Data disks containing the narrative and native schedule files.
- e. Unless the Contracting Officer requests otherwise, only add conformed Contract modifications into the Project schedule.

1.6 3-WEEK LOOK AHEAD SCHEDULE

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Construction Schedule including any required Government QA inspections. Key the work plans to activity numbers when a schedule is required and update each week to show the planned work for the current and following two-week period. Additionally, include upcoming outages, closures, preparatory meetings, and initial meetings. Identify critical path activities on the Three-Week Look Ahead Schedule. The detail work plans are to be bar chart type schedules, maintained separately from the Construction Schedule on an electronic spreadsheet program and printed on 8-1/2 by 11 inch sheets as directed by the Contracting Officer. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools, and equipment required to complete the work. Deliver three (3) hard copies and one electronic file of the 3-Week Look Ahead Schedule to the Contracting Officer no later than 8 a.m. each Monday, and review during the weekly CQC Coordination or Production Meeting.

1.7 Outage Schedule

The Contractor must provide a comprehensive Outage Schedule prior to the Pre-Construction meeting that includes all required Outages (Road, Rail, Utility etc) required to complete the work. The Schedule must include the dates the outage request packages will be submitted and must include a 15 calendar day Government Concurrence period. See Section 01 14 00.00 22 for

Outage Requirements. The Contractor must attend the Outage Coordination Meeting specified in 01 14 00.00 22 to ensure the Outage Package requirements are understood and to review the Outage Schedule with the Contracting Officer Representatives.

1.8 CORRESPONDENCE AND TEST REPORTS:

Correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) must reference Schedule Activities that are being addressed. Test reports (e.g., concrete, soil compaction, weld, pressure) must reference Schedule Activities that are being addressed.

1.9 ADDITIONAL SCHEDULING REQUIREMENTS

- a. Any references to additional scheduling requirements, including systems to be inspected, tested and commissioned, that are located throughout the remainder of the Contract Documents, are subject to all requirements of this Section.
- b. Government Activities: Government and other agency activities that could impact progress shall be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators, utility outages, Design Start, Construction Start, (including Design/Construction Start for each Fast-Track Phase), Notice(s) to proceed and delivery of Government Furnished Material/Equipment. The Government will provide a list of QA inspections and testing at the pre-construction meeting.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES

10/23

PART 1 GENERAL

This Section applies only to Design-Bid-Build projects at the Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittals requirements are specified in the technical Sections. Submittals are identified by Submittal Description (SD) numbers and titles as follows:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase Contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Submittals which are required prior to or commencing work on site.

Certificates of Insurance

Surety Bonds

List of Proposed Subcontractors

List of Proposed Products

Construction Progress Schedule

Outages Schedule

Construction Schedule

Submittal Register

Schedule of Prices or Earned Value Report

Accident Prevention Plan (APP)

Work Plans

Quality Control (QC) Plan

Environmental Management Plan

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the Contract requires extended product warranties.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product, or system identical to the material, product, or system to be provided has been tested in accordance with specified requirements. Unless specified in another Section, testing must have been within three years of date of Contract award for the project.

Report which includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report which includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports.

Daily logs and checklists.

Final acceptance test and operational test procedure.

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system, or material attesting that product, system, or material meets specification requirements. Must be dated after award of project Contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer, or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits.

Text of posted operating instructions.

The Contractor shall maintain submitted certifications and licenses through the duration of the project. Any certification or license that expires prior to or during the execution of the affected DFOV shall be resubmitted to the Contracting Officer for approval.

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system, or material, including special notices and Safety Data Sheets (SDS) concerning impedances, hazards, and safety precautions.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance, and repair of the item.

This data is intended to be incorporated in an operations and maintenance manual or control system.

Data incorporated in an operations and maintenance manual or control system.

eOMSI submittals per Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principal Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction Contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase Contract.

DD Form 1354 with cost breakout for all assets 30 days prior to facility turnover.

Red Zone documents per Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this Section, on- and off-site construction required by Contract Documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with this Section:

SD-01 Preconstruction Submittals

Submittal register; G

Other submittals required for Gov't approval prior to beginning any work include, but are not limited to, the following:

Certificates of Insurance
 Surety Bonds
 List of Proposed Subcontractors
 List of Proposed Products
 Construction Schedule)
 Construction Progress Schedule
 Schedule of Prices or Earned Value Report
 Outages Schedule
 List of Contact Personal
 Qualifications
 Work Plans
 Quality Control (QC) Plan
 Environmental Management Plan
 Solid Waste Management Plan and Permit
 Storm Water Pollution Protection Plan
 Accident Prevention Plan (APP)
 Activity Hazard Analysis (AHA)
 Crane Critical List Plan
 Crane Operator Qualifications
 Construction Site Plan
 Traffic Control Plan
 Dirt and Dust Control
 Construction Hazardous Material Inventory Log

The following Preconstruction submittals must be submitted to the Contracting Officer fifteen (15) calendar days prior to the pre-construction meeting:

Specification Section	SD #	SD Description	Item Submitted	Paragraph #
01 32 16.00 20	01	Preconstruction Submittals	Construction Schedule	1.3
01 35 26.00 22	01	Preconstruction Submittals	Accident Prevention Plan (APP)	1.7
01 45 00.00 22	01	Preconstruction Submittals	QC Plan	1.6

The following Preconstruction submittals must be submitted to the Contracting Officer at the pre-construction meeting:

Specification Section	SD #	SD Description	Item Submitted	Paragraph #
01 14 00.00 22	01	Preconstruction Submittals	List of Contact Personnel	1.2.1
01 20 00.00 22	01	Preconstruction Submittals	Schedule of Prices	1.3

Specification Section	SD #	SD Description	Item Submitted	Paragraph #
01 30 00.00 22	01	Preconstruction Submittals	NAVFAC Red Zone Checklist	1.6.1
01 30 00.00 22	01	Preconstruction Submittals	NAVFAC PWD ME Internal Service Requirements List	1.6.4
01 31 23.13 20	01	Preconstruction Submittals	List of Personnel (eCMS)	1.4.2
01 32 16.00 20	01	Preconstruction Submittals	Three-Week Look Ahead Schedules	1.6
01 32 16.00 20	01	Preconstruction Submittals	Outages Schedule	1.7
01 33 00	01	Preconstruction Submittals	Submittal Register	1.7
01 33 10.05 20	01	Preconstruction Submittals	Consolidated RFP Documents	2.1
01 33 00.05 20	01	Preconstruction Submittals	Submittal Register (DB)	1.5.5
01 50 00.00 22	01	Preconstruction Submittals	Construction Site Plan	1.4

The following Preconstruction submittals must be submitted to the Contracting Officer prior to the start of construction:

Specification Section	SD #	SD Description	Item Submitted	Paragraph #
01 50 00.00 22	01	Preconstruction Submittals	Traffic Control Plan	3.3.1
01 57 19.00 22	01	Preconstruction Submittals	Preconstruction	1.5.1
01 57 19.00 22	01	Preconstruction Submittals	Solid Waste Management Plan	3.4
01 57 19.00 22	01	Preconstruction Submittals	Regulatory Notifications	1.5.2
01 57 19.00 22	01	Preconstruction Submittals	Environmental Management Plan (EMP)	3.1
01 57 19.00 22	01	Preconstruction Submittals	Dirt and Dust Control Plan	3.14.1
01 57 19.00 22	01	Preconstruction Submittals	Contractor Hazardous Material Inventory Log	3.6

Specification Section	SD #	SD Description	Item Submitted	Paragraph #
01 57 19.00 22	01	Preconstruction Submittals	Storm Water Management/Erosion and Sedimentation Control Plan	3.2.1
01 57 19.00 22	01	Preconstruction Submittals	Spill Prevention, Control, and Countermeasures (SPCC) Plan	3.1.f.2
01 74 19	01	Preconstruction Submittals	Waste Management Plan	1.3

1.3 PACKAGING AND SUBMISSION OF SUBMITTALS

Prepare and submit submittals required by each individual Specification Section. Each required submittal as listed in the submittal register must be packaged and submitted individually so that it can be tracked, reviewed, and returned in a concise and orderly fashion.

1.4 SUBMITTAL CLASSIFICATION

Submittals are classified as follows:

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, deviations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government. The Government

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are considered to be "shop drawings."

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

1.5.1 Submittals Required from the Contractor

As soon as practicable after award of Contract, and before procurement of fabrication, forward to NAVFAC PWD ME submittals required in the technical Sections of this Specification, including shop drawings, product data, and samples. In addition, forward a copy of the submittals to the Contracting Officer.

The Architect-Engineer for this project and NAVFAC will review and approve

for the Contracting Officer those submittals reserved for Contracting Officer approval to verify submittals comply with the Contract requirements.

1.5.1.1 O&M Data

The Architect-Engineer for this project and NAVFAC will review and approve for the Contracting Officer O&M Data to verify the submittals comply with the Contract requirements; submit data specified for a given item within 30 calendar days after the item is delivered to the Contract site.

- a. In the event the Contractor fails to deliver O&M Data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M Data apply.

1.5.1.2 Submittals Reserved for NAVFAC MIDLANT Approval

- a. All fire protection and fire alarm systems submittals are to be reviewed by:

NAVFAC MIDLANT CI45 Fire Protection
Attn: NAVFAC FPE
Bldg Z-140, RM 126
9742 Maryland Avenue
Norfolk, VA 23511

1.6 PREPARATION

1.6.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

Use the transmittal form provided by the Contracting Officer at the Pre-Con for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms will be furnished to the Contractor. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the Contract drawings pertinent to the data submitted for each item.

1.6.2 Identifying Submittals

The Contractor's Quality Control Manager must prepare, review, and stamp submittals, including those provided by a Subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

- a. Project title and location.
- b. Construction Contract Number.

- c. Date of the drawings and revisions.
- d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.
- e. Section number of the specification by which submittal is required.
- f. Submittal description (SD) number of each component of submittal.
- g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission.
- h. Product identification and location in project.

1.6.3 Submittal Format

1.6.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the Contract Documents.

1.6.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the Contract drawings. Identify materials and products for work shown.

1.6.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government Contract Number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 3 by 4 inches on the right-hand side of each sheet for the Government disposition stamp.

1.6.3.3 Format of SD-03 Product Data

Present product data submittals for each Section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification Section number and paragraph number to which it pertains.

1.6.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract Documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in Contract Document units, on a separate sheet.

1.6.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.6.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will [not] be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.6.3.4 Format of SD-04 Samples

1.6.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by

11 inches and adequate to indicate color, texture, and material variations.

- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 4 by 4 feet.
- h. Sample Installation: 100 square feet.

1.6.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.6.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.6.3.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.6.3.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.6.3.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.6.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each Section. Include the manufacturer's name, trade name, place of manufacture, and catalog

model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.6.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.6.3.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.6.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format. Refer to Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI) for additional requirements.

1.6.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the Contract Documents.

1.6.4 Source Drawings for Shop Drawings

The entire set of Source Drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.6.4.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection

with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of duty to fully comply with the Contract Documents, including and without limitation, the need to check, confirm, and coordinate the work of all Contractors for the project. If the Contractor uses, duplicates, or modifies these electronic source drawing files for use in producing construction data related to this Contract, remove all previous indication of ownership (seals, logos, signatures, initials, and dates).

1.6.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to the electronic submittal, provide three (3) hard copies of the submittals. Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is both searchable and can be copied. If documents are scanned, Optical Character Resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or scan of a signature.

Email electronic submittal documents fewer than 10MB to an email address as directed by the Contracting Officer. Provide electronic documents over 10MB on an optical disc, or through an electronic file sharing system such as the DoD SAFE Web Application located at the following website:
<https://safe.amrdec.army.mil/safe/>.

Provide hard copies of submittals when requested by the Contracting Officer. Up to three (3) additional hard copies of any submittal may be requested at the discretion of the Contracting Officer, at no additional cost to the Government.

1.7 QUANTITY OF SUBMITTALS

Make use of electronic media for submittals to the greatest extent possible except for operation and maintenance manuals and associated submittals to be forwarded to the NAVFAC PWD ME located at Portsmouth Naval Shipyard, Kittery, Maine. Refer to Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION

AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (if applicable) for additional requirements.

1.7.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit three (3) sets of administrative submittals.

1.7.2 Number of Copies of SD-02 Shop Drawings

Submit six (6) copies of submittals of shop drawings requiring review and approval only by QC organization and seven (7) copies of shop drawings requiring review and approval by Contracting Officer.

1.7.3 Number of Copies of SD-03 Product Data

Submit in compliance with quantity requirements specified for shop drawings.

1.7.4 Number of Samples SD-04 Samples

- a. Submit two (2) samples, or two (2) sets of samples showing range of variation, of each required item. One (1) approved sample or set of samples will be retained by approving authority and one (1) will be returned to the Contractor.
- b. Submit one (1) sample panel or provide one (1) sample installation where directed. Include components listed in technical Sections or as directed.
- c. Submit one (1) sample installation, where directed.
- d. Submit one (1) sample of non-solid materials.

1.7.5 Number of Copies SD-05 Design Data

Submit in compliance with quantity requirements specified for shop drawings.

1.7.6 Number of Copies SD-06 Test Reports

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that must be submitted with QC reports.

1.7.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.7.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.7.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.7.10 Number of Copies of SD-10 Operation and Maintenance Data

Submit three (3) copies of O&M Data to the Contracting Officer for review and approval.

1.7.11 Number of Copies of SD-11 Closeout Submittals

Unless otherwise specified, submit two (2) sets of administrative submittals.

1.8 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the Contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.9 PROJECT SUBMITTAL REGISTER

1.9.1 Submittal Management

Prepare and maintain submittal register, as the work progresses. Do not change data which is output in columns (c), (d), (e), and (f) as delivered by Government; retain data which is output in columns (a), (g), (h), and (i) as approved. A submittal register showing items of equipment and materials for which submittals are required by the specifications is provided as an attachment. The attached Submittal Register may not be all inclusive and additional submittals may be required. **The Contractor must review the plans and specifications and ensure all required submittals are included in the Project Submittal Register which must be submitted to the Contracting Officer with the QC Plan and Project Schedule.**

Column (c): Lists specification Section in which submittal is required.

Column (d): Lists each submittal description (SD No. and type, e.g. SD-02 Shop Drawings) required in each specification Section.

Column (e): Lists one principal paragraph in specification Section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting project requirements.

Column (f): Indicate approving authority for each submittal.

The Contractor must track all submittals by maintaining a complete list, including completion of all data columns, including dates on which

submittals are received and returned by the Government.

1.9.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and project schedule. Verify that all submittals required for the project are listed and add missing submittals. The attached submittal register may not be complete. The Contractor must include all required submittals including any submittals that are required on the plans and specifications. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.9.3 Contractor Use of Submittal Register

Update the following fields with each submittal throughout Contract.

Column (b) Transmittal Number: List of consecutive Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.9.4 Approving Authority Use of Submittal Register

Update the following fields.

Column (b) Transmittal Number: List of consecutive Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.9.5 Action Codes

Entries for columns (j) and (o), are to be used as follows (others may be prescribed by Transmittal Form):

1.9.5.1 Government Review Action Codes

"A" - "Approved as submitted"; "Completed"

"B" - "Approved, except as noted on drawings"; "Completed"

"C" - "Approved, except as noted on drawings; resubmission required"; "Resubmit"

"D" - "Returned by separate correspondence"; "Completed"

"E" - "Disapproved (See attached)"; "Resubmit"

"F" - "Receipt acknowledged"; "Completed"

"G" - "Other (Specify)"; "Resubmit"

"X" - "Receipt acknowledged, does not comply with Contract requirements"; "Resubmit"

1.9.6 Delivery of Copies

Submit an updated submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.10 VARIATIONS

Variations from Contract requirements require both Designer of Record (DOR) and Contracting Officer approval pursuant to Contract Clause FAR 52.236-21 SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION and will be considered where advantageous to Government.

1.10.1 Considering Variations

Discussion of variations with the Contracting Officer before submission, after consulting with the DOR, will help ensure functional and quality requirements are met and minimize rejections and re-submittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the Contractor. When contemplating a variation which results in lower cost, consider submission of the variation as a Value Engineering Change Proposal (VECP).

Specifically point out variations from Contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.10.2 Proposing Variations

When proposing variation, deliver written request to the Contracting Officer, with documentation of the nature and features of the variation and why the variation is desirable and beneficial to the Government, including the DOR's written analysis and approval. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

1.10.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor, including its

Designer(s) of Record, warrants that this Contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.10.4 Review Schedule Extension

In addition to normal submittal Government review period of 20-30 working days depending on the quality of the submittal documents, a additional period of 14 working days will be allowed for the Government to consider submittals with variations.

1.11 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. The normal Government review period is 20-30 working days depending on the size & quality of the submittal documents. Incomplete submittals will be rejected for resubmission. Any delay claims for additional time or costs that are related to poor submittal quality will be rejected.

Allow an additional 20 calendar days for review and approval of submittals for refrigeration and HVAC control systems.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.
- b. Submittals required by the Contract Documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the Contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the Contract Documents but that have been omitted from the register or marked "N/A".
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further re-submittal is required.
- d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least 15 working days for submittals for QC Manager approval and 20 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.
- e. For submittals requiring review by a Government fire protection engineer, allow a review period, beginning when the Government receives the submittal from the QC organization, of 30 working days for return of submittal to the Contractor.

1.11.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with Contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification Sections, a notation "G," following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government approving authority.

1.11.2 Constraints

- a. Conform to provisions of this Section, unless explicitly stated otherwise for submittals listed or specified in this Contract.
- b. Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.
- c. When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.
- d. Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.

1.11.3 QC Organization Responsibilities

- a. Review submittals for conformance with project design concepts and compliance with Contract Documents.
- b. Process submittals based on the approving authority indicated in the submittal register.

(1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.

(2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.

- c. Ensure that material is clearly legible.
- d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.

(1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number [_____], is in compliance with the Contract

drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Certified by QC Manager _____, Date _____"
(Signature)

(2) When approving authority is the QC Manager, the QC Manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number [____], is in compliance with the Contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Approved by QC Manager _____, Date _____"
(Signature)

- e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.
- f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.
- g. Retain a copy of approved submittals and approved samples at the project site.

1.12 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received from the QC Manager.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with Contract Documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS herein and with comments and markings appropriate for action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Two (2) copies of the submittal will be retained by the Contracting Officer and the remaining copies of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.12.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved except as noted, resubmittal not required" authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved" or "disapproved," or "revise and resubmit" indicate an incomplete submittal or noncompliance with the Contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by the Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.13 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications; notice as required under the FAR clause entitled CHANGES, is to be given to the Contracting Officer. Notification shall be provided as indicated in para 1.13.1(b). Contractor is responsible for the dimensions and design of connection details and construction of work. Failure to point out deviations may result in the Government requiring rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and submission of the submittals in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

All revisions made to a previous submittal shall be clouded and specifically indicated in the transmittal to clearly delineate and identify what has changed. The Government has the right to reject revised submittals which do not cloud or delineate changes from previous submittals.

1.14 CONTRACTOR COORDINATION

The Contractor and their sub-contractors are responsible for confirming and correlating dimensions and weights at the project site for tolerances, clearances, quantities, fabrication processes, and techniques of

construction.

The contractor shall be responsible for the coordination of all submittals with related building elements. The contractor and their sub-contractors are responsible for coordination of the work with other trades and full compliance with contract documents.

Comprehensive coordination and integration of submittals between trades is required and must be completed. The Contractor will be responsible for all schedule delays and cost impacts resulting from changes and conflicts which have not been fully coordinated as part of the buy-out and shop drawing preparation.

1.14.1 COORDINATION DRAWINGS

Prepare Coordination Drawings in accordance to the requirements of this specification. Before work progresses, in addition to the shop drawings listed herein, coordination drawings shall be created and prepared by the HVAC and Sheetmetal Subcontractors in AutoCAD DWG electronic format. The Coordination Drawings, once completed by the HVAC and Sheetmetal contractors, shall be delivered to the Controls Contractor, Fire Protection Contractor, Structural Steel Contractor, Interior/Exterior Finishes Contractor, and lastly the Electrical Contractor for inclusion of their respective equipment and systems. Provide drawings in electronic format, one 3/8 inch scale reproducible and one 3/8 inch scaleable print of coordination drawings.

Prepare Coordination Drawings in accordance to the requirements of this specification and in addition at a minimum, adhere to the following:

1. Indicate temporary relocation, phasing, sequencing and moving of large equipment in the building during construction.
2. Floor plans and details, including the following:
 - a. HVAC Coordination Drawings shall include, as a minimum, all ductwork, insulation, filters, air-handlers, fans, terminal equipment, piping, pumps, isolation valves, control valves, thermostats, sensors, controllers, control cabinets, skids, platforms, and all other equipment installed under this Section showing the adjoining work of the other trades at all floor/enclosure levels. Refer to articles throughout this SECTION for additional coordination requirements.
 - b. HVAC Sub-contractor shall note apparent conflicts and suggest alternate solutions.
 - c. Composite systems coordination drawings showing how HVAC systems are to be installed where conflicts with the work of other trades may occur.
 - d. Access Door and Panel Coordination: Show sizes, locations, and required clearances of all access panels and doors on coordination drawings.
 - e. Conduit Layout Drawing: Show conduit routing for all systems (electrical, fire alarm, controls, security, etc.) on a plan. Identify any steel penetrations (as allowed in the contract documents) on this drawing. Sizes of conduit shall be color coded

and identified for each system. The Contractor is responsible for ensuring conduit is provided for all circuits/infrastructure as shown on the one-line diagrams, schematics including any circuits required for Contractor-selected equipment, RFI responses, ASI's, and Contractors means and methods.

- f. Piping Layout Drawing: Show piping layout for all systems including insulation and fittings and supports on a plan. Piping shall be color coded and identified for each system.

The Contractor, before transmittal of the Coordination Drawings to the Contracting Officer for approval, shall review with and obtain sign-off from the following contractors indicating that the work of their trade is fully coordinated (examples to be modified based on specific project requirements):

- a. General Contractor
- b. Metal Framing Sub-Contractors
- c. Mechanical Systems Sub-Contractors
- d. Sheetmetal Sub-Contractors
- e. Insulation Sub-Contractors
- f. ACT Sub-Contractors
- g. Testing and Balancing Sub-Contractors
- h. Air Duct and System Cleaning Sub-Contractors
- i. Piping/Compressed Air Contractors
- j. Electrical Sub-Contractors
- k. Low Voltage Wiring and Controls Sub-Contractors
- l. Fire Protection Sub-Contractors
- m. Fire Alarm Sub-Contractors
- n. Steel Sub-Contractor/Fabricators
- o. Doors and Openings Sub-Contractors
- p. Other Sub-Contractors as required

1.14.2 MEP Coordination Meetings

The Contractor shall schedule in person coordination meetings with the mechanical, electrical, piping, and fire protection trades on a monthly basis at a minimum. The intent of the meetings is to review the progress of the coordination drawings and coordinate work being installed.

Meeting minutes must be provided within 24 hours of completion of the MEP coordination meeting.

The QC organization will be responsible for leading the MEP Coordination Meetings with other trades encouraged to participate.

1.15 APPROVED SUBMITTALS

The Contracting Officer's approval or acceptance of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing and other information are satisfactory.

Approval or acceptance will not relieve the Contractor of the responsibility for any error which may exist, as the Contractor under the Contractor Quality Control (CQC) requirements of this contract is responsible for dimensions, the design of adequate connections and details,

and the satisfactory construction of all work.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.16 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any Contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those which may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the Contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this Contract, any further samples of the same brand or make of that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet Contract requirements will automatically void previous approvals. Replace such materials or equipment to meet Contract requirements.

Approval of the samples by the Contracting Officer does not relieve the Contractor of its responsibilities under the Contract.

1.17 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.18 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the Contract drawings and specifications except as otherwise stated."

_____NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 35 26.00 22

GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)

10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard and AOR Facilities. Specific information throughout the Section requires coordination and editing.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ASSP A10.34	(2001; R 2012) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2016) The Fall Protection Code
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ASSP Z359.12	(2009) Connecting Components for Personal Fall Arrest Systems
ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ASSP Z359.18	(2017) Safety Requirements for Anchorage

Connectors for Active Fall Protection Systems

- ASSP Z359.2 (2017) Minimum Requirements for a Comprehensive Managed Fall Protection Program
- ASSP Z359.3 (2019) Safety Requirements for Lanyards and Positioning Lanyards
- ASSP Z359.4 (2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
- ASSP Z359.6 (2016) Specifications and Design Requirements for Active Fall Protection Systems
- ASSP Z359.7 (2019) Qualification and Verification Testing of Fall Protection Products

ASME INTERNATIONAL (ASME)

- ASME B30.20 (2018) Below-the-Hook Lifting Devices
- ASME B30.22 (2016) Articulating Boom Cranes
- ASME B30.23 (2016) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
- ASME B30.26 (2015; R2020) Rigging Hardware
- ASME B30.3 (2020) Tower Cranes
- ASME B30.5 (2018) Mobile and Locomotive Cranes
- ASME B30.7 (2016) Winches
- ASME B30.8 (2015) Floating Cranes and Floating Derricks
- ASME B30.9 (2018) Slings

ASTM INTERNATIONAL (ASTM)

- ASTM F855 (2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 1048 (2016) Guide for Protective Grounding of Power Lines
- IEEE C2 (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2 (2011; R 2017) Environmental and Facility
Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10 (2022; ERTA 1-2 2018) Standard for
Portable Fire Extinguishers

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

NFPA 306 (2019) Standard for Control of Gas Hazards
on Vessels

NFPA 51B (2019) Standard for Fire Prevention During
Welding, Cutting, and Other Hot Work

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 70E (2021) Standard for Electrical Safety in
the Workplace

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222 (2018H; Add 1 2019) Structural Standard
for Antenna Supporting Structures and
Antennas and Small Wind Turbine Support
Structures

TIA-1019 (2012; R 2016) Standard for Installation,
Alteration and Maintenance of Antenna
Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20 Standards for Protection Against Radiation

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.146 Permit-required Confined Spaces

29 CFR 1910.147 Control of Hazardous Energy (Lock Out/Tag
Out)

29 CFR 1910.333 Selection and Use of Work Practices

29 CFR 1915 Confined and Enclosed Spaces and Other
Dangerous Atmospheres in Shipyard
Employment

29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1919	Gear Certification
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.1400	Cranes and Derricks in Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146
ATTACHMENT A	"CONTRACTOR CRANE, LHE, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS"
ATTACHMENT B	"NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM"
ATTACHMENT C	"CONTRACTOR DIG-SAFE CERTIFICATION FORM"
ATTACHMENT D	"CONTRACTOR UTILITY LOCATE - PRE-EXCAVATION SAFETY CHECKLIST"
ATTACHMENT E	"CONTRACTOR DAILY CHECKLIST FOR TRENCHING/EXCAVATION"

The attachments are included at the end of this Section. If the attachments are missing, notify the Contracting Officer.

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge, and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person

requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146 and 29 CFR 1926 Subpart AA, and designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the confined space program, who through training, knowledge, and experience in confined space entry is capable of identifying, evaluating, and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person, who has been designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the Crane and Rigging Program, who through training, knowledge, and experience in crane and rigging is capable of identifying, evaluating, and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the excavation/trenching program, who through training, knowledge, and experience in excavation/trenching is capable of identifying, evaluating, and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing, and monitoring of the fall protection program, who through training, knowledge, and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating, and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q, and designated in writing by the employer to be responsible for immediate supervising, implementing, and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge, and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented and include experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented, minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even though provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the

work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the requirements of EM 385-1-1 Appendix Q, and ASSP Z359.0, with a recognized degree or professional certificate and with extensive knowledge, training, and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);
- c. Restricted work;
- d. Transfer to another job;
- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above.

1.2.17 USACE Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload, and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, or bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap or accident using the NAVFAC prescribed Navy Crane Center (NCC) accident form.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

APP - Construction; G

Dive Operations Plan; G

SD-06 Test Reports

Monthly Exposure Reports; G

Notifications and Reports; G

Accident Reports; G

LHE Inspection Reports; G

SD-07 Certificates

Contractor Safety Self-Evaluation Checklist; G

Crane Operators/Riggers; G

Standard Lift Plan; G

Critical Lift Plan; G

Naval Architecture Analysis; G

Activity Hazard Analysis (AHA); G

Confined Space Entry Permit; G

Hot Work Permit; G

Certificate of Compliance; G

Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes; G

License Certificates; G

Radiography Operation Planning Work Sheet; G

1.4 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and Subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.5 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the pre-construction meeting. Complete the checklist monthly and submit with each request for payment voucher. An

acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Contractor Safety Self-Evaluation checklist can be found on the Whole Building Design Guide website at www.wbdg.org/ffc/dod/unifiedfacilities-guide-specifications-ufgs/ufgs-01-35-26

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and applicable Federal, State, and local laws, ordinances, criteria, rules, and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this Specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.6.1 Subcontractor Safety Requirements

For this Contract, neither Contractor nor any Subcontractor may enter into Contract with any Subcontractor that fails to meet the following requirements. The term Subcontractor in this and the following paragraphs means any entity holding a Contract with the Contractor or with a Subcontractor at any tier.

1.6.1.1 Experience Modification Rate (EMR)

Subcontractors on this Contract must have an effective EMR less than or equal to 1.10, as computed by the National Council on Compensation Insurance (NCCI) or if not available, as computed by the state agency's rating bureau in the state where the Subcontractor is registered, when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable EMR range cannot be achieved. Relaxation of the EMR range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain the certified EMR ratings for all Subcontractors on the project and make them available to the Government at the Government's request.

1.6.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate

Subcontractors on this Contract must have a DART rate, calculated from the most recent, complete calendar year, less than or equal to 3.4 when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The OSHA Dart Rate is calculated using the following formula:

$$(N/EH) \times 200,000$$

where:

N = number of injuries and illnesses with days away, restricted work, or job transfer

EH = total hours worked by all employees during most recent, complete

calendar year

200,000 = base for 100 full-time equivalent workers (working 40 hours per week, 50 weeks per year)

The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable OSHA Dart rate range cannot be achieved for a particular Subcontractor. Relaxation of the OSHA DART rate range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain self-certified OSHA DART rates for all Subcontractors on the project and make them available to the Government at the Government's request.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Personnel Qualifications

1.7.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one (1) person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the project site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs. Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents. The SSHO shall be a direct employee of the prime contractor, not an employee of a subcontractor.

If the SSHO is off-site for a period longer than 24 hours, an equally-qualified alternate SSHO must be provided and must fulfill the same roles and responsibilities as the primary SSHO. When the SSHO is temporarily (up to 24 hours) off-site, a Designated Representative (DR), as identified in the AHA may be used in lieu of an Alternate SSHO, and must be on the project site at all times when work is being performed. The DR may not relieve an Alternate SSHO. Note that the DR is a collateral duty safety position, with safety duties in addition to their full time occupation. The primary SSHO may not be absent from the site for more than two weeks at one time and not more than 30 work days in a calendar year.

1.7.1.2 Quality Control (QC) Manager:

The Quality Control Manager cannot be the SSHO on the project.

1.7.1.3 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Appendix Q and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the the Contracting Officer for information in consultation with the Portsmouth Naval Shipyard Safety Office.

1.7.1.3.1 Competent Person for Confined Space Entry

Provide a Confined Space Competent Person (CP) who meets the requirements of EM 385-1-1, Appendix Q, and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1, Section 34.

If work involves marine operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.147 for general industry.

1.7.1.3.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Section 22.B.02 and herein.

1.7.1.3.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Section 21.C.04 and herein.

1.7.1.4 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour Contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction, Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L. Instructors are required to:

- a. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five (5) years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.
- b. Update training course materials whenever an update of the EM 385-1-1 becomes available.
- c. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.
- d. Request, review, and incorporate student feedback into a continuous course improvement program.

1.7.1.5 Dredging Contract Requirements

1.7.1.5.1 SSHO Requirements for Dredging

- a. In addition to requirements stated elsewhere in this specification, an individual serving as a SSHO must be present at the project site, located so that they have full mobility and reasonable access to all major work operations, for at least one shift in each 24 hour period when work is being performed. The SSHO must be available during their shift for immediate verbal consultation and notification, either by phone or radio.
- b. The SSHO is a full-time, dedicated position, except as noted above, who must report to a senior project (or corporate) official. When the SSHO is permitted to be a collateral duty, the SSHO is not permitted to be in another position requiring continuous mechanical or equipment operations, such as equipment operators.
- c. The SSHO must inspect all work areas and operations during initial set-up and at least monthly observe and provide personal oversight on each shift during dredging operations for projects with many work sites, more often for those with less work sites.

1.7.1.5.2 Collateral Duty Safety Officer (CDSO) Requirements for Dredging

- a. A CDSO is an individual who is assigned collateral duty safety responsibilities in addition to their full-time occupation, and who supports and supplements the SSHO efforts in managing, implementing and enforcing the Contractor's Safety and Health Program. The assigned CDSO must be an individual(s) with work oversight responsibilities, such as master, mate, fill foreman, or superintendent. A CDSO must not be an employee responsible for continuous mechanical or equipment operations, such as an equipment operator.
- b. A CDSO performs safety program tasks as assigned by the SSHO and must report safety findings to the SSHO. The SSHO must document results of safety findings and provide information for inclusion in the CQC reports to the Contracting Officer.

1.7.1.5.3 Safety Personnel Training Requirements for Dredging

A SSHO and a CDSO for dredging Contracts must take either a formal classroom or online OSHA 30-hour Construction Safety Course, or an equivalent 30 hours of formal classroom or online safety and health training covering the subjects of the OSHA 30-hour Course in accordance with EM 385-1-1 Appendix A, paragraph 3.d.(3), applicable to dredging work, and given by qualified instructors. In exception to EM 385-1-1, Section 01.A.17, comply with the following:

- a. The SSHO must maintain competency through having taken 8 hours of formal classroom or online safety and health related coursework every year. Hours spent as an instructor in such courses will be considered the same as attending them, but each course only gets credit once (for example, instructing a 1-hour asbestos awareness course five times in a year provides one hour credit for training).
- b. The SSHO and a CDSO must have a minimum of three years of experience within the past five years in one of the following:

- (1) Supervising/managing dredging activities
- (2) Supervising/managing marine construction activities
- (3) Supervising/managing land-based construction activities
- (4) Work managing safety programs or processes
- (5) Conducting hazard analyses and developing controls in activities or environments with similar hazards

1.7.1.6 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1, Section 15.B for Riggers, and Section 16.B for Crane Operators and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 50,000 pounds or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.2 Personnel Duties

1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, and estimated and actual dates of corrections. Attach safety inspection logs to the Contractor's daily production report.
- b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and Subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300 on the site Safety Bulletin Board.
- d. Maintain applicable safety reference material on the job site.
- e. Attend the pre-construction meeting, pre-work meetings including preparatory meetings, and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement, and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure Subcontractor compliance with safety and health requirements.
- i. Maintain a list of hazardous chemicals on site and their associated Safety Data Sheets (SDS).

- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.
- k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, Subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out. The dismissal of the Superintendent, QC Manager, and/or SSHO will not be cause for claims for a Contract modification(s) for an extension to the Contract duration or for additional compensation.

1.7.3 Meetings

1.7.3.1 Preconstruction Meeting

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction meeting. This includes the Project Superintendent, Site Safety and Occupational Health Officer, Quality Control Manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program, and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed at the conference and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.7.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors on the project location. The SSHO, supervisors, foremen, or Collateral Duty Safety Officer's (CDSOs) must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8 ACCIDENT PREVENTION PLAN (APP)

[Note: Use the following paragraph for Design-Build projects only.]

1.8.1 APP - Design Team for Design-Build

Provide a site-specific Accident Prevention Plan (APP), including Activity Hazard Analyses (AHA), in accordance with EM 385-1-1 Appendix A, for the design team to follow during site visits and investigations. For subsequent visits, update the plan if there are changes in the personnel who will be attending, or the tasks to be performed. Submit the APP for review and acceptance by the Government at least 15 calendar days prior to the start of the design field work. Field work may not begin until the design APP is accepted by the Contracting Officer.

If the design scope includes borings or other subsurface investigations, include in the APP the type of field investigation and verification techniques, such as visual, local utility locating service scanning and third party/Subcontractor scanning, potholing, or hand digging within two feet of a known utility that will be required. Mark underground utilities before starting any ground-disturbing actions. Notify the Contracting Officer 15 days prior to the start of soil borings or sub-surface investigations.

Prior to the start of construction incorporate the Design APP into the Construction APP so that one site specific APP exists for the project and submit to the Contracting Officer for acceptance.

1.8.2 APP - Construction

A qualified person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, Appendix A, and as supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1, Appendix A. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential Subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of Subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the Subcontractors. Contractor's are responsible for informing their Subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one trade/craft from interfering with or creating hazardous working conditions for other trades/crafts, and inspecting Subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each Subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction meeting for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, Project Superintendent, SSHO, and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within twenty-four (24) hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.8.3 Names and Qualifications

Provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience, and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.8.4 Plans

Provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.8.4.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other Federal, State, and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.8.4.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed, and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three (3) months.

1.8.4.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. In addition, Critical Lift Plans are required for the following:

- a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.
- b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.
- c. For lifts with anticipated binding conditions.
- d. When erecting cranes.

1.8.4.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

1.8.4.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, Section 16.H.02, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1, Section 16.T.

1.8.4.4 Barge Mounted Mobile Crane Lift Plan

Provide a Naval Architecture Analysis and include an LHE Manufacturer's Floating Service Load Chart in accordance with EM 385-1-1, Section 16.L.03.

1.8.4.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.8.4.6 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Section 21.D and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 6 feet. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six (6) months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems, or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.8.4.7 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Section 21.N and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.8.4.8 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Section 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Section 12.A.02. Ensure that each employee is familiar with and complies with these procedures.

1.8.4.9 Excavation Plan

Identify the safety and health aspects of excavation and provide and prepare the plan in accordance with EM 385-1-1, Section 25.A and Section 31 23 00.00 22 EXCAVATION AND FILL (PWD ME).

1.8.4.10 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium, and chromium, including occupant protection and prepare in accordance with Section 02 83 00 LEAD, CADMIUM, AND CHROMIUM REMEDIATION (PWD ME).

1.8.4.11 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION (PWD ME).

1.8.4.12 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with

Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.8.4.13 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

1.8.4.14 Lighting Lamps and Ballasts Containing PCBs and Mercury Plan

Identify the safety and health aspects of Polychlorinated Biphenyls and Mercury work, and prepare in accordance with Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBs AND MERCURY.

1.8.4.15 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION and referenced sources. Include engineering survey as applicable.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or Subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, Subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager, and the Subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Section 1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel, and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task, or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure

a clear understanding of the AHA and its contents.

1.10 DISPLAY OF SAFETY INFORMATION

1.10.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1, Section 01.A.07. Additional items required to be posted include:

- a. Confined space entry permit.
- b. Hot work permit.

1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and must be updated daily, and provide the following information:

- a. Date deficiency identified;
- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;
- d. Projected resolution date; and
- e. Date actually resolved.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS AND REPORTS

1.13.1 Mishap Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four (24) hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Appendix Q, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the Contracting Officer as soon as practical but not more than four

(4) hours after the mishap.

The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving (if applicable). These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

1.13.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable NAVFAC Contractor Significant Incident Report (CSIR). The Contracting Officer will provide copies of any required or special forms.
- b. Near Misses: The Contracting Officer will provide the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.
- c. Conduct an accident investigation for any load handling equipment accident (including rigging gear accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Gear) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a FORM 16-1 Certificate of Compliance for LHE entering the [Activity] [Portsmouth Naval Shipyard] under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Section 16.H.03 using Form 16-2 Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15

calendar days in advance of planned lift.

Provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

1.13.5 Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes

Floating cranes and barge-mounted mobile cranes used to perform work under the terms of this Contract must be certified in accordance with 29 CFR 1919 by an OSHA accredited person prior to submitting the required Lift Plan. Include proof of certification with the initial Lift Plan submission.

1.14 HOT WORK

1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit no later than five (5) working days prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the [Activity][Portsmouth Naval Shipyard]Fire Department. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two (2) 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" performed at the Portsmouth Naval Shipyard. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Portsmouth Naval Shipyard Fire Department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE PORTSMOUTH NAVAL SHIPYARD FIRE DEPARTMENT AND THE CONTRACTING OFFICER IMMEDIATELY.

a. Duration of Hot Work

At the discretion of the [Portsmouth Naval Shipyard Fire Inspectors, a hot work permit may be written for up to one work week, (Monday through Friday), provided work is being performed at one specific location and will not hinder any life safety code.

Hot work permits must be issued for five (5) work days with a time period not to exceed twelve (12) hours. Request for weekend permits must be submitted at least two (2) working days in advance of the work to be performed and will only be valid for the weekend requested.

b. Hot Work Within a Confined Space

Hot work permits within a confined space will be issued on a daily basis with a time period not to exceed a twelve (12) hour shift. All requests for hot work permits within a confined space must be received

a minimum of two (2) complete working days prior to entry. A duplicate copy of the air monitoring results will be furnished to the fire inspector at time of issuance.

1.14.2 Work Around Flammable Materials

Obtain services from a NFPA Certified Marine Chemist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Section 06.H.

1.15 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO), and Contracting Oversight Technician (COT) for all specialized and licensed material and equipment proposed for use on the construction project. Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.15.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices for all equipment to be used on Portsmouth Naval Shipyard. (PNSY)

The Contracting Officer and COT will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.15.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer and COT for all radiological materials and equipment containing ionizing radiation that are proposed for use on a Government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Portsmouth Naval Shipyard. The Navy COT or Government authorized representative will meet the Contractor at a designated location outside the Portsmouth Naval Shipyard, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Portsmouth Naval Shipyard, to the job site, and off the Portsmouth Naval Shipyard.

Provide a copy of all calibration records, and utilization records to the

COT for radiological operations performed on the site.

1.15.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Portsmouth Naval Shipyard Security Department Emergency Number.

1.15.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.15.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in a Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.15.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.15.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must be scheduled only upon final approval from the local COTS or RSO Representative.

1.15.8 Transmitter Requirements

Adhere to the Portsmouth Naval Shipyard policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.16 CONFINED SPACE ENTRY REQUIREMENTS.

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive

CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. [Contractors entering and working in confined spaces while performing shipyard industry work are required to follow the requirements of OSHA 29 CFR 1915 Subpart B.]

1.16.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Section 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.16.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.16.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.16.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.17 HIGH NOISE LEVEL PROTECTION

Operations that involve the use of equipment with output of high noise levels (i.e. jackhammers, air compressors, and explosive-actuated devices) must be coordinated with the Contracting Officer. Use of any such equipment outside normal working hours must be approved in writing by the Contracting Officer prior to commencement of work.

1.18 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) for review and acceptance at least 30 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

The following documents are required for all Contractor diving operations. All documents must be reviewed and found acceptable by two (2) of the following: DDC/ADC/DSR, prior to start of diving operations. Contractors must submit the documents to the DDC through the Contracting Officer. Additional documentation may be required depending on the scope of the diving operation:

- a. Safe Practices Manual: See EM 385-1, Section 30.A.15.
- b. Dive Operations Plan(s): See EM 385-1, Section 30.A.16.

- c. AHA to cover all aspects of the job: See EM 385-1, Section 30.A.17.
- d. Emergency Management Plan: See EM 385-1, Section 30.A.18.
- e. Dive Personnel Qualifications: See EM 385-1, Sections 30.A.05 - 30.A.09.

1.19 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

2.1 CONFINED SPACE SIGNAGE

Provide permanent signs integral to or securely attached to access covers for new permit-required confined spaces. Signs for confined spaces must comply with NEMA Z535.2. Signs wording: "DANGER--PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" in bold letters a minimum of one inch in height and constructed to be clearly legible with all paint removed. The signal word "DANGER" must be red and readable from 5 feet.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and Portsmouth Naval Shipyard fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory minimum PPE includes:

- a. Hard Hat
- b. Long Pants
- c. Appropriate Safety Shoes
- d. Appropriate Class Reflective Vests
- e. Gloves
- f. Safety Glasses

g. Hearing Protection

h. Fall Protection

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones, or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. An employee check-in/check-out communication procedure must be developed to ensure employee safety.

3.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Officer or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

3.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury, polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on Portsmouth Naval Shipyard.

3.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If the material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If the material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

See Section 01 14 00.00 22 Paragraph 1.8 for Outage requirements.

3.3 UTILITY OUTAGE SAFETY REVIEW MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage safety review meeting in accordance with EM 385-1-1, Section 12.A, Hazardous Energy Control Program HEC. This meeting must include the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, and the NAVFAC PWD ME UEM representative. All parties must fully coordinate HEC activities with one another. During the safety review meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Section 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph entitled HAZARDOUS ENERGY CONTROL PROGRAM (HECP) herein.

Contracting Officer will, upon request, apply lockout/tag-out tags and take other actions that, because of experience and knowledge, are known to be necessary to make the particular equipment safe to work on.

No person, regardless of position or authority, must operate any switch, valve, or equipment that has an official lockout/tag-out tag attached to it, nor can such tag be removed except as provided in this Section.

No person must work on any equipment that requires a lockout/tag-out tag unless he/she, his/her immediate supervisor, project leader, or a subordinate has in his/her possession the stubs of the required lockout/tag-out tags.

When work is to be performed on electrical circuits, only qualified personnel are to perform work on electrical circuits.

A supervisor who is required to enter an area protected by a lockout/tag-out tag will be considered a member of the protected group provided he/she notifies the holder of the tag stub each time he/she enters and departs from the protected area.

Portsmouth Naval Shipyard and NAVFAC Personnel use a red lock and a red tag to indicate personnel are working on the systems. Use of a red lock and a red tag is highly encouraged to maintain continuity throughout the

Portsmouth Naval Shipyard. The use of another colored locks and tags (blue for Portsmouth Naval Shipyard workers and Yellow for NAVFAC personnel) indicate that the system is out of service for some reason.

Identification markings on building light and power distribution circuits must not be relied on for established safe work conditions.

Before clearance will be given on any equipment other than electrical (generally referred to as mechanical apparatus), the apparatus, valves, or systems must be secured in a passive condition with the appropriate vents, pins, and locks.

Pressurized or vacuum systems must be vented to relieve differential pressure completely.

Vent valves must be tagged open during the course of the work. (PNSY)

Where dangerous gas or fluid systems are involved, or in areas where the environment may be oxygen deficient, system or areas must be purged, ventilated, or otherwise made safe prior to entry.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECF. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Sections 05.I and 11.B. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Section 12.E.06.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECF. In accordance with EM 385-1-1, Section 12.E.08, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.4.4 Tag Placement (PNSY)

Lockout/tag-out tags must be completed in accordance with the regulations printed on the back thereof and attached to any device which, if operated, could cause an unsafe condition to exist.

If more than one group is to work on any circuit or equipment, the employee in charge of each group must have a separate set of lockout/tag-out tags completed and properly attached.

When it is required that certain equipment be tagged, the Government will review the characteristics of the various systems involved that affect the safety of the operations and the work to be performed; take the necessary actions, including voltage and pressure checks, grounding, and venting, to make the system and equipment safe to work on; and apply such lockout/tag-out tags to those switches, valves, vents, or other mechanical devices needed to preserve the safety provided. This operation is referred to as "Providing Safety Clearance."

3.4.5 Tag Removal (PNSY)

When any individual or group has completed its part of the work and is clear of the circuits or equipment, the supervisor, project leader, or individual for whom the equipment was tagged must turn in his/her signed lockout/tag-out tag stub to the Contracting Officer. That group's or individual's lockout/tag-out tags on equipment may then be removed on authorization by the Contracting Officer.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment, and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Sections 21.A and 21.D.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21.C. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21.C.04 and ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Section 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Section 21.I, 29 CFR 1926.500

Subpart M, ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, ASSP Z359.15, ASSP Z359.16, and ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection

In addition to the required fall protection systems, other protection measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Sections 21.0 through 21.0.06. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Harnesses

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabiners must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21.I.06.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

- (1) For work within 6 feet of an edge, on a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1, Section 21 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.
- (2) For work greater than 6 feet from the unprotected roof edge, in addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1, Section 21.L.

- b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with EM 385-1-1, Section 21.I.08.d.2. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified, and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install, and use guardrails and safety nets in accordance with EM 385-1-1, Section 21.F.01 and 29 CFR 1926 Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of EM 385-1-1, ASSP Z359.2, and ASSP Z359.4.

3.6 SHIPYARD REQUIREMENTS

All personnel who enter the Controlled Industrial Area (CIA) must wear mandatory personal protective equipment (PPE) at all times and comply with PPE postings of shops both inside and outside the CIA.

3.7 WORK PLATFORMS

3.7.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.

- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.
- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than six feet.
- k. Delineate fall protection requirements when working above six feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.7.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit a worker from climbing out of the basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWPs must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWPs must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.8 EQUIPMENT

3.8.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.8.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the [Activity] [Portsmouth Naval Shipyard] but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in EM 385-1-1, Section 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the [Activity,][Portsmouth Naval Shipyard,]in accordance with EM 385-1-1, Section 16.A.02, so that necessary quality assurance spot checks can be coordinated. [Prior to cranes entering Federal activities, a Crane Access Permit must be obtained from the Contracting Officer. A copy of the permitting process will be provided at the Preconstruction Meeting.]Contractor's operator must remain with the crane during the spot check. Rigging gear must comply with OSHA ASME B30.9 Standards safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.
- d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices, and ASME B30.26 for rigging hardware.
- e. Under no circumstance make a lift at or above 90 percent of the cranes rated capacity in any configuration.
- f. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Section 11, and ASME B30.5 or ASME B30.22 as applicable.
- g. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.

- h. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.
- i. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- j. Use cribbing when performing lifts on outriggers.
- k. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- l. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching, or crushing personnel.
- m. Maintain inspection records in accordance by EM 385-1-1, Section 16.D, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- n. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Section 16.F, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- o. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- p. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, the rigger, and the lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- q. Follow FAA guidelines when required based on project location.

3.8.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.
- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.8.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ASSP A10.22.
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.

- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 Section 16.B.05 and trained, at a minimum, in accordance with EM 385-1-1 Section 16.U and 16.T. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.
- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8.5 Use of Explosives

Use of explosives is not allowed.

3.9 EXCAVATIONS AND UTILITY LOCATING

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.9.1 Utility Locations

A. General

Excavation or ground penetrating work is defined as any operation in which earth, rock or other material below ground is moved or otherwise displaced, by means of power and hand tools, power equipment which includes grading, trenching, digging, boring, auguring, tunneling, scraping and cable or pipe driving except tilling of soil, gardening or displacement of earth, rock or other material for agricultural purposes. Removal of bituminous concrete pavement or concrete is not considered excavation. Ground penetrating work may include but is not limited to installing fence posts, probes, borings, piles, sign posts, stakes or anchor rods of any kind that penetrates the soil more than 3". The "Excavator" is defined as the person directly responsible for performing the excavation or ground penetrating work.

B. Underground Utilities Location

The Contractor/Excavator shall fully comply with the State of Maine "DIG SAFE" law (Title 23, MRSA 3360-A). Existing underground utilities shown on the plans are based on PNS total map and are shown in their approximate locations only.

The Excavator shall pre-mark the excavation area in "White Paint Only". (Field notes may be done in Pink paint). The Excavator shall notify "DIG SAFE" (1-888-344-7233) at least within 14 calendar days, but no more than 30 calendar days prior to the commencement of the excavation or ground penetrating activity.

The Prime Contractor / Excavator shall prepare a NAVFAC Dig Safe Utility Locate Request Form (Attachment B) at least within 14 calendar days prior to the commencement of the excavation or ground penetrating activity and submit the Form to the CM/ET. (The NAVFAC Dig Safe Utility Locate Request Form Attachment B is attached at the end of this Section.)

The Government will locate and mark the underground utilities within 14 calendar days of receiving the Dig Safe Notification.

Excavation or ground penetrating activities cannot commence until the utilities have been marked in the field and the NAVFAC Dig Safe Utility Locate Request Form has been returned indicating the PWD ME Dig Safe review process has been completed and excavation has been approved by the Contracting Officer and the PWD ME Dig Safe Coordinator.

If the excavation or ground penetrating activities do not commence within 27 days of Dig Safe notification or the excavation work is expanded outside the location originally specified in the notification, the Excavator shall renotify Dig Safe, the Contracting Officer and the PWD ME Dig Safe Coordinator.

The Contractor shall maintain the utility markings through out the excavation period at the Contractor's expense.

The Contractor shall contact the PWD ME Dig Safe Coordinator (DSC) if there are any questions regarding the underground utilities or the Dig Safe process.

C. Third Party Utility Locate

The Contractor must provide the services of a third party, qualified independent utility locating company/person(s) (Cannot be the Government's Utility Locating Firm) to positively identify underground utilities in the work area. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

The Third Party review must be completed in conjunction with the PWD ME Dig Safe marking process. Once the Third Party Locate Company has completed their review of the excavation area and the review of the Government markings are confirmed, the Third Party Locate Company & Contractor must sign the Contractor Dig-Safe Certification Form (Attachment C) and submit the form to the Contracting Officer/DSC prior to commencing excavation. If the Third Party Locate Company finds any discrepancies with the Government's utility markings, the Contractor must notify the Contracting Officer/DSC immediately and work with the DSC to resolve the discrepancy **PRIOR TO THE START OF EXCAVATION**.

3.9.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within three (3) feet of the underground system.

3.9.3 Utilities Within and Under Concrete (roofs, walls, floors, etc) Bituminous Asphalt, and Other Impervious Surfaces (Including Dry Dock and Berthing Walls).

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. This will require utility locating **both before and after** concrete slab removal when excavation continues below bottom of slab. Whenever Contract work involves chipping, saw cutting, drilling or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with Portsmouth Naval Shipyard UEM department in addition to location and depth verification by a third party, independent, private locating company.

The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement. Any markings made during the utility investigation must be maintained by the Contractor throughout the Contract.

3.9.4 Excavating with Hazardous or Mission Critical Utilities Within the Excavation Area (as identified by DSC or plans).

- a. The Contractor must employ supplemental daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan to confirm all utilities have been clearly marked and reviewed with the SSHO. The SSHO & the Excavator must complete and sign CONTRACTOR UTILITY LOCATE - PRE-EXCAVATION SAFETY CHECKLIST (Attachment D). This checklist must be reviewed and signed by the PWD ME ET prior to the commencing any excavation/trenching work.
- b. The Contractor must provide a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility.
- c. The SSHO must complete a pre-excavation walk as part of the morning procedure to help ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint. The Excavation/Trenching Competent person must complete the Contractor Daily Checklist for Trenching/Excavation included in Attachment E. The Daily Checklist must be completed prior commencing excavation/trenching work and must be submitted with the CM/ET daily.
- d. Contractor must provide additional danger signage, to mark areas of known live underground utilities.
- e. Contractor must ensure a 'spotter' accompanies the equipment operator during excavation work.
- f. Contractor to provide Construction CM/ET notification no later than 7 working days prior to holding the preparatory and initial pre-excavation/demo safety review meeting. The CM/ET, as well as the PWD ME DSC, **WILL** attend this meeting to ensure safety and preservation of critical utilities is discussed with adequate focus.
- g. The Contractor must confirm and identify the closest utility isolation

points and develop mitigation strategies with the utility owner (Coordinate with the PWD ME DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.

- h. All construction activities (including excavation adjacent to Critical Infrastructure) and outages directly impacting or within three feet of shipyard critical infrastructure are required to have a Critical Infrastructure Risk Analysis and Risk Mitigation plan (CI Risk Mitigation Plan). The Contract Documents identify the Critical Infrastructure that could be impacted by the work associated with this project.

The CI Risk Mitigation Plan (See Example below) is the Contractor's responsibility. General information related to Project Specific Outage Phasing & Sequencing may be provided in the Contract Documents. See Paragraph 1.8 for Outage Requirements.

The Contractor's proposed means and methods related to any work that may impact shipyard critical infrastructure are subject to Government approval. The anticipated phasing & sequencing for completing this work may be outlined in the Drawings. Any additional work and costs associated with any changes to the phasing & sequencing based on the Contractor's means & methods are the responsibility of the Contractor and will not be a basis for a claim for additional cost and time. The Contracting Officer will require approval of each CI Risk Mitigation Plan prior to work, and when associated with an outage, it must be submitted along with the Outage Request.

DEFINITION:

a. "Critical Infrastructure" are shipyard facilities (utilities and structures) that directly affect personnel safety, radiological controls, nuclear and reactor safety, and ship safety and require unhindered reliable operation. This definition is intended to be used in a conservative manner when determining if work impacts Critical Infrastructure, and must be reviewed by the Contracting Officer representatives through appropriate Shipyard personnel in those cases when there is any question as to the applicability.

i. Critical structures (and their foundation systems) include: dry dock structures, pump well structures, caissons as well as caisson seat and apron, submarine piers, reactor servicing complexes, crane and rail road rails, and bridges 1 and 2. Also, Radiological Material Storage facilities and Emergency Control Center. All work within three feet of a dry dock pump well and dry dock (when a submarine is docked) must follow critical shipyard infrastructure requirements.

ii. Critical utilities/services support dry docks and piers with submarines, reactor servicing complexes, Radiological Material Storage facilities, and Emergency Control Center including: dry dock pumping systems (drainage, dewatering, and ASW- both temporary and permanent systems) high voltage electrical systems and low voltage electrical systems, communication systems, high pressure air systems, low pressure air (breathing air), gas lines to the power plant, fresh water, firefighting water supply, and high pressure steam.

CRITICAL INFRASTRUCTURE RISK ANALYSIS (RA) AND RISK MITIGATION PLAN (RMP)

(Example additional information may be required to address the RA & RMP requirements)

1. Work: (describe in enough detail to allow someone not involved to have a good general understanding of scope):

2. Risks to Critical Infrastructure:

A. Describe what could go wrong with the work and the Impact to shipyard operations (Air, Water, Electrical Power, Telecom/IT, Rail or Structures).

i. Risk Assessment: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact. See RISK DETERMINATION TABLE - in the Outage Form.)

ii. Mitigated Risk Assessment: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below.

B. What could go wrong #2:

i. Risk Assessment #2: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact)

ii. Mitigated Risk Assessment #2: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below. State why or how the risk is reduced.

3. Work Control Plan:

A. Describe specific steps and actions (how, when, and by who) to mitigate the risks to critical infrastructure that were identified above. (examples only shown below)

i. Specific utilities/services in the area will be secured prior to work, or, potholing/GPR or other utility/service locating has been performed to identify specific utility/service location and depth, which is included in Preparatory Phase meeting agenda.

ii. Protective timber matting installed over crane and train rail to prevent damage to rails supporting train and crane movements.

iii. Noise/Dust/Vibration. Managed per contract requirements, and to be briefed during Initial Phase meeting.

iv. Detailed work plan.

v. Specific selection of equipment/tools for work.

vi. Foreman/safety pre-job reviews.

vii. Foreman pre-job work brief with crew.

4. Oversight: Describe specific oversight work leading up to and for the execution of work (how, when, and by who) to mitigate the risks to critical infrastructure that were identified above. (examples only shown below)

- A. Daily Contractor Production Reports.
 - B. Daily Contractor Quality Control (QC) Reports.
 - C. QC Manager field visits (include periodicity).
 - D. Periodic project Coordination Meeting (include parties involved).
- i. If there is a utility strike, the Contractor must stop all work and notify the Contracting Officer's Representatives immediately. The Contractor must complete the Underground Utility Strike Review Checklist Attachment F with 2 working days of the Utility Strike. See requirements in Paragraph 1.13.1 Mishap Notifications & 1.13.2 Accidents Reports for additional requirements.3.10 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Sections 11 and 12.

3.10.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Cables shall be verified as de-energized by using a spiking tool, or equivalent insulation displacement voltage verification method to confirm de-energized state. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all medium and high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves, and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

3.10.2 Qualifications

Electrical work must be performed by QP personnel with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National, and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State and Local requirements applicable to where work is being performed.

3.10.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.10.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.10.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification, and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

3.10.6 Portable Extension Cords

Size portable extension cords in accordance with manufacturer ratings for the tool to be powered and protected from damage. Immediately remove from service all damaged extension cords. Portable extension cords must meet the requirements of EM 385-1-1, NFPA 70E, and OSHA electrical standards.

-- End of Section --

SECTION 01 35 26 – ATTACHMENT A

CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1 CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.1 The following is a list of requirements that contractors shall comply with for all contracts that may result in the use of a category 1 or 4 crane, multi-purpose machines, forklifts, construction equipment, and rigging gear when used on Navy property to lift suspended loads. Non-compliance with the requirements of this instruction may result in denial of access, stopping of operations, or removal from Navy property.

1.2 References:

1.2.1 NOT USED

1.2.2 American Society of Mechanical Engineers (ASME) B30.3 (tower cranes), B30.5 (mobile cranes), B30.8 (floating cranes), B30.9 (slings), B30.20 (below the hook lifting devices), B30.22 (articulating booms), B30.26 (rigging hardware); ANSI/ITSDF B56.6 (rough terrain forklifts); Safety Standards for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings

1.2.3 CFR, Title 29, Chapter XVII, Part 1917, Marine Terminals.

1.2.4 CFR, Title 29, Chapter XVII, Part 1926, Safety and Health Regulations for Construction

1.2.5 CFR, Title 29, Chapter XVII, Part 1915, Occupational Safety and Health Standards for Shipyard Employment

1.2.6 OPNAVINST 5100.23, Navy Safety and Occupational Health Program Manual

1.2.7 EM 385-1-1, Safety and Health Requirements Manual, U.S. Army Corps of Engineers

1.2.8 NAVFAC Guide Specification NFGS-01525D, Safety Requirements

1.3 These requirements are solely intended to provide for the protection of Government property and personnel and are not intended to, and do not, in any manner whatsoever, relieve the contractor of its responsibility, including, without limitation, its responsibility for the protection of its equipment and personnel.

1.4 Notification Requirement: Contractor shall notify the Contracting Officer 7 calendar days in advance of the intent of bringing a non-Navy owned crane onto Navy property or of any multi-purpose machines, material handling equipment, or construction equipment that may be used in a crane-like application to lift suspended loads. The contractor shall also specify when crane entry onto Navy property is scheduled during back shift, weekend, or holiday hours of operation. All entries shall be through a prearranged entry point. The following documentation shall be provided along with notification: a copy of Form 16-1 and objective evidence of operator qualifications for cranes with rated capacities of 2,000 lbs. or greater. Failure to schedule or provide necessary documentation may result in the crane being denied access to the facility.

1.5 The contractor shall comply with applicable reference 1.2.2 standards (e.g., B30.3 for construction tower cranes, B30.5 for mobile cranes, B30.8 for floating cranes, B30.9 for slings, B30.20 for below the hook lifting devices, and B30.22 for articulating boom cranes), B30.26 for rigging hardware, and ANSI/ITSDF B56.6 for rough terrain forklifts). For barge mounted mobile cranes, require a third party certification from an OSHA accredited organization (or from a state accredited organization for those states with OSHA approved state plans), a load indicating device, a wind-indicating device, and a marine type list and trim indicator readable in one-half degree increments. Third party certification is not required for barge-mounted mobile cranes at naval activities in foreign countries.

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CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.6 Certification of Compliance (reference 1.2.1): The contractor shall complete a certificate of compliance that the crane (or other machine if used to lift suspended loads) and rigging gear meet applicable OSHA and ANSI/ASME regulations (with the contractor citing which OSHA regulations are applicable, e.g., cranes/multi-purpose machines used in cargo transfer shall comply with reference 1.2.3; cranes/multi-purpose machines used in construction, demolition, or maintenance shall comply with reference 1.2.4; cranes/multi-purpose machines used in ship repair shall comply with reference 1.2.5; slings shall comply with ASME B30.9; rigging hardware shall comply with ASME B30.26). For cranes (or other machine if used to lift suspended loads) and rigging equipment at naval activities in foreign countries, the contractor shall certify that the crane and rigging gear conform to the appropriate host country safety standards. The contractor shall also certify that all of its crane (or other machine) operators working on the naval activity have been trained not to bypass safety devices (e.g., anti-two block devices) during lifting operations, and that its operators, riggers, and company officials are aware of the actions required in the event of an accident as specified in the contract. Require that the certifications be posted on the crane. When a crane on Navy property is not authorized for use, the Certification of Compliance shall state, "Operation of this Crane is NOT Authorized."

1.7 The contractor shall certify (reference 1.2.1) that the crane or machine operator is qualified and trained for the operation of the crane to be used. For mobile and commercial truck mounted cranes with OEM rated capacities of greater than 2,000 pounds, the crane operator shall be designated as qualified by a source that qualifies crane operators (i.e., a union, a government agency, or an organization that tests and qualifies crane operators). Proof of current qualification shall be provided.

1.8 For multi-purpose machines, material handling equipment and construction equipment used to lift loads suspended by rigging equipment, the contractor shall have proof or authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. The contractor shall demonstrate that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.9 All hooks used on cranes, hoists, other machines, and rigging gear shall have self-closing latches or the throat opening shall be "moused" (secured with wire, rope, heavy tape, etc.) or otherwise secured to prevent the attached item from coming free of the hook under a slack condition. The following exceptions apply and shall be approved by the contractor's technical organization: items where the hook throat is fully obstructed and not available for manual securing and lifts where securing the hook throat increases the danger to personnel such as forge shop, dip tank, or underwater work.

1.10 Loading Limitations:

CAUTION: Piers and waterfront areas such as along dry docks and quay walls may have load restrictions.

1.10.1 The contractor shall notify the Contracting Officer prior to moving a crane on a pier, dry dock, or other waterfront area. Provide the Contracting Officer with the crane make, model, and configuration in which it is to be used.

1.10.2 The contractor shall comply with crane access routes and load limitations issued with the contract.

1.10.3 Allowable Surface Loads. Loads transferred to soils and pavements shall be minimized to a desired maximum of 3000 pounds per square foot, by placement of cribbing or steel pads under rubber-tired crane outriggers and trailer stanchions/sand shoes, or by placement of mats under treads of crawler cranes. Visually inspect areas adjacent to cribbing or plates and report any unusual bituminous pavement surface conditions, irregularities, or cracking to the Contracting Officer.

1.10.3.1 Outriggers of rubber-tired cranes shall be landed on two layers of timbers of appropriate thickness, oriented at right angles to each other, or landed on properly designed steel pads. Treads of crawler cranes shall run on appropriate mats. Use and design of cribbing, plates and mats shall be in a manner consistent with general construction industry standards.

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CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.10.3.2 Position loads that will remain on trailers detached from tractors to attain a distribution of 65 percent to rear axles and 35 percent to front support stanchions/sand shoes. For example, assuming an 83000 pound maximum gross weight and a soil bearing pressure of 3000 pounds per square foot, the required support under each sand shoe would be 2.5 feet x 2.5 feet. Accordingly, two tiers of timber cribbing at right angles, each 2.5 feet x 2.5 feet x 4 inches, or a properly designed 2.5 feet x 2.5 feet steel pad would be utilized under each trailer stanchion/sand shoe.

1.11 Prior to making any critical lift, the contractor shall provide a critical lift plan for each of the following lifts: lifts over 75 percent of the capacity of the crane, hoist, or other machine (50 percent of the capacity of a barge mounted mobile crane's hoists) at any radius of lift; lifts involving more than one crane, hoist, or other machine; lifts of personnel (lifts of personnel suspended by rigging equipment from multi-purpose machines, material handling equipment, or construction equipment shall not be permitted); lifts made in the vicinity of overhead power lines; erection of cranes; and lifts involving non-routine rigging or operation, sensitive equipment, or unusual safety risks. The plan shall include the following as applicable:

1.11.1 The size and weight of the load to be lifted, including crane (or other machine) and rigging equipment that add to the weight. The OEM's maximum load capacities for the entire range of the lift shall also be provided.

1.11.2 The lift geometry, including the crane (or other machine) position, boom length and angle, height of lift, and radius for the entire range of the lift. Applies to both single and tandem crane/machine lifts.

1.11.3 A rigging plan, showing the lift points, rigging equipment, and rigging procedures.

1.11.4 The environmental conditions under which lift operations are to be stopped.

1.11.5 For lifts of personnel, the plan shall demonstrate compliance with the requirements of reference

1.11.6 For barge mounted mobile cranes, barge stability calculations identifying crane placement/footprint; barge list and trim based on anticipated loading; and load charts based on calculated list and trim specific to the barge the crane is mounted on. The amount of list and trim shall be within the crane manufacturer's requirements.

1.11.7 For lifts in the vicinity of overhead power lines (i.e., if any part of the crane or other machine, including the fully extended boom of a telescoping boom crane or machine, or the load could approach the distances noted in figure 10-3 of reference 1.2.1 during a proposed operation), the plan shall demonstrate compliance to 29 CFR 1926.550(a)(15).

1.12 The following additional documentation is required for contractor provided tower cranes (those cranes defined by ASME B30.3).

1.12.1 Foundation design and requirements

1.12.2 Installation instructions 1.12.4

1.12.3 Assembly and disassembly instructions including climbing/jumping instructions if applicable

1.12.4 Operating manual, limitations, and precautions

1.12.5 Periodic inspection and maintenance requirements

1.13 Crane and Rigging Gear Accident Reporting and Record Keeping: Contractor's operating cranes on Navy property shall report all WHE accidents that occur incidental to an operation, project, or facility as

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CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

prescribed by paragraphs (1.10.1) through (1.10.3) requirements below. Contractors shall report directly to their respective Contracting Officer. There are two general categories of accidents as defined below. Crane accidents are those that occur during operation of a crane. Rigging gear accidents are those that occur when gear is used by itself in weight handling operation i.e., without a crane.

1.13.1 Crane Accident: For the purpose of this definition, it is assumed there is an "operating envelope" around any crane, and inside the envelope are the following elements:

- The crane
- The operator
- The rigger(s) and crane walker
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The rigging gear between the hook and the load
- The load
- The crane's supporting structure (ground, rail, etc.)
- The lift procedure

1.13.1.1 Definition: A crane accident occurs when any one or more of the six elements in the operating envelope fails to perform correctly during operation, including operation during maintenance, or testing resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process in lieu of these requirements.
- Material or equipment damage
- Dropped load
- Derailment
- Two-blocking
- Overload (This includes load tests when the test load tolerance is exceeded.)
- Collision, including unplanned contact between the load, crane, and/or other objects.

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, roll over, etc.). [Bullets] 3, 4, 5, 6, and 7 are considered crane accidents even though no material damage or injury occurs.

Exception: If a crane is used as an anchor point for a portable hoist/rigging gear, rigging gear accident as defined in paragraph 1.10.2 below is not considered a crane accident if the crane is not being operated (no functions are in motion) at the time of the rigging gear accident, unless the accident results in an overload or damage to the crane, in which case it shall be reported as a crane accident.

1.13.2 Rigging Gear Accidents: For the purpose of this definition, it is assumed there is an "operating envelope" around any weight handling operation, and inside the envelope are the following:

- Rigging gear and miscellaneous equipment
- The user of the gear or equipment
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The load
- The gear or equipment's supporting structure
- The load's rigging path

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CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

- The rigging procedure

1.13.2.1 Definition. A rigging gear accident occurs when any one or more of the five elements in the operating envelope fails to perform correctly during weight handling operations resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.
- Material or equipment damage that requires the damaged item to be repaired because it can no longer perform its intended function. This does not include superficial damage such as scratched paint, damaged lagging, or normal wear on rigging gear.
- Dropped load.
- Two-blocking of cranes and powered hoists.
- Overload. (This includes load tests when the test load tolerance is exceeded.)

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped load, damaged load, etc.). [Bullets] 3, 4, and 5 are considered accidents even though no material damage or injury occurs.

1.13.3 The contractor shall notify the Contracting Officer as soon as practical, but not later than four hours, after any WHE accident. The contractor shall secure the accident site and protect evidence until released by the Contracting Officer. The contractor shall conduct an accident investigation to establish the root cause(s) of the accident. Crane operations shall not proceed until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The contractor shall provide the Contracting Officer within 30 days of any accident a Crane and Rigging Gear Accident Report using the form provided in reference 1.2.1 consisting of a summary of circumstances, an explanation of causes(s), photographs if available, and corrective actions taken. These notifications and reporting requirements are in addition to those promulgated by reference 1.2.6 and related claimant instructions.

1.14 Each contractor shall perform the following actions prior to conducting crane operations on Navy property:

1.14.1 Inspection Requirements: It shall be the sole responsibility of the contractor to assure the Contracting Officer and/or designated Navy personnel that the crane and associated rigging gear are in good working order and safe for use.

1.14.1.1 Crane Inspection: Perform pre-operational inspection of the crane in the presence of a representative of the Contracting Office of the crane prior to starting work on Navy property. Inspection shall meet all applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.2 Wire Rope Inspection: Perform a Wire Rope Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.3 Rigging Gear Inspection: Perform a Rigging Gear Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

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CONTRACTOR CRANE ENTRY CHECKLIST

1	Crane Company:		Date of Entry:			
	Crane Manufacturer/Crane Model/Crane Number:		Time of Entry:			
2	Date of Annual Inspection Expiration					
3	Date of Quadrennial Inspection Expiration					
4	Name & phone number of Contracting Official (or designated local representative)		Contracting Official			
			Phone Number			
5	Does the package include a routine or critical lift plan?			Yes ☐	No ☐	
6	Location of lift site?					
7	Duration crane will be continuously on the job site (hrs, days, weeks...)					
8	Does plan include certification from contractor that the crane complies with ASME B30 standard [B30.5 (mobile cranes), B30.8 (floating cranes), B30.22 (articulating boom cranes), or B30.3 (construction tower cranes)] as applicable?			Yes ☐	No ☐	
9	Does plan include a certificate of compliance?			Yes ☐	No ☐	
10	Which OSHA regulations does the certificate of compliance indicate? (For cranes used in cargo transfer, 29 CFR 1917 applies; for cranes used in construction, demolition, or maintenance, 29 CFR 1926 applies; for cranes used in shipbuilding, ship repair, or ship breaking, 29 CFR 1915 applies).					
11	Does plan include valid medical certificate and proof of operator qualification from a source that qualifies crane operators (union, governmental agency, or an organization that tests and qualifies crane operators)? Verify qualification for each back-up operator (if provided) on the certificate of compliance.			Yes ☐	No ☐	N/A ☐
12	Does the plan designate a qualified Rigger-in-Charge			Yes ☐	No ☐	
13	What is the weight of the heaviest load to be lifted?			lbs.		
14	What is the weight of the rigging gear?			lbs.		
15	What are the crane components (and their weights) that add to the weight of the load (hook, jib, etc.)?		Main Block	lbs.		
			Aux. Block	lbs.		
			Jib (Stowed)	lbs.		
			Jib (Erected)	lbs.		
			Other	lbs.		
16	What is the maximum total crane lift (sum of 13, 14 & 15 above)?		TOTAL	lbs.		
17	What is the capacity of the crane as configured			lbs.		
18	What percentage of crane capacity does this lift represent?			%		

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CONTRACTOR CRANE ENTRY CHECKLIST

19	What is the main boom length? If a jib will be utilized, indicate the length and offset.		MAIN	JIB	OFFSET
20	What are the minimum and maximum load radii?	Min		Max	
21	Does the plan include the manufacturer's load chart for entire range of lift(s)?			Yes ☐	No ☐
22	Does plan include ground loading and outrigger reaction data to determine cribbing requirements, or a Waterfront Operational Permit?			Yes ☐	No ☐ N/A ☐
23	For crawler crane, does the plan indicate area restrictions for operation?			Yes ☐	No ☐ N/A ☐
24	For floating crane, does plan include maximum allowable list?			Yes ☐	No ☐ N/A ☐
25	For mobile crane mounted on barge, is crane equipped with load indicating device? Wind indicating device? Marine type list and trim indicator (readable in one-half degree increments)?			Yes ☐	No ☐ N/A ☐
26	For mobile crane mounted on barge, does plan include revised load chart?			Yes ☐	No ☐ N/A ☐
27	What are the environmental conditions under which crane operations are to be stopped?				
28	Will the crane perform critical lifts? (If no, skip items 29 –49.)			Yes ☐	No ☐
29	What circumstances require this lift to be classified as a critical lift? (Blind lift, 75% of chart, non-routine rigging, etc.)				
30	What are the exact dimensions of the load? (L x W x H)				
31	Does the plan indicate the crane position? (Overhead view)			Yes ☐	No ☐
32	What is the maximum lift height of the lift?				
33	What is the minimum boom angle?				
34	What is the maximum boom angle?				
35	What is the name of the operator?				
36	Indicate name(s) of backup operator (if required).				
37	Does the plan show lift points?			Yes ☐	No ☐
38	Does the plan describe the rigging procedures?			Yes ☐	No ☐
39	Does the plan indicate rigging hardware requirements?			Yes ☐	No ☐

Enclosure 2 Cont'd

SECTION 01 35 26 – ATTACHMENT A
CONTRACTOR CRANE ENTRY CHECKLIST

40	For personnel lifts, does the plan demonstrate compliance with 29 CFR 1926.550?	Yes ☐	No ☐	N/A ☐
41	Does EM 385-1-1 govern this lift?	Yes ☐	No ☐	N/A ☐
42	What are the coordination and communication requirements for the lift (e.g., radio and hand signals)?			
43	For tandem or tailing crane lifts, does the plan indicate the make and model of the crane, the line, boom, and swing speeds, and the requirement for an equalizer beam?	Yes ☐	No ☐	N/A ☐
44	For floating cranes, refer to questions 20-22?			
45	What is the name of the lift supervisor?			
56	Does the plan indicate the qualifications of the lift supervisor?	Yes ☐	No ☐	
47	What are the names of the riggers?			
48	Does the plan indicate the qualifications of the riggers?	Yes ☐	No ☐	
49	Did all involved personnel (Operator, Riggers, Lift Supervisor, etc.) sign the critical lift plan?	Yes ☐	No ☐	

Name	Organization	Signature	Date	Phone
Contracting Official:				
Wed By				

<u>FORM 16-1</u> <u>Certificate of Compliance for LHE and Rigging</u>	
This certificate shall be signed by an official of the company that provides LHE/cranes and rigging gear for any application under this contract.	
Contracting Officer's Point of Contact: (Government Designated Representative)	Phone #:
Prime Contractor/Phone #:	Contract Number:
SSHO/QC:	Phone #:
LHE Manufacturer/Type/Capacity:	
LHE Operator(s) Name(s):	
I certify that: 1. The above noted LHE and all rigging gear conform to the EM 385-1-1, applicable OSHA regulations (host country regulations in foreign countries) and applicable ASME standards. 2. The operator(s) noted above has been trained, qualified and designated in accordance with the requirements in Section 16, EM 385-1-1 for the operation of the above noted LHE. 3. The operator(s) noted above has been trained not to bypass safety devices during LHE operations. 4. The operator(s), rigger(s) and company official (staff) are aware that immediate notification to the GDA of any incident or accident involving this equipment is required.	
Company Official Signature:	Date:
Company Official Name/Title:	
Post on Crane/LHE. (In Cab and Contractor's Office for each LHE onto USACE Project/Property)	

EM 385-1-1
XX Jul 14

FORM 16-2
Standard Pre-Lift Crane Plan/Checklist

DATE: ___/___/___ Job Number: _____ Location: _____

TIME: _____ Completed By (Competent Person): _____

NOTE: Applies to Cranes, Derricks, Hoists and Power-Operated equipment that can be used to hoist, lower and/or horizontally move a suspended load (includes excavators, forklifts, Rough Terrain equipment, etc., when used with rigging).

Crane Considerations		Yes	No
1	Are the lifts within the crane's rated capacities? (based on boom height, radius)		
2	Boom deflections considered?		
3	Have all potential crane boom obstructions been identified?		
4	Have Environmental Considerations been addressed? (Wind, Weather-Lightning)		
5	Have electrical hazards been addressed (Overhead / Underground) - Clearance distances established? - Is a spotter required? - Public Utility contact required?		
6	Crane swing radius properly barricaded and personnel advised of hazards?		

Comments:

Load		Yes	No
1	Weights and Centers of Gravity (COG) have been Determined?		
2	Anything Inside / Outside the loads that could shift during the lift?		
3	Determine if the rigging needs protection from the loads?		
4	All anchor bolts, hold downs, or fasteners have been removed?		
5	Potential for binding – are load cells required to verify the loads are free?		
6	Attachment points rated to take load weight?		
7	Are the loads structurally capable of being lifted? (bending & twisting issues)		
8	Is a critical lift plan required per the EM section 16.H?		

Comments:

FORM 16-2 (cont'd)
Standard Pre-Lift Crane Plan/Checklist

Rigging		Yes	No
1	All rigging has been inspected by a Qualified Rigger?		
2	Have sling angles been calculated?		
3	Are shackles correctly sized for the sling eyes?		
4	Are softeners needed?		

Comments:

Personnel		Yes	No
1	The roles, responsibilities and qualifications for personnel have been defined? (Operator, Lift Supervisor, Rigger, Signal Person)		
2	A Pre-Lift meeting has been conducted?		
3	Personnel trained per the EM?		

Comments:

Area Preparation		Yes	No
1	The locations for the load landings has been selected and prepared?		
2	Blocking and or Cribbing is available to set the loads on?		
3	Travel paths have been determined and cordoned off?		
4	Other personnel in the area have been notified of the lifts?		
5	Have ground bearing support questions been addressed?		

Comments:

Crane Operator: _____ Date: _____

Riggers: _____ Date: _____

Signal Person: _____ Date: _____

Others: _____ Date: _____

NAVFAC MIDLANT CRITICAL LIFT PLAN																																																																				
Date:		Qualified Person:																																																																		
Location:																																																																				
<i>Contractor is required to fill sections A.thru F. prior to submitting plan. Sections G.- H to be completed by PWD Maine ET</i>																																																																				
A. TOTAL LOAD 1. Load Weight lbs 2. Wt. of Aux. Block lbs 3. Wt. of Main Block lbs 4. Wt. of Lifting Beam lbs 5. Wt. of Sling/Shackles lbs 6. Wt. of Jib/Ext. (erected/stowed) lbs 7. Wt. of Hoist Rope lbs 8. Other: lbs TOTAL WEIGHT lbs Note: Source of load weight (Drawings, Calcs., etc.) must be attached on Page 2.		E. CRANE PLACEMENT (Mobile Cranes Only) 1. Maximum Bearing Pressure PSF Note: Bearing Pressure Calculations must be attached on Page 3. 2. Ground Conditions Suitable for Load? ☐ YES / NO Note: Ground Condition Calculations must be attached on Page 3. 3. High Voltage or Electrical Hazards? ☐ YES / NO Note: If Electrical Hazards are present they must be shown on Page 4. 4. Obstructions to Lift or Swing? ☐ YES / NO Note: If Obstructions are present they must be shown on Page 4. 5. Travel with Load Required? ☐ YES / NO 6. Other?																																																																		
B. CRANE 1. Type of Crane <u>Mobile Hydraulic Truck</u> 2. Maximum Crane Capacity lbs. 3. Radius (Maximum) ft. 4. Radius (Minimum) ft. 5. Boom Length (Maximum) ft. 6. Boom Length (Minimum) ft. 7. Crane Capacity (Max Radius) lbs. 8. Crane Capacity (Min Radius) lbs. 9. Boom Angle (Maximum) deg. 10. Boom Angle (Minimum) deg. 11. Gross Load of Crane lbs. 12. Lift is % of the Crane's rated capacity 13. If Jib/Ext. is to be used: Length ft. Offset ft. 14. Rated Capacity of Jib/Ext. lbs		F. OPERATOR QUALIFICATIONS 1. Certified Operator? ☐ YES / NO 2. Option? 3. Certified for Type, Class & Capacity? ☐ YES / NO 4. Designated in writing by emp G. COMPLETED IN FIELD: <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 70%;"></th> <th style="width: 10%;">(YES)</th> <th style="width: 10%;">NO</th> <th style="width: 10%;">PWD ET</th> </tr> </thead> <tbody> <tr><td>1. Crane Inspected</td><td></td><td></td><td></td></tr> <tr><td>2. Rigging Inspected</td><td></td><td></td><td></td></tr> <tr><td>3. Crane Set-up</td><td></td><td></td><td></td></tr> <tr><td>4. Overhead Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>5. Swing Check</td><td></td><td></td><td></td></tr> <tr><td>6. Counterweight Check</td><td></td><td></td><td></td></tr> <tr><td>7. Operator Qualifications</td><td></td><td></td><td></td></tr> <tr><td>8. Signal Person Qualifications</td><td></td><td></td><td></td></tr> <tr><td>9. Rigger Qualifications</td><td></td><td></td><td></td></tr> <tr><td>10. Load Chart in Crane</td><td></td><td></td><td></td></tr> <tr><td>11. Load Test</td><td></td><td></td><td></td></tr> <tr><td>12. Tag Lines</td><td></td><td></td><td></td></tr> <tr><td>13. Wind Conditions</td><td></td><td></td><td></td></tr> <tr><td>14. Traffic Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>15. Site Control</td><td></td><td></td><td></td></tr> </tbody> </table>				(YES)	NO	PWD ET	1. Crane Inspected				2. Rigging Inspected				3. Crane Set-up				4. Overhead Hazard Check				5. Swing Check				6. Counterweight Check				7. Operator Qualifications				8. Signal Person Qualifications				9. Rigger Qualifications				10. Load Chart in Crane				11. Load Test				12. Tag Lines				13. Wind Conditions				14. Traffic Hazard Check				15. Site Control			
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C. HOIST ROPE <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 20%;"></th> <th style="width: 15%;">Main</th> <th style="width: 15%;">Aux 1</th> <th style="width: 15%;">Aux 2</th> <th style="width: 15%;"></th> </tr> </thead> <tbody> <tr> <td>1. # of Parts</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>2. Rope Diamter</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>3. Capacity</td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>			Main	Aux 1	Aux 2		1. # of Parts					2. Rope Diamter					3. Capacity					H. SIGNATURES IN FIELD: 1. Crane Operator 2. Rigger 3. Signal Person 4. Lift Supervisor 5. PWD ET 6. Other																																														
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D. RIGGING 1. Hitch Type(s) 2. No. of Slings: Size: 3. Sling Type: 4. Sling Assembly Capacity: lbs. 5. Shackle Size(s): 6. Shackle Rated Capacity(s) lbs.																																																																				

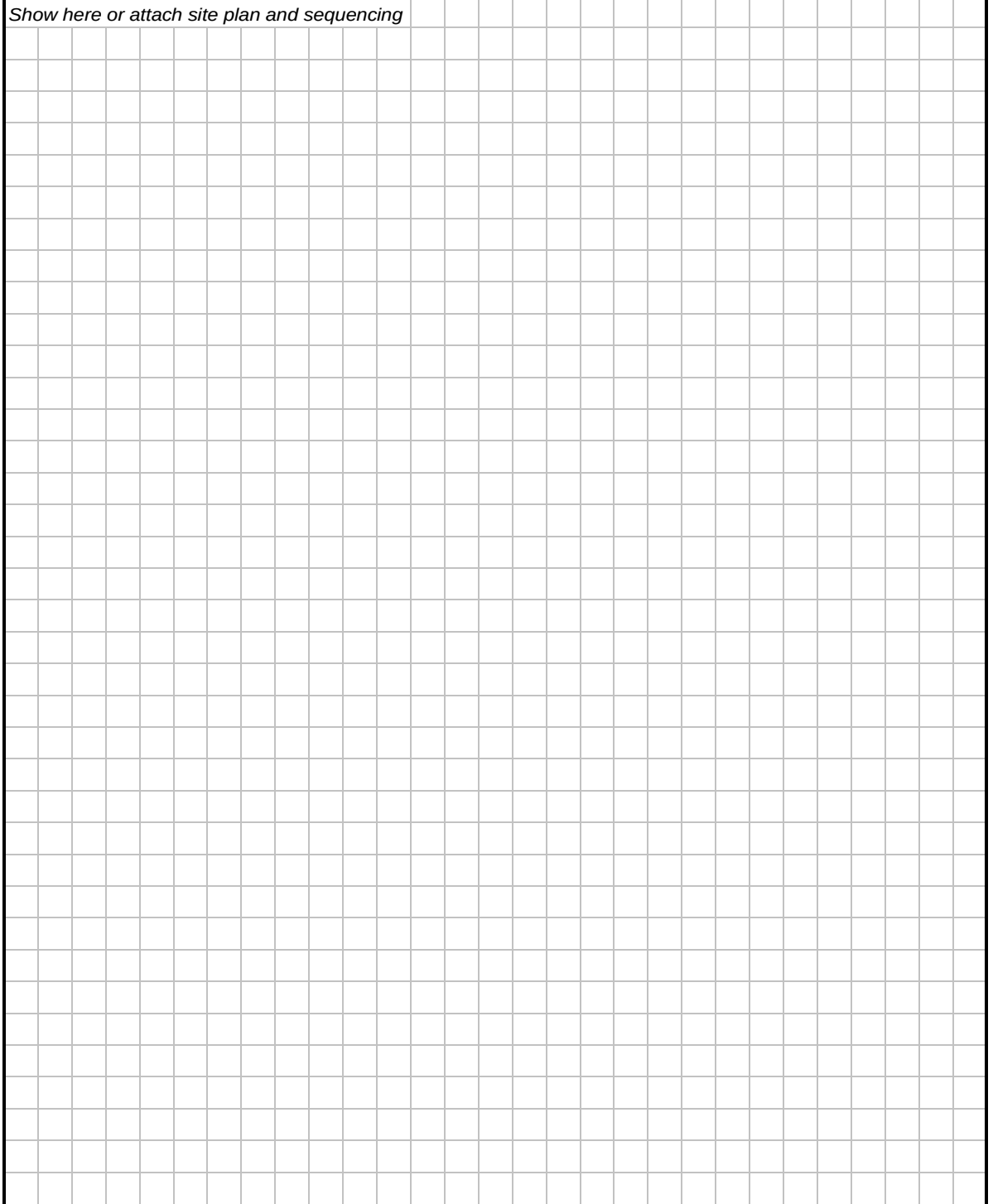
EM 385-1-1
XX Jul 14

U.S. Army Corps of Engineers
CRITICAL LIFT PLAN

For use of this form, see EM 385-1-1, Section 16. Proponent agency is Crane HHWG.

SITE PLAN

Show here or attach site plan and sequencing



P2 CONTRACTOR CRANE / MULTI PURPOSE MACHINE & RIGGING OPERATION CHECKLIST	
Prime Contractor: Phone Number:	Crane Operator's Name:
Crane Supplier: Phone Number:	Competent Riggers Name:
Crane Manufacturer: Model No.	Crane Capacity: Crane Serial No:
Inspection Due Date:	APP Rep: Print _____ Sign _____

CONTRACTOR CRANE OR RIGGING OPERATION CHECKLIST				
		YES	NO	N/A
1	Is the Certificate of Compliance, P-1, in the operator's cab (or in the contractor's on-site office for rigging operations) with the current operator's name listed?			
2	Is the crane/machine transited to and from the job site correctly? Are the OEM instructions for travel being followed?			
3	Does the operator know the weight of the load to be lifted?			
4	Is the load to be lifted within the crane/machine manufacturer's rated capacity in its present configuration?			
5	Are outriggers/stabilizers required and, if so, are they properly extended and down?			
6	If outrigger/stabilizers are used, and the wheels are not off the ground is this the correct setup in accordance with the OEM?			
7	Is the crane/machine level and on firm ground, or if the ground is not firm are adequate supporting materials provided?			
8	If supporting materials are provided, is the entire surface of the outrigger/stabilizer pad supported and is the supporting material of sufficient strength to safely support the loaded outrigger/stabilizer pad?			
9	If outriggers/stabilizers are not used, is the crane/machine rated for on-rubber lifts by the OEM's load chart?			
10	Is the swing radius of the crane counterweight clear of people and obstructions and are accessible areas within the swing area barricaded to prevent injury or damage?			

Figure P-2 (1 of 3)

		YES	NO	N/A
11	Is the Anti-two block safety device functional? When block is reconfigured has the device been retested to insure correct operation.			
12	Has the hook been centered over the load in such a manner to minimize swing?			
13	Is the load well secured and balanced in the sling or lifting device after it is lifted a few inches for verification?			
14	Is the lift and rotation path clear of obstructions?			
15	If rotation of the load being lifted is hazardous, is a tagline or other restraint being used?			
16	Are personnel prevented from standing or passing under a suspended load?			
17	Is the operator paying full attention to the signal person?			
18	Are proper signals being used? Is the operator responding properly to the signals? Are radios used for blind lifts?			
19	Are empty hooks lashed or otherwise secured during travel to prevent swinging?			
20	Does the operator remain at the controls while the load is suspended?			
21	Does the operator ensure that side loading is prohibited?			
22	Are personnel prevented from riding on a load?			
23	Are start and stop motions in a smooth fluid motion (no sudden acceleration or deceleration)?			
24	Is the lift a critical lift?			
25	If so, is a lift plan provided and understood and check-off sheets initialed and signed off?			
26	If overhead power lines are in the vicinity, is a critical lift plan provided addressing the requirements of 29 CFR 1926.1407-1411?			
27	If pick and carry operations are allowed and performed, are OEM directions followed (e.g. rotation lock engaged, boom centered over front or rear, etc.)?			
28	When the crane/machine is left unattended, is it in a safe condition?			
29	Is rigging gear undamaged and acceptable for the application?			
30	Does rigging gear meet applicable ASME or host nation standards (e.g., ASME B30.9 for slings, B30.10 for hooks, B30.26 for rigging hardware such as shackles, safety hoist rings, and eyebolts, B30.20 for below the hook lifting devices)?			

Figure P-2 (2 of 3)



		YES	NO	N/A
31	Was the rigging gear inspected prior to use?			
32	Is sling protection used to protect slings (especially synthetic slings) and equipment from damage due to abrasion and sharp corners and edges?			
33	Is the rigging gear used in accordance with its working load limit? Is the working load limit marked on the rigging gear?			
34	Are positive latching devices (or "moussing") used on crane and rigging hooks?			
35	If a mobile crane is used on a barge, are all rules of 29 CFR 1926.1437(construction) being followed?			
36	If a mobile crane is used on a barge are the outriggers/stabilizers blocked or are the crawlers traveling in a defined space as allowed by 29 CFR 1926.1437 (construction)?			
37	For floating cranes, are rules of 29 CFR 1915 (ship repair) or 29 CFR 1926.1437 (construction) being followed?			
38	If a multi-purpose machine, forklift, or construction equipment is being used, is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting and is there a load chart?			
39	If a personnel lift is being performed with a crane or base mounted hoist, are all requirements of NAVFAC P-307, paragraph 11.1.g (5) for a crane or 11.1.m for a base mounted hoist being followed?			
Contractor:		Subcontractor:		
Location:		Date:		
Notes:				
Location of Lift: _____ Material being lifted and Weight of lift: _____				
Signature of Government Representative:				

Figure P-2 (3 of 3)



Public Works Department, Maine

☐ RESET FORM

Form Revision Date 4/04/23

Attachment B

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. This Form shall be submitted to NAVFAC DSC at least fourteen (14) calendar days prior to Excavation.

INFORMATION THE EXCAVATOR WILL NEED TO OBTAIN A DIG SAFE TICKET (when the Excavator contacts DIG SAFE they will obtain a Dig Safe Ticket as proof of notification.)

Caller Details:

Caller Name _____ Title _____
Phone # _____ Fax # _____ Alt # _____
Email address _____ Business Hours _____ to _____
Company
Name _____
Address _____
City _____ State _____ Zip _____

Location Details (where excavation is planned)

(Check one) State: MA ☐ ME ☐ NH ☐ RI ☐ VT ☐

City/Town _____

(Optional) Latitude _____ Longitude _____

Address/Intersection _____

Nearest Cross Street _____

Additional Information _____

Type of Work _____

Excavation Planned Depth (feet) _____

Area (i.e. St to building, in the St., sidewalk area, right side of building)

Pre-marked? Yes ☐ No ☐

Excavation planned Start Month _____ Day _____ Year _____ Time (military) _____

Excavator Doing Work (if not same as above) _____



Form Revision Date 4/04/23

Attachment B cont'd

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. Submit this Form to PWD-ME DSC at least fourteen (14) calendar days prior to Excavation. See Note on ticket expiration.

Part I – Completed by the Excavator.

Today's Date: ____/____/____ DIG SAFE Ticket #: _____

Ticket Expiration Date:(60 days from date obtained): ____/____/____

Utility Companies Dig Safe will notify: _____

Requested by: _____ Phone #: _____

Code # / Company: _____ E-mail: _____

Contract #: _____ Project Title: _____

Shipyard POC: _____ Phone # _____

Excavation Location: _____ Area Pre-Marked YES ☐, NO ☐

Type of work: _____

Depth: (ft): _____ Anticipated Excavation Date: ____/____/____ Time: (military) _____

Attach a map or the contract drawings showing the excavation/ground penetrating area.

NAVFAC Construction Contractors: Complete Part 1 and Submit Form to NAVFAC CM/ET

AE Firms/Sub consultants: Complete Part 1 and Submit Form to PWD-ME DM

Shipyard Personnel: Complete Parts 1 & 2 and Submit Form to PWD-ME DSC

Part 2 – To be completed by the NAVFAC CM/ET/DM (If Shipyard work, Part 2 must be completed by Requestor)

Date: ____/____/____ Name: _____ Phone # _____ Locate

Priority: Routine (3 Business days) ☐; Emergency (<3 Business days) ☐

_____ (DSC Approval Req if <3 Business days)

Part 1 Reviewed and Complete: YES ☐, NO ☐ Initial: _____

Submit Completed Form to DSC. The DSC will review the Request and forward to the PWD-ME FSC Utility Locate Contract Rep who will review and forward to the Utility Locating Company for action.

Part 3 – To be completed by FSC Utility Locating Company

Approved by FSC Utility Locate Contractor: Initials _____

Date Utilities marked in the field: ____/____/____ Name: _____

Comments: _____

Utility Plan Discrepancies Noted: YES ☐, NO ☐ (PWD-ME FSC PAR & DSC must be Notified if discrepancies are found: Notification made: YES ☐ to: _____

Comments: _____

Submit to PWD-ME FSC Contract Rep who will forward to the DSC. If Plan Discrepancies are noted, provide a scaled markup plan showing the correct utility locations.



Form Revision Date 4/04/23

Attachment B cont'd

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

Part 4 – To be completed by PWD-ME DSC

Date ___/___/___ Logged Into Database: YES ☐ NO ☐ # _____

Any Hazardous/Mission Critical Utilities in Excavation Area:

YES ☐ NO ☐ (if YES, ensure Excavator is briefed to include additional controls of Appendix B.)

Comments: _____

Any unknown utilities, anomalies (including increased risk of passive detection modes), or discrepancies noted by FSC Utility Locate in Part 3 have been discussed with Project Team and adequate risk mitigations are in place.

DSC or DSC Representative Approval for excavation. (*See note below).

Signature/Date: _____

DSC return a copy of the completed Form to NAVFAC CM/ET/DM or Shipyard Personnel Requesting the Utility Locate.

NAVFAC CM/ET/DM/Requestor ensure the Excavator is aware if any Plan Discrepancies were found, establish risk mitigations, and to use extreme caution when excavating.

If Plan Discrepancies are noted, the PWD DSC shall coordinate any Utility Plan Updates with PWD-ME GIS lead.

Note:

If work has not begun within 30 days from the Dig Safe Ticket approval date or if the excavation area expands beyond the original limits, the Excavator must renew the Dig Safe ticket by notifying DIG SAFE (1-888-344-7233) and the Contracting Officer. If the proposed excavation is not completed within 60 calendar days after obtaining the Dig Safe ticket, the ticket expires. Tickets should be renewed by calling DIG SAFE (as needed) at least 72 hours prior to expiration so there is always an active ticket. The utility locators (not 3rd Party) must be notified to mark out each time a ticket is renewed.

**DSC approval does not waive the construction contractor's responsibility to perform a 3rd Party Locate (if specified by contract) prior to beginning excavation. The excavator is always responsible to ensure utilities are not damaged during excavation. DSC approval does not diminish or change that responsibility.*



Form Revision 4/4/23

Attachment C

Public Works Department, Maine

Reset Form



CONTRACTOR DIG-SAFE CERTIFICATION FORM

References:

1. Government Safety Requirements (PWD ME) Section 01 35 26.00 22

Project Title: _____

Prime Contractor: _____

Contract #: _____

Initial Dig-Safe Request #: _____

A. Third Party Utility Location Certification

In accordance with the Specification Section 01 35 26.00 22 Government Safety Requirements para 3.7.1, The Contractor must provide the services of a third party qualified independent utility locating company/person(s) (cannot be the Government's Utility Locating Firm) to positively identify underground utilities in the work area. The utility locating company shall indicate the utility type, size, and depth of all utilities within the proposed area of excavation including, but not limited to, voice/data, electrical, gas, water, air, sanitary sewer, and storm drainage. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

By signing below, the Contractor certifies that the Third Party Utility Locate Company has complete a review of the excavation area.

Third Party Utility Locate Company: _____

Date Utility Locate was completed: _____

Print Name of the Utility Locator: _____

Utility Locator Signature/Date: _____

Any Discrepancies Found with the Marked Utilities (Yes/No): Yes ☐ No ☐

(If Yes, notify the Contracting Officer Immediately)

Prime Contractor Signature/Date: _____



Public Works Department, Maine

CONTRACTOR DIG-SAFE CERTIFICATION FORM

B. Prime Contractor Utility Location Certification

Prime Contractor has performed a review of the following documents to ensure a comprehensive understanding of the sub-surface area to be excavated.

- Government's Dig-Safe report and associated drawings.
- Independent 3rd party Dig-Safe report and associated drawings.
- Project Contract drawings
- Government Total Map of approximate existing utility locations.

Prior to proceeding with excavation the prime contractor has the responsibility to access associated documentation of all items listed above and to mitigate the risk of a potential utility strike.

The final risk assessment shall be identified in the Contractor AHA and reviewed with the Contracting Officers Rep. Also, any hazardous/mission critical utilities identified by the DSC or plans, are covered in the Contractors AHA's and pre-job Safety Briefs prior to commencing excavation.

By signing below, the Contractor certifies the above documentation has been thoroughly reviewed and evaluated for inconsistencies, anomalies, and stated inadequacies, and the prime contractor has risk mitigation actions planned to ensure excavation does not damage underground utilities.

Prime Contractor Signature/date; _____



Reset Form



Public Works Department, Maine

Form Revision Date 4/04/23

Attachment D

**CONTRACTOR UTILITY LOCATE – PRE-EXCAVATION SAFETY
CHECKLIST**

Project Title: _____

Prime Contractor: _____

Contract No.: _____

Excavator: _____

Initial Gov't and Utility (if listed in Part 1 Appendix A) Dig Safe Markings Completed & Visible in the field.

Initial NAVFAC Dig Safe Utility Locate Request Form returned from DSC and any Noted Discrepancies are Resolved and risks mitigated.

Initial Contractor's Independent Third Party Utility Locate Completed and any discrepancies are resolved with NAVFAC (Attach Third Party Certification Form).

Are Hazardous or Mission Critical Utilities identified in the excavation area (as identified by DSC or plans)? Yes ☐ or No: ☐
If Yes complete the following:

- ☐ The Contractor must supplement daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan ensuring all utilities have been clearly marked and reviewed with the SSHO.
- ☐ The Contractor must have a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility.
- ☐ The SSHO must complete a pre-excavation walk as part of the morning procedure to ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint.
- ☐ Contractor must provide additional danger signage, to mark areas of known live underground utilities.
- ☐ Contractor must ensure a 'spotter' accompanies the equipment operator during excavation work.
- ☐ Contractor must provide Construction CM/ET notification no later than 7 working days prior to the preparatory and initial pre-excavation/ demo safety review meeting. (The CM/ET and the DSC will attend this meeting to ensure safety and preservation of Critical Utilities is discussed with adequate focus.)
- ☐ The Contractor must confirm & identify the closest utility isolation points & develop mitigation strategies with the utility owner (Coordinate with the DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.

***NOTE: The Hazardous or Mission Critical Utility requirements listed above are in addition to, and do not replace other requirements. ALL UTILITIES require hand excavation using wood or fiberglass tools when work is expected to come within 3' of the utility, per contract (Div 1 Specs, General Notes, Excavation specs).**

Prime Contractor SSHO Signature/Date: _____

Excavator Signature/Date: _____

NAVFAC ET Signature/Date: _____

Daily Worksite Checklist for Trenching/Excavation Sites

Project:		Date:	Weather:	Soil Type:
Trench Depth:	Length:	Width:	Type of Protective System:	

Signature of Competent Person:

Yes	No	N/A	Excavation
			Trench box extends at least 18 inches above the vertical wall of the excavation and to within 2 feet of the bottom of the trench (or less if soil collapsing behind or below trench box).
			Trench box installed in accordance with manufacturers specific instructions and use limitations.
			Trench box inspected for damage or defects and pins and spreaders are securely installed.
			If other soil protective systems are used, they are installed in accordance with manufacturers Instructions OR are approved by a Registered Professional Engineer.
			All employees at worksite trained in trenching safety procedures.
			Surface encumbrances such as utility poles, heavy equipment supported or removed.
			Heavy equipment safety zone at least 1½ times depth of trench for if not supported.
			Employees protected from loose rock or soil.
			Spoils, materials, and equipment set back a minimum of 2' from edge of excavation.
			Walkways and bridges over excavations 6' or more in depth are at least 20 inches wide and are equipped with required guardrails.
			Ladders placed no more than 25 feet apart.
			Employees prohibited from working or walking under suspended loads.
			Employees prohibited from working on faces of sloped or benched excavations above other employees.
			Warning system established and used when mobile equipment is operating near edge of excavation.
			Barriers provided if trench opening is not readily apparent.
			Barriers, fences available to secure area if left overnight.

Yes	No	N/A	Personal Protective Equipment
			Hard hats worn by all employees.
			Work boots or safety shoes worn by all employees.
			Eye protection worn by all employees (if applicable).
			Hearing protection worn by all employees (if applicable).
			Warning vests, or other highly visible PPE provided and worn by all employees exposed to vehicular traffic.

Daily Worksite Checklist for Trenching/Excavation Sites – pg 2

Yes	No	N/A	Utilities
			Utility companies contacted and/or utilities located.
			Exact location of utilities marked when near excavation.
			Underground installations protected, supported, or removed when excavation is open.

Yes	No	N/A	Wet Conditions
			Precautions taken to protect employees from accumulation of water.
			Water removal equipment monitored by Competent Person.
			Surface water controlled or diverted.
			Inspection made after each rainstorm.

Yes	No	N/A	Hazardous Atmosphere
			Atmosphere tested when there is a possibility of oxygen deficiency or build-up of hazardous gases. If yes, atmosphere will be tested every _____.
			Oxygen content is between 19.5% and 21%.
			Flammable gas build-up to 20% of lower explosive limit (LEL).
			Toxic Levels of gases are below limits set on gas monitor.
			Ventilation blowing into space and air intake placed away from vehicle exhaust.
			Program Manager contacted if atmosphere is above established limits. Source of contaminant to be determined and eliminated prior to entry or Program Manager will establish special procedures for entry.

Emergency Procedures for Trench Cave-Ins

- **GET ALL OTHER EMPLOYEES OUT OF THE TRENCH!!**
- **CALL 438-2333**
- **NOTIFY COMPETENT PERSON**

- **Note time**
- **Note location of trapped worker(s)**
- **Leave all victims hand tools in place**
- **Shut down all heavy equipment**
- **Stop nearby traffic that may cause vibration**
- **Keep everyone back from trench at least 50 feet**
- **Gather information for rescue team**
- **WAIT for rescue team. Do not attempt to rescue.**

Note-Do not attempt to dig the person out using hand tools or heavy equipment. This could cause the trench to collapse further and could cause further injuries!!!!



Form Reset

Cover Sheet for Contractor Dig Safe Utility Locate Request Form (PWD Maine & OICC Maine & other)

Rev. 4/03/23

Ktr		<i>Check blocks when action completed.</i>
☐		Prime Contractor Name:
☐		Contract Title:
☐		Requestor Name:
☐		Location of Excavation:
Ktr	ET	<i>Attach Items as Noted for DSC for Review:</i>
☐	☐	Dig Safe Ticket #:
☐	☐	Date of Excavation Start:
☐	☐	Renewal Date: (If work is planned to exceed 60 days from when State issued Ticket#)
☐	☐	If job includes digging <20' from a gas line, Unitil has been called by Prime Ktr.
☐	☐	If Potholing is required by contract, notified DSC with anticipated dates. Potholing is N/A. ☐
☐	☐	Ensure General Ktr has latest Total Map & Project Plans prior to mapping:
☐	☐	Navfac Dig Safe Utility Locate Request Form Attached: (Call in data page and Parts #1 thru #4)
☐	☐	Confirm white paint is marked and all items cleared out of the way:
☐	☐	Ensure Genral Ktr has latest Total Map & Project Plans prior to mapping:
☐	☐	Contractor Dig Safe Certification Form Completed and Attached:
☐	☐	Contractor Utility Locate Pre-Excavation Safety Check List Completed and Attached:
☐	☐	Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Preparatory Mtg:
☐	☐	Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Initial Phase Mtg:
☐	☐	Inside bldgs. for borings: (as needed based on design, and for PWD ME shop)

ET Validation all steps are completed. Sign/date_____



Public Works Department, Maine

PWD -ME Dig Safe Coordinator
gary.r.gagnon4.civ@us.navy.mil
207-994-9208



All fields are mandatory and must be filled in.
Prime Contractor to complete and submit within 2 working days of the utility strike. For detailed instructions, refer to Dig Safe Strike Checklist Instructions.

UNDERGROUND UTILITY STRIKE REVIEW CHECKLIST

1. General Information:

- a. Project & Contract#: _____
- b. Excavation Site: _____
- c. Utility Impacted: _____
- d. Depth of Utility Impacted (ft.): _____
- e. Date Utility was impacted: ____/____/____, Time Utility was impacted: _____
- f. Describe how and why damage occurred, what equipment was being used, what went wrong: (attach separate sheet as needed)

- g. Prime Contractor company name: _____
- h. Excavator company name: _____
- i. Project ET: _____
- j. PWD Locate Request Form Revision date used: ____/____/____ (Attach a completed copy)
- k. Is this identified as a Hazardous/Mission Critical utility on Part 4 of the Dig Safe Request Form? Yes: ☐, No: ☐

2. Utility Locate Information:

- a. **Dig Safe Locate Ticket;**
 - i. Original Dig Safe Ticket #: _____ and date pulled: _____
 - ii. Latest date Dig Safe Ticket was renewed: _____ N/A: ☐
 - iii. Date of latest FSC Utility Locate markings in excavation area: _____
 - iv. Was Prime Contractor maintaining visible markings? Yes: ☐, No: ☐
 - v. Was the impacted utility currently marked where it was struck? Yes: ☐
 - 1. If not, describe situation: _____

 - vi. Did FSC Utility Locate KTR flag anomalies or discrepancies on Field notes that apply to this utility? (Attach FSC Locate sketch and field notes) Yes: ☐, No: ☐
 - 1. Was DSC involved in resolution of anomalies or discrepancies? Yes: ☐, or why not? _____
 - 2. Was there a post FSC Locate Tailgate discussion? Yes: ☐, or why not? _____



Public Works Department, Maine

3. Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes: ☐, No: ☐

Describe how, or why not (include separate sheet if needed): _____

- vii. Did PWD ME Dig Safe Coordinator approve excavation on Utility Locate Request Form Part 4: Yes: ☐, or why not? _____

b. 3rd Party Utility Locate Information:

- i. 3rd Party Locate Company: _____

- ii. Date of the 3rd Party Locate: ____/____/____

- iii. Did the 3rd Party flag anomalies or discrepancies on field notes that apply to this utility? (Attach 3rd Party Locate sketch and field notes) Yes: ☐, No: ☐

1. Was DSC involved in resolution of anomalies or discrepancies? Yes: ☐, or why not? _____

2. Was there a post 3rd Party Tailgate discussion? Yes: ☐, or why not? _____

Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes: ☐, No: ☐

Describe how, or why not (include separate sheet if needed): _____

- iv. Did Prime Contractor complete the Contractor Dig Safe Certification Form prior to excavation? (attach completed copy): Yes: ☐, or why not? _____

3. Preparatory & Initial Meetings Information:

- a. Date of the Preparatory Meeting: ____/____/____
- b. Did PWD DSC attend the Preparatory meeting? Yes: ☐ or why not? _____
- c. Were Haz/Mission Critical Utilities reviewed at the meeting? Yes: ☐, No: ☐
- d. Did PWD UEM personnel attend the meeting? Yes: ☐, No: ☐
- e. Were the Utility Markings in the field reviewed? Yes: ☐, No: ☐
- f. Were FSC and 3rd Party Identified discrepancies or anomalies noted and discussed? Yes: ☐, If No, why not? ☐ _____
- g. Date of the Initial Meeting: _____
- h. Were Utility Risk Mitigation Strategies reviewed (attach minutes): Yes: ☐, if No, why not? ☐ _____
- i. Were Utility Outages part of the Risk Mitigation Plan? Yes: ☐, No: ☐
- j. Was excavation performed with expected precaution? Yes: ☐, No: ☐



Public Works Department, Maine

4. Utility Mapping Review (DSC input):

- a. Did the FSC Dig Safe KTR have the latest Total Map? (Explain): Yes: ☐, No: ☐

- b. Were inaccuracies between the field conditions & the Total Map found? Explain;
Yes: ☐, No: ☐, _____

- c. If Total Map inaccuracies were identified for this area, has CAD updated for most
current As-Built Info? Yes: ☐, No: ☐

- d. Was digging within 3' of marked utility? Yes: ☐, No: ☐

5. Items to submit along with this Review Checklist:

- a. _____ A copy of the Dig Safe Ticket.
b. ☐ Diagram/sketch of the area with specific location of the strike.
c. ☐ Appropriate Contractor's Daily Work Notes.
d. ☐ FSC Dig Safe Locate Report (sketch and field notes).
e. ☐ 3rd Party Locate Report (sketch and field notes).
f. ☐ Statement from Equipment Operator.
g. ☐ Preparatory Meeting Minutes.
h. ☐ Initial Meeting Minutes.

6. Project Specific Comments:

7. What could be done better?

- a.

b.

c.

8. Lessons Learned:

- a.

b.

c.



Public Works Department, Maine

ATTACHMENTS FOR DETAILS AND EXPLANATIONS AS NEEDED

SECTION 01 44 00.00 22

ARCHAEOLOGICAL MONITORING (PWD ME)
010/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard and at PWD ME AOR Facilities.

1.1 SUMMARY

Perform archaeological monitoring for all forms of soil disturbing activities and excavations **as identified in project drawings**. Work must include, but is not limited to: background research and review of previous surveys, monitoring during excavation, installation of components, regrading, and other work indicated or specified. Results of monitoring efforts must be summarized in Archaeologist Resource Reports, draft report, final draft report, and final report.

The Government will provide available existing documentation pertaining to the project area, which will include previous investigations, plans, photographs, and research.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 79

(2012) Curation Of Federally-Owned And Administered Archaeological Collections

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Archaeologist Qualifications; G

Archaeological Coordinator Designation Letter

Archeological Monitoring Plan; G

SD-06 Test Reports

Archaeological Monitoring Reports; G

Archaeologist Resource Reports - Progress Report; G

Archaeologist Resource Reports - Draft Report; G

Archaeologist Resource Reports - Draft Final Report; G

Archaeologist Resource Reports - Final Report; G

Curated Artifacts (if collected); G

1.4 QUALITY ASSURANCE

The Contractor must submit an Archeological Monitoring Plan that includes the proposed Archaeologist Qualifications. The Archaeologist must meet the Secretary of the Interior's Professional Qualification Standards for Archaeologist as outlined in 36 CFR Part 61 as required by the National Historic Preservation Act (Section 112, a, 1, A), OPNAVIST 5090.1E (Section 13-3.6), and the Principal Investigator must be an approved Archaeologist on the Maine Historic Preservation Commission's Level II Approved List. Field staff do not need to be named on the Level II list as long as they are supervised by an archaeologist with those qualifications. The Archaeologist must specialize in historic archaeology and have experience monitoring construction work in urban or industrial archaeological settings. The Archaeologist must have a demonstrated track record of success in carrying projects to completion, including the submission of all final project documentation and curated artifacts.

1.5 COORDINATION

The Contractor shall designate in writing the Archaeological Coordinator responsible for coordination with the archaeological subcontractor. This individual must have a thorough understanding of the project schedule. The Contractor shall notify Contracting Officer of any changes in the Archaeological Coordinator in writing.

The Contractor is responsible for obtaining permission and clearance from the Contracting Officer to enter areas of the required fieldwork.

The Contractor is responsible for recording minutes of all meetings and furnishing a copy of the minutes to the Contracting Officer within 15 cal days. Prior to the start of any ground disturbing/excavation work, the Archaeological Coordinator/archaeologist, Contractors QC Manager, PWD ME CM/ET and PWD-ME Cultural Resources Manager must meet to review the proposed approach to the work, establish a protocol for collection of artifacts, and review the proposed and review relevant documentation.

For projects with a duration exceeding one year, re-initial coordination meeting must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor; Archaeological Coordinator; QC Manager, Site Superintendent, Project Manager, etc.)

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 INSPECTION AND MONITORING

- a. The Archaeologist must observe excavation activities for archaeological resources. The Archaeologist must be present on the project site according to a schedule agreed upon by Contracting Officer until it is determined that project construction activities could have no effects on significant archaeological resources.
- b. The Archaeologist must advise all project Contractors to be on the alert for evidence of the presence of the expected resource(s), of how to identify the evidence of the expected resource(s), and of the appropriate protocol in the event of apparent discovery of an archaeological resource. GPS points, photographs, sketch maps, and measured drawings must be provided as applicable to document to the greatest extent possible resources encountered.
- c. If an intact archaeological resource is encountered, all soil disturbing activities in the vicinity of the resource must cease. The Archaeologist must be empowered to temporarily redirect excavation/ construction crews and heavy equipment until the resource is evaluated. The Archaeologist must immediately notify the Contracting Officer of the encountered archaeological resource. The Archaeologist must, after making a reasonable effort to assess the identity, integrity, and significance of the encountered archaeological resource, present the findings of this assessment to the Contracting Officer. Following verbal communication of this information, the assessment must be formalized in a Resource Report.
- d. The Contractor must be responsible for the safety of all work conducted and must insure that the work area is safe during any archaeological investigation. This includes providing appropriate shoring, access, and egress routes and any needed equipment. No one will be expected to enter the excavation until it has been deemed safe and it will be the responsibility of the Contractor to insure that the archaeological monitors have the ability to enter the excavation as needed.
- e. The Archaeologist must document the resource(s) encountered with photographs (including scale), mapping, measured drawings, collection of GPS points, written description of the resource; and collect artifacts, as appropriate.

3.2 ARCHAEOLOGICAL MONITORING REPORTS

Record findings of the archaeological monitoring at project completion in a written report. The report must include a map created by GPS points showing the location of any resources documented and must be illustrated by photographs. The report must include a recommendation for the future treatment of archaeological resources encountered during construction, but left in place. The Archaeologist must record and be authorized to collect archaeological material as warranted for analysis.

3.2.1 Report Requirements:

- a. Progress Report: Provide a brief Monthly Progress Report consisting of a table indicating the dates of archaeological monitoring activities, the names of the archaeological monitors on site each day, and a

summary of resources encountered/not encountered. The dates shall be keyed to an accompanying map showing the areas monitored.

If a potentially significant resource is encountered, summarize the results of the archaeological investigations in a Resource Report. The Archaeologist, in coordination with the Contractor must produce the Resource Report to include a map and written summary of the discovery to support PWD ME coordination with SHPO. The Report must include, at a minimum, a physical description of the resource and its context; the depth at which it was encountered; potential resource identification; a description of the proposed impacts to the resource; and recommended follow-on steps, including options to avoid or minimize impacts on the potential site. The Resource Report must be submitted to the Contracting Officer no later than 2 business days after the potentially significant resource is encountered.

- b. Provide two (2) hard copies and one (1) electronic copy of a draft report (99 percent complete) for review by the Government. Once the draft report has been reviewed by the Government, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of the final draft report must be within ten (10) business days of receipt of the Government comments.
- c. Provide two (2) hard copies and one (1) electronic copy of the final draft report for review by the Government and Maine SHPO. The final draft report must be completed in accordance with current State of Maine archaeological survey requirements. Draft GIS data must also be provided at this time.
- d. Once the final draft report has been reviewed by the SHPO, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of three (3) hard copies and three (3) electronic copies of the final reports must be within ten (10) business days of receipt of the Government comments.
- e. The final report must be completed in accordance with current State of Maine archaeological survey requirements and the associated materials.
- f. The Contractor must submit reports (e.g., progress reports, draft and final reports, etc.) directly to the Contracting Officer within 30 calendar days of completion of monitoring. Copies of submittal distribution letters must be forwarded to the Contracting Officer.
- g. The Contractor must submit all informal documents regarding archeological field site reconnaissance and archaeological site mapping (field notes, etc.) directly to the Contracting Officer. Upon request or after acceptance of the final report, all field notes, photographs, artifacts, and related data will be turned over to the Contracting Officer.
- h. The Contractor must submit all documentation in editable electronic format (Microsoft Word, Excel, ESRI ArcINFO, etc.) and in PDF upon completion of work. Three (3) hard copies and three (3) electronic copies of the report must also be submitted upon completion and acceptance of work.
- i. The Contractor is required to prepare and submit all data, reports, graphics, maps, charts, etc. in an electronic format consistent with the platform currently used by the PWD ME Cultural Resources GIS

database. Details and requirements will be provided to the Contractor by the Contracting Officer upon award of this Contract.

3.3 ARTIFACT CURATION

Artifacts recovered during the monitoring effort must be cleaned, recorded, and curated in accordance with 36 CFR 79. The Archaeologist must identify a repository for permanent curation and pay initial curation fees. If no suitable repository is identified, curated artifacts must be provided to the Government with the Final Report Submission.

-- End of Section --

SECTION 01 45 00.10 22

QUALITY CONTROL FOR MINOR CONSTRUCTION (PWD ME)

04/22

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

QC Plan; G,

SD-07 Certificates

Contractor QC Self-Evaluation; G

1.3 INFORMATION FOR THE CONTRACTING OFFICER

Prior to commencing work on construction, the Contractor can obtain a single copy set of the current report forms from the Contracting Officer. The report forms will consist of the Contractor Production Report, Contractor Production Report (Continuation Sheet), Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Rework Items List, and Testing Plan and Log.

Deliver the following to the Contracting Officer:

- a. CQC Report: Submit the report electronically by 10:00 AM the next working day that work is performed.
- b. Contractor Production Report: Submit the report electronically by 10:00 AM the next working day that work is performed;
- c. Preparatory Phase Checklist: Submit the report electronically in the same manner as the CQC Report for each Preparatory Phase held;
- d. Initial Phase Checklist: Submit the report electronically in the same

manner as the CQC Report for each Initial Phase held;

- e. Field Test Reports: Within two (2) working days after the test is performed, submit the report as an electronic attachment to the CQC Report;
- f. CQC Meeting Minutes: Within two (2) working days after the meeting is held, submit the report as an electronic attachment to the CQC Report; and
- g. QC Certifications: As required by the paragraph entitled QC CERTIFICATIONS.

1.4 QC PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this Section. This QC program is a key element in meeting the objectives of NAVFAC. The QC program consists of a QC Manager, a QC plan, a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, and QC certifications and documentation necessary to provide materials, equipment, workmanship, fabrication, construction, and operations which comply with the requirements of this Contract. The QC program must cover on-site and off-site work and must be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the Project site. "Project Site" is the specific area of construction within the limits of work as defined by the contract documents.

The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent, and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job. The QC Manager will be subject to removal by the Contracting Officer for non-compliance with the Quality requirements specified in the Contract, failure to perform the duties of the QCM specified herein, or for failure to manage the QC program. The removal and replacement of the QC Manager or other QC personnel will not be cause for claim of additional compensation or extensions of the Contract Completion Date (CCD). Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents.

1.4.1 Preliminary Work Authorized Prior to Acceptance

The only work that is authorized to proceed prior to the acceptance of the QC plan is mobilization of storage and office trailers, temporary utilities, and surveying.

1.4.2 Acceptance

Acceptance of the QC plan is required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC plan and operations as necessary, including removal of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time in order to verify the submitted qualifications.

1.4.3 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed change, including changes in the QC organization personnel, a minimum of seven (7) calendar days prior to a proposed change. Proposed changes must be subject to the acceptance by the Contracting Officer.

1.5 QC ORGANIZATION

1.5.1 QC Manager

1.5.1.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. In addition to implementing and managing the QC program, the QC Manager may **not** perform the duties of SSHO. The QC Manager **must not** perform the duties of Project Superintendent **nor** the duties of Project Manager. The QC Manager is required to attend the Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control, perform submittal review and approval, ensure testing is performed, and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by others.

1.5.1.2 Qualifications

An individual with a minimum of 5 years combined experience as a superintendent, inspector, QC Manager, project manager, or construction manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must be familiar with the requirements of the EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

The Contractor may propose individuals that do not meet the qualifications specified above by submitting a RFI to the Contracting Officer Representative (COR) along with the individual's qualifications. The COR will review the qualifications of the individuals being proposed and will make a determination to waive the qualifications specified based on the information submitted. If it is determined the proposed individual fails to perform the duties required herein, the Contractor must replace the individual with an individual that meets the qualifications specified. If the QC Manager is dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out. Any schedule delays or additional costs incurred are the Contractor's responsibility.

1.5.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager must have completed the course entitled "Construction Quality Management (CQM) for Contractors" and must have a current certificate. If the QC Manager does not have a current certification, they must obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

1.5.2 Alternate QC Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The period of absence may not exceed two weeks at one time, and not more than 30 workdays during a calendar year. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.5.3 Not Used

1.6 QC PLAN

Submit a QC plan within 15 calendar days after receipt of Notice of Award.

1.6.1 Requirements

Provide, for acceptance by the Contracting Officer, a QC plan, prior to start of construction, that covers both on-site and off-site work and includes the following with a table of contents listing the major sections identified, with pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project:

- I. QC ORGANIZATION: A chart showing the QC organizational structure and its relationship to the production side of the organization.
- II. NAMES AND QUALIFICATIONS: In resume format, for each person in the QC organization. Include the CQM for Contractors course certification required by the paragraphs entitled CONSTRUCTION QUALITY MANAGEMENT TRAINING and ALTERNATE QC MANAGER DUTIES AND QUALIFICATIONS.
- III. DUTIES, RESPONSIBILITY, AND AUTHORITY OF QC PERSONAL: Of each person in the QC organization.
- IV. OUTSIDE ORGANIZATIONS: A listing of outside organizations such as architectural and consulting engineering firms that will be employed by the Contractor and a description of the services these firms will provide.
- V. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager and stating that they are responsible for managing and implementing the QC program as described in this Contract. Include in this letter the QC Manager's authority to direct the removal and replacement of non-conforming work.
- VI. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, and managing submittals. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval.
- VII. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraphs ACCREDITED LABORATORIES or TESTING LABORATORY REQUIREMENTS, as applicable.
- VIII. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, referenced by the specification Section paragraph number requiring the test, the frequency, and the person

responsible for each test.

IX. PROCEDURES TO COMPLETE REWORK ITEMS: Procedures to identify, record, track, and complete rework items.

X. DOCUMENTATION PROCEDURES: Use Government form.

XI. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task, which is separate and distinct from other tasks, that has the same control requirements and work crews. The list must be cross-referenced to the Contractor's Construction Schedule and the specification Sections. For projects requiring a Progress Chart, the list of definable features of work must include, but not be limited to, all items of work on the schedule. For projects requiring a Network Analysis Schedule, the list of definable features of work must include, but not be limited to, all critical path activities.

XII. PROCEDURES FOR PERFORMING THREE PHASES OF CONTROL: For each DFOW provide Preparatory and Initial Phase Checklists. Each list must include a breakdown of quality checks that will be used when performing the quality control functions, inspections, and tests required by the Contract Documents. The preparatory and initial phases must be conducted with a view towards obtaining quality construction by planning ahead and identifying potential problems.

XIII. PERSONNEL MATRIX: Not Applicable.

XIV. PROCEDURES FOR COMPLETION INSPECTION: See the paragraph entitled "COMPLETION INSPECTIONS".

XV. TRAINING PROCEDURES AND TRAINING LOG: Not Applicable.

1.7 COORDINATION AND MUTUAL UNDERSTANDING MEETING

During the Pre-Construction meeting and prior to the start of construction, discuss the QC program required by this Contract. The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor will be required to explain how the three phases of control will be implemented for each DFOW. Contractor's personnel required to attend must include the QC Manager, Project Manager, and Superintendent. Minutes of the meeting must be prepared by the QC Manager and signed by both the Contractor and the Contracting Officer. The Contractor must provide a copy of the signed minutes to all attendees. Repeat the coordination and mutual understanding meeting when a new QC Manager is appointed.

1.8 QC MEETINGS

After the start of construction, the QC Manager must conduct QC meetings once every week at the work site with the Superintendent and the foreman responsible for the ongoing and upcoming work. The QC Manager must prepare the minutes of the meeting and provide a copy to the Contracting Officer within two working days after the meeting. As a minimum, the following must be accomplished at each meeting:

- a. Review the minutes of the previous meeting;

- b. Review the schedule and the status of work and rework;
- c. Review the status of submittals and RFIs;
- d. Review the work to be accomplished in the next two weeks and documentation required;
- e. Resolve QC and production items (RFIs, etc.);
- f. Address items that may require revising the QC plan;
- g. Review Accident Prevention Plan (APP);
- h. Review environmental requirements and procedures;
- i. Review Waste Management Plan;
- j. Review IAQ Management Plan;
- k. Review Environmental Management Plan;
- l. Review the status of training completion;
- m. Review the progress of As-Builts and O&M and eOMSI Documentation; and
- n. Review Non-Compliance Notices and any actions required to address non-complaint work/actions.
- o. Review Special Inspections (Section 01 45 35) and any deficiencies identified by the Special Inspectors. Review actions taken or planned actions to correct any deficiencies noted.

1.9 THREE PHASES OF CONTROL

The three phases of control must adequately cover both on-site and off-site work and must include the following for each Definable Feature of Work (DFOW).

1.9.1 Preparatory Phase

Notify the Contracting Officer at least five (5) work days in advance of each preparatory phase.

The meeting must be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent, the Project SSHO, the CxC, the Special Inspector, the Special Inspector of Record, and the foreman responsible for the DFOW or as approved by the Contracting Officer. When the DFOW will be accomplished by a Subcontractor, that Subcontractor's foreman must attend the preparatory phase meeting. Document the results of the preparatory phase actions in the daily Contractor Quality Control Report and in the Preparatory Phase Checklist. (Use Attachment A for the PWD ME Preparatory Phase Checklist) Perform the following prior to beginning work on each DFOW (Note: Preparatory Meeting shall only be held if the shop drawings or submittals have been approved, hard copies are printed for the meeting, the APP and appropriate AHA related to the DFOW have been submitted and the appropriate personnel as stated above are present for the meeting):

- a. Review each paragraph of the applicable specification Sections.
- b. Review the Contract drawings.
- c. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- d. Review the testing plan and ensure that provisions have been made to provide the required QC testing and applicable required A/E (DOR) Quality Assurance Inspections.
- e. Examine the work area to ensure that the required preliminary work has been completed.
- f. Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data.
- g. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- h. Discuss specific controls used and construction methods, construction tolerances, layout/survey controls (horizontal and Vertical controls) necessary, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOV.
- i. Discuss the QC documentation required to be collected as part of the work.
- j. Identify any changed conditions or modifications that may impact the execution of the work.
- k. Review the processes/strategies that will be implemented to address any issues if the work does not go as planned or if any unforeseen conditions are encountered or if any changes arise that may impact the successful execution of the work.
- l. Review the listing of Government QA inspections that may be implemented as part of the execution of the work.
- m. Review any Special Inspections (Section 01 45 35) required as part of the execution of the work.
- n. If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1 Section 23 Demolition, Renovation and Re-Occupancy.
- o. If work includes excavation, at a minimum the PWD ME Dig Safe Coordinator / CM / ET shall be invited to the prep/initial phase meetings and will review Utility Locating Requirements. (Note: Special attention is required if Hazardous or Mission Critical Utilities are within the excavation area.)

If Hazardous and Mission Critical Utilities (as identified in the Scope of Work & Drawings) are located within the excavation area,

the PWD ME Dig Safe Coordinator, PWD ME CM/ET, PWD ME Design Manager (DM) and the assigned PWD ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities is discussed with adequate focus. Contractor to provide Construction CM/ET notification no later than 7 working days prior to holding the preparatory and initial pre-excavation/demo safety review meeting. The CM/ET, as well as the PWD ME DSC, attend this meeting to ensure safety and preservation of critical utilities is discussed with adequate focus.

- p. If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or Hazardous Soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management processes and procedures.)
- q. If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/ approvals that are required to complete the work. (Note: Include PWD ME EV Representatives at this meeting to review the project specific requirements).
- r. If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.)

1.9.2 Initial Phase

Notify the Contracting Officer at least two (2) work days in advance of each initial phase. Conduct the Initial Phase meeting with the foreman responsible for that DFOW. When the DFOW will be accomplished by a Subcontractor, that Subcontractor's foreman must attend the initial phase meeting. Observe the initial segment of the work to ensure that it complies with Contract requirements. (See Attachment B for the Initial Phase Checklist). Document the results of the Initial Phase in the daily CQC Report and in the QC checklist. Perform the following for each DFOW:

- a. Establish the quality of workmanship required;
- b. Resolve conflicts;
- c. Ensure that testing is performed by the approved laboratory; and
- d. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.

1.9.3 Follow-Up Phase

Perform the following for on-going work daily, or more frequently as necessary, until the completion of each DFOW and document in the daily CQC Report and in the QC checklist:

- a. Ensure the work is in compliance with Contract requirements;

- b. Maintain the quality of workmanship required;
- c. Ensure that testing is performed by the approved laboratory;
- d. Ensure that rework items are being corrected;
- e. Assure manufacturers representatives have performed necessary inspections, if required; and
- f. Assure applicable required A/E (DOR) Quality Assurance Inspections are scheduled.

1.9.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phase meetings on the same Definable Feature of Work (DFOW) for all work that is on-going where one or more of the following circumstances apply:

- a. If there are any quality control issues or if the work is determined to be unacceptable as determined by Contracting Officer or QCM.
- b. The quality control personnel are not the same that were initially briefed at the original meetings.
- c. The on-site production supervision or Subcontractor/work crew have changed.
- d. Work on a DFOW is resumed after a substantial period (1 month) of inactivity.
- e. The Contracting Officer's Representative determines additional meetings are needed to address any other problems.
- f. If there are any deviations/changes/modifications to the scope of the DFOW elements modified from the original meeting.
- g. If there are any changes to the methods of execution, weather conditions that may impact the execution or quality of the work.
- h. If there are any changes to the work or conditions that would effect compliance with the terms & conditions of the state/federal permits.

1.9.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least two (2) weeks prior to the start of the preparatory and initial phases.

1.10 SUBMITTAL REVIEW AND APPROVAL

Procedures for submission, review, and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES.

1.11 TESTING

Except as stated otherwise in the specification Sections, perform sampling and testing required under this Contract. The testing and retesting must

be performed at no additional cost to the Government.

1.11.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and will be required to submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory's scope of accreditation must include the appropriate ASTM standards (i.e.; E 329, C 1077, D 3666, D 3740, A 880, E 543) listed in the technical Sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the "Corporate Office."

1.11.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology, the American Association of State Highway and Transportation Officials (AASHTO), International Accreditation Services, Inc. (IAS), U. S. Army Corps of Engineers Materials Testing Center (MTC), and the American Association for Laboratory Accreditation (A2LA).

1.11.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract.

1.11.4 Test Results

Cite applicable Contract requirements, tests, or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer.

1.12 QC CERTIFICATIONS

1.12.1 Contractor Quality Control Report Certification

Each CQC Report must contain the following statement: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge except as noted in this report."

1.12.2 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current and attesting that the work for which payment is requested, including stored

material, is in compliance with Contract requirements.

1.12.3 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested, and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the eOMSI preparer for inclusion into the eOMSI documentation.

1.13 COMPLETION INSPECTIONS

1.13.1 Punch-Out Inspection

Near the completion of all work or any increment thereof established by a completion time stated in the Contract clause "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a punch list of items which do not conform to the approved drawings and specifications. Include in the punch list any remaining items of the "Rework Items List", which were not corrected prior to the Punch-Out inspection. The punch list must include the estimated date by which the deficiencies will be corrected. A copy of the punch list must be provided to the Contracting Officer. The QC Manager or staff must make follow-on inspections to ascertain that all deficiencies have been corrected. Once this is accomplished, the Contractor must notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.13.2 Pre-Final Inspection

The Government and QC manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government pre-final punch list may be developed as a result of this inspection. The QC Manager must ensure that all items on this list are corrected prior to notifying the Government that a "Final" inspection with the customer can be scheduled. Any items noted on the "Pre-Final" inspection must be corrected in a timely manner and must be accomplished before the Contract Completion Date (CCD) for the work or any particular increment thereof if the project is divided into increments by separate completion dates.

1.13.3 Final Acceptance Inspection

The QC Manager, the superintendent, or other Contractor management personnel and the Contracting Officer will be in attendance at this inspection. Additional Government personnel may be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the "Pre-Final Inspection". Notice must be given to the Contracting Officer at least 14 days prior to the final inspection. The notice must state that all specific items previously identified to the Contractor as being unacceptable will be complete by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract clause "Inspection of Construction".

1.14 DOCUMENTATION

Maintain current and complete records of on-site and off-site QC program

operations and activities. The forms identified under the paragraph "INFORMATION FOR THE CONTRACTING OFFICER" must be used. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. The reporting of work must be identified by terminology consistent with the construction schedule. Indicate in the "remarks" section of this report pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, and a record of visitors to the work site. For each remark given, identify the Schedule Activity No. that is associated with the remark.

1.14.1 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting. This information must be readily available to the Contracting Officer during all business hours.

- a. All completed Preparatory and Initial Phase Checklists, arranged by specification Section.
- b. All milestone inspections, arranged by Activity/Event Number.
- c. A current up-to-date copy of the Testing and Plan Log with supporting field test reports, arranged by specification Section.
- d. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- e. A current up-to-date copy of the Rework Items List.
- f. Maintain up-to-date copies of all punch lists issued by the QC Staff on the Contractor and Sub-Contractors and all punch lists issued by the Government.
- g. All Special Inspection Daily Reports and Biweekly Reports.

1.14.2 As-Built Drawings

The QC Manager is required to review the as-built drawings, required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME), to ensure they are kept current on a daily basis and marked to show deviations, which have been made from the Contract drawings. Ensure each deviation has been identified with the appropriate modifying documentation, e.g. PC number, modification number, RFI number, etc. The QC Manager must initial each deviation or revision. Upon completion of work, the QC Manager must submit a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.15 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected

non-compliance with the foregoing requirements. The Contractor must take immediate corrective action. If the Contractor fails or refuses to correct the non-compliant work, the Contracting Officer will issue a non compliance notice. Such notice, when delivered to the Contractor at the work site, must be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. The Contractor must make no part of the time lost due to such stop orders the subject of claim for extension of time, for excess costs, or damages.

1.16 Contractor QC Self-Evaluation

Contracting Officer will provide a "Contractor QC Self-Evaluation checklist" to the Contractor at the pre-construction meeting. Complete the checklist monthly and submit with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Govt. will randomly perform QC inspections following the provided checklist. If the Govt inspection score is less than 90% the Govt will withhold 10% from the next invoice.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

PREPARATORY PHASE CHECKLIST			SPEC SECTION	DATE
CONTRACT NO.	DEFINABLE FEATURE OF WORK (DFOW)	SCHEDULE ACT NO.	INDEX #	
PERSONNEL PRESENT		GOVERNMENT REP NOTIFIED	HOURS IN ADVANCE: ☐ YES ☐ NO	Add Del
NAME	POSITION	COMPANY/GOVERNMENT		
SUBMITTALS				
REVIEW SUBMITTALS AND/OR SUBMITTAL REGISTER. HAVE ALL SUBMITTALS BEEN SUBMITTED AND APPROVED? (IF NO, WHAT ITEMS HAVE NOT BEEN SUBMITTED OR APPROVED) ☐ YES ☐ NO				
ARE ALL MATERIALS ON HAND? ☐ YES ☐ NO IF NO, WHAT ITEMS ARE MISSING.				
CHECK APPROVED SUBMITTALS AGAINST DELIVERED MATERIAL. (THIS SHOULD BE DONE AS MATERIAL ARRIVES.) COMMENTS:				
MATERIAL STORAGE				
ARE MATERIALS STORED PROPERLY? ☐ YES ☐ NO IF NO, WHAT ACTION IS TAKEN?				
SPECIFICATIONS				
REVIEW EACH PARAGRAPH OF SPECIFICATIONS.				
DISCUSS PROCEDURE FOR ACCOMPLISHING THE WORK.				
CLARIFY ANY DIFFERENCES.				
PRELIMINARY WORK				
ENSURE PRELIMINARY WORK IS CORRECT AND PERMITS ARE ON FILE. IF NOT, WHAT ACTION IS TAKEN?				
TESTING				
IDENTIFY TEST TO BE PERFORMED, FREQUENCY, AND BY WHOM.				
WHEN REQUIRED?				
WHERE REQUIRED?				
REVIEW TESTING PLAN.				
HAS TEST FACILITIES BEEN APPROVED?				
SAFETY				
ACTIVITY HAZARD ANALYSIS APPROVED? ☐ YES ☐ NO REVIEW APPLICABLE PORTION OF EM 385-1-1.				
MEETING COMMENTS				
NAVY/ROICC COMMENTS DURING MEETING.				
OTHER ITEMS				
OTHER ITEMS OR REMARKS.				
QC MANAGER			DATE	

****Note: Preparatory Meeting shall only be held if the shop drawings or submittal have been approved, the APP and appropriate AHA related to the DFOV have been submitted and the appropriate personnel (QC Manager, QC Specialists, Project Sup, CxC, Special Inspector, Special Inspector of Record, Foreman responsible for the DFOV, subcontractor foreman) are present for the meeting.****

PREPARATORY PHASE CHECKLIST

Part 2, DIV 1 Compliance

Date	Contract No.	Project Title
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- ☐ Identify any changed conditions or modifications that may impact the execution of the work.
- ☐ Examine the work area to ensure that the required preliminary work has been completed.
- ☐ Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data.
- ☐ Review each paragraph of the applicable specification Sections.
- ☐ Review the Contract drawings
- ☐ Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- ☐ Discuss specific controls used and construction methods, constructions tolerances, layout/survey controls (horizontal and vertical controls) necessary, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential programs for each DFOV.
- ☐ Review the testing plan and ensure that provisions have been made to provide the required QC testing and applicable required A/E (DOR) Quality Assurance Inspections.
- ☐ Review the listing of Government QA inspections that may be implemented as part of the execution of the work.
- ☐ Review any Special Inspections (Section 01 45 35) required as part of the execution of the work.
- ☐ Discuss the QC documentation required to be collected as part of the work. (e.g. Special inspector reports within 24 hours, material testing reports as soon as available.)
- ☐ If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1 Section 23 Demolition, Renovation and Re-Occupancy.
- ☐ If work includes excavation, review Utility Locating Requirements. (Note: special attention is required if Hazardous or Mission Critical Utilities are within the excavation area. (If Hazardous and Mission Critical Utilities (as identified in the Scope of Work & Drawings) are located within the excavation area, the PWD ME Dig Safe Coordinator, PWD ME CM/ET, PWD ME Design Manager (DM) and the assigned PWD-ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities as discussed with adequate focus.)

- ☐ If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or hazardous soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management process and procedures.)
- ☐ If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/approvals that are required to complete the work. (Note: Include PWD-ME EV Representatives at this meeting to review the project specific requirements.
- ☐ If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.
- ☐ Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- ☐ Review the processes/strategies that will be implemented to address any issues if the work does not go as planned or if any unforeseen conditions are encountered or if any changes arise that may impact the successful execution of the work.

QC Manager

[illegible]

SECTION 01 50 00.00 22

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME)
10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2007) Reduced-Pressure Principle Backflow Prevention Assembly

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH (FCCCHR)

FCCCHR List (continuously updated) List of Approved Backflow Prevention Assemblies

FCCCHR Manual (10th Edition) Manual of Cross-Connection Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction Marking and Lighting

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual of Uniform Traffic Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction site plan; G

Traffic control plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Employee Parking Plan; G

SD-03 Product Data

Backflow Preventers; G

SD-06 Test Reports

Backflow Preventer Tests; G

SD-07 Certificates

Backflow Tester Certification; G

Backflow Preventers Certificate of Full Approval; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site construction plan showing the locations and dimensions of temporary facilities (including layouts and details and equipment and material onsite and offsite storage areas), access and haul routes, avenues of ingress/egress to the fenced area and details of the temporary construction safety fencing/barriers systems that complies with EM 385-1-1 Sections 4 and 8.

Identify any areas where vehicle track pads will be installed to prevent the tracking of mud onto the pavement outside the project site limits. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and dewatering system storage tanks and infiltration pits.

Prior to the start of Construction, furnish and erect temporary project safety fencing at the work site in accordance with the plans and the following requirements:

- a. Temporary project fencing (or a substitute acceptable to the Contracting Officer and delineated in the APP) must be provided on all projects. See also EM 385-1-1 Sections 4 and 8.

- b. Fencing must extend from grade to a minimum of 4 feet above grade and must have a maximum mesh size of 2 inches. Fencing must remain rigid/taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.
- c. Signs warning of the presence of construction hazards and requiring unauthorized persons to keep out of the construction area must be posted on the fencing. At a minimum, signs must be posted every 150 feet. Fenced sides of projects that are less than 150 feet must, at minimum, have at least one warning sign.
- d. Depending upon the nature and location of the project site, the Contractor may request not to install temporary fencing in some sections of the project site. The SSHO must submit a risk analysis (AHA) to the Contracting Officer for review. This must be based on a risk analysis of public exposure and other project specific considerations, and must be included in the applicable AHA.
- e. Access gates to the job site shall remain closed when not actively in use to prevent none essential personnel from entering the construction area.

If the Contracting Officer approves the request and has determined fencing is not required, install signs and other acceptable barrier systems, warning of construction hazards, and must be conspicuously posted.

If at any time it is determined the risk to the public changes based on the work, take immediate action to address any risk to the public.

1.4 BACKFLOW PREVENTERS CERTIFICATE

1.4.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in any other phase of this Contract.

1.4.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.5 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Section 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.
- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general [Portsmouth Naval Shipyard][Activity] areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.6 CYBERSECURITY DURING CONSTRUCTION

The Contractor must meet the following requirements throughout the construction process.

1.6.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, Contractor computers must meet the following requirements along with the requirements of paragraph SECURITY RESPONSIBILITIES (PNSY) of Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME):

Note: Any Computers (including tablet, laptop, or other computers) must not be introduced into nor removed from the Portsmouth Naval Shipyard. Obtain approvals from the PNSY Security Officer via the Contracting Officer. Cameras, video equipment, or similar photographic equipment installed in computers must be disabled. Proof of computer Portsmouth Naval Shipyard Security Officer approval must be with the tablet, laptop, or other computers at all times.

1.6.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.6.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.6.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number, and one special character.

1.6.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using Contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.6.2 Temporary IP Networks

Temporary Contractor-installed IP networks may be used during construction. When used, temporary Contractor-installed IP networks must meet the following requirements:

1.6.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.6.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.6.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.6.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum of 8 characters with a minimum of one uppercase letter, one lowercase letter, one number, and one special character.

1.6.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.7 Employee Parking Plan

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. Contractor's employees & Subcontractor's employees must park privately owned vehicles in the Jamaica Island Parking Lot only. Parking is not allowed in any other parking lot or area on the Portsmouth Naval Shipyard. Employee parking must not interfere with existing and established parking requirements of the Portsmouth Naval Shipyard. The Contractor & subcontractors must not violate any Town of Kittery Parking regulations. If a privately owned vehicle is found parked outside the Jamaica Island parking lot or found violating any Shipyard or Town of Kittery Parking Regulations, the vehicle will be subject to towing, a non-compliance notice will be issued and may reflect poorly on the Prime Contractor's performance rating (CPARS) and noted in their final performance evaluation. It should be noted that there may not be sufficient on-shipyard parking available for all contractor & subcontractor employees. The Contractor may need to make arrangement for off shipyard parking and shuttling employees on to the Shipyard. Limited employee parking will not be a basis for a claim for additional time or costs. The Contractor must provide an employee parking plan that must include the estimated amount employee parking required. There is a Maine DOT Park & Ride located in Kittery (<https://www.kitteryme.gov/home/news/maine-department-transportation-launches-new-park-a>). The Contractor must provide the anticipated number of contractor employees and subcontractor employees with they anticipated number of Contractor/Subcontractor POV parking that will be required to support the construction effort. The Contractor's plan must identify their plan to ensure employees will park at the Jamaica Island Parking Lot only. The Plan must include proposed any off-yard parking and shuttling plans that may be required to support the construction activities. The Plan must provide anticipated parking requirements vs the planned construction schedule.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Section 01, and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as indicated or as specified in Section 01 58 00 PROJECT IDENTIFICATION (if included in the specifications). Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Section 04. Attach signs to the perimeter fencing a maximum of every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

At no expense to the Government, construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1 Section 04. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and haul roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas, or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.2.3 Temporary Security Lighting

Provide temporary security lighting if any existing permanent security lighting is disturbed during the execution of the work.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized people. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work. Fencing must be installed to be able to restrain a force of at least 250 pounds against it.

2.3.1 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.3.2 TEMPORARY SECURITY LIGHTING

Provide temporary security lighting if any existing permanent security lighting is disturbed during the execution of the work. Any CIA lighting outages must be approved through code 1122 and temporary lighting must be provided that is equal to or greater than the current foot-candle rating of the area. Allow 5 working days for review.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Section 11, NFPA 241, and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size, and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements of AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron mounted gate valve for water lines greater than 2 inches and a lead free, brass body valve on water lines 2 inches and smaller, strainer, and 304 stainless steel or bronze internal parts.

Backflow Preventers must be reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged mounted gate valve, stainless steel or bronze, internal parts. The particular make, model/design, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies issued by the FCCCHR List and be accompanied by a Certificate of Full Approval from FCCCHR List. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the

Government installation. Refer to Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) paragraph EMPLOYEE PARKING for additional requirements.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.2.2 Government Provided Utility Services

- a. The Government will make all reasonably required utilities available from existing outlets and supplies, as specified in the Contract. Reasonable amounts of the following utilities will be made available without charge. Carefully conserve utilities furnished without charge.

Electricity
Water

- b. The point at which the Government will deliver such utilities and the quantity available must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing backflow-preventing devices on connections to domestic water lines; providing meters; and providing transformers; and make disconnections. Provide the backflow devices as specified above and NAVFAC PWD ME shop personnel (at the Portsmouth Naval Shipyard only) will install the backflow preventer. Do not operate any Portsmouth Naval Shipyard water system valves. Notify the Contracting Officer a minimum of 15 calendar days prior to the desired date of the installation of the backflow device.

3.2.2 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities approved in accordance with EM 385-1-1 Section 02 and by the Contracting Officer. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or Portsmouth Naval Shipyard sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Maintain these conveniences at all times without nuisance. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.3 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.4 Obstruction Lighting of Cranes

Provide a minimum of two (2) aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 100 feet above ground level. Light construction and installation must comply with FAA AC 70/7460-1. Lights must be operational during periods of reduced

visibility, darkness, and as directed by the Contracting Officer.

3.2.5 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

3.3.1 Restricted Access Areas

The Government will monitor work in areas indicated. Notify the Contracting Officer at least 14 calendar days prior to starting work in these areas.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide to the Contracting Officer a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the State of Maine Department of Transportation have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct vehicle or pedestrian traffic.
- c. Provide, erect, and maintain, at no expense to the Government, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage and overhead protection authority having jurisdiction.
- d. The area adjacent to the project site will be an active Shipyard work site. All work shall be coordinated to avoid impacting Shipyard operations. Work shall not impact Shipyard Fire Department access in any way unless otherwise approved by the Contracting Officer, Government and Shipyard Fire Department via outage request process.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and

property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. The Contractor is responsible for the repair of damage to roads caused by construction operations.

3.4.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.4.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

3.5 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

3.6 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable State or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1 Section 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.6.1 Safety Systems

Protect the integrity of any installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.6.2 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to Contractor personnel.

3.6.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Include plastic strip inserts, colored green, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the boundaries of the [Portsmouth Naval Shipyard][Activity]. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.6.4 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the [Portsmouth Naval Shipyard][Activity] boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.6.5 Maintenance of Storage Area(s)

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, must be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles; gravel gradation will be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.6.6 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate State and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Portsmouth Naval Shipyard property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.6.7 Laydown Space

Parking and laydown space on the site is limited to the area shown on the plans. Manage the on-site work including equipment, storage trailers, material, material deliveries to allow the work to be completed within the specified Contract duration. This may require the Contractor to locate suitable storage off-site and multiple equipment mobilizations to allow the work to be completed. Equipment or materials not used to complete the work must be removed from the site. If additional offsite storage, additional

mobilization or demobilizations are required, all these costs must be included in the base bid.

Failure to plan the work based on the space limitations must not be the basis for any claim nor an equitable price or Contract time adjustment.

3.6.8 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the temporary facilities are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the temporary facilities from damage.

3.6.9 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools, and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6.10 Temporary Partitions

Provide "No Dust" temporary partitions of wood or metal frame, heavy duty plastic sheathing and negative pressure HEPA filtered ventilation. Provide access door and vestibules as required to prevent dust from escaping the enclosure. Refer to Section 02 41 00 [DEMOLITION] [AND] [DECONSTRUCTION] for additional information and requirements if included as part of the specifications.

3.7 PLANT COMMUNICATION

Whenever there are individual elements of the plant located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and made available for use by Government personnel.

3.8 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site as specified in paragraph entitled CONSTRUCTION SITE PLAN herein. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.9 CLEANUP

Remove construction debris, waste materials, packaging material, and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.10 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and any other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the areas to their original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION 01 57 19.00 22

TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME)
10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard only.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Note: This is not an all-inclusive list of publications and other references may be applicable.

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846 (Third Edition; Update IV) Test Methods
for Evaluating Solid Waste:
Physical/Chemical Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.120	Hazardous Waste Operations and Emergency Response
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 82	Protection of Stratospheric Ozone
40 CFR 112	Oil Pollution Prevention
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of

	Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 270	EPA Administered Permit Programs: The Hazardous Waste Permit Program
40 CFR 272	Approved State Hazardous Waste Management Programs
40 CFR 273	Standards for Universal Waste Management
40 CFR 279	Standards for the Management of Used Oil
40 CFR 280	Technical Standards and Corrective Action Requirements for Owners and Operators of Underground Storage Tanks (UST)
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 302	Designation, Reportable Quantities, and Notification
40 CFR 355	Emergency Planning and Notification
40 CFR 372-SUBPART D	Specific Toxic Chemical Listings
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

STATE OF MAINE REGULATIONS

The following STATE OF MAINE REGULATIONS are available on the Internet at:
<http://www.maine.gov/dep/permits.htm>

STATE OF MAINE Statutes are available on the internet at
<https://legislature.maine.gov/statutes/38/title38ch0sec0.html>

STATE OF MAINE RULES are available on the internet at
<https://www.maine.gov/sos/cec/rules/06/chaps06.htm>

MAINE DEP AIR BUREAU CHAPTER 101 Visible Emissions Regulations;
<http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP AIR BUREAU CHAPTER 151 Architectural and Industrial Maintenance (AIM) COATINGS; <http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP AIR BUREAU CHAPTER 159 Control of Volatile Organic Compounds from Adhesives and Sealants; <http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP 38 MSRA 420-C Erosion and Sedimentation Control Law and Rules

MAINE DEP 38 MSRA 420-D Stormwater Management

MAINE 38 MRSA 439-B Contractors Certified in Erosion Control (Latest Edition)

MAINE DEP MSRA 481-490 Site Location of Development

MAINE 38 MSRA 850 Identification of Hazardous Waste

MAINE 38 MSRA 851 Standards for Generators of Hazardous Waste

MAINE 38 MSRA 852 Land Disposal Restrictions

MAINE 38 MSRA 853 Licensing of Transporters of Hazardous Waste

MAINE 38 MSRA 857 Hazardous Waste Manifest Requirements

MAINE 38 MSRA 858 Universal Waste Rules

MAINE 38 MSRA 860 Waste Oil Management Rules

MAINE 88 MRSA 480A-480Z Natural Resources Protection Act

CODE OF MAINE RULE 06-096 Chapter 500 Stormwater Management

CODE OF MAINE RULE 06-096 Chapter 573 Snow Dumps: Best Management Practices for Pollution Prevention

MAINE DEP Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition.

MEPDES Permit #MER042000 General Permit for the Discharge of Stormwater from State or Federally Owned Municipal Separate Storm Sewer Systems

PORTSMOUTH NAVAL SHIPYARD INSTRUCTIONS

NAVSHIPY PTSMH INST 5090.8 Environmental Sampling Manual

NAVSHIPY PTSMH INST 5090.30 Hazardous Waste Generator Standards

1.2 DEFINITIONS

1.2.1 Chemical Wastes

This includes salts, acids, alkalies, herbicides, pesticides, and organic chemicals.

1.2.2 Garbage

Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.3 Hazardous Debris

As defined in Solid Waste paragraph, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) per 40 CFR 261; or debris that exhibits a characteristic of hazardous waste per 40 CFR 261.

1.2.4 Hazardous Materials

Hazardous materials as defined in 49 CFR 171 and listed in 49 CFR 172.

Hazardous material is any material that:

- a. Is regulated as a hazardous material per 49 CFR 173, or
- b. Requires a Safety Data Sheet (SDS) per 29 CFR 1910.120, or
- c. During end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D.

Designation of a material by this definition, when separately regulated or controlled by other instructions or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs). Nonetheless, the exposure may occur incident to manufacture, storage, use and demilitarization of these items.

1.2.5 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibit a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D or as identified by Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860. Hazardous waste controls also apply to Universal Wastes.

1.2.6 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by installations to obtain NPDES permit coverage for their stormwater discharges.

1.2.7 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.8 Non-Emergency Spill Event

A non-emergency spill event is a discharge of a known material or any hazardous substance that does not pose an immediate threat to human health or the environment, can be cleaned up as part of normal housekeeping by the personnel who discovered the spill, and is not released on the soil or into any waterway inlet (for example, storm drain) or outside Navy property boundaries.

1.2.9 Oil or Oily Waste

Oil: Oil of any kind or in any form, including, but not limited to: fats, oils, or greases of animals, fish or marine mammal origin; vegetable oils, including oils from seeds, nuts, fruits, or kernels; and, other oils and greases, including petroleum fuel oil, sludge, synthetic oils, mineral oils, oil refuse, or oil mixed with wastes other than dredged oil.

Oily Waste: Those materials which are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by POLs.

All oily waste must be collected and turned in to Building 357 for disposal. A hazardous waste determination form may be required prior to disposal. All waste characterization sampling must be conducted in accordance with Paragraph 3.13.4.2 herein entitled SAMPLING AND ANALYSIS OF HAZARDOUS WASTE. Oily waste must be minimized through good housekeeping practices and employee education.

1.2.10 Ozone Depleting Substance (ODS)

Chlorofluorocarbons (CFCs), halons or chlorinated hydrocarbons (such as carbon tetrachloride and methyl chloroform), and hydrochlorofluorocarbon (HCFCs) which have been linked to depletion of the earth's ozone layer are all substances collectively known as ozone depleting substances or ODSs. Class I or Class II ODS substances are defined and listed in the Clean Air Act Section 602 and 40 CFR 82.

1.2.11 Regulated Waste

Those solid wastes that have specific additional Federal, State, or local controls for handling, storage, or disposal.

1.2.12 Reportable Release

A reportable release means any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment of a known or unknown material or

hazardous substance that poses an immediate threat to human health or the environment to the air, soil, or water. Reportable releases are: a sheen of oil on the water; a violation of the Installation's or project's water permit (NPDES permit(s)); A sewage spill that threatens human health or the environment; a Comprehensive Environmental Response, Compensation, and Liability Act reportable quantity for hazardous/toxic substances (40 CFR 302); an air or hazardous substance release that is a threat to human health or the environment or released outside the facility boundaries.

1.2.13 Sediment

Soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.14 Solid Waste

Garbage, refuse, debris, sludge, or other discharged material, including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations. Types of solid waste typically generated at construction sites may include:

- a. Debris: Non-hazardous solid material generated during the construction, demolition, or renovation of a structure which exceeds 2.5 inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories, and weldments.
- b. Green waste: The vegetative matter from landscaping, land clearing, and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps, and plant roots. Marketable trees, grasses, and plants that are indicated to remain, be re-located, or be re-used are not included.
- c. Hazardous Waste: By definition, to be a hazardous waste a material must first meet the definition of a solid waste. Hazardous waste and hazardous debris are special cases of solid waste. They have additional regulatory controls and must be handled separately. They are thus defined separately in this document.
- d. Non-Hazardous Waste: Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 263.
- e. Paint cans: Metal cans that are RCRA empty of paints, solvents, thinners, and adhesives. NOTE: Aerosol (paint) cans are Hazardous Wastes and must not be disposed of as solid waste or be considered in any definition of "empty", "paint", or "metal" cans.
- f. Recyclables: Materials, equipment, and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing, and mirrors that are recovered and sold as recyclables. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.
- g. Scrap metal: Scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the

definition of hazardous material or hazardous waste is not included. Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

- h. Surplus soil: Existing non-hazardous soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving.
- i. Wood: Dimension and non-dimension lumber, plywood, chipboard, and hardboard. Treated and/or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included.
- j. Material not regulated as solid waste are: nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.
- k. Special Waste: "Special waste" means any solid waste generated by sources other than household and typical commercial establishments that do not meet the definition of a hazardous waste but exists in such an unusual quantity or in such a chemical or physical state, or any combination thereof, that may disrupt or impair effective waste management or threaten the public health, human safety, or the environment and requires special handling, transportation, and disposal procedures. Special waste includes, but is not limited to:
 - (1) Ash;
 - (2) Industrial and industrial process waste;
 - (3) Sludge and dewatered septage;
 - (4) Debris from nonhazardous chemical spills and cleanup of those spills;
 - (5) Contaminated soils and dredge materials;
 - (6) Asbestos and asbestos-containing waste;
 - (7) Sand blast grit and non-liquid paint waste;
 - (8) PCB and PCB contaminated waste;
 - (9) High and low pH waste;
 - (10) Spent filter media residue;
 - (11) Shredder residue; and
 - (12) Railroad ties with or without creosote.

1.2.15 Sewage

Liquid waste designated by the Government as "domestic sanitary sewage" and normally discharged through domestic sanitary sewage systems. Liquids designated as "sewage" include human body waste, and wastewater from

sinks, showers, laundries, dishwashers, and garbage disposals when these liquids use only chemicals approved by the Government for discharge into the sanitary sewer.

1.2.16 Spill Event

A spill is any release of oil or hazardous substances to the water or ground that is not controlled or permitted. This includes any spilling, leaking, pumping, emitting, discharging, injecting, escaping, leaching, disposing, or dumping of liquid or solid material that is not authorized in writing by the Contracting Officer.

1.2.17 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, and mercury-containing equipment (e.g., thermostats) and lamps (e.g., fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.2.18 Waste Hazardous Material (WHM)

Any waste material which because of its quantity, concentration, or physical, chemical, or infectious characteristics may pose a substantial hazard to human health or the environment and which has been so designated. Used oil not containing any hazardous waste, as defined above, falls under this definition.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with [Section 01 33 00 SUBMITTAL PROCEDURES][Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES]:

SD-01 Preconstruction Submittals

Preconstruction Survey; G

Solid Waste Management Plan; G

Regulatory Notifications; G

Environmental Management Plan (EMP); G

Dirt and Dust Control Plan; G

Contractor Hazardous Material Inventory Log; G

Stormwater Management/Erosion and Sedimentation Control Plan; G

Spill Prevention, Control, and Countermeasures (SPCC) Plan; G

SD-06 Test Reports

Laboratory Analysis; G

Disposal Requirements; G

Erosion and Sediment Control Inspection and Corrective Action; G

Solid Waste Management Report; G

SD-11 Closeout Submittals

Some of the records listed below are also required as part of other submittals. For the "Records" submittal, maintain on-site a separate three-ring Environmental Records binder and submit at the completion of the project. Make separate parts to the binder corresponding to each of the applicable sub items listed below.

Stormwater Management and Erosion Control Compliance Notebook; G

Waste Determination Documentation; G

Disposal Documentation for Hazardous and Regulated Waste; G

Contractor 40 CFR Employee Training Records; G

Solid Waste Management Report; G

Contractor Hazardous Material Inventory Log; G

Hazardous Waste/Debris Management; G

Regulatory Notifications; G

Asbestos Free Certification Form; G

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined herein. Plan for and provide environmental protective measures to control pollution that develops during normal construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Comply with Federal, State, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Conduct tests and procedures for the purpose of assessing whether construction operations are in compliance with Applicable Environmental Laws. Analytical work must be done by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Environmental Compliance Assessment Training and Tracking System (ECATTS)

The QC Manager is responsible for environmental compliance on projects unless an Environmental Manager is named. The QC Manager (and alternative QC Manager) or Environmental Manager must complete ECATTS training prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete ECATTS training within 14 days of

assignment to the project.

Submit an ECATTS certificate of completion for personnel who have completed the required "Environmental Compliance Assessment Training and Tracking System (ECATTS)" training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Training and Tracking System, by logging on to <https://environmentaltraining.ecatts.com/start>. Obtain the password for registration from the Contracting Officer.

This training has been structured to allow personnel to receive credit under this Contract and also to carry forward credit to future contracts. Ensure the QC Manager (and alternate QC Manager) or Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work. Some training modules are tailored for specific State regulatory requirements; therefore, working in multiple States will require personnel to re-take modules tailored to the State where the Contract work is being performed.

ECATTS is available for use by all personnel associated with this project. These other personnel are encouraged (but not required) to take the training and may do so at their discretion.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Survey

Perform a Preconstruction Survey of the project site with the Contracting Officer, and when requested, take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record with a copy provided to the Contracting Officer. Obtain a camera pass from Portsmouth Naval Shipyard Security (via Contracting Officer) prior to use of a camera at Portsmouth Naval Shipyard. Digital cameras can only be used. All computer discs must be turned over to Portsmouth Naval Shipyard Security (via Contracting Officer) for review and clearance prior to use.

1.5.2 Regulatory Notifications

Prepare all regulatory notification requirements in accordance with Federal, State, and local regulations. Regulatory notifications must be submitted by the Government unless otherwise directed by the Contracting Officer. Submit copies of all regulatory notifications to the Contracting Officer prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all inclusive): demolition, renovation, remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.5.3 Environmental Brief

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. The Brief will be conducted by the Contracting Officer's Representative (COR) with PWD ME & Shipyard C106 Environmental Representatives. The Brief will be part of the Pre-Construction Meeting agenda or held as a separate meeting depending on the size and scope of work and potential environmental impact, as determined by the Government.

The meeting will discuss details for all the submittals that are required

in Section 01 57 19.00 22 including review of the submittal requirements for the Environmental Management Plan, important notes regarding Installation Restoration (IR), Historic, and Cultural Resources, Wetlands, and Waste Generation, Concrete Wash Water/Wastewater, Hazard Materials Management including sampling, testing and disposal requirements and other applicable topics.

At the Brief, Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the Portsmouth Naval Shipyard; types and quantities of wastes/wastewater that may be generated during the Contract; types and quantities of oil that will be brought onto the activity; and pollution control measures for spill prevention and control, and any bulk oil storage container information including quantity and type of product stored. Discuss the results of the Preconstruction Survey at this time.

Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural resources, historic properties, required reports, required permits, specific permit requirements, and other measures to be taken. Identify additional environmental concerns specific to the site (e.g., historic, archaeological and natural resources, IR, erosion and sediment control, spill prevention and control, soil management, testing and disposal requirements, etc.

1.5.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager will be directly responsible for coordinating compliance with Federal, State, local, and Portsmouth Naval Shipyard requirements. The Environmental Manager cannot perform the duties of the Project Superintendent or the SSHO. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, daily turn-in, manifesting, and disposal as applicable); implement the Environmental Management Plan; ensure that all environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program Management requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB containing building materials, PCB Universal Waste/e-Waste, etc.). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure all personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal or turn-in of waste containers; implement, inspect, and maintain erosion and sediment controls as required by State law; and maintain the Environmental Records binder and required documentation, ensure compliance with all Spill Prevention, Control, and Countermeasures (SPCC) requirements, including but not limited to, the proper storage of tanks and containers and their secondary containment, inspections, spill procedures, etc. including environmental permits compliance and close-out.

1.5.5 Contractor 40 CFR Employee Training Records

Prepare and maintain employee training records throughout the term of the Contract meeting applicable 40 CFR requirements. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with

Federal, State, and local regulatory requirements for a RCRA Large Quantity Generator. Provide a Position Description for each employee, by Subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements as defined in 40 CFR 265 for a Large Quantity Generator facility. Submit these training records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

1.6 SOLID WASTE DISPOSAL PLAN

Provide a Solid Waste Disposal Plan to Portsmouth Naval Shipyard Environmental Division (Code 106.3) and NAVFAC Environmental (PWD-ME EV) Staff in accordance with Paragraph entitled SOLID WASTE MANAGEMENT PLAN in Part 3 of this Section.

1.7 SITE WASTE REMOVAL (SWR) MEETING

A SWR meeting must be held with the PWD ME CM and Code 106.3 fourteen (14) calendar days prior to the removal of wastes from a project site to ensure all pertinent requirements of the Standard Operating Procedure for Site Waste Removal(SWR)(Attachment B) have been met. No transport of wastes will be allowed until full concurrence is provided by the PWD ME CM and Code 106.3 following the satisfactory completion of the SWR.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ENVIRONMENTAL MANAGEMENT PLAN (EMP)

Prior to initiating any work on site, meet with the Contracting Officer, Code 106.3, and PWD-ME EV Staff to discuss the proposed Environmental Management Plan (EMP) and develop a mutual understanding relative to the details of environmental protection required to be addressed in the EMP, including measures for protecting natural resources and other measures to be taken.

The EMP must be submitted in the following format and include all sections specified below, unless a separate plan submittal is accomplished for the EMP section as identified in a.(1)(b). All sections not applicable to the work shall be marked N/A.

a. Description of the Environmental Management Plan

(1) General overview and purpose

(a) A brief description of the work to be performed to include each type of work activity that will potentially impact the environment (e.g. soil excavation, hazardous waste, abatement (PCB, Lead, Asbestos), concrete slurry/wastewater, emergency generator installation, paint blasting, 55 gallons and greater oil/fuel storage etc.).

(b) A list of other plans required by environmental permit or being submitted separately (e.g. Solid Waste Management Plan, Hazardous Waste Management Plan, Storm Water Pollution Prevention

Plan (SWPPP), Spill Prevention Controls and Countermeasures (SPCC) plan, etc).

(c) A copy of any standard or project specific operating procedures that will be used to effectively manage and protect the environment on the project site.

(d) Communication and training procedures that will be used to convey environmental management requirements to employees and Subcontractors.

(e) Emergency contact information for senior construction project contractor personnel and those responsible for environmental management and spill response (office phone number, cell phone number, and e-mail address).

- (2) General site information including a site plan showing haul routes, construction material laydown and storage areas, construction trailer locations, sanitary facilities, excavation and soil stockpile locations, dewatering and infiltration areas, location map showing selected erosion and sediment controls, and all other construction facilities required for the work.
- (3) A letter signed by an officer of the firm appointing the Environmental Manager and stating that he/she is responsible for managing and implementing the Environmental Program as described in this Contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

b. Management of Natural Resources

- (1) Land resources
- (2) Tree protection
- (3) Replacement of damaged landscape features
- (4) Temporary construction
- (5) Stream crossings
- (6) Fish and wildlife resources
- (7) Wetland areas

c. Protection of Historical and Archaeological Resources

- (1) Objectives
- (2) Methods

d. Dirt and Dust Control

- (1) Describe how dust will be controlled in construction zones, along haul routes, and at construction entrances/exits.
- (2) List of control measures, operational practices, and BMPs to be employed for dirt and dust control. Selected BMPs must adhere to

the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Examples include construction entrance/exit design, dust suppressants, stabilization areas, using covers on dump trucks, and sweeping and vacuuming.

e. Stormwater Management and Control

Additional information/requirements for stormwater management and control are found in the Erosion and Sediment Control Measures section and SOLID WASTE MANAGEMENT PLAN section of this specification and may be necessary based on the work to be performed. This plan section must include:

- (1) A statement that dumpsters and rollofs will be equipped with a secure lid or covering. Watertight dumpsters and rollofs are preferred. All dumpsters and rollofs must be kept closed at all times, except when being loaded with trash or construction debris.
- (2) Erodible soil areas.
- (3) Temporary measures for runoff control.
- (4) Best Management Practices (BMPs) selected for use from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition.
- (5) A location map of selected erosion and sediment controls and manufacture data sheets of products to be used (e.g. erosion control sock, silt sacks, filter fabric etc.).
- (6) A statement that erosion and sediment control inspections will be performed for all projects with ground disturbances greater than or equal to 1 cubic yard and comply with erosion and sediment control practices within a shoreland zone. Inspections must be completed at least once per week as well as BEFORE and AFTER a storm event that has the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater.
- (7) The erosion and sediment control inspector(s) name(s) and current certificate for Maine Certification in Erosion Control Practices.
- (8) A statement that a certified individual will be responsible for management of erosion and sediment control practices at the site each day earth-moving activities occur.
- (9) Erosion and sediment control BMP maintenance and corrective action procedures, see Erosion and Sediment Control Inspection and Corrective Action section of this specification.
- (10) Include a draft (or blank) inspection form that will be used for Erosion and Sediment Control Inspections.
- (11) Concrete wastewater/slurry containment and turn-in procedures. All concrete wastewater/slurry for disposal must be turned in to building 357 by the end of the shift generated.

f. Protection of the Environment from Waste Derived from Contract Operations

- (1) Control and disposal of solid and sanitary waste.
- (2) Control and disposal procedures for hazardous materials and hazardous waste management. Additional information for requirements below are found in the Hazardous Waste/Debris Management Section of this specification and may be necessary based on the work to be performed.

All hazardous waste generation must comply with the procedures in NAVSHIPY PTSMH INST 5090.30 Hazardous Waste Generator Standards. PTSMH INST 5090.30 will be provided by the Contracting Officer.

This section must include:

(a) A list and projected amount of potentially hazardous waste, universal waste or otherwise regulated waste streams that may be generated. Hazardous waste, universal waste, and other regulated waste must be turned in to Building 357, Hazardous Waste Storage Facility (HWSF) by the end of the shift generated, unless otherwise approved by Code 106.3 (e.g. used oil, concrete wastewater/slurry, paint and paint related debris, used aerosol cans, etc.).

(b) Procedures to be employed to complete a written waste determination for wastes which are to be generated. Written waste determinations and procedures will be as requested by the Government;

(c) Sampling/analysis plan, if a separate sampling/analysis plan is not required elsewhere in the contract for the work to be performed, see CONTROL AND MANAGEMENT OF HAZARDOUS WASTES section of this specification for details;

(d) Management procedures for hazardous materials and hazardous waste daily accumulation, storage (if approved by Code 106.3), labeling, transportation, turn-in and disposal of waste. All waste turned in to Building 357 must be in containers provided by HWSF personnel and have the contents clearly identified. Disposal or treatment of hazardous waste by the Contractor is not allowed unless specifically noted in contract documents or approved by Code 106.3;

(e) Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, as applicable (40 CFR 268);

(f) Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar;

(g) Used oil management procedures in accordance with 40 CFR 279 and PTSMH INST 5090.30. All used oil must be turned in to Building 357 by the end of the shift generated;

(h) Pollution prevention/hazardous waste minimization procedures;

(i) Plans for the disposal of hazardous and other regulated waste by permitted facilities (as approved by Code 106.3); and

(j) Procedures to be employed to ensure all required employee training records are maintained.

g. Prevention of Releases to the Environment

This plan section must include:

- (1) Procedures to prevent releases to the environment, emergency response procedures and notifications to make in the event of a release to the environment, see RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES section of this specification for items to include in this plan section.
- (2) A Spill Prevention, Control, and Countermeasures (SPCC) Plan is required if work is anticipated to extend beyond 6 months, AND will use bulk oil storage containers 55 gallons or greater, in accordance with 40 CFR 112. All SPCC plans must be approved by Code 106.3. Plans need not be certified by a Professional Engineer but must clearly demonstrate proper management of all tanks and containers on site.
- (3) Spill plans must include:
 - (a) Type of tank or container, quantity stored, type of product stored, and location.
 - (b) Secondary containment required for tanks/containers 55 gallons or greater; double-wall tanks preferred.
 - (c) Tank inspection forms (industry standard forms are acceptable, but the use of Portsmouth Naval Shipyard inspection forms is preferred). Records must be kept for 3 years or for the duration of the project. Tanks must be inspected monthly.
 - i) Bulk storage containers, to include those on equipment, 55 gallons or greater require monthly inspection.
 - ii) Inspection procedures and an inspection sheet for the release of retained stormwater from secondary containment used for bulk storage tanks and containers.
 - (d) Where spill kits are located and a description of the spill kit contents for the type of spill anticipated.
 - (e) If transferring fuel: how often, what type of fuel, and where. Coordinate with the Contracting Officer's Representative (COR) and Code 106.3 prior to transferring fuel over water.
 - (f) Who to notify in case of ANY spill (Portsmouth Naval Shipyard Fire Department: 207-438-2333, Contracting Officer's Representative, NRC, MEDEP, etc.). All regulatory notification must be coordinated with Code 106.3.
 - (g) How to clean up a spill safely and a statement that all spill cleanup waste will be brought to Building 357 by the end of the shift generated.

h. Air Emissions/Air Permitting

- (1) Installation of a permanent emergency generator for use by the government requires coordination with Code 106.3 for air permitting determination prior to the installation of any portion of the generator or generator support infrastructure.
- (2) Provide details of painting and paint blasting operations to include the following, as applicable.
 - (a) Spray painting and blasting must be performed within containment.
 - (b) Tarps used to contain the material must prevent visible emissions from exceeding 20% opacity.
 - (c) Unless otherwise approved, debris collection systems must be suspended from the structure and extend a minimum of 10' wider in all directions than the area in which work is done. Collection systems must be positioned in a manner such that no loss of paint residue occurs, emptied at least once a day and must not remain in place overnight without being emptied.
 - (d) Painting and Blasting operations must not be performed when the direction and velocity of the wind is such that residue falls outside the limits of the collector.
 - (e) Spray painting dry fall paint will not be allowed if wind speeds exceed 30 feet/s (20mph), unless approved by Code 106.3.

i. Regulatory Notification and Permits

List of notifications made and permits obtained for the work. Demonstrate that those permits were obtained by including copies of all applicable environmental permits.

3.1.1 Environmental Management Plan Review

Within 30 calendar days after the Contract award date, submit the proposed Environmental Management Plan for further discussion, review, and approval.

Commencement of work must not begin until the environmental management plan has been approved by the Contracting Officer, Code 106.3, and PWD-ME EV Staff.

3.1.2 Licenses, State and Federal Permits

Copies of the approved permit(s) are available from the Contracting Officer. Maintain copies of all permits at the project site. Comply with all terms and conditions of the approved permits.

Where required by the State regulatory authority, the inspections and certifications will be provided through the services of a Professional Engineer (PE), registered (licensed) in the State of Maine. Where a PE is not required, the individual must be otherwise qualified by other current State licensure, specific training, and prior experience (minimum 5 years). As a part of the quality control plan, which is required to be submitted for approval by the Quality Control section, provide a sub item containing the name, appropriate professional registration or license number, address, and telephone number of the professionals or other

qualified persons who will be performing the inspections and certifications for each permit.

3.2 PROTECTION OF NATURAL RESOURCES

Preserve the natural resources within the project boundaries and outside the limits of permanent work and as specified in the permits issued for the work. Restore to an equivalent or improved condition upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

Do not disturb fish and wildlife. Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as indicated or specified.

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor will be responsible for any resultant damage.

Protect existing trees which are to remain and which may be injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. By approved excavation, remove trees with 30 percent or more of their root systems destroyed. Remove trees and other landscape features scarred or damaged by equipment operations, and replace with equivalent, undamaged trees and landscape features. Obtain Contracting Officer's approval before replacement. Tree wound paint must not be used for tree cuts or stumps.

3.2.1 Erosion and Sediment Control Measures

The State of Maine Erosion and Sediment Control Law requires persons undertaking activity involving filling, displacing, or exposing soil or other earthen materials to take measures to prevent unreasonable erosion of soil or sediment beyond the project site or into a protected natural resource.

At the Portsmouth Naval Shipyard, the Piscataqua River, Upper Meade Pond, and Lower Meade Pond are protected natural resources under State Law. Erosion control measures must be in place before the activity begins, maintained and must remain in place and functional until the site is permanently stabilized.

Temporary and permanent erosion control measures must meet, at a minimum, the Best Management Practices (BMPs) presented in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Other techniques may be employed if approved by Code 106.3 and PWD-ME EV and demonstrated to the Contracting Officer that the practice will achieve the required result of no release of sediment per State law.

Site work including any filling, excavation, landscaping, and/or other earthwork in excess of one cubic yard of disturbance, must comply with State of Maine requirements for certification in erosion and sediment control practices within a shoreland zone. A certified individual must be responsible for management of erosion and sediment control practices at the site each day earth moving activities occur. A certified individual is

required to visit the site every day to ensure proper erosion and sediment control practices are followed. As an alternative, the Contractor may choose to contract with a certified individual to supervise the work in shoreland areas.

The shipyard is entirely within the federal coastal zone as administered under the Coastal Zone Management Act. The shoreland zone is any area within 250 feet of mean high water.

a. Stormwater Management/Erosion and Sedimentation Control Plan

- (1) Submit a Stormwater Management/Erosion and Sedimentation Control Plan to the Contracting Officer, for review and approval by Code 106.3 and PWD-ME Environmental, for all earthwork in excess of one cubic yard of disturbance. The Plan must demonstrate effective selection, implementation and maintenance of Best Management Practices (BMPs) demonstrating compliance with the Portsmouth Naval Shipyard's Maine Pollutant Discharge Elimination System Municipal Storm-Separate Sewer System Permit (MS4).

CMR 06-096 Chapter 500 Stormwater Management and terms and conditions specified in other approved permits (e.g., Maine Construction General Permit (MCGP), U.S. Army Corps of Engineers (ACOE)) issued for the work.

Provide details of chosen temporary erosion and sediment controls to be employed specific to the work site to include a site plan showing the locations of controls. Ensure proposed controls comply with MEDEP approved plans and State regulations.

Submit Stormwater Management and Erosion Control Compliance Notebook at project completion or as directed by the Contracting Officer.

The Plan must:

- (a) Identify potential sources of pollution which may be reasonably expected to affect the quality of stormwater discharge from the site.
- (b) Describe and ensure implementation of practices which will be used to reduce the pollutants in stormwater discharge at the manufacturing, storage and lay down, and construction sites.
- (c) Describe and ensure full compliance with all MEDEP General Permits (MS4/MCGP) and any other regulatory permits (e.g., ACOE, USEPA) specific to the project.
- (d) Describe and ensure compliance with MEDEP over winter stabilization and construction requirements, as applicable.
- (e) Identify inspections and maintenance schedules for Best Management Practices (BMPs) demonstrating compliance with the State of Maine standards. Maintenance procedures must address regular cleaning of drainage structures and repair of temporary erosion control structures, as well as a final cleaning of all drainage structures and removal and reclamation of temporary erosion and sediment control BMP's upon completion of the project.

- (f) Select applicable best management practices from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Submit a site plan showing locations of controls and provide manufacturer's product sheets for each control to be used.
- (g) Include documentation that the individual responsible for management of erosion and sediment control practices at the site is certified in accordance with the State of Maine DEP regulations.
- (h) Control of Manufactured Concrete Product Waste Plan. At a minimum, must include a description of concrete manufacturing activities and the management of concrete wastewater/wash water as described in Paragraph 3.19 herein entitled CONCRETE WASH WATER.

3.2.2 Stockpiles

Assumed Non-Hazardous soil stockpiles require erosion and sediment controls around the downhill leading edge of the stockpile, or if a leading edge cannot be found, around the entire stockpile perimeter; a MEDEP approved erosion control cover on top of the stockpile when inactive 7 or more days, when stockpile operations are complete and while awaiting soil characterization test results; while awaiting test results, placarded as Pending Characterization and based on those sample results placarded as either Non-Hazardous or Hazardous; immediately cover with 6 mil plastic sheeting if soil characterization test results return as Hazardous.

Known Hazardous (IR Soil) or suspect contaminated soil stockpiles require 6 mil plastic sheeting underneath; erosion control sock around the entire stockpile perimeter; a 6 mil plastic sheeting cover unless actively being worked or while obtaining soil characterization test samples; at all times placarded as Hazardous for IR soil stockpiles. Suspect contaminated soil stockpiles must be placarded as Pending Characterization while awaiting soil characterization test results and based on those results placarded as either Non-Hazardous or Hazardous.

3.2.3 Erosion and Sediment Control Inspection and Corrective Action

Inspection reports must be kept on file at the project site and submitted electronically to the Contracting Officer upon request. The State of Maine requires inspections of disturbed and impervious areas, erosion and sediment control measures, areas used for storage that are exposed to precipitation, and locations where vehicles enter or exit the site.

Inspections must be performed at least once per week as well as BEFORE and AFTER a storm event. A storm event is any precipitation event with the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater. Inspection reports must document compliance with State of Maine requirements. If erosion and sediment control BMPs need to be repaired, the repair work should be initiated upon discovery of the problem but no later than the end of the next workday. If additional BMPs or significant repair of BMPs are necessary, implementation must be completed within 7 calendar days and prior to any storm event (rainfall). For each BMP requiring maintenance, BMP needing replacement, and location needing additional BMPs, note in the inspection report the corrective action taken and when it was taken.

3.2.4 Burnoff

Burnoff of the ground cover is not permitted.

3.3 HISTORICAL AND ARCHAEOLOGICAL RESOURCES

Identify presence of historic properties impacted by the project and outline any required protection methodologies. Carefully protect in-place and report immediately to the Contracting Officer historical and archaeological items or human skeletal remains discovered in the course of work. Upon discovery, stop work in the immediate area of the discovery and protect the area from further disturbance. Notify the Contracting Officer immediately. Do not resume activities within the area until directed by the Contracting Officer to resume work. The Government retains ownership and control over historical and archaeological resources. If historical, archaeological items, or human remains are discovered during excavations for the project, a certified Maine Archaeologist must be on site to evaluate the discovery and monitor excavation work. The qualifications of the Archaeologist must be submitted and approved by the Contracting Officer. A site monitoring report prepared by the Archaeologist must be submitted to the Contracting Officer within 21 calendar days of completing site excavation work. Refer to Section 01 44 00.00 22 ARCHAEOLOGICAL MONITORING (PWD ME) for additional requirements.

3.4 SOLID WASTE MANAGEMENT PLAN

Provide a written Solid Waste Disposal Plan (SWDP) to the Contracting Officer, of intended licensed disposal sites for Government approval and for submission to State regulatory agencies. At a minimum, the SWDP must contain, but not be limited to, the following wastes: stumps and grubblings, excess soil, construction debris, demolition debris, household solid waste, special waste, and industrial solid waste. The submission must contain the name of the disposal facility, address, facility phone number, and the waste type and quantity to be disposed of at the facility.

If waste from the site is taken to a transfer station, identify the facility or facilities at which the waste is ultimately disposed. Government approval for the facility must be obtained prior to transporting wastes off Government property.

Provide to the Contracting Officer written notification of the quantity of solid waste/debris that is anticipated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance or as applicable, submit one copy of a State license showing such agency's approval of the disposal plan before transporting wastes off Government property.

3.4.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report must state the classification (using the definitions provided in this Section), amount, location, and name of the business receiving the solid waste.

Include copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, submit a statement indicating the disposal location for the solid waste which is signed by an officer of the firm authorized to legally obligate or bind the firm. The sales documentation or certification must include the receiver's tax identification number and business, EPA, or State registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for use,

submit on the solid waste disposal report the information previously described in this paragraph. Prices paid or received will not be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.4.2 Control and Management of Solid Wastes

Pick up solid wastes and place into containers which are regularly emptied. Containers must be kept covered except when being loaded with trash and debris. Watertight covered dumpsters and roll-offs are preferred. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Recycling is encouraged and can be coordinated with the Contracting Officer and the Portsmouth Naval Shipyard Recycling Coordinator. Remove all solid waste (including non-hazardous debris) from Government property and dispose off-site at an approved landfill. Solid waste disposal off-site must comply with the most stringent local, State, and Federal requirements including 40 CFR 241, 40 CFR 243, and 40 CFR 258. Discharges from dumpsters and roll-offs are prohibited.

Manage spent hazardous material used in construction including, but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, as per environmental law and Portsmouth Naval Shipyard requirements. Note: Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

3.4.2.1 Dumpsters and Roll-offs

Dry weather discharges from dumpsters and roll-offs are prohibited. Equip dumpsters and roll-offs with a secure lid or covering. Keep covers closed at all times, except when being loaded with trash and debris, or provide secondary containment to ensure that discharges have a control. Locate dumpsters and roll-offs behind the construction fence or out of the public view and away from storm drains. Empty site dumpsters and roll-offs at least once a week or as needed to keep the site free of debris and trash. If necessary, provide 55-gallon trash containers to collect debris in the construction site area. Locate the trash containers behind the construction fence or out of the public view. Empty trash containers at least once a day. For large demolitions, dumpsters and roll-offs must use a cover and watertight containers are preferred; debris must not be higher than the sides before emptying. Water must not be collected in watertight roll-offs or dumpsters. Any water collected in roll-offs, dumpsters, or secondary containment could become contaminated with oil or other contaminants and may require waste characterization by the Contractor prior to release and/or removal from PNS.

3.5 WASTE DETERMINATION DOCUMENTATION

Complete a Waste Determination form (provided by the Contracting Officer at the pre-construction conference) for derived wastes to be generated, as requested by Code 106.3. Base the waste determination upon either a constituent listing from the manufacturer used in conjunction with consideration of the process by which the waste was generated, EPA approved analytical data, and/or laboratory analysis (Safety Data Sheets (SDS) by themselves are not adequate). Attach all support documentation to the Waste Determination form. A Waste Determination form may be requested for the following types of wastes (this listing is not all inclusive): oil and

latex based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and all containers of the original materials.

3.6 CONTRACTOR HAZARDOUS MATERIAL INVENTORY LOG

Submit the "Contractor Hazardous Material Inventory Log" (found at: <http://www.wbdg.org/ccb/NAVGRAPH/graphtoc.pdf>), which provides information required by EPCRA Sections 312 and 313 along with corresponding Safety Data Sheets (SDS) to the Contracting Officer at the start and at the end of construction (30 days from final acceptance), and update no later than January 31 of each calendar year during the life of the Contract. Documentation for any spills/releases, environmental reports, or off-site transfers must be submitted to the Contracting Officer.

3.7 POLLUTION PREVENTION/HAZARDOUS WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of hazardous waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the Environmental Management Plan.

3.8 WHM/HW MATERIALS PROHIBITION

Do not improperly dispose of hazardous materials/hazardous waste on Government property. No hazardous material will be brought onto Government property that does not directly relate to requirements for the performance of this Contract.

Incidental materials used to support the contract including, but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, rags, clothing, empty containers, etc. may be considered hazardous wastes and must be turned in to Building 357 for disposal by the Government as described in the paragraph entitled HAZARDOUS MATERIAL MANAGEMENT of this Section. The list is illustrative rather than inclusive. Universal wastes must be managed with controls similar to those for hazardous waste.

Do not discharge any materials to sanitary sewer, storm drain, or to the Piscataqua River or conduct waste treatment or disposal on Government property without written approval by the Contracting Officer and Code 106.3.

3.9 HAZARDOUS MATERIAL MANAGEMENT

No hazardous material will be brought onto Government property that does not directly relate to requirements for the performance of this Contract. Hazardous materials for disposal may be considered hazardous wastes and must be turned in to Building 357 for hazardous waste determination by the end of the shift generated.

Include hazardous material control procedures in the Environmental Management Plan as described in Paragraph 3.1 herein entitled ENVIRONMENTAL MANAGEMENT PLAN (EMP) and the Safety Plan in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. All hazardous material must be in active use or properly stored. Hazardous materials must not be left unattended or uncontrolled while on Portsmouth Naval Shipyard.

Submit a SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the

Portsmouth Naval Shipyard. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. At the end of the project, provide the Contracting Officer with the maximum quantity of each material that was present at the site at any one time, the dates the material was present, the amount of each material that was used during the project, and how the material was used. Ensure that hazardous materials are utilized in a manner that will minimize the amount of hazardous waste that is generated. Ensure that all containers of hazardous materials have NFPA labels or their equivalent. Keep copies of the SDS for hazardous materials on site at all times and provide them to the Contracting Officer at the end of the project. Certify that all hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste per 40 CFR 261.

3.10 PETROLEUM PRODUCTS AND REFUELING

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation, away from storm drains and surface waters. Manage all used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while on-site exhibits a characteristic of hazardous waste. All used oil will be turned in to Building 357 and disposed of by the Government.

3.10.1 Oily and Hazardous Substances

Provide secondary containment and overfill protection for oil and hazardous substance storage tanks. Prevent oil or hazardous substances from entering the ground, drainage areas, storm drains, or navigable waters. In accordance with 40 CFR 112, surround all temporary fuel oil or petroleum storage tanks with a temporary berm or containment of sufficient size and strength to contain the contents of the tanks, plus 10 percent freeboard for precipitation. The berm must be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Provide procedures and equipment to prevent overfilling of tanks. Monitor and remove any rainwater that accumulates in open containment dikes, berms, or spill pallets. Inspect the accumulated rainwater prior to draining from a containment dike, berm or spill pallet to the environment, to determine there is no oil sheen present.

3.10.2 Inadvertent Discovery of Petroleum Contaminated Soil or Hazardous Wastes

If petroleum contaminated soil or suspected hazardous waste is found during construction that was not identified in the Contract Documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.11 FUEL TANKS

Petroleum products and lubricants required to sustain up to 30 days of construction activity may be kept on site. Storage and refilling practices must comply with 40 CFR Part 112. Secondary containment must be provided and be no less than 110 percent of the tank volume plus five inches of free-board. If a secondary berm is used for containment then the berm must be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Drip pans are required. Cover tanks and drip pans during

inclement weather to prevent spills, as necessary.

3.12 RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES

Exercise due diligence to prevent, contain, and respond to **ALL** spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of **ANY** releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Contracting Officer's Representative, and the Contracting Officer. If the response is inadequate, the Navy may respond. If this should occur, reimbursement to the Government for spill response assistance and analysis will be required. Contractor will provide SDSs, Waste Profile Sheets, and/or other technical data on the material spilled to emergency response personnel.

The Contractor is responsible for verbal and written notifications as required by the Federal 40 CFR 300.125 and 40 CFR 355, State, and local regulations and Navy Instructions. These notifications will be done in coordination with Code 106.3. Spill response must be in accordance with 40 CFR 300 and applicable State and local regulations. Contain and clean up these spills without cost to the Government. If Government assistance is requested or required, reimburse the Government for such assistance. Provide copies of the written notification and documentation that a verbal notification was made within 20 days.

Maintain spill cleanup equipment and materials at the work site. Clean up all hazardous and non-hazardous (WHM) waste spills. Reimburse the Government for all material, equipment, and clothing generated during any spill cleanup. Reimburse the Government for all costs incurred including sample analysis materials, equipment, and labor if the Government must initiate its own spill cleanup procedures, for spills, when:

- a. Spill cleanup procedure has not begun within one hour of spill discovery/occurrence, or
- b. If, in the Government's judgment, the spill cleanup is not adequately abating a life-threatening situation and/or is a threat to any body of water or environmentally sensitive areas.
- c. All discharged material and other spill cleanup wastes (e.g., absorbent, disposable clothing, brushes, rags, brooms, containers, decontamination water etc.), must be collected, containerized, properly marked and brought to Building 357 by the end of the shift generated. Decontamination water must be collected and turned in separately from other generated wastes or as directed by Code 106.3.

3.13 CONTROL AND MANAGEMENT OF HAZARDOUS WASTES

At the time of the pre-construction conference, the Contractor will be briefed and provided written information regarding hazardous waste management. The Government will provide technical and oversight guidance in all aspects of hazardous waste management.

3.13.1 General

All hazardous wastes generated within the confines of the Portsmouth Naval Shipyard must be disposed of by the Government. Accordingly, all hazardous wastes generated to accomplish the requirements of this Contract must be considered Government-generated, and must be disposed of by the Government. Do not bring hazardous wastes onto Government property. Hazardous wastes must be handled in compliance with 40 CFR 260-268, 273, 279 and State of Maine MEDEP Regulations Chapter 850 to 860. For hazardous waste spills, immediately notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, Contracting Officer's Representative and the Contracting Officer.

3.13.2 Containers

Use only Government-furnished, Government-labeled containers for the packaging of hazardous soils and hazardous wastes. Containers are requested and picked up at Building 357 following approval of the Management Plan required above.

- a. Segregate hazardous and non-hazardous soils/wastes. Hazardous soils/wastes must be placed into containers provided by the Government. All containers of hazardous waste must have a Code 106.3 tracking number affixed to the container, as applicable.
- b. Hazardous soils must be turned over to the Government, at Building 357, as coordinated with Code 106.3. All other containers of waste hazardous materials and hazardous waste, full or partially full, must be turned in to Building 357 by the end of the shift generated.

While in control of hazardous soils and hazardous wastes, the soils/wastes must be handled in accordance with Portsmouth Naval Shipyard requirements.

- c. Prior to Government acceptance of the containers, provide the certification required by the "Submittals" paragraph of this Section, and such additional information regarding contents of the containers. Submittal of a Waste Determination form may be required for proper waste characterization as requested by Code 106.3.

3.13.3 Facility Hazardous Waste Generator Status

Portsmouth Naval Shipyard is designated as a Large Quantity Generator. All work conducted within the boundaries of the Portsmouth Naval Shipyard must meet the regulatory requirements of this generator designation. Comply with all provisions of Federal, State, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of all construction derived wastes.

3.13.4 Hazardous Waste/Debris Management

Identify all construction activities which will generate hazardous waste/debris and universal wastes. Provide a documented waste determination for all resultant waste streams as requested by Code 106.3 or other Government personnel. Hazardous waste/debris must be identified, labeled, handled, stored, and disposed of in accordance with all Federal, State, and local regulations including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860, and NAVSHIPY PTSMH

INST 5090.30, latest edition.

Hazardous wastes and universal wastes must also be managed in accordance with the approved Hazardous Waste Management Section of the Environmental Management Plan. Daily worksite accumulation of hazardous wastes and universal wastes must be in Government-provided containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities must be identified as being generated by the Government.

Prior to removal of any hazardous waste from Government property, all hazardous waste manifests must be signed by Portsmouth Naval Shipyard personnel from the Portsmouth Naval Shipyard Environmental Office. No hazardous waste will be brought onto Government property. Provide to the Contracting Officer a copy of all waste determination documentation that was requested by the Government for hazardous waste or solid wastes with chemical constituents listed in 40 CFR 372-SUBPART D. For hazardous wastes spills, notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, Contracting Officer's Representative and the Contracting Officer.

3.13.4.1 Regulated Waste Storage/Satellite Accumulation/90 Day Storage Areas

If the work requires the temporary storage/collection of regulated or hazardous wastes, request the establishment of a Regulated Waste Storage Area, a Satellite Accumulation Area, or a 90 Day Storage Area at the point of generation. Regulated waste and Hazardous Waste Accumulation Areas (HWAAs) are approved by Code 106.3.

Submit a request in writing to the Contracting Officer providing the following information:

<u>Contract Number</u>	_____	<u>Contractor</u>	_____
<u>Haz/Waste or Regulated Waste POC</u>	_____	<u>Phone Number</u>	_____
<u>Type of Waste</u>	_____	<u>Source of Waste</u>	_____
<u>Emergency POC</u>	_____	<u>Phone Number</u>	_____

Location of the Site: _____
(Attach Site Plan to the Request)

Attach a waste determination form. Allow ten (10) working days for processing this request. The designated area where waste is being stored must be barricaded and a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.13.4.2 Sampling and Analysis of Hazardous Waste

a. Waste Sampling

Sample waste in accordance with EPA SW-846 and NAVSHIPY PTSMH INST 5090.8, latest revision. Each sampled drum or container must be clearly marked with the Contractor's identification number and cross referenced to the chemical analysis performed. All sampling events

will require a Code 106.3 reviewed and approved sampling plan and availability for Code 106.3 oversight.

b. Laboratory Analysis

Follow the analytical procedure and methods in accordance with 40 CFR 261. Provide all analytical results and reports performed to the Contracting Officer and Code 106.3 Environmental Sampling Project Manager for review and approval.

All laboratory analysis for hazardous waste identification must be performed by a laboratory compliant with OPNAVINST 5090.1 Chapter 7-3.3. Proof of compliance must be made available upon request. All analyses provided by laboratories that are not compliant with the stated requirements will be rejected.

c. Analysis Type

Identify waste material/hazardous waste by analyzing for properties that are reasonably suspected of the waste. Soil and other materials may require specific analysis for acceptance to a disposal facility - contact Code 106.3 personnel at the Hazardous Waste Storage Facility, Building 357, before choosing parameters.

d. Typical Hazardous Waste and PCB Sampling Plan Requirements

- i. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.
- ii. The timing of waste characterization sampling and analysis. PCB sampling must be conducted prior to the accomplishment of any work. To the extent feasible, all other waste sampling and analysis should be completed prior to the start of work. Waste sampling after the start of work requires Code 106.3 approval.
- iii. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.
- iv. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.
- v. The proposed date(s) for sampling and expected duration of the sampling event(s). All hazardous waste and PCB sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.
- vi. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.

- vii. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
- viii. A list of all equipment to be used for sampling. Examples include:
 - 1. Individual sample jars and vials;
 - 2. Field log;
 - 3. Site diagram;
 - 4. Chain-of-Custody (COC) forms;
 - 5. Nitrile gloves;
 - 6. Stainless steel or plastic soil scoops;
 - 7. Metal chisels of assorted sizes;
 - 8. Razor knives and replacement blades;
 - 9. Water and spray bottle;
 - 10. Trash bags;
 - 11. Disposable cleaning wipes;
 - 12. Paper towels;
 - 13. Duct tape;
 - 14. Flashlight;
 - 15. Personnel lifts; and
 - 16. Other equipment as necessary.
- ix. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers.
- x. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods, as applicable.
- xi. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
- xii. Confirmation that the sampling results report will be transmitted to the Contracting Officer and Code 106.3 for each material sampled. The report must include quality control data, be reported down to the Method Detection Limit (MDL) and include:
 - 1. Laboratory analytical results
 - 2. Field log book entries
 - 3. Copy of the Chain of Custody (COC)
 - 4. Pertinent diagrams and maps
 - 5. All other supporting data

3.13.4.3 Asbestos Certification

Items, components, or materials disturbed by or included in work under this Contract may involve asbestos. Other materials in the general area around where work will be performed may contain asbestos. All thermal insulation,

in all work areas, should be considered to be asbestos unless positively identified by conspicuous tags or previous laboratory analysis certifying them as asbestos free.

Inadvertent discovery of non-disclosed asbestos that will result in an abatement action requires a change in scope before proceeding. Upon discovery of asbestos containing material not identified in the Contract documents, immediately stop all work that would generate further damage to the material, evacuate the asbestos exposed area, and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area. Do not remove or perform work on any asbestos containing materials without the prior approval of the Contracting Officer. Do not engage in any activity, which would remove or damage such materials or cause the generation of fibers from such materials.

Asbestos containing waste must be managed and disposed of in accordance with applicable environmental law. Asbestos containing waste must be manifested and the manifest provided to the Contracting Officer. Disposal of asbestos-containing waste must be coordinated with PWD-ME EV and Code 106.3.

Provide the attached Asbestos Free Certification Form (Attachment A) prior to the Government taking beneficial occupancy certifying that all materials, including those supplied by third parties, are asbestos free.

3.13.4.4 Polychlorinated Biphenyls (PCBs)

Polychlorinated Biphenyl's (PCBs) were manufactured between 1950 and 1979 and widely used in building construction and renovation materials (e.g., paints, varnishes, caulks, rubber/felt gaskets, adhesives, siding, mastics, roofing, tar paper, fireproofing etc.). Work may include the disturbance of items, components, or materials containing PCBs as identified in project documents. Suspect PCB containing materials must be analytically tested prior to disturbance and managed per 40 CFR 761 and applicable environmental law. If adequate PCB testing for industrial hygiene and waste disposal was not provided in the contract documents, the Contractor shall be responsible for PCB suspect materials testing prior to the start of work.

- Sampling of suspect PCB material requires a Code 106.3 reviewed and approved sampling plan and Code 106.3 oversight.
- Disposal of all PCB containing waste(s) ≥ 1 ppm must be coordinated with B357/Code 106.3 personnel.

Inadvertent discovery of undisclosed PCBs resulting in a new or expanded abatement action requires a change in scope review before proceeding. Upon discovery of PCB containing material not identified in the Contract documents, immediately stop all work that could impact PCB containing materials and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area.

3.13.4.5 Hazardous Waste Disposal

Control of stored waste, packaging, sampling, analysis, and disposal will be determined by the details in the Contract. The requirements for jobs in the following paragraphs will be used as the guidelines for disposal of any hazardous waste generated and disposed of by the Contractor. All hazardous waste disposed of by the Contractor must be approved by and coordinated with Code 106.3 prior to transporting wastes off PNS property.

a. Responsibilities for Disposal

Responsibilities include any generation of WHM/HW requiring disposal of solid waste or liquid.

- (1) Agree to provide all service necessary for the final treatment/disposal of the hazardous material/waste in accordance with all local, State, and Federal laws and regulations, and the terms and conditions of the Contract within sixty (60) days after the materials have been generated. These services must include all necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal, and/or transportation, including manifesting or completing waste profile sheets, equipment, and the compilation of all documentation is required).
- (2) Contain all waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, 40 CFR 270, 40 CFR 272, 40 CFR 273, 40 CFR 279, 40 CFR 280, and 40 CFR 761.
- (3) Obtaining a representative sample of the material generated for each job done to provide waste stream determination.
- (4) Analyzing for each sample taken and providing analytical results to the Contracting Officer. Provide two copies of the results.
- (5) Determine the DOT proper shipping names for all waste (each container requiring disposal) and will demonstrate how this determination is developed and supported by the sampling and analysis requirements contained herein to the Contracting Officer.

Disposal Turn-In Requirements

For any waste hazardous materials or hazardous waste generated which requires disposal, the following conditions must be met in order to be acceptable for disposal:

- a. Drums compatible with waste contents and drums meet DOT requirements for 49 CFR 173 for transportation of materials.
- b. Drums banded to wooden pallets. No more than three (3) 55-gallon drums to a pallet, or two (2) 85-gallon over packs.
- c. Band using 1-1/4 inch minimum band on upper third of drum.
- d. Contents of drum identified on the outside of the drum, as well as the volume of material, name of material manufacturer, and other vendor information as available.
- e. Always have three (3) to five (5) inches of empty space above volume of material. This space is called 'outage'.
- f. Provide disposal documentation for hazardous and regulated waste.

3.13.5 Class I ODS Prohibition

Class I ODS must not be used in the performance of this Contract, nor be

provided as part of the equipment. This prohibition will be considered to prevail over any other provision, specification, drawing, or referenced documents. Class I and II ODSs are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant. Regulations related to the protection of stratosphere ozone may be found in 40 CFR 82.

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

Heating and air conditioning technicians must be certified through an EPA-approved program. Copies of certifications must be maintained at the employees' place of business and be carried as a wallet card by the technician, as provided by environmental law. Accidental venting of a refrigerant is a release and must be reported to the Contracting Officer.

3.13.6 Universal Waste/e-Waste Management

Universal waste including, but not limited to, some mercury containing building products such as florescent lamps, mercury vapor lamps, high pressure sodium lamps, CRTs, batteries, aerosol paint containers, electrical equipment containing PCBs, and consumed electronic devices, must be managed in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 273, MAINE DEP 38 MSRA 858, and applicable environmental law.

3.14 DUST CONTROL

Dust control must meet the requirements of the BMPs in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Keep dust down at all times, including during nonworking periods. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming or sweeping. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Runoff from cutting water is prohibited from entering storm drains or leaving the work site. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

When temporary dust control measures are employed, repetitive treatment must be applied as needed to accomplish control.

Visible emissions from a fugitive emission source (including stockpiles and roadways) must not exceed an opacity of 20 percent, except for no more than five (5) minutes in any 1-hour period.

Dust suppression during demolition work may include using a manned water hose. Runoff from the site is prohibited.

3.14.1 Dirt and Dust Control Plan

Submit truck and material haul routes along with a plan for controlling dirt, debris, and dust on Portsmouth Naval Shipyard roadways. As a minimum, identify in the plan the Subcontractor and equipment for cleaning along the haul route and measures to reduce the dirt, dust, and debris from roadways.

3.15 ABRASIVE AND/OR WET BLASTING AND WATER CUTTING

3.15.1 Blasting Operations

a. Abrasive Blasting

The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive, agent, paint chips, and other debris.

Abrasive blasting must take place in containments with emissions vented through bag house filters and emissions must be limited to 10% opacity on a six minute block average. The bag houses must be used to control PM emission and operated properly at all times abrasive blasting is being performed.

b. Wet Blasting and Water Cutting

The use of wet blasting requires the capture and proper disposal of all wastes, including the blasting water, associated with the process. Provide enclosed containments to confine and collect wastewater and debris.

3.15.2 Disposal Requirements

Submit analytical results of the wastes and/or debris generated from blasting operations per paragraph entitled LABORATORY ANALYSIS of this Section. Hazardous waste generated from blasting operations must be managed in accordance with paragraph entitled HAZARDOUS WASTE/DEBRIS MANAGEMENT of this Section and the approved EMP. Concrete wash water and oily waste generated from blasting operations must be disposed of in accordance with the policy outlined in these specifications.

3.16 SPRAY PAINTING OPERATIONS

Spray painting operations must take place in containment. Emissions from spray painting must vent through air filters and are limited to 10% opacity on a six-minute block average. The air filters are used to control particulate emissions.

3.17 NOISE

Make the maximum use of low-noise emission products, as certified by the EPA and/or sound deadening enclosures to limit noise within the limits of work. Blasting or the use of explosives will not be permitted. Confine any operations that may generate excessive noise to the period between 7 a.m. and 5 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified or approved by the Contracting Officer.

The maximum permissible sound pressure levels, as measured at the limits of the Navy Property boundary, must not exceed the maximum noise levels as specified in the Town of Kittery's Ordinance and all applicable OSHA Regulations. The Kittery Ordinance specifies the maximum permissible sound level limit at the Navy Property Boundary to be 65 dBA from 0700 to 2100 and 60 dBA from 2100 to 0700.

For any construction or demolition operations that may generate noise levels in excess of 80 dBA in adjacent facility work areas outside the limits of work, use low-noise emission equipment, sound deadening enclosures or schedule work outside the normal work hours (with approval by the Contracting Officer and if the noise levels will not exceed the Kittery Noise Ordinance levels stated above) to ensure the adjacent work areas are not impacted.

Provide all noise monitoring equipment necessary to record and document noise levels associated with any operations generating excessive noise to ensure these noise levels specified are not exceeded.

3.18 MERCURY MATERIALS

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed. Immediately (within 15 minutes) notify instances of breakage or mercury spillage to the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, Contracting Officer's Representative and the Contracting Officer. Clean the mercury spill area(s) to the satisfaction of the Contracting Officer. Cleanup of a mercury spill must not be recycled and must be managed as a hazardous waste for disposal.

3.19 CONCRETE WASH WATER

Concrete wash water is water, pressure washing water, or storm water that has come into contact with cement, uncured concrete, concrete dust, or other material of a similar nature generated during construction activities including, but not limited to, washing down ready-mix trucks, mixers, wheelbarrows, pre casting equipment, forms, manufactured cast concrete sections, tools, concrete areas; masonry cutting operations; cleaning up of split mortar or block fill; and hosing away excess materials.

Water or stormwater that has come into contact with pre casting equipment, forms, tools, etc. which have been subjected to oil based form release agents will be considered an oily waste if a visual inspection indicates any signs of oily residue. Oily wastes must be collected and disposed of in accordance with Portsmouth Naval Shipyard policy.

Concrete wash water must be collected and either placed in a concrete wash out structure for settling solids and evaporating liquid or turned-in as wastewater to Building 357 by the end of the shift generated. As approved by Code 106.3, large volumes of concrete wastewater may be disposed of at a facility designed and approved for the disposal or recycling of concrete wastewater. Under no circumstances will clean water be added to concrete wash water/wastewater for dilution purposes or any other reason.

Containment wash out structures must be watertight, designed to promote evaporation, and provide adequate freeboard to contain the wash water, solids, and rainfall to prevent overflow. Washout must only occur in designated areas that have been approved by the Contracting Officer's Representative.

Inspect all concrete wash out structures daily to determine filled capacity. Remove all materials from containment wash out structures when 75% fill capacity has been reached. Concrete wash out structure operations must be conducted to reduce the amount of concrete wastewater generated for disposal. Remove liquids or cover structures before predicted rainstorms to prevent overflows and infiltration of rainwater. Inspect structures for holes and tears daily and repair to maintain watertight conditions.

Hardened solids can be removed from containment structures and recycled, reused, or disposed of per regulatory requirements. Liquids remaining in the containment structure must be vacuumed and turned-in to Building 357 or, as approved by Code 106.3, disposed of at a facility designated for disposal/recycling of concrete wastewater.

3.19.1 Pollution Prevention

Store dry and wet concrete supplies under cover away from drainage areas. Concrete wash water must not be released to the storm drain system, sewer system, roadways, other uncontained impervious surfaces, or to natural waterways including the Piscataqua River and its tributaries. Take all precautions necessary to prevent rainwater or stormwater runoff from coming into contact with concrete wash water. Divert clean stormwater and roof runoff from contact with concrete wash water. Take all measures necessary to minimize the volume of concrete wash water generated. Protect all waterways, catch basins, and storm drain structures from potential discharges of concrete wash water. Collect and control concrete wash water separately from wastewater determined to be oily waste.

3.19.2 On-Shipyard Disposal

ALL CONCRETE WASTEWATER MUST BE COLLECTED AND TURNED-IN TO BUILDING 357 BY THE END OF THE SHIFT GENERATED. CONCRETE WASH WATER THAT IS COLLECTED AND DEPOSITED IN WASH OUT STRUCTURES MUST BE TURNED IN TO BUILDING 357 AT OR BEFORE THE CONCRETE WASH OUT STRUCTURE REACHES 75% CAPACITY.

3.19.3 Off-Shipyard Disposal

Concrete wastewater may be disposed of off-site as approved by Code 106.3. The Contractor must submit for approval the disposal facility designated for concrete wash water/wastewater and provide careful oversight to prevent improper dumping of concrete wash water. Any clean up resulting from improper control of concrete wash water will be at no expense to the Government.

3.20 DISPOSAL OF CHLORINATED WATER AND DECHLORINATION REQUIREMENTS

Chlorinated water created during disinfection procedures must not be directly discharged to storm drains or sanitary sewers without prior dechlorination and approval by Code 106.3. Chlorinated water must be neutralized by the controlled addition of a reducing chemical such as sodium thiosulfate, sodium bisulfate, sodium sulfite, sulfur dioxide, or ascorbic acid (commonly known as Vitamin C), as approved by Code 106.3. Dechlorination must be sufficiently effective to reduce total residual

chlorine concentration to 1 mg/L or below.

3.21 SNOW STORAGE AND DISPOSAL

Site and operate snow storage and disposal areas to prevent the discharge of snow directly into surface waters and minimize discharges of pollutants from snow maintenance activities. As necessary, the Contractor must employ applicable snow dump BMPs from CMR 06-096 Chapter 573 Snow Dumps: Best Management Practices for Pollution Prevention for snow storage areas.

-- End of Section --



Public Works Department, Maine

ATTACHMENT A

ASBESTOS FREE CERTIFICATION FORM

References:

1. NAVFAC P-502 Asbestos Program Management

A. Asbestos Certification

The Contractor shall not use any asbestos containing material (ACM) at any time during the Work. Contractor shall certify that all material/equipment installed in any portion of the Work is asbestos free. The Government may request that the Contractor provide Safety Data Sheets, manufacturer certifications, laboratory analytical results, and/or other documentation to support the Contractor's certification that all suspect material/equipment is asbestos free. If any material/equipment is found to contain asbestos, the Contractor shall pay for the lawful and proper removal and disposal of product(s), and re-install acceptable material/equipment all at its sole expense.

By signing below, the Contractor certifies that all material/equipment installed in any portion of the Work is asbestos free:

Project Title:

Contract No.:

Company:

Date:

Print Name:

Contractor Signature:



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STANDARD OPERATING PROCEDURE FOR SITE WASTE REMOVAL (SWR) PROCESS

Applicability: This SOP applies to any special waste and excavated soil removed from Portsmouth Naval Shipyard (PNSY) including dredged materials. Herein the term “waste” represents both landside and dredged excavated materials, and bulk special waste that exceeds the storage capacity of the Hazardous Waste Facility.

Overview: Wastes removed from PNSY shall be characterized and disposed in accordance with the regulatory requirements applicable to the waste. NAVFAC PWD-Maine, as the Contracting Officer Representative (COR), owns the overall responsibility for ensuring this removal and disposal process is followed and shall ensure Contractor compliance. The NAVFAC PWD-Maine COR (e.g. CM or ET) shall also ensure required documentation is complete and stored in the project file.

A critical step in this process is the scheduling and execution of a Site Waste Removal (SWR) meeting held at the project site. Wastes cannot be removed from a project site before this meeting is held and documented by a signed Site Waste Removal Form [*Enclosure (1)*]. Three parties must sign the form to complete the process – the NAVFAC COR, the C106 representative, and the contractor.

NOTE: Waste characterization must be completed in accordance with C106 requirements before the SWR meeting can be scheduled. The Contractor shall prepare and submit a Sampling and Analysis Plan (SAP) to C106.3 for approval prior to implementation in accordance with the Contract Documents. The SAP must demonstrate compliance with the sampling requirements of the proposed disposal facility. Following C106 approval of the SAP, waste sampling can be performed by C106 or by the contractor with direct oversight by C106. Next, the analytical results (complete laboratory report) are submitted to C106 for review. C106 will provide direction on the disposal requirements for the waste (e.g. non-hazardous or hazardous). The sampling plan must include a site diagram clearly showing the source location of the waste and the stored waste stockpile location. This diagram must be reviewed at the SWR meeting.

SWR Process:

1. Contractor requests SWR meeting via Microsoft Outlook sent to project CM or ET and the C106 representative a minimum of three (3) business days for non-hazardous wastes and ten (10) business days for hazardous wastes prior to the desired SWR meeting date. The invite shall include a proposed hauling schedule. The meeting must be held at the removal site or area where the waste is stored and will be hauled from.
2. Prior to the meeting, the Contractor shall inspect the site and confirm the field conditions (e.g. source of wastes, location and characterization of stockpiled wastes designated for disposal, etc.) match the information presented on the site diagram previously provided to C106 in the waste sampling plan. Wastes shall not be removed from PNSY if the sampling plan documentation does not match the field conditions found at the removal site.
3. Hold on-site SWR meeting. Contractor, NAVFAC CM/ET, and C106 representative shall be present for discussion and signing of the Initial SWR Form [*Enclosure (1)*]. If one of these three parties is not in attendance at the SWR it must be rescheduled.

**STANDARD OPERATING PROCEDURE FOR
SITE WASTE REMOVAL (SWR) PROCESS****SWR Process-Continued:**

4. The SWR meeting discussion shall *review* the following:
 - a. Source of the waste.
 - b. Location of removal/stock pile areas designated for removal.
 - c. Analytical characterization results of the samples taken for the waste designated for disposal demonstrating non-hazardous or hazardous nature of the waste.
 - d. Estimated tonnage of waste to be removed.
 - e. Anticipated calendar time frame and duration of waste removal activities.
 - f. Phases of waste removal that will require additional C106 review and SWR signoff. 'Phases' of waste removal will be established for each specific project. The SWR meeting group will establish how phases are defined for that particular project. Phase examples include:
 - i. One stockpile of soil on site or in one bin from excavation activity.
 - ii. One barge load of dredge spoils from dredge material excavation.
 - iii. One single area or multiple areas for excavation/demolition within one project site as illustrated on a site diagram for live loading projects.
5. SWR group will complete a site walk to identify the waste source area(s) to ensure in-field conditions match the site diagram.
6. The contractor will fill out the descriptive fields of the Initial SWR form. C106 representative and NAVFAC CM/ET will review the information and provide signatures if the information is complete and everyone is in concurrence with the waste removal requirements.
7. The contractor will scan both the Initial SWR Form and corresponding site diagram and send to the NAVFAC CM/ET via email.
8. NAVFAC CM/ET will send the Initial SWR form and corresponding site diagram to the C106 SWR group email **pnsyswr@us.navy.mil ('PNSY SWR' on the GAL)** for final approval.
9. The Contractor will email the SWR group on a daily or weekly basis detailing hauling specifics including the specific site the waste will be removed from (specific pile or profile if multiple for an activity), hauling dates, transporter company name and the number of loads anticipated. The SWR group must be contacted before 1400 of the day prior to shipping.
10. On the day of hauling, the NAVFAC CM/ET shall double check that the correct stockpile was loaded for disposal, sign the DRR form, and submit to the Contractor to file.
11. NAVFAC will sign a DRR Form for each truck of each day of removal activity. A copy of the DRR form will be given to each truck hauling waste for the activity, and the DRR form will be delivered with the each truck's first truck load to B-357 as proof of DRR form approval for the day. The DRR does not need to be resubmitted for multiple loads from the same truck in a single day. The DRR forms will include the hauler name and truck numbers as well as the information provided in the Initial SWR Form signed by C106.
12. The Contractor will email DRR forms for daily waste removal activities to the SWR group.
13. The NAVFAC CM/ET will continue daily surveillance, site condition reviews, and execution of DRR forms for each shipping day in the date range indicated on the initial SWR form. The NAVFAC CM/ET will also review the Contractor Daily Production Reports to ensure all loads are recorded so they can be traced back to disposal manifests.
14. If suspected contaminated wastes (identified by odor or visual examination) are discovered during removal, removal activities shall immediately cease *and contact* C106 to evaluate the situation.



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**STANDARD OPERATING PROCEDURE FOR
SITE WASTE REMOVAL (SWR) PROCESS**

INITIAL SITE WASTE REMOVAL PASS - ACTIVITY DESCRIPTION		
1. Contract No:		
2. Contract title:		
3. Prime contractor company name:		
4. Hauling contractor company name:		
5. Anticipated hauling date range:		through
6. Waste type: ☐ Non-hazardous soils ☐ Hazardous soils ☐ Special waste _____ ☐ Other _____		
7a. Waste source:		
7b. Current waste location:		
8. Anticipated total hauling amount (tons):		
9. PNS approved waste disposal facility:		Profile No:
WASTE REMOVAL APPROVALS – The signatures below verify that an onsite visual inspection of the material has been performed, and that this form is a record of the facts agreed upon.		
10. Prime contractor signature:		
11. NAVFAC CM/ET signature:		
12. Code 106.3 signature:		
DAILY REMOVAL RECORD - to be filled each morning of waste removal		
13. Hauling date:		
14. Anticipated hauling amount:		tons truckloads
15. Hauling contractor Name & truck ID number(s):		
16. Prime contractor signature:		
17. NAVFAC CM/ET signature:		

**STANDARD OPERATING PROCEDURE FOR
SITE WASTE REMOVAL (SWR) PROCESS**

WASTE REMOVAL PASS ATTRIBUTE DESCRIPTIONS

1. *Contract No.* The unique NAVFAC contract identification number applied to the project requesting waste removal.
2. *Contract title.* The unique contract title/description applied to the project requesting waste removal.
3. *Prime contractor company name.* The project's prime contractor company name.
4. *Hauling contractor company name.* The subcontractor hired to haul waste from the project site to the PNSY approved waste disposal company.
5. *Anticipated hauling date(s).* Hauling date range agreed upon at SWR meeting for duration of waste removal in the approved phase of removal.
6. *Waste description.* Description of waste in terms of hazardous/non-hazardous designation and any other characteristic determined at the SWR meeting.
- 7a. *Waste source.* Detailed written description of the source of the waste to be removed. Include removal areas, buildings, streets, land marks, coordinates, and other information as available.
- 7b. *Current waste location.* Detailed written description of the waste location if displaced from the original source during removal. Include removal areas, bin designations, stockpile locations, buildings, streets, land marks, coordinates and other information as available.
8. *Anticipated total hauling amount (tons).* Total amount of waste anticipated for removal in the approved phase of removal.
9. *PNSY approved waste disposal facility and profile number.* Name of the PNSY approved waste disposal facility. Waste disposal facility provided profile number for the waste.
10. *Prime contractor signature.* Written signature of project's prime contractor.
11. *NAVFAC CM/ET signature.* Written signature of NAVFAC CM or ET in attendance at the SWR meeting.
12. *Code 106.3 signature.* Written signature of Code 106.3 representative in attendance at the SWR meeting.
13. *Hauling date.* Date of the waste removal.
14. *Anticipated hauling amount.* Entire hauling amount (total tons) anticipated for said hauling date.
15. *Hauling contractor truck ID number(s).* The identification number of the contractor's truck.
16. *Prime contractor signature.* Written signature of prime contractor.
17. *NAVFAC CM/ET signature.* Written signature of project's NAVFAC CM or ET overseeing daily removal of waste from the site.

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 1: 08/20

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 273	Standards for Universal Waste Management
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill, by means of eliminating or minimizing waste, or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a construction waste management program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all Subcontractors, vendors, and suppliers to participate in the construction waste management program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the construction waste management program will be implemented in a construction waste management plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or

deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.4.2 Oversight

The Environmental Manager, as specified in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME, is responsible for overseeing and documenting results from executing the construction waste management plan for the project and is responsible for all waste being disposed of in accordance with the contract documents and state & Federal regulations. If any waste materials are removed from the Shipyard or Facility not in compliance with the requirements stated in the Contract Documents, the person assigned as the Environmental Manager is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor. The Contractor will be responsible to take all required actions to address and correct any non-compliance as well as paying any fines as result of improper handling or disposal. This will reflect poorly on the Prime Contractor's performance rating (CPARS) and noted in their final performance evaluation.

1.4.3 Special Programs

Implement any special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by Federal, State, and local regulations.

1.4.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.4.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the construction waste management plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Masonry and CMU
- d. Concrete
- e. Metals (e.g. banding, stud trim, ductwork, piping, rebar, roofing,

other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze, etc.)

- f. Wood (nails and staples allowed)
- g. Glass
- h. Paper
- i. Plastics (PET, HDPE, PVC, LDPE, PP, PS, Other)
- j. Gypsum
- k. Non-hazardous paint and paint cans
- l. Carpet
- m. Ceiling Tiles
- n. Insulation
- o. Beverage Containers

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL]:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; G

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed construction waste management plan and to develop a mutual understanding relative to the management of the construction waste management program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00.00 22 QUALITY CONTROL (PWD ME) or 01 45 00.10 22 QUALITY CONTROL FOR MINOR CONSTRUCTION (PWD ME). At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction & Pre-demolition meeting.
- b. Regular Quality Control meetings.
- c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 15 calendar days after Contract Award. Revise and resubmit Construction Waste Management Plan until it receives final approval from the Contracting Officer, in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved construction waste management plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all Subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation, including coordination with Subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and/or incinerator to be used.
- f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.
- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which any materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for Subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and

waste materials.

- l. Facilities or Subcontractors offering construction waste transport on-site or off-site must ensure that proper shipping orders, bill of lading, manifests, or other shipping documents containing waste diversion information meet requirements of 40 CFR 273 Universal Waste Management, 49 CFR 173 Shippers - General Requirements for Shipments and Packagings, and 49 CFR 178 Specifications for Packaging. Individuals signing manifests or other shipping documents should meet the minimum training requirements.
- m. List each supplier who deliver construction materials, in bulk, or package products in returnable containers or returnable packaging, or have take-back programs. List each program and the applicable material to actively monitor and track to assist in meeting waste diversion requirements on the project.

Distribute copies of the waste management plan to each Subcontractor, Environmental Manager, and the Contracting Officer.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted through re-use, recycling and/or sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for any materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions, or minor deconstruction activities.

1.9 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide the Contracting Officer the Final Construction Waste Diversion Report 60 days prior to the Beneficial Occupancy Date (BOD).

1.10 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train Subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the construction waste management plan. Handle hazardous waste and hazardous materials in

accordance with applicable regulations and coordinate with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME). Separate materials by one of the following methods described herein:

1.10.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the construction waste management plan.

1.10.2 NOT USED

1.10.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.11 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other Sections of the specifications, dispose of in accordance with the following:

1.11.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved construction waste management plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials [is] [is not] allowed on the [Portsmouth Naval Shipyard][Activity].

1.11.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.11.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

1.12 ADDITIONAL REPORTING AND RECORDING REQUIREMENTS

Provide monthly cost and revenue data to the NAVFAC Midlant Integrated Solid Waste Management office. Submit report by e-mail to: IntegratedSolidWasteManagement@navy.mil no later than the 3rd of each month. Data must be reported on an excel document provided by the Contracting Officer. Comply with the requirements specified in Appendix 01 74 19-1, CONSTRUCTION AND DEMOLITION SOLID WASTE REPORT.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

CONSTRUCTION AND DEMOLITION SOLID WASTE REPORT				
SITE: _____		Month: _____		
Contractor's Company Name: _____		Contract # _____		
Contractor's POC and Telephone or Email Address: _____				
Project Description: _____				
SECTION 1	Tons	Cost	Revenue	Remarks
Recycled (tons)				
Concrete(incl: brick & block)				
Wood				
Metal				
Asphalt				
Green waste(clearing debris)				
Dirt				
Sand				
Gravel/Rock				
Mixed				
Misc				
Subtotal - Recycled	0.00	\$ -	\$ -	
SECTION 2				
Landfilled (tons)				
Concrete(incl: brick & block)				
Wood				
Metal				
Asphalt				
Green Waste(clearing debris)				
General C&D				
Dirt				
Sand				
Gravel/Rock				
Mixed				
Misc				
Subtotal - Landfilled	0.00	\$ -	\$ -	
Solid Waste (tons)				
Total Solid Waste	0.00	\$ -	\$ -	

REPORTING DEADLINE IS NO LATER THAN THE 3RD OF EACH MONTH
--

SECTION 01 78 00.00 22

CLOSEOUT SUBMITTALS (PWD ME)
05/21

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2005; R 2011) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
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GREEN SEAL (GS)

GS-37	(2017) Cleaning Products for Industrial and Institutional Use
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U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N	(2014; with Change 6, 2021) Navy and Marine Corps Design
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UFC 1-300-08	(2009, with Change 2, 2011) Criteria for Transfer and Acceptance of DoD Real Property
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1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: Contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor; and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation. Submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan

One (1) paper and one (1) pdf set of the warranty management plan containing information relevant to the warranty of materials and equipment incorporated into the construction project, including the starting date of warranty of construction. Furnish with each warranty the name, address, and telephone number of each of the guarantor's representatives nearest to the project location.

Warranty Tags

Two (2) paper record copies and one pdf copy of the warranty tags showing the layout and design.

Final Cleaning

Two (2) copies of the listing of completed final clean-up items.

Spare Parts Data

Two (2) paper copies and one pdf copy of the list that indicates manufacturer's name, part number, nomenclature, and stock level recommended for maintenance and repair. List those items that may be standard to the normal maintenance of the systems.

SD-08 Manufacturer's Instructions

Preventative Maintenance; G and Condition Monitoring (Predictive Testing); G and Inspection; G schedules with instructions that state when systems should be retested.

Define within the schedule the anticipated length of each test, test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements. On each test feature; e.g., gpm, rpm, psi, provide a signoff blank for the Contractor and Contracting Officer. Within a remarks column of the testing validation procedure include references to operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventative maintenance, condition monitoring (predictive testing) and inspection, adjustment, lubrication, and cleaning necessary to prevent failure.

Posted Instructions

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G in accordance with paragraph entitled OPERATION AND MAINTENANCE MANUALS of this Section. Submit three (3) paper copies, and two (2) pdf copies on CD-Rom.

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format. Refer to Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI) for additional requirements.

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

Drawings showing final as-built conditions of the project. The final CADD record drawings must consist of one (1) set of electronic CADD drawing files in the specified electronic format saved on a CD, one (1) set of mylar drawings, two (2) sets of blue-line prints of the mylars, and one (1) set of the approved working Record drawings.

As-Built Record of Equipment and Materials

Two (2) paper copies and one pdf copy of the record listing the as-built materials and equipment incorporated into the construction of the project.

Certification of EPA Designated Items; G

Certification Of USDA Designated Items; G

Interim DD FORM 1354; G

Checklist for DD FORM 1354; G

Red Zone Documents per Section 01 30 00.00 22; G

eOMSI, Final Submittal per Section 01 78 24.00 20; G

Post Installation Sanitary System Survey per Section 22 00 00; G

1.5 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

- a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

Supply two (2) items of each part for spare parts inventory. Provision of spare parts does not relieve the responsibilities listed under the Contract

warranty provisions.

1.6 WARRANTY MANAGEMENT

1.6.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this Contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

- a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, Subcontractors, manufacturers or suppliers involved.
- b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.
- c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.
- d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:
 - (1) Name of item.
 - (2) Model and serial numbers.
 - (3) Location where installed.
 - (4) Name and phone numbers of manufacturers or suppliers.
 - (5) Names, addresses and telephone numbers of sources of spare parts.
 - (6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
 - (7) Cross-reference to warranty certificates as applicable.
 - (8) Starting point and duration of warranty period.
 - (9) Summary of maintenance procedures required to continue the warranty in force.
 - (10) Cross-reference to specific pertinent Operation and Maintenance manuals.
 - (11) Organization, names and phone numbers of persons to call for warranty service.

(12) Typical response time and repair time expected for various warranted equipment.

- e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.
- f. Procedure and status of tagging of equipment covered by warranties longer than one year.
- g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.6.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

- a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
- b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.
- c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.6.3 Pre-Warranty Conference

Prior to Contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this Section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.6.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also,

submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

1.7 PROJECT CLOSEOUT DOCUMENTS

1.7.1 As-Built Record of Equipment and Materials

Furnish one (1) copy of preliminary record of equipment and materials used on the project 15 working days prior to final inspection. This preliminary submittal will be reviewed and returned 5 working days after final inspection with Government comments. Submit two (2) sets of final record of equipment and materials 10 working days after final inspection. Key the designations to the related area depicted on the Contract Drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA

Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used
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1.7.2 Final Approved Shop Drawings

Furnish final approved project shop drawings 30 calendar days after transfer of the completed facility.

1.7.3 Construction Contract Specifications

Furnish final record (as-built) construction Contract Specifications, including modifications thereto, 30 calendar days after transfer of the completed facility.

1.7.4 Real Property Equipment

Furnish a list of installed equipment furnished under this Contract. Include all information usually listed on a manufacturer's name plate. In the "EQUIPMENT-IN-PLACE LIST" include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Furnish a draft list at time of transfer. Furnish the final list 30 calendar days after transfer of the completed facility.

1.7.5 Red Zone Documents

Submit Red Zone Documents in accordance with Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME). Failure to provide acceptable and timely Red Zone Documents as specified may reflect poorly on the Prime Contractor's performance evaluation.

1.7.6 eOMSI, Final Submittal

Submit eOMSI, Final Submittal in accordance with Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Failure to provide an acceptable and timely final eOMSI submittal as specified may reflect poorly on the Prime Contractor's performance evaluation.

1.7.7 Post Installation Sanitary System Survey

Submit Post Installation Sanitary System Survey CD in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

1.8 PREVENTATIVE MAINTENANCE

Submit Preventative Maintenance and Condition Monitoring (Predictive Testing) and Inspection schedules with instructions that state when systems should be retested.

Define the anticipated length of each test, test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a signoff blank for the Contractor and the Contracting Officer for each test feature; e.g., gpm, rpm, psi. Include a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventative maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize corrective

maintenance and repair.

Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.9 COMMISSIONING

1.9.1 Building Commissioning

All Contract requirements for building commissioning must be completed prior to Contract completion. Failure to provide acceptable and timely commissioning services as specified may reflect poorly on the Prime Contractor's performance evaluation.

1.9.2 HVAC Commissioning

All Contract requirements of Commissioning of HVAC Systems must be fully completed, including all testing concurrent with Building Commissioning. All Contract requirements of Testing, Adjusting, and Balancing of HVAC Systems must be fully completed, including testing and inspection, prior to HVAC commissioning, except as noted otherwise in the Section for Testing, Adjusting, and Balancing for HVAC. All Contract requirements of Space Temperature Control Systems, Direct Digital Control for HVAC and Other Control Systems, or BACnet Direct Digital Control for HVAC and Other Building Control Systems must be fully completed, including all testing, prior to HVAC commissioning. The time required to complete all work and testing as prescribed in the specifications is included in the allotted calendar days for completion.

1.10 FINAL CLOSEOUT DOCUMENTS

Failure to provide acceptable closeout documents as specified above may reflect poorly on the Prime Contractor's performance and noted in their final performance evaluation.

PART 2 PRODUCTS

2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the Certification of EPA Designated Items as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

- a. The product does not meet appropriate performance standards;
- b. The product is not available within a reasonable time frame;
- c. The product is not available competitively (from two or more sources);

- d. The product is only available at an unreasonable price (compared with a comparable non-recycled content product)."

Record each product used in the project that has a requirement or option of containing recycled content, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, exemptions (a, b, c, or d, as indicated), and comments. Recycled content values may be determined by weight or volume percent, but must be consistent throughout.

2.2 CERTIFICATION OF USDA DESIGNATED ITEMS

Submit the Certification of USDA Designated Items as required by FAR 52-223-1 Bio-based Product Certifications and FAR 52.223-2 Affirmative Procurement of Biobased Products Under Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current USDA standards for biobased materials content. The following exemptions may apply to the non-procurement of biobased content materials:

- a. The product does not meet appropriate performance standards;
- b. The product is not available within a reasonable time frame;
- c. The product is not available competitively (from two or more sources);
- d. The product is only available at an unreasonable price (compared with a comparable bio-based content product)."

Record each product used in the project that has a requirement or option of containing biobased content, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, [total value of biobased content,] exemptions (a, b, c, or d, as indicated), and comments. Biobased content values may be determined by weight or volume percent, but must be consistent throughout.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF Contract Drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and red-lined PDF files.

Provide Monthly Progress As-Builts. Provide one (1) set of working as-built drawings (CADD) in the specified software and format, hard copy and electronic, to the Contracting Officer with the Monthly Progress Invoice.

Submit As-Built Drawings 30 days prior to Beneficial Occupancy Date (BOD).

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing.

Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:
 - (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
- c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
- d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
- e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
- f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
- g. For deletions, cross out all features, data and captions that relate to that revision.
- h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
- i. Indicate one of the following when attaching a print or sketch to a markup print:
 - 1) Add an entire drawing to Contract Drawings
 - 2) Change the Contract Drawing to show
 - 3) Provided for reference only to further detail the initial design.
- j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Show on the as-built drawings, but not limited to, the following information:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become

covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, manholes, utility structures, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from Contract plans.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, invert elevations and grades of drainage structures and pipes installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.
- h. Where Contract Drawings or Specifications present options, show only the option selected for construction on the working as-built markup drawings.
- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- l. Modifications and compliance with FC 1-300-09N procedures.
- m. Actual location of anchors, construction and control joints, etc., in concrete.
- n. Unusual or uncharted obstructions that are encountered in the Contract work area during construction.
- o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2 RECORD DRAWINGS

Prepare and provide Record Drawings and Source Documents in accordance with FC 1-300-09N. Provide four copies of Record Drawings and Documents on separate CDs or DVDs 30 days after BOD.

3.3 PROJECT RECORD DOCUMENTS

3.3.1 Record Drawings

This paragraph covers Record Drawings complete, as a requirement of the Contract. The terms "drawings," "Contract Drawings," "drawing files," "working as-built record drawings," and "final record drawings" refer to Contract Drawings (hard copy and CADD) which are revised to be used for final record drawings reflecting current project as-built conditions.

3.3.1.1 Government Furnished Materials

One (1) set of electronic CADD files in the specified software and format of the Contract Drawings will be provided by the Government at the preconstruction conference for projects requiring Final Record Drawings in CADD format.

3.3.1.2 Working Record and Final Record Drawings

Revise two (2) sets of hard copy paper Contract Drawings by red-line process described herein to reflect the current as-built conditions during the prosecution of the project. Keep the working as-built drawings current and keep at least one set available on the jobsite for review at all times. Changes from the Contract plans which are made in the work or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. **After the completion of each definable feature of work as listed in the Contractor Quality Control Plan (Foundations, Utilities, Structural Steel, etc., as appropriate for the project) provide one (1) set of working as-built drawings (CADD) in the specified software and format, hard copy and electronic, to the Contracting Officer.** The working as-built drawings, hard copy and (CADD), will be jointly reviewed for accuracy, completeness, and format by the Contracting Officer and the Contractor prior to submission of each monthly pay estimate. Failure to maintain the working as-built drawings, hard copy and (CADD) as specified herein, will result in the Contracting Officer deducting from the monthly progress payment an amount representing the estimated cost of maintaining the record drawings. Items to be shown on the working as-built drawings, hard copy and (CADD) are, but are not limited to, the following information:

- a. The actual location (horizontal and vertical position based on the datum established for the contract drawings), kinds, and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, as a back-up to the horizontal and vertical position, feature must also be shown by offset dimensions to two (2) permanently fixed surface features at the end of each run including each change in direction. Locate valves, splice boxes, and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run of pipe, fittings, valves, manholes, utility structures, etc.
- b. The actual location (horizontal and vertical position based on the datum established for the contract drawings), kind, and size of any sub-surface feature uncovered not accurately represented on the Contract Drawings.
- c. The location and dimensions of any changes within the building

structure.

- d. Changes in grade, elevations, cross section, or alignment of roads, earthwork, structures, or utilities.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared and/or furnished by the Contractor; including, but not limited to, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment foundations, etc.
- f. The topography, invert elevations, and grades of drainage installed or affected as part of the project construction.
- g. Changes or modifications which result from the final inspection.
- h. Where Contract Drawings or specifications present options, identify the option selected for construction on the working as-built prints.
- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Modifications (include within change order price the cost to change working and final record drawings to reflect modifications) and compliance with the following procedures:
 - (1) Both sets of the hard copy paper Contract working as-built drawings must be neat, legible, and accurate. Any drawings damaged, lost or corrupted by the Contractor must be satisfactorily replaced by the Contractor at no expense to the Government.
 - (2) For text deletions/revisions; strikeout existing drawing text with a single line as to not obscure or make the original text unreadable. Place the new text adjacent, clearly annotating the intent of the change.
 - (3) For line work; strikeout entities with parallel lines drawn at 45 degrees to the object, not to obscure or make the original object unreadable. Place the new object in its correct location and clearly annotate the intent of the change.
 - (4) Place a Revision Symbol at the location of each modification on the drawing sheet along with descriptive annotations of the revision.
 - (5) For details, sections, or schedules which are added to a drawing sheet, place a Revision Symbol by the detail, section, or schedule title.
 - (6) For major changes to a drawing, place a Revision Symbol by the title of the affected plan, section, or detail at each location.
 - (7) For changes within schedules, place a Revision Symbol by the change in the schedule.

- (8) The Revision Symbol must be a Delta sized to allow for a capital letter to fit within. The letter must have a height of not less than 1/8-inch when plotted.
- (9) The revision symbol letter must be consistent for all drawing modifications for each monthly billing cycle. Drawing modifications for the first monthly billing cycle must be designated as "A" for all modifications throughout the drawing package. The next month's revisions must be designated as "B" throughout the drawing package, and so on.

3.3.1.3 Drawing Preparation

At project completion, provide two (2) sets of the approved hard copy paper Contract Drawings modified to reflect the final as-built conditions of the project to the Contracting Officer. Modify the Contract Drawings as necessary to correctly show the features of the project as it has been constructed by bringing the Contract Drawings into agreement with the second set of approved working as-built drawings. The second set of approved working as-built drawings are also part of the permanent records of this project and must be returned to the Contracting Officer after final approval of the Record Drawings by the Government. Any drawings or drawing files damaged, lost or corrupted must be satisfactorily replaced at no expense to the Government.

3.3.1.4 Computer Aided Design and Drafting (CADD) Drawings

Only employ personnel proficient in the preparation of CADD drawings to modify the Contract Drawings or prepare any additional drawings sheets required. Modifications, to the Record Drawings must be equal in quality and detail to that of the original Contract Drawings and per PWD ME CADD Standards and As-Built CADD Standards. (For information on PWD Maine CADD standards, please e-mail Preston Gowen at preston.gowen@navy.mil.)

Line colors, line weights, lettering, layering conventions, and symbols must remain consistent throughout the record drawing set, regardless of either as-built or record drawing. Modify the original Contract Drawing files to reflect the Construction Contract as-built conditions reviewed and accepted by **the Contracting Officer**. Each as-built condition added to a drawing file must be encapsulated by a closed polygon or "revision cloud." A revision symbol must be placed outside the "revision cloud" with the appropriate letter designating the revision sequence. Annotate in the "revision block" of each drawing file modified as to the type of revisions made to the drawing file. The Contract Drawings are to be edited to reflect the as-built conditions only. No part of the original drawings must be deleted, erased or rendered illegible. Parts of the Contract Drawing found to be in error or modified during construction, must be over struck using methods described not to obscure the original drawing, and annotations must be added adjacent that clearly explain the modification, including accurate dimensions locating the feature. If additional drawings are required, the drawings must be prepared using the specified electronic file format applying, the same graphic standards specified for the original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used to create the Contract Drawings. Modifications, additions, and corrections to the Contract Drawings must be made to the electronic AutoCAD file(s). The original Contract Drawing files in the AutoCAD software format currently in use by PWD-ME will be furnished on a compact disc (CD). Provide all computer

software and hardware necessary to prepare the final record drawing set. The Contracting Officer will review the final record drawing set for accuracy and return them for required corrections, changes, additions, and deletions.

- a. Provide Record Drawings (CADD) in the following format:
 - (1) As-built Layering; follow original drawing layer naming conventions followed by "-AB".
 - (2) Deletions (Cyan) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Cyan) - Added items, lettering in notes and leaders.
 - (4) Special (Cyan) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (5) Furnish the Contract record drawing files in the AutoCAD software format currently in use by PWD-ME.
- b. Drawing files modified for as-built condition must be renamed by adding an underscore and the letters "AB" to the end of the existing file name. Drawing files where no modifications were required must be renamed by adding an underscore and the letters "RD" to the end of the existing file name.
- c. When final revisions have been completed to the record drawing set, add the wording "RECORD DRAWINGS / AS-BUILT CONDITIONS" followed by the name of the Contractor in letters at least 3/16 inch high in the lower left hand corner of the cover sheet drawing. Mark all other Contract Drawings in the same location and manner as either "Record Drawing" denoting no revisions on the sheet or "As built Drawing" denoting modifications, additions, or corrections have been made to the drawing sheet. Modify the revision block to reflect either "record drawing", for no changes or "as built drawing", for changes and date for submittal.
- d. Within 20 working days after Government approval of all of the working record drawings for a phase of work, prepare the CADD electronic files for that phase of work and submit for Government review and approval. The Government will promptly return one set of prints annotated with any necessary corrections. Within 10 working days revise the CADD files accordingly at no additional cost to the Government and submit one set of final prints for the completed phase of work to the Government.
- e. Within 20 working days of substantial completion of all phases of work, submit the final record drawing package for the entire project. Submit one (1) set of electronic files on compact disc, read-only memory CD-ROM, one (1) set of mylars and one set of the approved working record drawings. They must be complete in all details and identical in form and function to the Contract Drawing files supplied by the Government. Any transactions or adjustments necessary to accomplish this is the responsibility of the Contractor. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CADD system. Paper prints, drawing files, and storage media submitted will become the property of the Government upon final approval. Failure to submit final record drawing files and marked

prints as specified will be cause for withholding any payment due under this Contract. Approval and acceptance of final record drawings must be accomplished before final payment is made.

3.4 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave site and surrounding premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

3.5 REAL PROPERTY RECORD

Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354.

3.5.1 Interim DD FORM 1354

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. Include any additional assets, improvements, and alterations from the Draft DD FORM 1354.

3.5.2 Completed DD FORM 1354

For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link:

www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

Submit the completed Checklist for DD FORM 1354 of Installed Building Equipment items. Attach this list to the updated DD FORM 1354.

-- End of Section --

27. CONSTRUCTION DEFICIENCIES <i>(Attach blank sheet for continuations)</i>	28. PROJECT REMARKS <i>(Attach blank sheet for continuations)</i>
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INSTRUCTIONS	
GENERAL. This form has been designed and issued for use in connection with the transfer of military real property between the military departments and to or from other government agencies. It supersedes ENG Forms 290 and 290B (formerly used by the Army and Air Force) and NAVDOCKS Form 2317 (formerly used by the Navy). Existing instructions issued by the military departments relative to the preparation of DD Form 1354 are applicable to this revised form to the extent that the various items and columns on the superseded forms have been retained. The military departments may promulgate additional instructions, as appropriate. For detailed instructions on how to fill out this form, please refer to Unified Facilities Criteria (UFC) 1-300-08, dated 16 April 2009 or later. SPECIFIC DATA ITEMS. 1. From. Name of the transferring agency. 2. Date Prepared. Date of actual preparation. Enter all dates in YYYYMMDD format (Example: March 31, 2010 = 20100331). 3. Project/Job Number. Project number on a DD Form 1391 or Individual Job Order Number. 4. Serial Number. Sequential serial number assigned by the preparing organization (e.g., 2010-0001). 5. To. Name and address of the receiving installation, activity, and Service of the Real Property Accountable Officer (RPAO). 6. RPSUID/SITENAME/INSTCODE/INSTNAME. Site Unique Identifier and name or installation code and name where the constructed facility is located. 7. Contract Number(s). Contract number(s) for this project. 7a. Placed-In-Service Date. RPA Placed In Service Date. This is the date the asset is actually placed-in-service. 8. Transaction Details. a. Method of Transaction. Mark (X) as many boxes as apply. b. When/Event. When or event causing preparation of DD Form 1354. X only one box. c. Type. Draft, interim, or final DD Form 1354. X only one box. 9. Item Number. Use a separate item number for each facility, no item number for additional usages.	10a. Facility Number. Assigned in accordance with the Installation/Base Master Numbering Plan. 10b. RPUID. Identified in Real Property Inventory. 11. Category Code. The category code describes the facility usage. 12. Catcode Description. The category code name which describes the facility usage. 13. Type. Type of construction: P for Permanent; S for Semi- permanent; T for Temporary. 14. Sustainability Code. Reports whether or not an asset meets the sustainability guidelines set forth in Section 2(g) of Executive Order 13514. Valid values are: 1 (asset meets the guidelines); 2 (asset does not meet the guidelines); 3 (asset not evaluated); 4 (asset not subject to guidelines). 15. Area: UM 1. Area unit of measure; use the unit of measure associated with the category code selected in 11. 16. Total Quantity UM 1. The total area for the measure identified in Item 15. Use negative numbers for demolition. 17. Other: UM 2. Unit of Measure 2 is the capacity or other measurement unit (e.g., LF, MB, EA, etc.). 18. Total Quantity UM 2. The total capacity/other for the measure identified in Item 17. 19. Cost. Cost for each facility; for capital improvements to existing facilities, show amount of increase only. If there is no increase for the capital improvement, enter N/A. 20. Fund Source. Enter the Fund Source Code for this item. 21. Funding Organization. Enter the code for the organization responsible for acquiring this facility. 22. Interest Code. Enter the code that reflects government interest or ownership in the facility. 23. Item Remarks. Remarks pertaining only to the item number identified in Item 9; show cost sharing. 24. Statement of Completion. Typed name, signature, title, and date of signature by the responsible transferring individual or agent. 25. Accepted By. Typed name, signature, title, and date of signature by the RPAO or accepting official. 26. Property Voucher Number. Next sequential number assigned by the RPAO in voucher register. 27. Construction Deficiencies. List construction deficiencies in project during contractor turnover inspection. 28. Project Remarks. Project level remarks and continuation of blocks.

NAVFAC MID-ATLANTIC

**Public Works Department
Maine**

PORTSMOUTH NAVAL SHIPYARD

AS-BUILT CADD STANDARDS

June 2014



NAVFAC Midlant PWD makes these documents available on an “as is” basis. This document including attached file(s) and contained information is (are) provided as guidance NAVFAC Midlant PWD. All warranties and representations of any kind with regard to said documents are disclaimed, including the implied warranties of merchantability and fitness for a particular use. NAVFAC Midlant PWD does not warrant the documents against deficiencies of any kind and makes no claims, promises or guaranties about the accuracy, completeness, or adequacy of the contents of the files, and expressly disclaims liability for error and omissions thereof.

STANDARDS FOR EDITING EXISTING CONTRACT DRAWINGS TO REFLECT AS-BUILT CONDITIONS

Following are the *Standards* for editing existing contract drawings to reflect the construction as-built conditions for **NAVFAC MID-ATLANTIC; Public Works Department Maine (PWD-ME)**. The purpose of these *Standards* is to provide a uniform system of drawing formats. This system will be used in retrieving information from the drawings in the future. Also, these *Standards* were established to eliminate various recurring problems encountered when transferring electronic files from non-government sources. All drawings modifications found to be generated outside of these standards will be returned to the drawing provider for correction at no additional cost to the **GOVERNMENT**.

As-built Drawings:

Drawing Sets:

The contractor shall maintain at the jobsite one set of full-size hard copy prints of the contract drawings, accurately marked in red with adequate dimensions, to show all variations between the construction actually provided and that indicated or specified in the contract documents, including buried or concealed construction. **Special attention shall be given to recording the horizontal and vertical location of all buried utilities that differ from the contract drawings.** Existing utility lines and features revealed during the course of construction shall also be accurately located and dimensioned, using permanent existing features and station coordinates as a reference. Acceptable features include; building corners, centers of utility manhole covers, fire hydrants, etc. Existing topographic features that differ from those shown on the contract drawings shall also be accurately located and recorded. Where a choice of materials or methods is permitted herein or where variations in scope or character of work from that of the original contract are authorized, the drawings shall be marked to define the construction provided. The representations of such changes shall conform to standard drafting practices and shall include such supplementary notes, legends, and details as necessary to clearly portray the as-built construction. These drawings shall be available for review by the Contracting Officer at all times. Upon completion of the work, the marked up prints shall be certified as correct, signed by the Contractor, and delivered to the Contracting Officer for review by **PWD-ME**. After acceptance of the as-builts, the contractor will add the as-built information to the original contract drawing files and plot new Mylar reflecting the as-built conditions.

Requests for partial payments will not be approved if the marked prints are not kept current. Request for final payment will not be approved until Contracting Officer receives delivery of the original electronic contract drawing CADD files, modified to reflect the as-built conditions and new Mylar plotted to reflect the as-built conditions.

Computer Aided Drafting Design (CADD) As-built Drawings:

File formats:

Although all methods of CADD drawing are permitted, the final product of all computer-aided drawings shall be made compatible with the **AutoCAD** currently in use by **PWD-ME** Drawings

created using non-AutoCAD programs that do not support AutoCAD's DWG file format, shall be transferred to the Drawing Interchange Format (DXF).

If the drawing provider's CADD software does not support the DXF format, it is the responsibility of the drawing provider to contact the Contracting Officer who will in turn contact the **PWD-ME** to make special arrangements for file transfer.

Each drawing file must not depend on any other drawing file to completely represent the finished product (*Mylar*). All drawings found to use dependent external reference drawings, **will be returned to the drawing provider for correction.**

Each drawing file must not depend on any **"THIRD PARTY"** utility or software to represent the finished Mylar. Before using any third party software to create a finished product, the drawing provider shall verify that no additional cost or effort is required by the **PWD-ME** to completely represent the finished product (*Mylar*).

Drawing Sheets:

Electronic files of the contract drawings shall be provided to the Contractor following award of the contract. The contractor will add the accepted as-built conditions to the original electronic file of the drawing sheet. If additional sheets need to be added to the contract drawing package because of insufficient sheet space, it is the responsibility of the Contractor to ensure the PWD-ME CADD Drawing Standards are obtained and followed. Additional drawing numbers will be assigned by **PWD-ME** personnel prior to submittal of as-built drawings.

Basic Drawing Standards:

Electronic As-builts:

The contractor shall modify the original contract drawing files to reflect the construction contract as-built conditions reviewed and accepted by **the Contracting Officer**. Each as-built condition added to a drawing file shall be encapsulated by a closed polygon or "revision cloud. A revision symbol shall be placed outside the "revision cloud" with the appropriate letter designating the revision sequence. The contractor shall annotate in the "revision block" of each drawing file modified as to the type of revisions made to the drawing file. The contract drawings are to be edited to reflect the as-built conditions only. No part of the original drawings shall be deleted, erased or rendered illegible. Parts of the contract drawing found to be in error or modified during construction, shall be over struck using methods described not to obscure the original drawing, and annotations will be added adjacent that clearly explain the modification, including accurate dimensions locating the feature.

Drawing Sheet Management:

Where construction projects encompass multiple facilities (buildings) or systems (utilities), all plans, details, legends, notes etc., for an individual facility or system must appear on the drawing sheets for each facility or system. Standardized details must appear on the drawing sheets for each facility or system where they apply. Cross-referencing details between facilities or systems is not permitted.

Drawing Units:

All drawings of buildings, structures, floor plans, etc. are to be drawn full size, with one drawing unit equal to one inch. All site plans, location plans, etc. are to be drawn full size, with one drawing unit equal to one foot. The original drawing's origin shall not be modified.

Entity Management:

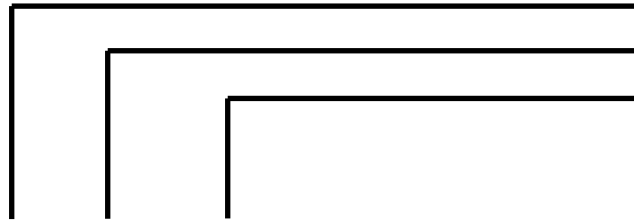
All additions to the original contract drawings will be made on additional layers as designated below. The minimum text height for the "D" size sheet is 1/8in plotted and 3/32in plotted for the "B" size sheet. All as-built modification layers are to be **CYAN** in, including all annotations, dimensions, leaders and callouts.

Entities discovered during construction but not represented on the original construction drawings shall be assigned to layers as described below. This information shall be added to the drawing in such a way not to obscure any original drawing entity when plotted.

Level/layer Management:

All drawing entities shall be grouped together by layer. Drawing entities shall be assigned to layers by discipline code for that entity, see table page D4. This includes non-text entities such as strikeouts, lines, circles, symbols, etc. See Table-1 for typical layer-naming.

Table-1

	As-built Code (Always "Z")
	Discipline Code (Non-text entities)
	Annotations (Text describing the As-built Condition)
Z – ARCH - TEXT	

Identify entities by discipline. For instance: the example below shows layering configuration for all architectural as-built entities **EXCLUDING TEXT** that deviate from the original design:

Z-ARCH.

Annotations, dimensions, leaders and callouts for those architectural entities shall be assigned to layer:

Z-ARCH-TEXT.

All annotations shall be assigned to layers by discipline and designated as **TEXT**.

Each layer name will follow the discipline code for that entity. See Table-2 for Discipline Descriptions:

Table-2

<i>Discipline Code:</i>	<i>Discipline:</i>	<i>Description:</i>
ARCH	Architectural	Architectural design, building's interior/exterior, walls, doors, windows etc.
CIVIL	Civil	Site work, external to buildings and structures. Typically surface work.
ELEC	Electrical	Electrical work pertaining to buildings/structures internal/external and distribution.
FP	Fire Protection	Fire Protection systems pertaining to buildings/structures internal/external
GEN	General	Typically drawing information, general notes, symbols, etc....
MECH	Mechanical	Mechanical work, HVAC components, compressed air, etc. pertaining to buildings/structures internal/external
PLUMB	Plumbing	Plumbing, piping and fixtures, etc.. pertaining to buildings/structures internal/external
UTIL	Utilities	Utility distribution systems, duct work, manholes and piping/cabling underground/direct buried
STRUC	Structural	All structural work, building framing, conc. duct reinforcement, misc. steel work, etc.
TELE	Telecommunications	Phone lines, cable TV, computer data lines, etc. pertaining to buildings/structures internal/external

Electronic Deliverables:

Submit all as-built documentation and drawing files to the Contracting Officer via recordable CD in the file formats discussed in this and other sections. Electronic transfer of files via E-mail or other methods is not permitted.

Standards Deviation:

Contact the Contracting Officer if questions arise about these standards or these drawing standards cannot be followed. Otherwise, all drawing files will be returned to the drawing provider for correction at no additional cost to the **PWD-ME**.

Last updated June 18, 2014

NAVFAC MID-ATLANTIC

**Public Works Department
Maine**

PORTSMOUTH NAVAL SHIPYARD

CADD STANDARDS

June 2014



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1. CADD STANDARDS

All design documents and deliverables prepared by In-House forces and/or contracted A/E firms including IDIQ, Design Bid-Build and Design-Build process shall conform to the following NAVFAC MIDLANT and PWD-ME CADD standards areas not addressed on this standard shall follow:

- UFC 1-300-09N

U.S. DEPARTMENT OF DEFENSE (DOD) [UFCs available on <http://dod.wbdg.org/>]

These are NAVFAC MIDLANT PWD-MAINE specific guidelines that supersede all others for contract documents utilized for execution. Refer to the U.S. National CAD Standard Version 3.0 for any items not covered in this or the referenced NAVFAC UFCs.

1. A. General: (for both Design-Build and Design-Bid-Build)

For drawing templates, pen tables, sample symbology, and instructions on sheet arrangement & procedures, see attached NAVFAC MIDLANT PWD-MAINE; Sheet Border Use and Procedures. Provide *.PDF format of all drawing sheets at each submittal phase. In addition, provide *.PDF and *.DWG format of all drawing sheets at the final design submission. CADD files shall be made available upon request, in addition to the full size, stamped Mylar and half-size hard copies or as directed by the specific contract specifications.

1. B. Design Drawings

Prepare, organize, and present design drawings in accordance with the requirements of UFC 1-300-09N.

All drawings and their associated PDFs will maintain a “PRELIMINARY Not for Construction” stamp across the signature areas of the title block, until the actual final design submittal. The stamp shall be in translucent lettering across the Project Title area of the drawing title block and shall be displayed on layer “G-ANNO-TTLB-PRLM”. The “G-ANNO-TTLB-PRLM” shall be off for final submittal.

Design revisions shall be tracked by number and date in the revision block of each drawing sheet during the design process. Design revisions shall be assigned to layer “G-ANNO-TTLB-PRLM” and shall be frozen for final submittal.

1. C. Drafting Conventions

1. C.1. Fonts

The standard text height for a plotted D-size drawing shall be **1/8”** for typical text, and **1/4”** for titles, and **1”** maximum for project titles on cover sheets. The standard text height for a plotted B-size drawing shall be **3/32”** for typical text, and **1/8”** for titles, and **1/2”** maximum for project titles on cover sheets. For typical text, use the “**RomanS**” font. For titles, use the “**Swis 721**”

BT” font. Both fonts shall have a width ratio of no less than **0.8**.

ROMANS – 1/8” TALL

Swis 721 BT – 1/4” TALL (up to 1”)

These are the only fonts that should be used within the drawing area, with one exception: on civil site plans, the Design Manager (EIC/AIC) may use text (“RomanS” font, 1/8” height {Leroy 100}, 0.8 width factor, with an oblique angle of 12 degrees) to annotate existing site features only. Typical (or Normal) text as defined above must be used for all other annotations.

1. C. 2. Symbols

Approved symbology in use by PWD-ME is provided on the PWD-ME title sheet. On the PWD-ME title and standard sheet, borders and plot styles shall delivered at each project design kickoff MTG.

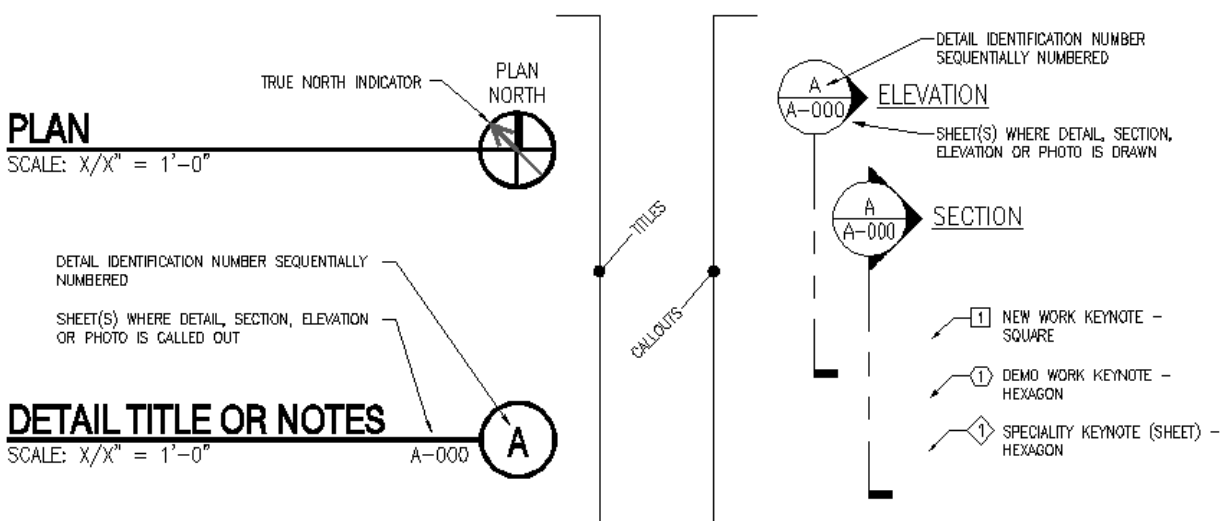


Figure 3-1 PWD-ME Approved Symbols

Note that detail, section, elevation and photograph callouts utilize two-part bubbles which indicate a detail identification number and identify sheet where the detail, section, elevation or photo is shown. Additional sheets(s) on which the detail is called out is (are) displayed to the bottom left of the bubble.

1. C. 3. Line Weights / Pen Tables

NAVFAC’s Electronic Documents and Deliverables Working Group (EDDWG) developed a comprehensive pen table that utilized the National Cad Standard 255-pen table as a basis, but added much needed thinner lines and additional greyscales (shaded or screened pens) that had not made the transition from the original US Coast Guard standard. The NAVFAC-specific pen

table is provided as filename **NavFacStd.ctb**.

The pen tables can be thought of in groups of 20. After pen 19, pens 20 thru 39 are in the “Rust” range when displayed on screen. The first 10 (pens 20 – 29) look “Rust” and print shades of “Rust”; while the second 10 (pens 30 – 39) look “Rust” but print Black. Typically, design drawings for contract documents utilize the pens that print Black (currently colored pens are used for renderings and some other instances, but not normally on contract drawings).

Since these pen tables are established to be legible when printed at full-size (22”x34”), the corresponding text height is 1/8”. To be consistent, the related B-size (11”x17”) & A-size sheet (8-1/2”x11”)—normally used for sketches, uses fonts and line weights that are ½ the size of the large format (D-size) documents—otherwise the fonts are too big and the line weights are too bold. The associated pen table for B-size & A-size documents is **NavFacStd-Sketch.ctb**

1. D. NAVFAC Logo

A NAVFAC Publications/Logo Guide will be published at a later date, but in the interim, all deliverables shall have the NAVFAC Brand Logo on the title block area of the drawing border (as displayed in the Templates). The logo should not be removed, deleted, or turned off and should appear on all PDFs to be utilized for the electronic bid solicitations. If contract drawings are produced by an A/E firm and that firm desires to have their logo on the drawings then the A/E must apply that logo to all sheets consistently in the same location. Typically that is in the lower- or upper-right hand corner of the drawform, adjacent to the title block.

1. E. Units of Measure

Unless otherwise directed, for all projects the default shall be English units of measurement (U.S. Customary System of units). All drawings, specifications, cost estimates, and design calculations shall reference this system of units.

1. F. Datums and Coordinate Systems

VERTICAL DATUM: All elevations, grades, or profiles shall be represented in U.S. feet and referenced to NAVD88 vertical datum and will be provided as required.

HORIZONTAL DATUM: All Portsmouth Naval Shipyard mapping, utility information, site drawings and surveys are represented in U.S. feet and referenced to the state plane coordinate system, NAD83 Maine west, zone 1802. All electronic mapping and utility information shall be maintained to this system.

1. G. Drawing Numbers/File Naming

In addition to NAVFAC drawings numbers discussed in UFC 1-300-09N, Chapter 4-3, “Drawing Numbers” the Government Project Manager will provide PWD (Public Works Department) Drawing Numbers for each drawing sheet, similar to NAVFAC drawing numbers. The PWD Drawing Number to be requested by the drawing provider at the 100% design stage. The CADD

files shall be named by the PWD drawing number in accordance with the following guidance: The CADD file for each ANSI D (22" x 34") size drawing sheet shall be named by the corresponding PWD Number, minus the design year designator. Where the assigned PWD Number is: 59-08-501 the corresponding CADD file shall be named 059-501.DWG. Where a single drawing file is used to create multiple drawing sheets via layout tabs, the drawing file shall be named by the PWD Number of the first drawing sheet of the drawing set. Drawing sets shall be arranged by discipline.

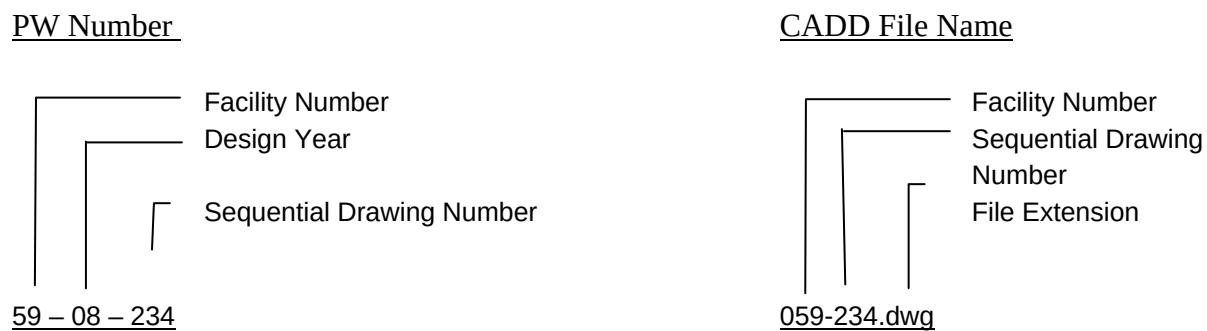


Figure 7-1 DRAWING NUMBERS/FILE NAMING

Files for sketches, drawings other than ANSI D (22" x 34") size shall be named by construction contract number and sheet number of the first sheet of the sheet set.

1. G. 1. File Naming Convention for PDF Files:

ADOBE PDF file naming shall follow CADD file naming methods.

1. H. Electronic Submittals

Submit all CADD files for the final drawings in Autodesk Drawing (*.DWG) compatible with current PWD-ME CADD format. Drawing files shall be full files, uncompressed and unzipped. All submitted CADD files shall maintain their original and complete data structure as produced in their native Autodesk application (i.e. Architectural Desktop, Land Development Desktop, Civil 3D, Building Systems...). All files utilizing AutoCAD's "Xref" feature shall be delivered with all external files "bound" to the drawing file. There shall be no external references in the final drawing file.

Multiple layout tabs are allowed, the tabs shall be annotated by PWD number for each sheet as described in section PWD-ME 7.0 Drawing Numbers/File Naming. The electronic file(s) shall be named by the PWD number of the first sheet of the sheet set. When required, the drawing file shall be split by sheet discipline. Corresponding PDF files shall be grouped into one file and named by the project title.

1. I. Record Documents ...1.10.1 Record Drawings (As-Built drawings, G)

The Quality Control Manager shall deliver the marked-up As-Built drawings to the [Contractor's](#) Designer of Record who shall incorporate all as-built modifications.

The as-built modifications shall be accomplished by electronic drafting methods on the original CADD (*.DWG) design drawing files to create a complete set of record drawings. For each Record drawing, provide a CADD drawing identical to signed [A/E- or] Contractor-originated *.PDF drawings, that incorporates modifications to the as-built conditions. In addition, copy initials and dates from the Contracting Officer approved *.PDF documents to the title block of the Record CADD (*.DWG) drawings. The Record electronic files shall use the file name of the original signed CADD drawing file name with the suffix “-RD” before the file extension, “.DWG” (example; 059-501-RD.dwg – see Drawing File Naming). The original design, RFP, reference, or definitive drawings are not required for inclusion in the Record set of drawings.

After all as-built conditions are recorded on the CADD (*.DWG) files, produce a PDF file of each individual record drawing in conformance with UFC 1-300-09N. Generate PDF drawing files using a PDF page size that corresponds to the original document sheet size (NAVFAC utilizes an ANSI D-size, 22”x34” frame and a 0” border all around) and a PDF print resolution that results in clear detail of all drawing features.

Provide one set of signed and stamped record drawings, plotted on Mylar that fully represent the PDF file. Provide all project final submittals including as-builts in accordance with UFC 1-300-09N Chapter 7-5.2. Each CD-ROM shall be marked with Project Name, Construction Contract Number, Project Number, Specification Number, and Record Drawing date.

1. I. 1. Source Documents

In addition to the drawings provide the specifications, design analysis, reports, survey data, calculations, and any other contract documents utilized in creating the design package (drawings, specifications, and cost estimate) on the CD-ROM disk(s) as specified in preceding paragraph.

1. J. Forwarding Submittals

The [A/E- or] Contractor shall provide all submittals, that lend themselves to paper format (i.e. 8-1/2”x11” documents, 11”x17” or tabloid sheets, or 22”x34” drawings), in electronic *.PDF (Portable Document Format) to the Contracting Officer. The appropriate forwarding e-mail address(es) will be provided at the Pre-Construction meeting. Provision of electronic submittals with appropriate electronic transmittal may alleviate some of the paper copies required herein but must be verified at Pre-Construction meeting. In such instances a minimum of one hard copy shall be forwarded to the Contracting Officer.

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

08/15, CHG 1: 11/20

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with [Section 01 33 00 SUBMITTAL PROCEDURES][Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and Section 01 33 10.05 22 DESIGN SUBMITTAL PROCEDURES]:

SD-10 Operation and Maintenance Data

Training Plan; G

Training Outline; G

Training Content; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

1.3 OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data for the provided equipment, product, or system, defining the importance of system interactions, troubleshooting, and long-term preventive operation and maintenance. Compile, prepare, and aggregate O&M data to include clarifying and updating the original sequences of operation to as-built conditions. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this Section and Section 01 33 00 SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES.

Coordinate the work of this Section with Section 01 78 24.00 20 FACILITY

ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

1.3.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.3.2 Package Content

Provide data package content in accordance with paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES. Comply with the data package requirements specified in the individual technical Sections, including the content of the packages and addressing each product, component, and system designated for data package submission, except as follows. Use Data Package [3][4][5] for commissioned items without a specified data package requirement in the individual technical Sections. Provide a Data Package [3][4][5] instead of Data Package 1 or 2, as specified in the individual technical Section, for items that are commissioned.

1.3.3 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.4 OPERATION AND MAINTENANCE MANUAL FILE FORMAT

Assemble data packages into electronic Operation and Maintenance Manuals. Assemble each manual into a composite electronically indexed file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance, and record files, project record documents, and training videos. Include a complete electronically linked operation and maintenance directory.

1.4.1 Organization

Bookmark Product and Drawing Information documents using the current version of CSI Masterformat numbering system, and arrange submittals using the specification Sections as a structure. Use CSI Masterformat and UFGS numbers along with descriptive bookmarked titles that explain the content of the information that is being bookmarked.

1.4.2 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Building Number
- b. Project Title
- c. Activity and Location
- d. Construction Contract Number/eProjects Number

- e. Prepared For: (Contracting Agency)
- f. Prepared By: (Name, title, phone number, and email address)
- g. Include the disk content on the disk label
- h. Date
- i. Virus scanning program used

1.5 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

The following are a detailed description of the data package items listed in paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES.

1.5.1 Operating Instructions

Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.5.1.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Provide recommended safeguards for each identified hazard.

1.5.1.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.5.1.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown, and Post-shutdown operating procedures including the control sequence for each procedure.

1.5.1.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.5.1.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones, or portions of systems controlled.

1.5.1.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.5.1.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component, or system. Describe conditions under which the item equipment should not be allowed to run.

1.5.1.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.5.1.9 Additional Requirements for HVAC Control Systems

Provide Data Package 5 and the following for control systems:

- a. Narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Full as-built sequence of operations.
- c. Copies of checkout tests and calibrations performed by the Contractor (not Cx tests).
- d. Full points list. Provide a listing of rooms with the following information for each room:
 - (1) Floor
 - (2) Room number
 - (3) Room name
 - (4) Air handler unit ID
 - (5) Reference drawing number
 - (6) Air terminal unit tag ID
 - (7) Heating or cooling valve tag ID
 - (8) Minimum cfm
 - (9) Maximum cfm
- e. Full print out of all schedules and set points after testing and acceptance of the system.
- f. Full as-built print out of software program.
- g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.5.2 Preventive Maintenance

Provide the following information for preventive and scheduled maintenance

to minimize repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.5.2.1 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.
- c. A Lubrication Schedule showing service interval frequency.

1.5.2.2 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests), and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including trade/craft requirements by type of trade/craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication, and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.5.2.3 Cleaning Recommendations

Provide environmentally preferable cleaning recommendations in accordance with ASTM E1971.

1.5.3 Repairs

Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.5.3.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.5.3.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.5.3.3 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards.

1.5.3.4 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings, and adjustments required. Use a combination of text and illustrations.

1.5.3.5 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.5.3.6 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of trade/craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.5.4 Appendices

Provide information required below and information not specified in the preceding paragraphs but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.5.4.1 Product Submittal Data

Provide a copy of SD-03 Product Data submittals documented with the required approval.

1.5.4.2 Manufacturer's Instructions

Provide a copy of SD-08 Manufacturer's Instructions submittals documented with the required approval.

1.5.4.3 O&M Submittal Data

Provide a copy of SD-10 Operation and Maintenance Data submittals documented with the required approval.

1.5.4.4 Parts Identification

Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master parts catalog.

1.5.4.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or Contract Documents in order to keep warranties in force. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

1.5.4.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification Section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

1.5.4.7 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.5.4.8 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.5.4.9 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.5.4.10 Field Test Reports

Provide a copy of Field Test Reports (SD-06) submittals documented with the required approval.

1.5.4.11 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name, address, and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical Sections. The information required in each type of data package is as follows:

1.6.1 Data Package 1

- a. Safety precautions and hazards
- b. Cleaning recommendations
- c. Maintenance and repair procedures
- d. Warranty information
- e. Extended warranty information
- f. Contractor information
- g. Spare parts and supply list

1.6.2 Data Package 2

- a. Safety precautions and hazards
- b. Normal operations
- c. Environmental conditions
- d. Lubrication data
- e. Preventive maintenance plan, schedule, and procedures
- f. Cleaning recommendations
- g. Maintenance and repair procedures
- h. Removal and replacement instructions
- i. Spare parts and supply list
- j. Parts identification

- k. Warranty information
- l. Extended warranty information
- m. Contractor information

1.6.3 Data Package 3

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Environmental conditions
- g. Operating log
- h. Lubrication data
- i. Preventive maintenance plan, schedule, and procedures
- j. Cleaning recommendations
- k. Troubleshooting guides and diagnostic techniques
- l. Wiring diagrams and control diagrams
- m. Maintenance and repair procedures
- n. Removal and replacement instructions
- o. Spare parts and supply list
- p. Product submittal data
- q. O&M submittal data
- r. Parts identification
- s. Warranty information
- t. Extended warranty information
- u. Testing equipment and special tool information
- v. Testing and performance data
- w. Contractor information
- x. Field test reports

1.6.4 Data Package 4

- a. Safety precautions and hazards

- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Operator service requirements
- g. Environmental conditions
- h. Operating log
- i. Lubrication data
- j. Preventive maintenance plan, schedule, and procedures
- k. Cleaning recommendations
- l. Troubleshooting guides and diagnostic techniques
- m. Wiring diagrams and control diagrams
- n. Maintenance and repair procedures
- o. Removal and replacement instructions
- p. Spare parts and supply list
- q. Maintenance and repair work-hours
- r. Product submittal data
- s. O&M submittal data
- t. Parts identification
- u. Warranty information
- v. Extended warranty information
- w. Personnel training requirements
- x. Testing equipment and special tool information
- y. Testing and performance data
- z. Contractor information
- aa. Field test reports

1.6.5 Data Package 5

- a. Safety precautions and hazards
- b. Operator prestart

- c. Start-up, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Environmental conditions
- f. Preventive maintenance plan, schedule, and procedures
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring and control diagrams
- i. Maintenance and repair procedures
- j. Removal and replacement instructions
- k. Spare parts and supply list
- l. Product submittal data
- m. Manufacturer's instructions
- n. O&M submittal data
- o. Parts identification
- p. Testing equipment and special tool information
- q. Warranty information
- r. Extended warranty information
- s. Testing and performance data
- t. Contractor information
- u. Field test reports
- v. Additional requirements for HVAC control systems

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the PWD ME Facilities Management Specialist (FMS), building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME). Address aspects of the eOMSI Manual, as submitted in Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Training must include classroom or field lectures based on the system operating

requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) and prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC Manager and . Include within the plan the following elements:

- a. Equipment included in training
- b. Intended audience
- c. Location of training
- d. Dates of training
- e. Objectives
- f. Outline of the information to be presented and subjects covered including description
- g. Start and finish times and duration of training on each subject
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)
- i. Instructor names and instructor qualifications for each subject
- j. List of texts and other materials to be furnished by the Contractor that are required to support training
- k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The [QC Manager][] is responsible for overseeing and approving the content and adequacy of the training. Provide a brief summary of the FACILITY INFORMATION manual, and a more detailed presentation of the PRODUCT AND DRAWING MANUAL, specified in Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual files (bookmarked PDF) and eOMSI Manual files as specified in Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI), and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two (2) copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two (2) copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the and QC Manager in accordance with or 01 45 00.10 22 QUALITY CONTROL FOR MINOR CONSTRUCTION (PWD ME.

-- End of Section --

SECTION 01 78 24.00 20

FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI)
02/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N

(2014; with Change 6, 2021) Navy and
Marine Corps Design

1.2 DEFINITIONS AND ABBREVIATIONS

1.2.1 eOMSI Manual

Manual (PDF file) provided by the Contractor that includes, but is not limited to, product information, a facility description with photos, and a list of primary facility systems.

1.2.2 eOMSI Facility Data Workbook (FDW)

A Microsoft Excel file containing required facility information populated by the Contractor.

1.2.3 Systems

The words "system", "systems", and "equipment", when used in this document refer to as-built systems and equipment.

1.2.4 Computer Assisted Design and Drafting (CADD)

Electronic Computer Assisted Design and Drafting graphic software program that is used to create facility design contract documents and Record Drawings.

1.2.5 KTR

An abbreviation for "Contractor."

1.3 eOMSI MEETINGS

1.3.1 Discussion

Be prepared to discuss the following during this meeting:

- a. eOMSI Manual and eOMSI Facility Data Workbook Development Meetings
- b. Processes and methods of gathering eOMSI Manual and eOMSI Facility Data Workbook information during construction.
- c. The eOMSI Submittals schedule. Include the eOMSI submittal schedule on

the Baseline Construction Schedule.

- d. Electronic eOMSI Facility Data Workbook file for Contractor's use and completion.

1.3.2 eOMSI Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Pre-eOMSI Development Meeting prior to the submission of the eOMSI Progress Submittal. Meeting attendance must include the Contractor's eOMSI Manual and Facility Data Workbook Preparer, and Quality Control Manager, and the Government's Design Manager (DM), Contracting Officer's Representative, and NAVFAC Public Works (PW) Facilities Management Division (FMD). Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for eOMSI and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of eOMSI.

1.3.3 Facility Turnover Meeting

Include eOMSI in NAVFAC Red Zone (NRZ) facility turnover meetings as specified in Section 01 30 00.00 ADMINISTRATIVE REQUIREMENTS.

1.4 SUBMITTAL SCHEDULING

1.4.1 eOMSI, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide eOMSI Manual Files (Bookmarked PDF) and eOMSI Facility Data Workbook (Excel). Include the elements and portions of system construction completed up to this point.

The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the eOMSI Coordination Meeting. Field verify a portion of the eOMSI information in accordance with paragraph FIELD VERIFICATION.

1.4.2 eOMSI, Prefinal Submittal

Submit the 100 percent submittal of the eOMSI Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of eOMSI Progress submittal must be corrected prior to the Prefinal submission.

The eOMSI Prefinal Submittal must include eOMSI Manual Files (Bookmarked PDF) and eOMSI Facility Data Workbook (Excel).

1.4.3 eOMSI, Final Submittal

Submit completed eOMSI Manual Files (Bookmarked PDF) and eOMSI Facility Data Workbook (Excel). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal eOMSI submittal, including the Field Verification, must be corrected prior

to the Final eOMSI submission.

1.5 UNITS OF MEASURE

Provide eOMSI utilizing the units of measure

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-11 Closeout Submittals

eOMSI, Progress Submittal; G

eOMSI, Prefinal Submittal; G

eOMSI, Final Submittal; G

PART 2 PRODUCTS

2.1 eOMSI FILES FORMAT

Format eOMSI manuals and files in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Include a complete electronically linked operation and maintenance directory. Provide four electronic copies of the eOMSI Manuals to the Contracting Officer for approval.

Provide eOMSI Facility Data Workbook on compact disks (CD) or data digital versatile disk (DVD) disks in (EXCEL) format. Scan eOMSI Manual Files and eOMSI Facility Data Workbook for viruses, malware, and spyware using a commercially available scanning program that is routinely updated to identify and remove current virus threats.

2.1.1 eOMSI Manual Organization

Organize the eOMSI Manuals into two parts: 1) Product and Drawing Information, and 2) Facility Information. Bookmark the PDF files for easy access to the information.

- a. Bookmark Product and Drawing Information documents in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- b. Bookmark Facility Information to at least one level lower than the major system.

2.1.2 eOMSI Manual CD or DVD Disk Label and Disk Holder or Case

Provide disks in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

2.2 eOMSI MANUAL

2.2.1 Product and Drawing Information

Provide an organized record of the facility products, materials, equipment, and minimum information necessary to operate the facility. Provide Product

and Drawing Information for the systems in the final constructed facility.

2.2.1.1 O&M Data

As a minimum, provide the approved O&M Data, submitted in the technical specification sections, in accordance with paragraph TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES in Section 01 78 23 OPERATION AND MAINTENANCE DATA.

2.2.1.2 Record Drawings

Provide an electronic, PDF copy of the Record Drawings, prepared in accordance with FC 1-300-09N and 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME). Bookmark drawings using the sheet title and sheet number.

Include Record Drawings as part of the Red-Zone specified in Section 01 30 00.00 ADMINISTRATIVE REQUIREMENTS.

2.2.1.3 Utility Record Drawings

Using Record Source Drawings, show and document details of the actual installation of the utility systems; annotate and highlight the eOMSI information. Provide Utility Record Drawings in PDF format. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

- a. Utility Schematic Diagrams - Provide a one line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the five-foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.
- b. Enlarged Connection and Cutoff Plans - Provide enlarged floor plans that provide information between the five foot utility connection point and where utilities connect to facility distribution. Enlarge floor plans/ elevations of the rooms where the utility enters the building and indicate on these plans locations of the main interior and exterior connection and cutoff points for the utilities. Also enlarge floor plans / elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number, of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

2.2.2 Facility Information

Provide the following in Facility Information:

2.2.2.1 General Facility and System Description

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code. List and generally describe all the facility systems and any special building features (for example, HVAC Controls, Cranes, and Generators). Include

photographs marked up and labeled to show key operating components and the overall facility appearance.

2.2.2.2 Floor Plans

Provide uncluttered, legible 11 by 17 inches floor plans. Include room numbers, type or function of spaces, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

2.2.2.3 Floor Coverings, Wall Surfaces, and Ceiling Surfaces

Provide a table that lists by room number (including hallways and common spaces), the type, and area of finish, manufacturer's product name, identifying number, and color. Include a facility summary of the total area for each type of space and floor, wall, or ceiling finish in the table.

2.2.2.4 Roofing

Provide the total area of each type of roof surface and system. Provide the name of the roofing product and system; manufacturer's, supplier's, and installer's names, addresses, and phone numbers; manufacturer's product name, identifying number, and color. For each type of roof, provide a recommended inspection, maintenance and repair schedule that details checkpoints, frequencies, and prohibited practices. List roof structural load limits.

2.2.2.5 HVAC Filters

Provide a table that lists the quantity, type, size, and location of each HVAC filter, manufacturer's product name, and identifying number.

2.2.2.6 Lighting Fixtures

Provide a table that lists by room number (including hallways and common spaces), the type of lighting fixture, ballast, number of lighting fixtures, type of lamps and number of lamps, and the manufacturer's product name and the identifying number. The table must include a facility summary of the total number of fixtures of each type and number of lamps of each type.

2.2.2.7 Equipment Listing

Provide a table that lists the major equipment shown on the design equipment schedules. Show the item descriptions, locations, model numbers; and the names, addresses, and telephone numbers of the manufacturers, suppliers, contractors, and subcontractors.

2.2.2.8 System Flow Diagrams

Provide a flow diagram indicating system liquid, air or gas flow during normal operations. Integrate the system components into the diagram. A compilation of non-integrated, flow diagrams for the individual system components are not acceptable.

2.2.2.9 Valve List

Provide a list of all valves associated with the system. Show valve type, identification number, function, location and normal operating position.

2.2.2.10 Riser Diagrams

Provide riser diagrams and settings of equipment.

2.3 eOMSI FACILITY DATA WORKBOOK

The following tabs are included in the eOMSI Facility Data File Workbook and serve the purpose stated:

- a. Instructions Tab: Instructions for completing Model & Facility Data Matrix Tab and KTR Facility Data File Tab. If a discrepancy exists between what is required in this section and the Workbook, the instructions within the workbook take precedence.
- b. Model & Facility Data Matrix Tab: - The Matrix lists Required Facility Asset Fields for each SYSTEM and SUBSYSTEM. The Designer of Record selects SYSTEMS and SUBSYSTEMS that are within the project scope, which the Contractor needs to include and populate in KTR Facility Data File tab. The "Required Facility Asset Field Position Numbers," one through seventeen, are pre-populated, and are not editable.
- c. Required Facility Asset Fields Tab: Defines the 17 Required Facility Asset Field Position Numbers used in Model and Facility Data Matrix and KTR Facility Data File tabs.
- d. KTR Sample Facility Data File Tab: Sample KTR eOMSI facility data file. This tab provides an example of the mandatory fields of equipment installed by the Contractor, and populated in the KTR eOMSI Facility Data File Tab, along with their descriptions.
- e. KTR Facility Data File Tab: Required eOMSI facility data file deliverable provided to the Government. Provide a separate and unique new row for each facility component or piece of equipment installed.

PART 3 EXECUTION

3.1 FIELD VERIFICATION

Field verify eOMSI Facility Data Workbook information with Contractor and Government personnel. Include the following personnel in this meeting: Contractor's eOMSI Manual and Facility Data Workbook Preparer and Quality Control Manager, and the Government's Contracting Officer's Representative and NAVFAC PW FMD. Request, and provide, an eOMSI Field Verification Meeting no sooner than 14 calendar days after submission of the Progress eOMSI submittal, and another, no sooner than 14 calendar days after submission of the Prefinal eOMSI submittal. During this meeting, the Government and Contractor will verify that the eOMSI Facility Data Workbook is complete and accurate.

Field verify that at least 5 Subsystems under each of the Mastersystems are accurate, for a total of 25 Subsystems. For each of these items, verify that the required facility asset field, as defined in the "Model & Facility Data Matrix" tab, contains the specified data and it is accurate (i.e. item description, manufacturer, model no., serial no.). 100 percent accuracy of eOMSI information is required for successful field verification. If data discrepancies are discovered amongst the 25 Subsystems verified, resubmit an updated eOMSI FDW, and request a make-up field verification meeting. At the make-up field verification meeting 25 new Subsystems and their

associated required facility asset fields will be field verified; the 25 new Subsystems must be 100% accurate. Any discrepancies discovered must be corrected prior to next eOMSI Facility Data Workbook Submittal.

- (1) A10 - FOUNDATIONS
- (2) B20 - EXTERIOR ENCLOSURE
- (3) D30 - HVAC
- (4) D40 - FIRE PROTECTION
- (5) D50 - ELECTRICAL

3.2 eOMSI TRAINING

Provide training on eOMSI Manuals and Facility Data Workbook in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

-- End of Section --

SECTION 02 82 00.00 22

ASBESTOS REMEDIATION
(PNS PROJECTS)
09/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM D4397 (2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications

ASTM E1368 (2014) Visual Inspection of Asbestos Abatement Projects

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7 (2014) Compressed Air for Human Respiration; 6th Edition

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1 (2015) Occupational and Educational Personal Eye and Face Protection Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2019) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM (2016; 5th Ed) NIOSH Manual of Analytical Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 340/1-90/018 (1990) Asbestos/NESHAP Regulated Asbestos Containing Materials Guidance

EPA 560/5-85-024 (1985) Guidance for Controlling
Asbestos-Containing Materials in Buildings
(Purple Book)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147 Control of Hazardous Energy (Lock Out/Tag
Out)

29 CFR 1926.103 Respiratory Protection

29 CFR 1926.1101 Asbestos

29 CFR 1926.200 Accident Prevention Signs and Tags

29 CFR 1926.51 Sanitation

29 CFR 1926.59 Hazard Communication

40 CFR 61-SUBPART A General Provisions

40 CFR 61-SUBPART M National Emission Standard for Asbestos

40 CFR 763 Asbestos

42 CFR 84 Approval of Respiratory Protective Devices

49 CFR 107 Hazardous Materials Program Procedures

49 CFR 171 General Information, Regulations, and
Definitions

49 CFR 172 Hazardous Materials Table, Special
Provisions, Hazardous Materials
Communications, Emergency Response
Information, and Training Requirements

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

NAVFAC P-502 (2017) Asbestos Program Management

ND OPNAVINST 5100.23 (2005; Rev G) Navy Occupational Safety and
Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586 (2009; Reprint Dec 2017) UL Standard for
Safety High-Efficiency Particulate, Air
Filter Units

STATE OF MAINE, DEPARTMENT OF ENVIRONMENTAL PROTECTION (DEP)

ME L&C PROGRAM State of Maine Licensing and Certification
Program
<http://www.mainelegislature.org/legis/statutes/38/title3>

06 096 CMR 425 Asbestos Management Regulations

<http://www.maine.gov/sos/cec/rules/06/096/096c425.doc>

1.2 DEFINITIONS

1.2.1 ACM

Asbestos Containing Materials.

1.2.2 Amended Water

Water containing a wetting agent or surfactant with a maximum surface tension of 0.00042 psi.

1.2.3 Area Sampling

Sampling of asbestos fiber concentrations which approximates the concentrations of asbestos in the theoretical breathing zone but is not actually collected in the breathing zone of an employee.

1.2.4 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, and actinolite asbestos and any of these minerals that has been chemically treated or altered. Materials are considered to contain asbestos if the asbestos content of the material is determined to be equal to or greater than one percent.

1.2.5 Asbestos Control Area

That area where asbestos removal operations are performed which is isolated by physical boundaries which assist in the prevention of the uncontrolled release of asbestos dust, fibers, or debris.

1.2.6 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5 micrometers as determined by National Institute for Occupational Safety and Health (NIOSH) Method 7400.

1.2.7 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average measured in the breathing zone as defined by 29 CFR 1926.1101 or other Federal legislation having legal jurisdiction for the protection of workers health.

1.2.8 Authorized Person

Any person authorized by the Contractor and required by work duties to be present in the regulated areas.

1.2.9 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

1.2.10 Competent Person (CP)

A person meeting the requirements for competent person as specified in 29 CFR 1926.1101 including a person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for project designer or supervisor, or its equivalent. The competent person must have a current State of Maine asbestos contractor's or supervisor's license.

1.2.11 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work.

1.2.12 Disposal Bag

A 6 mil thick, leak-tight plastic bag, pre-labeled in accordance with 29 CFR 1926.1101, used for transporting asbestos waste from containment to disposal site.

1.2.13 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 60 inches in length and width in order to access a building component.

1.2.14 NOT USED

1.2.15 NOT USED

1.2.16 Friable Asbestos Material

A term defined in 40 CFR 61-SUBPART M and EPA 340/1-90/018 meaning any material which contains equal to or greater than 1 percent asbestos, as determined using the method specified in 40 CFR 763, Polarized Light Microscopy (PLM), that when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

1.2.17 Glovebag Technique

Those asbestos removal and control techniques put forth in 29 CFR 1926.1101.

1.2.18 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

1.2.19 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.20 Model Accreditation Plan (MAP)

USEPA training accreditation requirements for persons who work with asbestos as specified in 40 CFR 763.

1.2.21 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.22 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.23 Nonfriable Asbestos Material

Material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating, binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.24 Permissible Exposure Limits (PELs)

1.2.24.1 PEL-Time Weighted Average (TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.24.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.25 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.26 Private Qualified Person (PQP)

That qualified person hired by the Contractor to perform the herein listed tasks.

1.2.27 Qualified Person (QP)

A Registered Architect, Professional Engineer, Certified Industrial Hygienist, consultant or other qualified person who has successfully completed training and is therefore accredited under a legitimate State

Model Accreditation Plan as described in 40 CFR 763 as a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer; and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The QP must be qualified to perform visual inspections as indicated in ASTM E1368. The QP must be appropriately licensed in the State of Maine.

1.2.28 TEM

Refers to Transmission Electron Microscopy.

1.2.29 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

1.2.30 Transite

A generic name for asbestos cement wallboard and pipe.

1.2.31 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most 0.00042 psi.

1.2.32 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed or by the state where the work is to be performed. The worker must be appropriately licensed in the State of Maine.

1.3 REQUIREMENTS

1.3.1 Description of Work

The work covered by this Section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The asbestos work includes the demolition and removal and as indicated on the Drawings, which is governed by MEDEP Chapter 425: Asbestos Management Regulations, 40 CFR 763, NAVFAC P-502, and other applicable Federal, State, and local requirements. Refer to the report located at the end of this Section. Under normal conditions non-friable or chemically bound materials containing asbestos would not be considered hazardous; however, this material may release airborne asbestos fibers during disturbance, demolition, and/or removal and therefore must be handled in accordance with the removal and disposal procedures as specified herein. Provide negative pressure enclosure and glovebag techniques as outlined in this specification. The building(s) work

area(s) will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

1.3.1.1 Wallboard/Joint Compound

Both composite samples of the wallboard and discrete samples of the components (wallboard and joint compound) have been tested and results are attached.

1.3.2 Unexpected Discovery of Asbestos

Notify the Contracting Officer if any previously untested building components suspected to contain asbestos are impacted by the work.

1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101.

1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or other pertinent State or local directives. This requirement must have been satisfied within the 12 months prior to the start of work on this Contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of indefinite time after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Assistant Secretary of Labor for Occupational Safety and Health (OSHA), or authorized representatives of them, and an employee's physician upon the request of the employee or former employee.

1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763 and 49 CFR 172; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification. Post appropriate evidence of compliance with the training requirements of 40 CFR 763. Train personnel involved in the asbestos control work in accordance with United States Environmental Protection Agency (USEPA) Asbestos Hazard Emergency Response Act (AHERA) training criteria or State training criteria whichever is more stringent.

Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures. Asbestos workers must have a current State of Maine asbestos worker's license. Comply with the ME L&C PROGRAM.

1.3.5 Permits, Licenses and Notifications

Prior to the start of work, obtain necessary permits and licenses in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by Federal, State, regional, and local authorities. Notify the Maine Department of Environmental Protection, the NAVFAC Asbestos Program Manager, and the Contracting Officer in writing 20 working days prior to commencement of work in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M and for approval by NAVFAC NPT. Notify the Contracting Officer and other appropriate Government agencies in writing 20 working days prior to the start of asbestos work as indicated in applicable laws, ordinances, criteria, rules, and regulations. Submit copies of all Notifications to the Contracting Officer. Notify the Shipyard fire department 3 days prior to removing fire-proofing material from the building including notice that the material contains asbestos.

1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of Federal, State, regional, and local authorities regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, 40 CFR 763, 49 CFR 171, and ND OPNAVINST 5100.23. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply. The following laws, ordinances, criteria, rules and regulations regarding removal, handling, storing, transporting and disposing of asbestos materials apply:

- a. U.S. EPA, Title 40, CFR Part 61, National Standards for Hazardous Air Pollutants (NESHAP), Subparts A and M.
- b. OSHA, 29 CFR, Part 1926.1101, Asbestos in Construction.
- c. State of Maine, Department of Environmental Protection (ME DEP), 06 096 CMR Asbestos Management Regulations, Chapter 425, effective date 4-3-2011.

1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health NIOSH, under the provisions of 42 CFR 84, for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained using the model Contractor accreditation plan as indicated in the Federal statutes for all portions of the herein listed work.

1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the removal and demolition of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the Contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection and if reusable coveralls are to be employed decontamination methods (operations and quality control plan), the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, removal method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust

equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in writing prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor, CP, and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the Contracting Officer prior to starting work.

The Plan must be accompanied by a completed Asbestos Hazard Abatement Plan Checklist included as Attachment A at the end of this specification and referenced in Paragraph entitled SUBMITTALS herein. The Checklist must be prepared and signed by the PQP.

1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the sampling, analysis, and reporting of airborne concentrations of asbestos fibers along with evidence that each laboratory selected holds the appropriate State license and permits and certification that each laboratory is American Industrial Hygiene Association (AIHA) accredited and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR) and successful participation of the laboratory in the Proficiency Analytical Testing (PAT) Program. Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited by the National Institute of Science and Technology (NIST) under National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.3.12 Landfill Approval

Landfill approval is not required as Portsmouth Naval Shipyard takes responsibility for disposal of all asbestos waste generated on the Shipyard.

1.3.13 Transporter Certification

Transporter certification is not required as Portsmouth Naval Shipyard takes responsibility for transport and disposal of all asbestos waste generated on the Shipyard.

1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in 29 CFR 1926.1101 and 29 CFR 1926.103 as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Amended Water; G

Safety Data Sheets (SDS) for All Materials; G

Encapsulants; G

Respirators; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Vacuums; G

Glovebags; G

SD-06 Test Reports

Air Sampling Results; G

Pressure Differential Recordings for Local Exhaust System; G

Clearance Sampling; G

Asbestos Disposal Quantity Report; G

SD-07 Certificates

Employee Training; G

Notifications; G

Respiratory Protection Program; G

Asbestos Hazard Abatement Plan; G

Asbestos Hazard Abatement Plan Checklist; G

Testing Laboratory; G

Medical Certification; G

Private Qualified Person Documentation; G

Competent Person; G

Worker's License; G

Contractor's License; G

Federal, State or Local Citations on Previous Projects; G

Equipment Used to Contain Airborne Asbestos Fibers; G

Water Filtration Equipment; G

Vacuums; G

Ventilation Systems; G

SD-11 Closeout Submittals

Permits and Licenses; G

Notifications; G

Respirator Program Records; G

Rental Equipment; G

1.5 QUALITY ASSURANCE

1.5.1 Private Qualified Person Documentation

Submit the name, address, and telephone number of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer as described by 40 CFR 763 and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The PQP must be appropriately licensed in the State of Maine as a Project Monitor. The PQP and the asbestos contractor must not have an employee/employer relationship or financial relationship which could constitute a conflict of interest. The PQP must be a first tier subcontractor.

1.5.2 Competent Person Documentation

The Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, and notification of other employees onsite. The Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current State of Maine Asbestos Contractor's and Supervisor's License. Submit evidence that the Competent Person has a minimum of 2 years of on-the-job asbestos abatement experience relevant to OSHA competent person requirements.

1.5.3 Worker's License

Submit documentation that workers meet the requirements of 29 CFR 1926.1101, 40 CFR 61-SUBPART M and have a current State of Maine Asbestos Workers

License.

1.5.4 Contractor's License

Submit a copy of the asbestos contractor's license issued by the State of Maine. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART M, EM 385-1-1, and the Federal, State and local requirements for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the Company's President or Chief Executive.

1.5.5 Air Sampling Results

Complete fiber counting and provide results to the PQP and GC for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer and Asbestos Program Manager immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and Asbestos Program Manager and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor, the Contracting Officer and Asbestos Program Manager immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.010 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.10 fibers per cubic centimeter. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.6 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the asbestos control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer and Asbestos Program Manager within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.7 Federal, State or Local Citations on Previous Projects

Submit a statement, signed by an officer of the company, containing a record of any citations issued by Federal, State or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations). If there are none, a negative declaration signed by an officer of the company must be provided.

1.5.8 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan, Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The safety preconstruction conference must include the Contractor and their Designated Competent Person, NAVFAC Asbestos Program Manager, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 SECURITY

A log book must be kept documenting entry into and out of the regulated area. Entry into regulated areas must only be by personnel authorized by the Contractor and the Contracting Officer. Personnel authorized to enter regulated areas must be trained, medically evaluated, and wear the required personal protective equipment.

1.7 EQUIPMENT

1.7.1 Rental Equipment

Provide a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

PART 2 PRODUCTS

2.1 NOT USED

2.2 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

2.3 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

- a. Disposal Bags

2.4 SHEET PLASTIC

Sheet plastic must be polyethylene of 6 mil minimum thickness and must be provided in the largest sheet size necessary to minimize seams. Film must be clear and conform to ASTM D4397, except as specified below.

2.4.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided. Film must be frosted and must conform to the requirements of NFPA 701.

2.4.2 Reinforced

Reinforced sheets must be provided where high skin strength is required,

such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film. Film must meet flame resistant standards of NFPA 701.

2.5 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 140 degrees F.

2.6 LEAK-TIGHT WRAPPING

At least two layers of 6 mil minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

2.7 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 1/8 inch thick, acrylic sheet, 18 by 24 inches, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

2.8 WETTING AGENTS

Removal encapsulant (a penetrating encapsulant) must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

PART 3 EXECUTION

3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least two complete sets of personal protective equipment as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all equipment used to contain airborne asbestos fibers.

3.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

- a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.
- b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5

liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter surface.

- c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.

3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services.

3.1.2.1 Respirators for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, encapsulation, removal and demolition of asbestos materials with respiratory protection as indicated in 29 CFR 1926.1101 and 29 CFR 1926.103. Breathing air must comply with CGA G-7.

3.1.3 Exterior Whole Body Protection

3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable," or reusable "non-breathable" whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.

3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as recommended by the PQP after each use.

3.1.3.3 Personal Decontamination Unit

Provide a temporary, negative pressure unit with a separate decontamination locker room and clean locker room with a shower that complies with 29 CFR 1926.51(f)(4)(ii) through (V) in between for personnel required to wear whole body protective clothing. Provide two separate lockers for each asbestos worker, one in each locker room. Keep street clothing and street shoes in the clean locker. HEPA vacuum and remove asbestos contaminated disposable protective clothing while still wearing respirators at the boundary of the asbestos work area and seal in impermeable bags or containers for disposal. HEPA vacuum and remove asbestos contaminated reusable protective clothing while still wearing respirators at the boundary of the asbestos work area, seal in two impermeable bags, label outer bag as asbestos contaminated waste, and transport for decontamination. Do not wear work clothing between home and work. Locate showers between the decontamination locker room and the clean locker room and require that all employees shower before changing into street clothes.

Collect used shower water and filter with approved water filtration equipment to remove asbestos contamination. Wastewater filters must be installed in series with the first stage pore size 20 microns and the second stage pore size of 5 microns. Dispose of filters and residue as asbestos waste. Discharge clean water to the sanitary system. Dispose of asbestos contaminated work clothing as asbestos contaminated waste or properly decontaminate as specified in the Contractor's Asbestos Hazard Abatement Plan. Keep the floor of the decontamination unit's clean room dry and clean at all times. Proper housekeeping and hygiene requirements must be maintained. Provide soap and towels for showering, washing and drying. Cloth towels provided must be disposed of as ACM waste or must be laundered in accordance with 29 CFR 1926.1101. Physically attach the decontamination units to the asbestos control area. Construct both a personnel decontamination unit and an equipment decontamination unit onto and integral with each asbestos control area.

3.1.3.4 Eye Protection

Provide eye protection that complies with ANSI/ISEA Z87.1 when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area and access tunnel. Attach the load-out unit in a leak-tight manner to each regulated area.

3.1.6 Warning Signs and Labels

Provide warning signs printed in English and at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable.

3.1.6.1 Warning Sign

Provide vertical format conforming to 29 CFR 1926.200, and 29 CFR 1926.1101 minimum 20 by 14 inches displaying the following legend in the lower panel:

<u>Legend</u>	<u>Notation</u>
DANGER	one inch Sans Serif Gothic or Block

<u>Legend</u>	<u>Notation</u>
ASBESTOS	one inch Sans Serif Gothic or Block
MAY CAUSE CANCER	one inch Sans Serif Gothic or Block
CAUSES DAMAGE TO LUNGS	1/4 inch Sans Serif Gothic or Block
AUTHORIZED PERSONNEL ONLY	1/4 inch Sans Serif Gothic or Block
WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA	1/4 inch Sans Serif Gothic or Block

Spacing between lines must be at least equal to the height of the upper of any two lines.

3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST AVOID CREATING DUST

3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSP Z9.2 and 29 CFR 1926.1101 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the asbestos control area. Filters on exhaust equipment must conform to ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSP Z9.2 and UL 586. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

3.1.10 Glovebags

Glovebag operations must be performed in accordance with the requirements of the State of Maine Department of Environmental Protection. Submit written manufacturers proof that glovebags will not break down under expected temperatures and conditions.

3.1.11 Decontamination Area Exit Procedures

Note: Whenever possible, decontamination area must be contiguous with, and connected to, the negative pressure enclosure.

Ensure that the following procedures are followed:

- a. Before leaving the regulated area, remove all gross contamination and debris from work clothing using a HEPA vacuum.
- b. Employees must remove their protective clothing in the equipment room and deposit the clothing in labeled impermeable bags or containers for disposal or laundering.
- c. Employees must not remove their respirators until showering.
- d. Employees must shower prior to entering the clean room. If a shower has not been located between the equipment room and the clean room or the work is performed outdoors, ensure that employees engaged in Class I asbestos jobs:
 - a) Remove asbestos contamination from their work suits in the equipment room or decontamination area using a HEPA vacuum before proceeding to a shower that is not adjacent to the work area; or
 - b) Remove their contaminated work suits in the equipment room, without cleaning worksuits, and proceed to a shower that is not adjacent to the work area.

3.2 WORK PROCEDURE

Perform asbestos related work in accordance with MEDEP Chapter 425: Asbestos Management Regulations, 29 CFR 1926.1101, 40 CFR 61-SUBPART M, 49 CFR 107, 49 CFR 172, NAVFAC P-502 and as specified herein. Use wet removal procedures and negative pressure enclosure techniques. Wear and utilize protective clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the encapsulation removal and demolition of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel. Seal all rooftop penetrations, except plumbing vents, prior to asbestos roofing work.

Shut down the building heating, ventilating, and air conditioning system, cap the openings to the system, and provide temporary heating and ventilation, prior to the commencement of asbestos work. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147. Disconnect electrical service when encapsulation wet removal is performed and provide temporary electrical service with

verifiable ground fault circuit interrupter (GFCI) protection prior to the use of any water encapsulant. All electrical work must be performed by a licensed electrician. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.010 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the Contracting Officer, including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer. Corrective actions must be documented. If an asbestos fiber release or spill occurs, stop work immediately, correct the condition to the satisfaction of the Contracting Officer including clearance sampling, prior to resumption of work.

3.2.1 Building Ventilation System and Critical Barriers

Building ventilation system supply and return air ducts in a regulated area must be shut down and isolated by lockable switch or other positive means in accordance with 29 CFR 1910.147. The airtight seals must consist of air-tight rigid covers for building ventilation supply and exhaust grills where the ventilation system is required to remain in service during abatement. Edges to wall, ceiling and floor surfaces must be sealed with industrial grade duct tape.

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.2 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection and air sampling results are obtained from the PQP work may proceed at the discretion of the Contracting Officer.

3.2.3 Furnishings

Furniture and equipment will remain in the building. Cover and seal furnishings and equipment with 6-mil plastic sheet or remove from the work area and store in a location on site approved by the Contracting Officer.

3.2.4 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

3.2.5 Asbestos Control Area Requirements

3.2.5.1 Negative Pressure Enclosure

Removal of Asbestos Contaminated Materials (ACM) require the use of a negative pressure enclosure. Block and seal openings in areas where the release of airborne asbestos fibers can be expected. Establish an asbestos negative pressure enclosure with the use of curtains, portable partitions, or other enclosures in order to prevent the escape of asbestos fibers from the contaminated asbestos work area. Negative pressure enclosure development must include protective covering of uncontaminated walls, and ceilings with a continuous membrane of two layers of minimum 6-mil plastic sheet sealed with tape to prevent water or other damage. Provide two layers of 6-mil plastic sheet over floors and extend a minimum of 12 inches up walls. Seal all joints with tape. Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

3.2.5.2 Glovebag

If the construction of a negative pressure enclosure is infeasible for the removal of ACM, use alternate techniques as indicated in MEDEP Chapter 425: Asbestos Management Regulations and 29 CFR 1926.1101. Establish designated limits for the asbestos regulated area with the use of rope or other continuous barriers, and maintain all other requirements for asbestos control areas. The PQP must conduct personal samples of each worker engaged in asbestos handling (removal, disposal, transport and other associated work) throughout the duration of the project. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers at any time exceeds background or 0.010 fibers per cubic centimeter whichever is greater, stop work, evacuate personnel in adjacent areas or provide personnel with approved protective equipment at the discretion of the Contracting Officer. This sampling may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those obtained by the Contractor, the Government will determine which results predominate. If adjacent areas are contaminated as determined by the Contracting Officer, clean the contaminated areas, monitor, and visually inspect the area as specified herein.

3.2.6 Removal Procedures

Wet asbestos material with a fine spray of amended water during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 6 mil plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 6 mil plastic bags, submit an alternate proposal for containment of asbestos fibers to the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in MEDEP Chapter

425: Asbestos Management Regulations and 40 CFR 61-SUBPART M.

3.2.6.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos lock-down encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants must comply with the performance requirements specified herein.

3.2.6.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 1/4 inch thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 4 inches. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

3.2.7 Methods of Compliance

3.2.7.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

- a. Vacuum cleaners equipped with HEPA filters.
- b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.
- c. Prompt clean-up and disposal.
- d. Inspection and repair of polyethylene.
- e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

3.2.7.2 Control Methods

Use the following control methods:

- a. Local exhaust ventilation equipped with HEPA filter;
- b. Enclosure or isolation of processes producing asbestos dust;
- c. Where the feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PELs, use them

to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

3.2.7.3 Unacceptable Practices

The following work practices must not be used:

- a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.
- b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.
- c. Dry sweeping, shoveling, or other dry clean up.
- d. Employee rotation as a means of reducing employee exposure to asbestos.

3.2.8 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the installation and operation of the control methods.
- b. Place critical barriers over all openings to the regulated area.
- c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.
- d. Impermeable dropcloths (6 mil or greater thickness) must be placed on surfaces beneath all removal activity.
- e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

3.2.9 Specific Control Methods for Class I Work

Use Class I work procedures, control methods and removal methods for the following ACM:

- a. Spray Applied Fireproofing
- b. Gypsum Wallboard and Joint Compound
- c. Thermal System Insulation and Mudded Pipe Fittings
- d. Plaster and Textured Ceilings and Walls
- e. Vermiculite

3.2.9.1 Negative Pressure Enclosure (NPE) System

The system must provide at least four air changes per hour inside the containment. The local exhaust unit equipment must be operated 24-hours per day until the containment is removed. The NPE must be smoke tested for leaks at the beginning of each shift and be sufficient to maintain a

minimum pressure differential of minus 0.02 inch of water column relative to adjacent, unsealed areas. Pressure differential must be monitored continuously, 24-hours per day, with an automatic manometric recording instrument and Records must be provided daily on the same day collected to the Contracting Officer and Asbestos Program Manager. The Contracting Officer and Asbestos Program Manager must be notified immediately if the pressure differential falls below the prescribed minimum. The building ventilation system must not be used as the local exhaust system for the regulated area. The NPE must terminate outdoors unless an alternate arrangement is allowed by the Contracting Officer and Asbestos Program Manager. All filters used must be new at the beginning of the project and must be periodically changed as necessary and disposed of as ACM waste. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.9.2 Glovebag Systems

Glovebags must be used without modification, smoke-tested for leaks, and completely cover the circumference of pipe or other structures where the work is to be done. Glovebags must be used only once and must not be moved. Glovebags must not be used on surfaces that have temperatures exceeding 150 degrees F. Prior to disposal, glovebags must be collapsed using a HEPA vacuum. Before beginning the operation, loose and friable material adjacent to the glovebag operation must be wrapped and sealed in 2 layers of plastic or otherwise rendered intact. At least two persons must perform glovebag removal. Asbestos regulated work areas must be established for glovebag abatement. Designated boundary limits for the asbestos work must be established with rope or other continuous barriers and all other requirements for asbestos control areas must be maintained, including area signage and boundary warning tape.

- a. Attach HEPA vacuum systems to the bag to prevent collapse during removal of ACM.
- b. The negative pressure glove boxes must be fitted with gloved apertures and a bagging outlet and constructed with rigid sides from metal or other material which can withstand the weight of the ACM and water used during removal. A negative pressure must be created in the system using a HEPA filtration system. The box must be smoke tested for leaks prior to each use.

3.2.9.3 Mini-Enclosure

Single bulkhead containment, double bulkhead containment, or mini-containment (small walk-in enclosure) to accommodate no more than two persons, may be used if the disturbance or removal can be completely contained by the enclosure. The mini-enclosure must be inspected for leaks and smoke tested before each use. Air movement must be directed away from the employee's breathing zone within the mini-enclosure.

3.2.9.4 Wrap and Cut Operation

Prior to cutting pipe, the asbestos-containing insulation must be wrapped with polyethylene and securely sealed with duct tape to prevent asbestos becoming airborne as a result of the cutting process. The following steps must be taken: install glovebag, strip back sections to be cut 6 inches from point of cut, and cut pipe into manageable sections.

3.2.9.5 Class I Removal Method

Class I ACM must be removed using a control method described above. Prepare work area as previously specified. Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area. Spread one layer of 6-mil seamless plastic sheeting on the floor below the work area. Using the following techniques are applicable:

- a. Remove asbestos containing spray applied fireproofing using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. After removal of the material use a wire brush to clean the exposed substrate to remove residual material. Continue wet cleaning until surfaces are free of visible debris.
- b. Cut manageable sections of gypsum wallboard and joint compound and immediately place into a 6-mil minimum thickness disposal bag or other approved container. Make every effort to keep the material from falling to the floor of the work area. Use a wire brush and wet clean to remove residual material from studs. Continue wet cleaning until the surface is clean of visible material and encapsulate stud walls.
- c. Remove ACM thermal system insulation and mudded pipe fittings using mechanical means and wet methods and immediately place into 6-mil thickness disposal bag. Continue wet cleaning until surfaces are free of visible debris.
- d. Remove ACM plaster ceilings or walls using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.
- e. Remove ACM textured ceiling finish using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. Floors are considered contaminated from fallen textured ceiling finish. Clean up debris on floor and dispose of carpet as asbestos contaminated material. After removal of the material use a wire brush to clean the exposed concrete ceiling to remove residual material. Continue wet cleaning until surfaces are free of visible debris.
- f. Remove ACM vermiculite using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.

Bag all asbestos debris which has fallen to the floor as asbestos-containing debris. Place all debris in plastic disposal bags of 6-mil minimum thickness. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the quantity of airborne asbestos fibers monitored at the breathing zone of

the workers or the designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work, and immediately correct the situation.

3.2.10 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.11 Specific Control Methods for Class II Work

3.2.11.1 Suspect Fire Doors

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. Spread 6-mil plastic sheeting on the ground beneath the work area and around the perimeter of the work area extending out in all directions. Remove door intact from hinges and wrap with 6-mil plastic sheeting. Inspect the interior areas of the door to determine if ACM is present. If ACM is not present the door may be disposed of as general construction debris. If ACM is present place whole door in enclosed container for disposal. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.2 Roofing Materials

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing roofing materials which contain ACM as described in 29 CFR 1926.1101 (g)(8)(ii), use the following practices. Roofing material must be removed in an intact state. Wet methods must be used to remove roofing materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards. When removing built-up roofs, with asbestos-containing roofing felts and an aggregate surface, using a power roof cutter, all dust resulting from the cutting operations must be collected by a HEPA dust collector, or must be HEPA vacuumed by vacuuming along the cut line. Asbestos-containing roofing material must not be dropped or thrown to the ground, but must be lowered to the ground via covered, dust-tight chute, crane, hoist or other method approved by the Contracting Officer. Any ACM that is not intact must be lowered to the ground as soon as practicable, but not later than the end of the work shift. While the material remains on the roof it must be kept wet or placed in an impermeable waste bag or wrapped in plastic sheeting. Intact ACM must be lowered to the ground as soon as practicable, but not

later than the end of the work shift. Unwrapped material must be transferred to a closed receptacle. Critical barriers must be placed over roof level heating and ventilation air intakes. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.3 Cementitious Siding and Shingles or Transite Panels

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing cementitious asbestos-containing siding, shingles or Transite panels use the following work practices. Intentionally cutting, abrading or breaking is prohibited. Each panel or shingle must be sprayed with amended water prior to removal. Nails must be cut with flat, sharp instruments. Unwrapped or unbagged panels or shingles must be immediately lowered to the ground via covered dust-tight chute, crane or hoist, or placed in an impervious waste bag or wrapped in plastic sheeting and lowered to the ground no later than the end of the work shift. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.4 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken within a glovebag. Any scraping to remove residue must be performed wet. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.12 NOT USED

3.2.13 Abatement of Asbestos Contaminated Soil

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Asbestos contaminated soil must be removed from areas to a minimum depth of 2 inches. Soil must be thoroughly dampened with amended water and then removed by manual shoveling into labeled containers. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.14 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101 must be performed by the PQP. Sampling performed for environmental and quality control reasons must be performed by the PQP. Unless otherwise specified, use NIOSH Method 7400 for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer and Asbestos Program Manager. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and made available to the Contracting Officer and Asbestos Program Manager. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.14.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each demolition or removal[encapsulation] site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

3.2.14.2 Sampling During Asbestos Work

The PQP must provide personal sampling as indicated in 29 CFR 1926.1101. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and 8 hours (environmental/clearance monitoring) after completion of a sampling period. At the same time, the GC will provide area sampling close to the work inside the enclosure, outside the clean

room entrance to the enclosure, at the waste load-out, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.010 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer and Asbestos Program Manager immediately. Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels. The written results must be signed by testing laboratory analyst, testing laboratory principal and the Contractor's PQP. The air sampling results must be documented on a Contractor's daily air monitoring log. Copies of the air monitoring results must also be submitted directly to the Asbestos Program Manager. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.14.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using NIOSH NMAM Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.010 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.010 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using NIOSH NMAM Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.010 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.14.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in 40 CFR 763, abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.14.5 Sampling After Final Clean-Up (Clearance Sampling)

Provide area sampling of asbestos fibers using aggressive air sampling techniques as defined in the EPA 560/5-85-024 and establish an airborne asbestos concentration of less than 0.010 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos work control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the PQP and GC must perform a visual inspection in accordance with ASTM E1368 to ensure that the asbestos control and work area is free of any accumulations of dirt, dust, or debris. Prepare a written report signed and dated by the PQP documenting that the asbestos control area is free of dust, dirt, and debris and all

waste has been removed. Collect the appropriate number of samples in accordance with Maine DEP regulations. Use transmission electron microscopy (TEM) to analyze clearance samples and report the results in accordance with current NIOSH criteria. The asbestos fiber counts from these samples must be less than 0.010 fibers per cubic centimeter or be not greater than the background, whichever is greater. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling and TEM analysis at the Contractor's expense.

3.2.14.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

3.2.15 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the PQP must conduct a visual inspection of all areas affected by the removal in accordance with ASTM E1368. Inspect for any visible fibers[and to ensure that encapsulants were applied evenly and appropriately]. Spray apply a post removal (lock-down) encapsulant to ceiling, walls, floors and other areas exposed in the removal area. The exposed area includes but is not limited to plastic barriers, furnishings and articles to be discarded as well as dirty change room, air locks for bag removal and decontamination chambers.

3.2.16 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3 CLEAN-UP AND DISPOSAL

3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The PQP and CP will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos

airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in applicable local, State, and Federal regulations and herein.

The Portsmouth Naval Shipyard Hazardous Waste Facility takes title and is responsible for all asbestos waste disposal.

3.3.3 Disposal of Asbestos

3.3.3.1 Procedure for Disposal

Coordinate all waste disposal with the Contracting Officer and the Portsmouth Naval Shipyard Hazardous Waste Facility (Code 106, Building 357). Collect asbestos waste, contaminated waste water filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 6 mils thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M. All waste materials, including, but not limited to, piping, must be less than 5 feet in length. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 6 mils thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete.

The Contractor must carefully remove material and immediately place it in approved asbestos disposal bags/containers. If the bags/containers are not pre-marked, a Contractor furnished caution label must be permanently attached to the bags/containers. The Contractor must also furnish and affix a label permanently attached to the bags which provides the following information:

- * Abatement Contractor Name
- * Construction Contract Number
- * Building Number
- * Portsmouth Naval Shipyard, Kittery, Maine
- * Date

Contractor must ensure all asbestos materials are properly packaged prior to placement into government-furnished dumpsters.

3.3.3.2 Asbestos Disposal Quantity Report

Direct the PQP to record and report, to the Contracting Officer and Asbestos Program Manager, the amount of asbestos containing material removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day reported in linear feet or square feet as described initially in this specification and in cubic feet for the amount of asbestos containing material released for disposal.

Allow the GC to inspect, record and report the amount of asbestos containing material removed and released for disposal on a daily basis.

-- End of Section --

Specification Section 1.3.10 - Asbestos Hazard Abatement Plan Checklist - Building XX

Prepared by: _____ Date: _____

Spec Section(s)	Item Required	Included?	Section/ Page #	Notes
	Confirm that the Asbestos Hazard Abatement Plan contains the following:			
1.3.10	- Signature and seal to document that Plan was prepared by a PQP			
1.3.10	- Site sketch with all project details (control areas, critical barriers, physical boundaries, decon facilities, viewing ports, and mechanical ventilation system (negative air machines, etc.))			
1.3.10, 1.5.8	- Confirmation that meeting will be scheduled w/ ktr, supervisor, CP, PQP, CO, and APM to review Plan			
3.2.2, 3.2.3, 3.2.4	- Description of preparatory activities - protecting existing conditions, moving furniture, pre-cleaning			
1.3.10, 3.2	- Description of safety precautions including, but not limited to, LO/TO, CSE, fall protection			
1.3.10	- Description of work procedures for each activity			
1.3.10, 3.2.6	- Description of asbestos removal method(s) for each activity			
3.1.10, 3.2.5.2	- Description of glovebag equipment and process (as appropriate)			
1.3.10	- Description of controls for each activity			
1.3.11	- Description of job responsibilities for each activity			
1.3.10	- Description of respirators and personal protective equipment			
1.3.10, 1.4	- Description of equipment for each activity. Include all equipment spec sheets			
1.3.9, 1.3.10, 1.4	- Description of materials for each activity. Include all SDS			
1.3.10	- Description of hygiene facilities and sanitary procedures (including no eating, drinking, smoking)			
1.3.10	- Description of interface of trades			
1.3.10	- Description of work sequencing			
3.1.4	- Description of the method and construction of containment for each activity			
3.1.6	- Description of required signage for each activity			
1.3.10, 3.1.3.3, 3.1.11	- Description of decontamination equipment and process			
1.3.10, 3.1.7, 3.2.1	- Description of local exhaust system and critical barrier equipment and process, including:			
	- Confirmation that Pressure Differential Recordings will be submitted w/in 24 hours			
	- Confirmation that negative pressure will be maintained continuously until work area cleared			
3.3	- Description of cleanup procedures during and at the end of work.			
3.2.15	- Description of method, material, etc. for application of lock-down encapsulant.			
3.2.14	- Description of personal sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all personal sampling results will be submitted to NAVFAC w/in 24 hours			
3.2.14	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.2.14	- Detailed description of area sampling and analysis strategy and methodology, frequency of sampling, sample locations, duration of sampling, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all area sampling results will be submitted to NAVFAC w/in 16 hours			
3.2.14	- Listing of specific sampling locations. Include locations in Notes column (to the right)			
3.2.14	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.2.14	- Detailed description of clearance sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all clearance sampling results will be submitted to NAVFAC w/in 8 hours			
3.2.14	- Listing of specific sampling locations. Include locations in Notes column (to the right)			
1.3.10, 1.4	- Detailed plan for Asbestos Waste Management, including:			
1.3.10	- Asbestos Materials			
1.3.10	- Ancillary waste, including poly sheeting			
1.3.10	- Wastewater - washwater, decon water, etc. treated and/or disposal			
3.1.5	- Detailed description of waste load-out set-up and procedure			
1.3.8	- Name, qualifications, and certification of Asbestos Hazard Control Supervisor			
1.3.10	- Copies of Fire and Medical Emergency Response Plans and AHA			
1.3.4, 1.4	- Copies of Training Certificates for Workers and Supervisors			
1.3.4, 1.4, 1.5.3	- Copies of Licenses for Workers and Supervisors			
1.3.4, 1.3.7	- Copies of Respiratory Protection Program, Including Training and Fit Testing Records			
1.3.3, 1.3.14	- Copies of Medical Examinations, Surveillance and Records			
1.3.5	- Copies of Abatement Licenses, Permits and Notifications			
1.3.11, 1.4	- Documentation of Testing Laboratory Qualifications/Certificates, including documentation that the lab is:			
1.3.11	- Maine licensed/permitted/certified			
1.3.11	- AIHA accredited			
1.3.11	- NIST/NVLAP accredited			
1.3.11	- Independent of contractor			
1.4, 1.5.1	- Documentation of Private Qualified Person Qualifications			
1.4, 1.5.2	- Documentation of Competent Person Qualifications			
1.4, 1.5.4	- Copy of Contractor's License			
1.4, 1.5.7	- Details on Federal, State, Local Citations w/in 5 Years (if any)			
1.4	- Documentation of Third Party Consultant Qualifications			
1.4, 3.1.9	- Copy of Statement Regarding Use of Rental Equipment			
CHECKLIST COMPLETED BY: _____ DATE: _____				
NOTE: This checklist is not intended to list all of the Asbestos Hazard Abatement Plan components required per Specification Section 02 82 00.00 22, Asbestos Remediation. The Contractor is required to identify and meet all applicable Specification requirements.				

SECTION 02 83 00.00 22

MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION,
REMOVAL, AND ABATEMENT
(PNS PROJECTS)
10/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613 (2012) Standard Test Method for Determination of Lead by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES) (IH DB-DBB SoW only identifies AAS), Flame Atomic Absorption Spectrometry (FAAS), or Graphite Furnace Atomic Absorption Spectrometry (GFAAS) Techniques

ASTM E1644 (2017) Standard Practice for Hot Plate Digestion of Dust Wipe Samples for the Determination of Lead

ASTM E1726 (2001; R 2009) Preparation of Soil Samples by Hotplate Digestion for Subsequent Lead Analysis

ASTM E1727 (2016) Standard Practice for Field Collection of Soil Samples for Subsequent Lead Determination

ASTM E1728 (2016) Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination

ASTM E1792 (2003; R 2016) Standard Specification for Wipe Sampling Materials for Lead in Surface Dust

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2019) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780 (1995; Errata Aug 1996; Rev Ch. 7 - 1997)
Guidelines for the Evaluation and Control
of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium (VI)
29 CFR 1926.1127	Cadmium
29 CFR 1926.21	Safety Training and Education
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

49 CFR 178

Specifications for Packagings

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23

(2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586

(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

PORTSMOUTH NAVAL SHIPYARD (PNSY)

5090.6F

(4-21) Solid Waste Operations

5090.30

(4-05) Hazardous Waste Generator Standards

1.2 DEFINITIONS

1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.3 Area Sampling

Sampling of lead, cadmium, and chromium (VI) concentrations within the lead, cadmium, and chromium (VI) control area and inside the physical boundaries which is representative of the airborne lead, cadmium, and chromium (VI) concentrations but is not collected in the breathing zone of personnel (approximately 5 to 6 feet above the floor).

1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.5 Certified Industrial Hygienist (CIH)

As used in this Section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current Federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of Industrial

Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.2.6 Child-Occupied Facility

Real property which is a building or portion of a building constructed prior to 1978 visited regularly by the same child, six-years of age or under, on at least two different days within any week (Sunday through Saturday period), provided that each day's visit lasts at least 3-hours, and the combined annual visits last at least 60-hours. Child-occupied facilities include but are not limited to, day-care centers, preschools and kindergarten classrooms.

1.2.7 Chromium (VI) Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.8 Competent Person (CP)

As used in this Section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current Federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium (VI) hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer in the State of Maine.

1.2.9 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.10 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.11 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium (VI) or to plan such activities in commercial buildings, bridges or other structures.

1.2.12 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, and chromium (VI) to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.2.13 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, and chromium (VI)-contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.14 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds. The use of the term Lead in this Section also refers to paints which contain detectable concentrations of Cadmium and Chromium (VI). For the purposes of the Section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium (VI).

1.2.15 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.16 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

1.2.17 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

1.2.18 Lead, Cadmium, and Chromium (VI) Control Area

A system of control methods to prevent the spread of lead, cadmium, and chromium (VI) dust, paint chips or debris to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.19 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

1.2.20 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this Section do not apply if no detectable levels of lead are found using a

quantitative method for analyzing paint or MCL using laboratory instruments with specified limits of detection (usually 0.01 percent). An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.21 Personal Sampling

Sampling of airborne lead, cadmium, and chromium (VI) concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and centered at the nose or mouth of an employee.

1.2.22 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, and chromium (VI) control area to limit unauthorized entry of personnel.

1.2.23 Target Housing

Residential real property which is housing constructed prior to 1978, except housing for the elderly or persons with disabilities (unless any one or more children age 6-years or under resides or is expected to reside in such housing for the elderly or persons with disabilities) or any zero bedroom dwelling.

1.3 DESCRIPTION

This specification covers all project-related activities, including, but not limited to, renovation, demolition, removal, and/or remediation, impacting PWL and/or material containing lead, cadmium, and chromium (VI) located as indicated on the drawings. The work covered by this Section includes work tasks and the precautions specified in this Section for the protection of building occupants and the environment during and after the performance of the hazard abatement activities in accordance with State and Federal regulations and as specified herein.

1.3.1 Protection of Existing Areas To Remain

Project work including, but not limited to, lead, cadmium, and chromium (VI) hazard abatement work, storage, transportation, and disposal must be performed without damaging or contaminating adjacent work and areas. Where such work or areas are damaged or contaminated, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, and Chromium (VI) Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium (VI) exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium (VI) activities.

1.3.3 Sampling and Analysis

Submit a log of the analytical results from sampling conducted during the abatement. Keep the log of results current with project activities and

brief the results to the Contracting Officer as analytical results are reported. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728 and ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

1.3.3.2 Soil Sampling and Analysis

Sampling must conform to ASTM E1727. Analysis must conform to ASTM E1613 and ASTM E1726.

1.3.3.3 Clearance Monitoring

- a. Collect dust wipe samples inside, and, as appropriate, outside, the lead, cadmium and chromium (VI) hazard control area after the final visual inspection.
- b. Collect exterior bare soil samples inside the lead, cadmium and chromium (VI) hazard control area after the final visual inspection.

1.3.4 Clearance Requirements

See Project Scope of Work for required Clearance Requirements.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Competent Person Qualifications; G

Training Certification; G

Occupational and Environmental Assessment Data Report; G

Medical Examinations; G

Lead, Cadmium, and Chromium (VI) Waste Management Plan; G

Licenses, Permits and Notifications; G

Occupant Protection Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist; G

Waste Characterization Sampling Plan; G

Sample Results; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Pressure Differential Log; G

SD-06 Test Reports

Sampling and Analysis; G

Occupational and Environmental Assessment Data Report; G

Sampling Results; G

Pressure Differential Recordings For Local Exhaust System; G

SD-07 Certificates

Testing Laboratory; G

Third Party Consultant Qualifications; G

[Occupant Notification; G

] Notification of the Commencement of Work Impacting Lead, Cadmium,
or Chromium (VI); G

Clearance Certification; G

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in paragraph COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, and chromium (VI); experience with the use of respirators, personal protective equipment and other exposure

reduction methods to protect employee health. Demonstrate a minimum of 5 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and licensed and certified in accordance with Federal, State and local laws. The competent person must possess the qualifications detailed in Paragraph entitled COMPETENT PERSON herein.

1.5.1.2 Training Certification

Submit a current certificate for each worker and supervisor, signed and dated by the accredited training provider, stating that the employee has received the required lead, cadmium and chromium (VI) training specified in 29 CFR 1926.62, 29 CFR 1926.1126, [and]29 CFR 1926.1127 [and 40 CFR 745] and is certified to perform or supervise deleading, lead removal or demolition activities in the State of Maine.

1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air[soil][and wipe] analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium (VI). Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved. Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification, and certifications for all corresponding EPA test methods are required for waste determination testing, as applicable.

1.5.1.4 Third Party Consultant Qualifications

Submit the name, address and telephone number of the third party consultant selected to perform the wipe sampling for determining concentrations of lead, cadmium and chromium (VI) in dust. Submit proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the USEPA authorized State (or local) certification and accreditation program.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all Federal, State, and local requirements.
- b. Review and approve Lead, Cadmium, and Chromium (VI) Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect LBP/PWL or MCL work for conformance with the approved plan.
- d. Perform (or oversee performance of) air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.

- e. Ensure work is performed in strict accordance with specifications at all times.
- f. Control work to prevent hazardous exposure to human beings and to the environment at all times.
- g. Supervise final cleaning of the lead, cadmium, and chromium (VI) control area, take clearance wipe samples if necessary; review clearance sample results and make recommendations for further cleaning.
- h. Certify the conditions of the work as called for elsewhere in this specification.
- i. The CP must be certified pursuant to 40 CFR 745, Section 226 and is responsible for development and implementation of the occupant protection plan, the abatement report and supervision of lead, cadmium and chromium (VI) hazard abatement work activities.

1.5.2.2 Lead, Cadmium, and Chromium (VI) Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium (VI), LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, and chromium (VI) control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity from which lead, cadmium, and chromium (VI) is emitted. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, and chromium (VI) related work, collected waste water and dust containing lead, cadmium, and chromium (VI) and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, and chromium (VI) is not released outside of the lead, cadmium, and chromium (VI) control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the air sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksites to inform affected employees and to clarify responsibilities to control exposures. If work will occur on a historic building, then include in the plan a description of how the proposed activities will comply with other relevant specification Sections to avoid or minimize impacts to historic fabric (for example, but not limited to, specification Sections related to historic masonry).

The plan must be developed and signed by a certified Lead Project Designer in the State of Maine. The plan must include the name and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to notify and protect the building occupants.

1.5.2.3 Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist

The Contractor must complete the Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist included as Attachment A at the end of this

Section. The Checklist must be signed by the properly-certified person preparing the Plan and must be submitted to the Government for review and approval with the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.5.2.4 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer and PD&C HAZMAT POC within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, and Chromium (VI) Compliance Plan.

- a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, and chromium (VI) for stated work. The initial monitoring must represent each task and/or type of activity. The initial monitoring must represent each/every PWL and/or material containing lead, cadmium, and chromium (VI).
- b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 with a complete process description. This includes, but is not limited to, manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium (VI) containing coatings are present.
- c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, and chromium (VI) compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.2.5 Medical Examinations

Submit a signed letter from an officer of the company confirming each worker engaged in heavy metals related activities in the contract has current medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Where exposure is determined to be above a lead, cadmium, or chromium action level the contractor must provide pre-work and post-work blood levels as requested by the government. All medical records submitted shall not contain any Protected Health Information (PHI) (e.g's home address, social security number, physicians name, medical record number, etc.).

Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, and chromium (VI) at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium (VI) in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, and chromium (VI) as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.6 Training

Train each employee performing work that disturbs lead, cadmium, and chromium (VI), who performs LBP/MCL/PWL disposal, and air sampling operations prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, [40 CFR 745,] 49 CFR 172, and State of Maine and local regulations where appropriate.

1.5.2.7 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and 29 CFR 1926.55.
- c. Provide training and fit test certificates for each employee performing LBP/MCL/PWL work.

1.5.2.8 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.9 Lead, Cadmium, and Chromium (VI) Waste Management

The Lead, Cadmium, and Chromium (VI) Waste Management Plan must comply with applicable requirements of federal, State, and local hazardous waste regulations and address:

- a. Identification and classification of wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.
- d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- e. Spill prevention, containment, and cleanup contingency measures

including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.

- f. Work plan and schedule for waste containment and management. Proper containment of the waste includes using acceptable waste containers, as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.
- g. Include any process that may alter or treat waste. Note: Altering or treatment of a hazardous waste is not allowed to include any process that dilutes the waste or renders a hazardous waste non-hazardous.
- h. All HW must be placed in a Shipyard-permitted HWAA or Contractors must turn HW in to a HWAA not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this Contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

1.5.2.10 Waste Characterization Sampling Plan

A sampling plan must be developed that details practices to be followed for the collection, management, and analysis of waste characterization samples to determine if waste materials generated by the referenced project will be classified as hazardous or non-hazardous to ensure proper disposal. The plan must propose sampling that meets the requirements of all applicable local, State, and Federal statutes and regulations, Navy and PNSY requirements, and project specifications. At a minimum, the sampling plan must include:

- a. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.
- b. The timing of waste characterization sampling and analysis. To the extent feasible, waste sampling and analysis must be completed prior to the accomplishment of any project work to properly characterize waste and ensure hazardous waste is not commingled with non-hazardous waste.
- c. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.
- d. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.
- e. The proposed date(s) for sampling and expected duration of the sampling event(s). All waste sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.

- f. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
- g. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers. At a minimum, the waste characterization must include laboratory analysis for RCRA 8 Metals via the Toxicity Characteristic Leaching Procedure (TCLP). Depending on the methods used in project implementation, additional waste characterization analysis may be required (for example, waste characterization analysis for corrosivity may be required if high pH solutions are used for paint removal).
- h. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods.
- i. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
 - 1. Laboratory analytical results.
 - 2. Field log book entries.
 - 3. Copy of the Chain of Custody (COC).
 - 4. Pertinent diagrams and maps.
 - 5. All other supporting data.
- j. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
- k. Confirmation that the sampling procedure will include, but not be limited to, the following:
 - 1. The Contractor will notify the Contracting Officer's Representative and PNSY Code 106.3 Sampling Program Manager seven (7) calendar days prior to each sampling event to ensure the area and materials to be sampled are witnessed by all parties.
 - 2. The EFT taking the samples will inspect the work area/waste with Code 106.3 to visually and tactilely assess the materials to be sampled.
 - 3. Using a field log, the EFT will record the locations, types, quantities and other salient details related to materials to be sampled.

4. The EFT will collect sufficient sample size of the materials from the project work area to complete all required analytical testing. Samples must be representative of all materials that comprise the waste for disposal.
5. The EFT will place the sample in a clean jar or vial, secure the jar or vial closed and label it with a unique sample number.
6. The EFT will record the sample number on the COC and note the location of the sample in the field log and on the site drawing.
7. The EFT will verify that the sample number(s) on the jar/vial are accurate and consistent with the COC.
8. The EFT will place the secured and labeled jar/vial into a cooler and arrange for prompt delivery of bulk samples to the analytical laboratory.
9. The EFT will clean up any debris generated by the sampling process.
10. The EFT will decontaminate the collection tool using wet wipes, water and paper towels after each sample is collected, or other approved method. Razor blades will be replaced on a periodic basis as determined by the EFT.
11. All samples will be collected in a manner to eliminate cross-contamination by the previous sample.
12. All materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
13. The EFT will obtain the following sampling equipment to perform the bulk sampling.
 - (a) Individual sample jars and vials;
 - (b) Field log;
 - (c) Site diagram;
 - (d) Chain-of-Custody (COC) forms;
 - (e) Nitrile gloves;
 - (f) Stainless steel or plastic soil scoops;
 - (g) Metal chisels of assorted sizes;
 - (h) Razor knives and replacement blades;
 - (i) Water and spray bottle;
 - (j) Trash bags;
 - (k) Disposable cleaning wipes;
 - (l) Paper towels;
 - (m) Duct tape;
 - (n) Flashlight;
 - (o) Personnel lifts; and
 - (p) Other equipment as necessary.

1.5.2.11 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of Federal, State, and local authorities regarding lead, cadmium, and chromium (VI). Comply with the

applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, and ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply. The following local laws, ordinances, criteria, rules and regulations regarding removing, handling, storing, transporting, and disposing of lead, cadmium, and chromium (VI)-contaminated materials apply:

- a. Shipyard Instruction 5090.6F Solid Waste Operations (included as Attachment B at the end of this Section)
- b. Shipyard Instruction 5090.30 Hazardous Waste Generator Standards

Licensing and certification in the State of Maine is required.

1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, and chromium (VI) control area is removed. Submit pressure differential recordings for each work day to the Private Qualified Person (PQP) and General Contractor (GC) for review and to the Contracting Officer and PD&C HAZMAT POC within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Maine Department of Environmental Protection and the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits, and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the Contract price.

1.5.5 Occupant Protection Plan

The certified project designer must develop and implement an Occupant Protection Plan describing the measures and management procedures to be taken during lead, cadmium and chromium (VI) hazard abatement activities to protect the building occupants/building facilities and the outside environment from exposure to any lead, cadmium and chromium (VI) contamination while lead, cadmium and chromium (VI) hazard abatement activities are performed.

1.5.6 Pre-Construction Conference

Along with the CP, meet with the Contracting Officer to discuss in detail the Lead, Cadmium, and Chromium (VI) Waste Management Plan and the Lead, Cadmium, and Chromium (VI) Compliance Plan, including procedures and precautions for the work.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for

Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium (VI) dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, and chromium (VI) contaminated dust must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during PWL or MCL handling and disposal, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

UL 586 labeled HEPA filters.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of the lead, cadmium and chromium (VI) removal work within the lead, cadmium and chromium (VI) controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machine for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ANSI/ASSP Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of minus 0.02 inch of water column relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a

clean first stage filter, an intermediate filter and a primary HEPA filter in place.

- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 2.5 inches of water static pressure differential on a manometric gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.02 inch of water and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.
- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.
- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.
- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 1,100 degrees F. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 6 mils thick (4 mils thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, and chromium (VI) contaminated debris must be polyethylene plastic that is a minimum of 6 mils thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 6 mil, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out of the ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 2 inches wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT and OSHA standards. For containerization of hazardous waste, the Contractor must use containers provided by Building 357.

2.1.1.6 Chemical Paint Strippers

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate

materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium (VI) control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

- a. Notify the Contracting Officer 20 days prior to the start of any lead, cadmium and chromium (VI) work.

- b. Occupant Notification

Submit occupant written acknowledgment of the delivery of lead hazard information pamphlet (EPA 747-K-99-001 "Protect Your Family From Lead in Your Home") prior to commencing the renovation work for each affected unit using language provided in 40 CFR 745 Subpart E.

- c. Notification of the Commencement of Work Impacting Lead, Cadmium, or Chromium (VI)

3.1.1.2 Lead, Cadmium, and Chromium (VI) Control Area

- a. Physical Boundary - Provide physical boundaries around the lead, cadmium, and chromium (VI) control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium (VI) will not escape outside of the lead, cadmium and chromium (VI) control area. Prohibit the general public from accessing the lead, cadmium, and chromium (VI) control areas.
- b. Warning Signs - Provide warning signs at approaches to lead, cadmium, and chromium (VI) control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Furnishings

Furniture and equipment will remain in the lead,

cadmium, and chromium (VI) control area. Protect and cover furnishings and equipment or remove furnishings and equipment from the work area and store in a location approved by the Contracting Officer.

3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, and chromium (VI) control areas. Seal intake and exhaust vents in the lead, cadmium, and chromium (VI) control area with 6 mil plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, and chromium (VI) control area. Provide temporary HVAC system for areas in which HVAC has been shut down outside the lead, cadmium, and chromium (VI) control area.

3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ANSI/ASSP Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the lead, cadmium, and chromium (VI) control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of minus 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ANSI/ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.6 Negative Air Pressure System Containment

- a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium (VI) contaminated waste.
- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during work activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Dismantling of the negative air pressure systems must be as presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental

contamination.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.1.1.7 Decontamination Shower Facility

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.9 Mechanical Ventilation System

- a. Use adequate ventilation to control personnel exposure to lead, cadmium and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. To the extent feasible, use local exhaust ventilation or other collection systems, approved by the CP. Evaluate and maintain local exhaust ventilation systems in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.
- b. Vent local exhaust outside the building and away from building ventilation intakes or ensure system is connected to HEPA filters.
- c. Use locally exhausted, power actuated tools or manual hand tools.

3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, and chromium (VI) control area. No one will be permitted in the lead, cadmium, and chromium (VI) control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, and Chromium (VI) Control Area Requirements

Establish a lead, cadmium, and chromium (VI) control area by completely establishing barriers and physical boundaries around the area or structure where PWL or MCL removal operations will be performed that will not create airborne, dust containing lead, cadmium, chromium (VI) (such as carefully unfastening sheets containing lead, cadmium, chromium (VI) from walls).

Full containment - Contain removal operations by the use of critical barriers and HEPA filtered exhaust, or a negative pressure enclosure system with decontamination facilities and with HEPA filtered exhaust where removal practice will create airborne, dust containing lead, cadmium, chromium (VI) (such as sanding, sawing, grinding, thermal cutting, or digging or demolition activities) and as required by the CP]. For containment areas larger than 1,000 square feet install a minimum of two 18 inch square viewing ports. Locate ports to provide a view of the required

work from the exterior of the enclosed contaminated area. Glaze ports with laminated safety glass.

3.3 APPLICATION

3.3.1 Lead, Cadmium, and Chromium (VI) Work

Perform lead, cadmium, and chromium (VI) work in accordance with approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Use procedures and equipment required to limit occupational exposure and environmental contamination with lead, cadmium, and chromium (VI) when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 [or 40 CFR 745] and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with Federal, State, and local requirements.

3.3.2 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) Removal

Manual or power sanding or grinding of lead, cadmium, and chromium (VI) surfaces or materials is not permitted unless tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead, cadmium, and chromium (VI) is prohibited. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Provide methodology for removing lead, cadmium, and chromium (VI) in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Select lead, cadmium, and chromium (VI) removal processes to prevent contamination of work areas outside the control area with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, and chromium (VI). Describe this removal process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

Provide methodology for lead, cadmium and chromium (VI), LBP/PWL removal, abatement/control and processes to prevent contamination of work areas outside the control area with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris/waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, and chromium (VI). Describe this lead, cadmium and chromium (VI), LBP/PWL removal/control process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

3.3.2.1 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Indoor Removal

Perform manual or mechanical removal and thermal cutting in the lead, cadmium, and chromium (VI) control areas using enclosures, barriers or containments and powered locally exhausted tools equipped with HEPA filters. Collect residue and debris for disposal in accordance with Federal, State, and local requirements.

3.3.2.2 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Outdoor Removal

Perform outdoor removal as indicated in Federal, State, and local regulations and in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

The worksite preparation (barriers or containments) must be job dependent and presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, and chromium (VI) controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing before entering the contaminated change room.
- b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.
- c. Shower.
- d. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air and Wipe Sampling

Conduct sampling for lead, cadmium, and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and as specified herein. Air and wipe sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing the air and wipe sampling and inspecting the project work to ensure that the requirements of the Contract, local, State and Federal statutes and regulations, Navy and PNSY requirements, and project specifications have been satisfied during all work impacting lead, cadmium, and/or chromium (VI).
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.
- c. Conduct area air sampling daily, on each shift in which lead, cadmium and chromium (VI) and lead-based paint removal operations are performed, in areas immediately adjacent to the lead, cadmium and chromium (VI) control area at exhaust points for negative air machines, at entrances to the negative pressure enclosure (NPE), and at other locations determined appropriate for the project and site conditions. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air is reached or exceeded, stop work, correct the condition(s) causing the increased levels. Notify the Contracting Officer and PD&C HAZMAT POC immediately. Determine if condition(s) require any further change in work methods. Resume removal work only after the CP and the Contracting Officer give approval.
- d. Submit sample results of air samples, signed by the CP, to the Contracting Officer within 72-hours after the air samples are taken.

- e. Before any work begins, a third party consultant must collect and analyze baseline wipe, air, and soil (if bare soil is present during exterior remediation operations) samples in accordance with methods defined by Federal, State, and local standards inside and outside of the physical boundary to assess the degree of dust contamination in the facility prior to lead, cadmium and chromium (VI) disturbance or removal. Provide Initial Sample Results to the Contracting Officer before work begins.
- f. Surface Wipe Samples - Collect surface wipe samples on floors at a location no greater than 10 feet outside the lead, cadmium, and chromium (VI) control area at a frequency of once per day while lead, cadmium, and chromium (VI) removal work is conducted in occupied buildings. Surface wipe samples or Micro Vacuum surface sample results must meet criteria in paragraph CLEARANCE CERTIFICATION.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.4.1.2 Sampling After Removal

After the visual inspection, conduct soil sampling if bare soil is present during external removal operations and collect wipe and soil samples according to the HUD protocol contained in HUD 6780 to determine the lead, cadmium and chromium (VI) content of settled dust in micrograms per square meter foot of surface area and parts per million (ppm) for soil.

3.4.1.3 Testing of Material Containing Lead, Cadmium, and Chromium (VI) Residue

Test residue in accordance with 40 CFR 261 for hazardous waste.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, and chromium (VI) control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead, cadmium, and chromium (VI) operation has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area as indicated by the Lead, Cadmium, and Chromium (VI) Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become contaminated at any time during the work, clean, visually inspect, and then wipe sample all contaminated areas. The CP must then certify in writing to the Contracting Officer that the area has been cleaned of lead, cadmium and chromium (VI) contamination before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing to the Contracting Officer and PD&C HAZMAT POC that air samples collected outside the lead, cadmium, and chromium (VI) control area during paint removal operations are less than 30 micrograms of

lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; and that there were no visible accumulations of material and dust containing lead, cadmium, and chromium (VI) left in the work site. Do not remove the lead, cadmium, and chromium (VI) control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt of the CP certification. Clearance sampling results must be submitted to NAVFAC within 72 hours after the samples are taken.

The third party consultant must perform surface wipe sampling for clearance certification pursuant to Paragraph entitled SAMPLING AND ANALYSIS herein. The third party consultant must certify surface wipe sample results collected inside and outside the work area are less than the thresholds detailed in Paragraph entitled CLEARANCE REQUIREMENTS herein.

For exterior work, soil samples taken at the exterior of the work site must be used to determine if soil lead, cadmium, and chromium (VI) levels have increased at a statistically significant level (significant at the 95 percent confidence limit) from the soil lead, cadmium, and chromium (VI) levels prior to the operation. If soil lead, cadmium, and chromium (VI) levels either show a statistically significant increase above soil lead, cadmium, and chromium (VI) levels prior to work or soil lead, cadmium, and chromium (VI) levels above any applicable federal or state standard for lead, cadmium, and chromium (VI) in soil, the soil must be remediated.

For lead, cadmium and chromium (VI)-based paint hazard abatement work, surface wipe and soil sampling must be conducted and clearance determinations made according to the requirements detailed in Paragraph entitled SAMPLING AND ANALYSIS herein and to the work practice standards presented in 40 CFR 745.227.

3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with local, State, and Federal statutes and regulations, Navy and PNSY requirements, and project specifications. Ensure all waste is properly characterized in accordance with the Waste Characterization Sampling Plan developed under Paragraph entitled WASTE CHARACTERIZATION SAMPLING PLAN herein. The result of each waste characterization will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, and chromium (VI) contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, and chromium (VI)-contaminated clothing that may produce airborne concentrations of lead, cadmium, and chromium (VI) particles. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.
- c. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with 40 CFR 262 and corresponding state regulations.
- d. Manage lead, cadmium, and chromium (VI) or lead, cadmium, and chromium

(VI) contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265.

- e. All lead, cadmium, and chromium (VI) waste generation, management, and disposal will be coordinated with the Shipyard environmental function (Code 106, Building 357) and the Contracting Officer.

All HW must be turned in to Building 357 not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this Contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

Containers positioned within the work area boundaries must have covers in place whenever containers are not in use.

3.5.2.1 Disposal Documentation

For non-hazardous waste, provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Non-Hazardous Waste

Payment for disposal of non-hazardous waste will not be made until a signed copy of the manifest from the disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of non-hazardous waste delivered to the disposal facility.

-- End of Section --

Specification Section 1.5.2.3 - Pb, Cr, Cd Compliance Plan Checklist - Building XX

Prepared by: _____ Date: _____

Spec Section(s)	Item Required	Included Yes?	N/A?	Notes / Reason N/A
	Confirm that the Pb/Cr/Cd Compliance Plan contains the following:			
1.5.2.2	- Signature and seal to document plan prepared by Maine Lead Project Designer			
1.5.2.2	- Site sketch with all project details (control areas, critical barriers, physical boundaries, decon facilities, viewing ports, and mechanical ventilation system)			
	- Confirmation that meeting will be scheduled w/ ktr, supervisor, CP, PQP, and CO to review Plan NAVFAC PD&C HAZMAT POC, NAVFAC CM/ET.			
3.1	- Description of preparatory activities - protecting existing conditions, moving furniture, pre-cleaning			
1.5.2.2	- Description of safety precautions including, but not limited to, LO/TO, CSE, fall protection			
1.5.2.2	- Description of work practices for each activity, including:			
	- If abrasive removal is proposed, document approval by NAVFAC Cultural Resources MGR			
1.5.2.2	- Description of controls for each activity			
1.5.2.2	- Description of job responsibilities for each activity			
1.5.2.2	- Description of respirators and personal protective equipment			
1.5.2.2	- Description of equipment for each activity. Include all equipment spec sheets			
1.5.2.2	- Description of materials for each activity. Include all Safety Data Sheets (SDS)			
1.5.2.2	- Description of hygiene facilities and sanitary procedures (including no eating, drinking, smoking)			
1.5.2.2	- Description of interface of trades			
1.5.2.2	- Description of work sequencing			
1.5.2.2	- Description of the method of containment of the operation to ensure that metals are not released outside of the control area. Negative Pressure Enclosure (NPE) or Regulated Area.			
3.1.1.2	- Description of required signage for each activity			
3.3.3, 3.1.1.7	- Description of decontamination equipment and process			
	- Rationale for not including shower (if applicable)			
1.5.3	- Description of local exhaust system and critical barrier equipment and process, including:			
	- Confirmation that Pressure Differential Recordings will be submitted within 24 hours			
	- Confirmation that negative pressure will be maintained continuously until work area cleared			
1.5.2.2, 3.5, 3.1.1.6	- Description of cleanup procedures during and at the end of work to include a description of the negative air pressure systems dismantling.			
3.4.1	- Description of personal sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
	- Confirmation that all personal sampling results will be submitted to NAVFAC w/in 72 hours			
	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.4.1, 3.4.1.1.e	- Detailed description of area sampling and analysis strategy and methodology, frequency of sampling, sample locations, duration of sampling, and qualifications of sampling personnel, including:			
	- Confirmation that all area sampling results will be submitted to NAVFAC w/in 72 hours			
	- Listing of specific sampling locations.			
	- Specific thresholds for work stoppage, PPE upgrade, etc.			
	- Confirmation a third party consultant will collect and analyze bare soil during exterior remediation operations to assess the degree of dust contamination prior to the accomplishment of work.			
3.4.1	- Detailed description of clearance sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
	- Confirmation that all clearance sampling results will be submitted to NAVFAC w/in 72 hours			
	- Listing of specific sampling locations.			
	- Listing of clearance thresholds.			
1.4, 1.5.2.8, 3.5.2	- Detailed plan for Pb, Cd, and Cr Waste Management, including:			
	- Painted Materials			
	- Paint chips, dust, debris			
	- Ancillary waste, including poly sheeting			
	- Wastewater (e.g., decontamination shower water)			
	- Note: Statement in plan all hazardous waste will be turned in to 8357 by the end of the shift generated.			
1.4, 1.5.1.2	- Copies of Training Certificates for Workers and Supervisors			
1.4	- Copies of Licenses for Workers and Supervisors			
1.4, 1.5.2.7	- Provide training and fit test certificates for each employee performing LBP/MCL/PWL work.			
1.4, 1.5.2.4	- Letter from officer of the company confirming medical surveillance and training is current for each employee performing LBP/MC/PWL work			
1.4, 1.5, 1.5.3	- Copies licenses, permits and notifications for demolition, abatement, remediation, etc.			
1.4, 1.5.1.3	- Documentation of Testing Laboratory Qualifications/Certificates			
1.4, 1.5	- Copy of Occupant Protection Plan			
1.4, 1.5.1.5	- Documentation of Certified Risk Assessor Qualifications			
1.4	- Sample Copy of Clearance Certification			
3.1.1.1	- Copy of Notification to Contracting Officer			
3.1.1.1	- Copy of Notification to Occupants			
1.4	- Name, qualifications, and certification of Supervisor			
1.4, 1.5.1.1	- Documentation of Competent Person Qualifications			
1.4, 1.5.1.1	- Certified Industrial Hygienist?			
1.4, 1.5.1.1	- Certified Safety Professional?			
1.4, 1.5.1.1	- Licensed Lead-Based Abatement Supervisor/Project Designer in the State of Maine?			
1.3.3	- Description of approach for maintaining and submitting log of analytical results			
1.4, 1.5	- Copy of Occupational and Environmental Assessment Data Report			
1.4, 1.5.1.4	- Documentation of Third Party Consultant Qualifications			
CHECKLIST COMPLETED BY: _____		DATE: _____		
NOTE: This checklist is not intended to list all of the Pb, Cr, Cd Compliance Plan components required per Specification Section 02 83 00.00 22, Pb, Cr and Cd Remediation. The Contractor is required to identify and meet all applicable Specification requirements.				

SECTION 02 83 00.00 22

MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION,
DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT
(PNS PROJECTS)
02/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP/SAFE Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613 (2012) Standard Test Method for Determination of Lead by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES), Flame Atomic Absorption Spectrometry (FAAS), or Graphite Furnace Atomic Absorption Spectrometry (GFAAS) Techniques

ASTM E1644 (2017) Standard Practice for Hot Plate Digestion of Dust Wipe Samples for the Determination of Lead

ASTM E1726 (2001; R 2009) Preparation of Soil Samples by Hotplate Digestion for Subsequent Lead Analysis

ASTM E1727 (2016) Standard Practice for Field Collection of Soil Samples for Subsequent Lead Determination

ASTM E1728 (2016) Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination

ASTM E1792 (2003; R 2016) Standard Specification for Wipe Sampling Materials for Lead in Surface Dust

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2019) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(Latest Revision) Safety and Health
Requirements Manual

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780

(1995; Errata Aug 1996; Rev Ch. 7 - 1997)
Guidelines for the Evaluation and Control
of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium (VI)
29 CFR 1926.1127	Cadmium
29 CFR 1926.21	Safety Training and Education
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures

49 CFR 172 Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

49 CFR 178 Specifications for Packaging

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23 (2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586 (2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

STATE OF MAINE, DEPARTMENT OF ENVIRONMENTAL PROTECTION (DEP)

ME L&C PROGRAM State of Maine Licensing and Certification Program
<http://www.mainelegislature.org/legis/statutes/38/tit13>

STATE OF MAINE STATUTES AND REGULATIONS

Laws
Title 38, Chapter 12-B; Lead Abatement
Emergency Provision §1296
Title 38, Chapter 12-B: Lead Abatement
§1291- §1297
Title 22, Chapter 252: Lead Poisoning
Control Act §1314-§1327
Title 14, Chapter 710:Environmental Lead
Hazards Disclosure §6030-B
Rules
Chapter 424- Lead Management Regulations

PORTSMOUTH NAVAL SHIPYARD (PNSY)

5090.6F (4-21) Solid Waste Operations

5090.30 (4-05) Hazardous Waste Generator Standards

5090.8 Environmental Sampling Manual

1.2 DEFINITIONS

1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.3 Area Sampling

Sampling of lead, cadmium, and chromium (VI) concentrations within the lead, cadmium, and chromium (VI) control area and inside the physical boundaries which is representative of the airborne lead, cadmium, and chromium (VI) concentrations but is not collected in the breathing zone of personnel (approximately 5 to 6 feet above the floor).

1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.5 Category A LCC Work

Work operations involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is less than the AL.

1.2.6 Category B LCC Work

Work operation involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is equal to or greater than the AL but less than the PEL.

1.2.7 Category C LCC Work

Work operation involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is likely to exceed the PEL.

1.2.8 Certified Industrial Hygienist (CIH)

As used in this section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of Industrial Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.2.9 Child-Occupied Facility

Real property which is a building or portion of a building constructed prior to 1978 visited regularly by the same child, six-years of age or under, on at least two different days within any week (Sunday through

Saturday period), provided that each day's visit lasts at least 3 hours, and the combined annual visits last at least 60 hours. Child-occupied facilities include but are not limited to, day-care centers, preschools and kindergarten classrooms.

1.2.10 Chromium (VI) Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.11 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium (VI) hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer in the State of Maine.

1.2.12 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.13 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.14 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium (VI) or to plan such activities in commercial buildings, bridges or other structures.

1.2.15 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, and chromium (VI) to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.2.16 Engineering and Work Practice Controls

Controls that are structural or functional. Ventilation is an example of an engineering control. No eating or smoking in a LCC work area is a work practice control.

1.2.17 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, and chromium (VI)-contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.18 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium (VI). For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium (VI).

1.2.19 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.20 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

1.2.21 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

1.2.22 Lead, Cadmium, and Chromium (VI) Control Area

A system of control methods to prevent the spread of lead, cadmium, and chromium (VI) dust, paint chips debris, and/or any other associated materials to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.23 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

1.2.24 Material Containing Lead, Cadmium and Chromium (VI)/Paint with Lead, Cadmium, and Chromium (VI), (MCLCC/PWLCC)

Any material, including paint, which contains lead, cadmium or chromium (VI) as determined by the testing laboratory using a valid test method.

The requirements of this section do not apply if no detectable concentrations of lead, cadmium, or chromium (VI) are found using an appropriate quantitative method for analyzing PCL or MCL using laboratory instruments with specified limits of detection. An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.25 Personal Sampling

Sampling of airborne lead, cadmium, and chromium (VI) concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and centered at the nose or mouth of an employee.

1.2.26 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, and chromium (VI) control area to limit unauthorized entry of personnel.

1.2.27 Routes of Exposure

The way by which workers can come into contact, or be exposed, to a hazardous substance. Routes of exposure include inhalation (breathing in contaminated air, dust, vapors, or aerosols), ingestion (eating, drinking, chewing), injection (entering the body through a needle or sharp object puncturing the skin), absorption through the skin or eyes through direct contact.

1.2.28 Target Housing

Residential real property which is housing constructed prior to 1978, except housing for the elderly or persons with disabilities (unless any one or more children age 6 years or under resides or is expected to reside in such housing for the elderly or persons with disabilities) or any zero bedroom dwelling.

1.3 DESCRIPTION

This specification covers all project-related activities, including, but not limited to, renovation, construction, demolition, removal, and/or abatement, impacting PWL and/or material containing lead, cadmium, and chromium (VI). The work covered by this section includes work tasks and the precautions specified in this section for the protection of workers, building occupants, the building interior and exterior, furniture and equipment, and the environment during and after the performance of all project-related activities in accordance with state and federal regulations and as specified herein.

1.3.1 Protection of Existing Areas To Remain

Any and all work performed under these specifications must be performed without damaging or impacting adjacent work and areas. Where such work or areas are damaged or impacted, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, and Chromium (VI) Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium (VI) exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium (VI) activities.

1.3.3 Sampling and Analysis

Submit all analytical results from sampling conducted during any and all work performed under these specifications, within the time-frames mandated by these specifications. Keep the log of results current with project activities and brief the results to the Contracting Officer as analytical results are reported.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject , or any component of any request for additional time or compensation.

1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728 and ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

1.3.3.2 Soil Sampling and Analysis

Sampling must conform to ASTM E1727. Analysis must conform to ASTM E1613 and ASTM E1726.

1.3.3.3 Clearance Monitoring

Clearance sampling via air, wipe, and soil sampling must be performed in accordance with the requirements of Section entitled FIELD QUALITY CONTROL.

1.3.4 Clearance Requirements

Clearance requirements for air, wipe, and soil sampling are included in Section entitled in Section entitled FIELD QUALITY CONTROL.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Statement of Competent Person Assignment, and Competent Person Qualifications; G

Statement of CIH Assignment, and CIH Qualifications; G

Statement Identifying and Designating Roles and Responsibilities for CP, CIH, SSHO, and all Other Parties Involved in Heavy Metals Management; G

Training Certification; G

Occupational and Environmental Assessment Data Report; G

Medical Examinations; G

Lead, Cadmium, and Chromium (VI) Waste Management Plan; G

Licenses, Permits and Notifications; G

Occupant Protection Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist; G

Waste Characterization Sampling Plan; G

Analytical Results; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Pressure Differential Log; G

SD-06 Test Reports

Sampling and Analysis; G

Occupational and Environmental Assessment Data Report; G

Pressure Differential Recordings For Local Exhaust System; G

SD-07 Certificates

Testing Laboratory; G

Occupant Notification; G

Notification of the Commencement of Work Impacting Lead, Cadmium, or Chromium (VI); G

Clearance Certification; G

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in Section entitled COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, and chromium (VI); experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 5 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and licensed and certified in accordance with federal, State and local laws. The competent person must possess the qualifications detailed in Section entitled COMPETENT PERSON herein.

1.5.1.2 Training Certification

Submit a current certificate for each worker and supervisor, signed and dated by the accredited training provider, stating that the employee has received the required lead, cadmium and chromium (VI) training specified in 29 CFR 1926.62, 29 CFR 1926.1126, [and]29 CFR 1926.1127 [and 40 CFR 745] and is certified to perform or supervise any and all work performed under these specifications.

1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air[soil][and wipe] analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium (VI). Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved. Maine Department of Environmental

Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification, and certifications for all corresponding EPA test methods are required for waste determination testing, as applicable.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all federal, State, and local requirements.
- b. Review and approve Lead, Cadmium, and Chromium (VI) Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect LBP/PWL or MCL work for conformance with the applicable specifications and approved plan. Direct Contractor to immediately correct any non-conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.
- d. Perform air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.
- e. Ensure work is performed in strict accordance with specifications at all times. Direct Contractor to immediately correct any non-conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.
- f. Control work to prevent hazardous exposure to human beings, to adjacent areas, and to the environment at all times.
- g. Supervise final cleaning of the lead, cadmium, and chromium (VI) control area, take clearance wipe samples; review clearance sample results with respect to established clearance thresholds, determine requirements for additional cleaning, and advise the Contractor and Contracting Officer of the results and need for cleaning within the time-frames established in these specifications. Direct the Contractor to Stop Work in subject area pending achievement of clearance thresholds.
- h. Certify the conditions of the work as called for elsewhere in this specification.
- i. The CP must be certified pursuant to 40 CFR 745, Section 226 and is responsible for development and implementation of the occupant protection plan, all summary and status reports, and supervision of any and all work performed under these specifications.

1.5.2.2 Lead, Cadmium, and Chromium (VI) Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium (VI), LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, and chromium (VI) control areas, critical barriers, physical boundaries, location and details of decontamination facilities,

viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity of any and all work performed under these specifications. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, and chromium (VI) related work, collected waste water and dust containing lead, cadmium, and chromium (VI) and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, and chromium (VI) is not released outside of the lead, cadmium, and chromium (VI) control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksite's to inform affected employees and to clarify responsibilities to control exposures. If work will occur on a historic building, then include in the plan a description of how the proposed activities will comply with other relevant specifications to avoid or minimize impacts to historic fabric (for example, but not limited to, specification Sections related to historic masonry).

The plan must be developed and signed by a CIH, CSP, or, as appropriate, a certified Lead Project Designer in the State of Maine. The plan must include the name and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to notify and protect the building occupants.

1.5.2.3 Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist

The Contractor must complete the Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist included as Attachment A at the end of this specification. The Checklist must be signed by the properly-certified person preparing the Plan and must be submitted to the Government for review and approval with the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.5.2.4 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer and PD&C HAZMAT POC within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, and Chromium (VI) Compliance Plan.

- a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to

estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, and chromium (VI) for stated work. The initial monitoring must represent each task and/or type of activity. The initial monitoring must represent each/every PWL and/or material containing lead, cadmium, and chromium (VI).

- b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 with a complete process description. This includes, but is not limited to, manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium (VI) containing coatings are present.
- c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, and chromium (VI) compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

1.5.2.5 Medical Examinations

Submit a signed letter from an officer of the company confirming each worker engaged in heavy metals related activities in the contract has current medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Where exposure is determined to be above a lead, cadmium, or chromium (VI) action level the contractor must provide pre-work and post-work blood levels as requested by the Government. All medical records submitted shall not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, and chromium (VI) at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium (VI) in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, and chromium (VI) as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.6 Training

Train each employee performing any and all work performed under these specifications prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, [40 CFR 745,] 49 CFR 172, and State of Maine and local regulations where appropriate.

1.5.2.7 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and 29 CFR 1926.55.\
- c. Provide training and fit test certificates for each employee performing LBP/ MCL/PWL work. Records submitted shall not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

1.5.2.8 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.9 Lead, Cadmium, and Chromium (VI) Waste Management

The Lead, Cadmium, and Chromium (VI) Waste Management Plan must comply with applicable requirements of federal, State, and local hazardous and non-hazardous waste regulations and address:

- a. Identification and classification of wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with waste materials.
- d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- e. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.
- f. Work plan and schedule for waste characterization, containment and management. Proper containment of the waste includes using acceptable waste containers, as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.
- g. Include any process that may alter or treat waste. Note: altering or treatment of a hazardous waste is not allowed, including any process that dilutes the waste or renders a hazardous waste non-hazardous waste.

- h. All hazardous waste must be placed in a Shipyard-permitted Hazardous Waste Accumulation Area or Contractors must turn hazardous waste in to a Hazardous Waste Accumulation Area not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor shall not bring hazardous wastes onto Government property.

1.5.2.10 Waste Characterization Sampling Plan

A sampling plan must be developed that details practices to be followed for the collection, management, and analysis of waste characterization samples to determine if waste materials generated by the referenced project will be classified as hazardous or non-hazardous to ensure proper disposal. The plan must propose sampling that meets the requirements of all applicable local, state, and federal statutes and regulations, Navy and PNSY requirements, and project specifications. At a minimum, the sampling plan must include:

- a. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.
- b. The timing of waste characterization sampling and analysis. To the extent feasible, waste sampling and analysis must be completed prior to the accomplishment of any project work to properly characterize waste and ensure hazardous waste is not commingled with non-hazardous waste.
- c. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.
- d. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.
- e. The proposed date(s) for sampling and expected duration of the sampling event(s). All waste sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.
- f. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.

- g. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers. At a minimum, the waste characterization must include laboratory analysis for RCRA 8 Metals via the Toxicity Characteristic Leaching Procedure (TCLP). Depending on the methods used in project implementation, additional waste characterization analysis may be required (for example, waste characterization analysis for corrosivity may be required if high pH solutions are used for paint removal).
- h. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods.
- i. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
 - 1. Laboratory analytical results.
 - 2. Field log book entries.
 - 3. Copy of the Chain of Custody (COC).
 - 4. Pertinent diagrams and maps.
 - 5. All other supporting data.
- j. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
- k. Confirmation that the sampling procedure will include, but not be limited to, the following:
 - 1. The Contractor will notify the Contracting Officer's Representative and PNSY Code 106.3 Sampling Program Manager seven (7) calendar days prior to each sampling event to ensure the area and materials to be sampled are witnessed by all parties.
- l. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
 - 2. The EFT taking the samples will inspect the work area/waste with Code 106.3 to visually and tactilely assess the materials to be sampled.
 - 3. Using a field log, the EFT will record the locations, types, quantities and other salient details related to materials to be sampled.

4. The EFT will collect sufficient sample size of the materials from the project work area to complete all required analytical testing. Samples must be representative of all materials that comprise the waste for disposal.
5. The EFT will place the sample in a clean jar or vial, secure the jar or vial closed and label it with a unique sample number.
6. The EFT will record the sample number on the COC and note the location of the sample in the field log and on the site drawing.
7. The EFT will verify that the sample number(s) on the jar/vial are accurate and consistent with the COC.
8. The EFT will place the secured and labeled jar/vial into a cooler and arrange for prompt delivery of bulk samples to the analytical laboratory.
9. The EFT will clean up any debris generated by the sampling process.
10. The EFT will decontaminate the collection tool using wet wipes, water and paper towels after each sample is collected, or other approved method. Razor blades will be replaced on a periodic basis as determined by the EFT.
11. All samples will be collected in a manner to eliminate cross-contamination by the previous sample.
12. All materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
13. The EFT will obtain the following sampling equipment to perform the bulk sampling.
 - (a) Individual sample jars and vials;
 - (b) Field log;
 - (c) Site diagram;
 - (d) Chain-of-Custody (COC) forms;
 - (e) Nitrile gloves;
 - (f) Stainless steel or plastic soil scoops;
 - (g) Metal chisels of assorted sizes;
 - (h) Razor knives and replacement blades;
 - (i) Water and spray bottle;
 - (j) Trash bags;
 - (k) Disposable cleaning wipes;
 - (l) Paper towels;
 - (m) Duct tape;
 - (n) Flashlight;
 - (o) Personnel lifts; and
 - (p) Other equipment as necessary.

1.5.2.11 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of federal, State, and local

authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, and ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply.

1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, and chromium (VI) control area is removed. Submit pressure differential recordings for each work day to the Contracting Officer and PD&C HAZMAT POC within 24-hours from the end of each work day.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Maine Department of Environmental Protection and the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits, and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the Contract price.

1.5.5 Occupant Protection Plan

The CP must develop and implement an Occupant Protection Plan describing the measures and management procedures to be taken during any and all work performed under these specifications to protect the building occupants/building facilities and the outside environment from exposure to any lead, cadmium and chromium (VI) impacts while any and all work is performed under these specifications.

1.5.6 Pre-Construction Conference

Conduct a safety preconstruction conference to discuss the details of the Lead, Cadmium, and Chromium (VI) Waste Management Plan including procedures and precautions for the work, roles and responsibilities, monitoring and sampling requirements, and Stop Work conditions. Also, to discuss the Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The safety preconstruction conference must include the Contractor and their Designated Competent Person, CP, CIH, Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium (VI) dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, and chromium (VI) must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during any and all work performed under these specifications, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

Utilize UL-Labeled HEPA filters for any and all work performed under these specifications.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of any and all performed under these specifications work within the lead, cadmium and chromium (VI) controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machines for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the 'Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must

meet the requirements of ASSP/SAFE Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of negative 0.02 inch of water column relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.
- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 2.5 inches of water static pressure differential on a magnehelic gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.02 inch of water and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.
- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.
- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.
- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 1,100 degrees F. Heat blower must be

(grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that contamination may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection including all necessary wipe and air sampling results are submitted to the Contracting Officer directly from the CP, work may proceed at the discretion of the Contracting Officer. All cleanup and testing must be completed in accordance with the requirements outlined herein this specification. All costs for the required clearance sampling and clean up efforts are the responsibility of the Contractor. Any delays associated with the clean up efforts are the responsibility of the Contractor. Achievement of restoration must be documented via air, wipe and/or soil sampling, as required under the Section entitled FIELD QUALITY CONTROL.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 6 mils thick (4 mils thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, and chromium (VI) contaminated debris must be polyethylene plastic that is a minimum of 6 mils thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 6 mil,

or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out-of-the-ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 2 inches wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT and OSHA standards. For containerization of hazardous waste, the Contractor must use containers provided by B357.

2.1.1.6 Chemical Paint Strippers

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium (VI) control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

- a. Notify the Contracting Officer 20 calendar days prior to the start of any and all work performed under these specifications.
- b. Occupant Notification

Provide building occupants with copy of Occupant Protection Plan prepared under Section entitled OCCUPANT PROTECTION PLAN. Where appropriate, submit occupant written acknowledgment of the delivery of lead hazard information pamphlet (EPA 747-K-99-001 "Protect Your Family From Lead in Your Home") prior to commencing the renovation work for each affected unit using language provided in 40 CFR 745 Subpart E.

3.1.1.2 Lead, Cadmium, and Chromium (VI) Control Area

- a. Physical Boundary - Provide physical boundaries around the lead, cadmium, and chromium (VI) control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium (VI) will not escape outside of the lead, cadmium and chromium (VI) control area. Prohibit all personnel, except those specifically authorized to enter, from accessing the lead, cadmium, and chromium (VI) control areas.
- b. Warning Signs - Provide warning signs at approaches to lead, cadmium, and chromium (VI) control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Furniture and Equipment:

Furniture and equipment will remain in the lead, cadmium, and chromium (VI) control area. Protect and cover furnishings and equipment or remove furnishings and equipment from the work area and store in a location approved by the Contracting Officer.

3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, and chromium (VI) control areas. Seal intake and exhaust vents in the lead, cadmium, and chromium (VI) control area with 6 mil plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, and chromium (VI) control area. Provide temporary HVAC system for areas in which HVAC has been shut down outside the lead, cadmium, and chromium (VI) control area.

3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of

water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.6 Negative Air Pressure System Containment

- a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until all wipe testing clearance thresholds have been achieved and the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor, under the oversight of the CP, is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Stop work and notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium (VI) contaminated waste.
- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during work activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Protocols for dismantling of the negative air pressure systems must be as presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination. Failure to submit the required data within the time frame detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

3.1.1.7 Decontamination Shower Facility

If the potential for exceedance of the PEL for lead, cadmium and/or chromium (VI) exists during work performed under these specifications, the decontamination facility must include a shower.

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.9 Mechanical Ventilation System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, and chromium (VI) control area. No one will be permitted in the lead, cadmium, and chromium (VI) control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, and Chromium (VI) Control Area Requirements

Establish a lead, cadmium, and chromium (VI) control area by completely establishing barriers and physical boundaries as appropriate around the area or structure where PWLCC or MCLCC activities will be performed. .

Full containment - Contain operations by the use of critical barriers and HEPA filtered exhaust, or a negative pressure enclosure system with decontamination facilities and with HEPA filtered exhaust where work practices will create airborne dust containing lead, cadmium, chromium (VI) (such as sanding, sawing, grinding, thermal cutting, or digging or demolition activities) as required by the CP. For containment areas larger than 1,000 square feet install a minimum of two 18-inch square viewing ports. Locate ports to provide a view of the required work from the exterior of the enclosed contaminated area. Glaze ports with laminated safety glass.

3.3 APPLICATION

3.3.1 Lead, Cadmium, and Chromium (VI) Work

Perform any and all work under these specifications in accordance with approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Use procedures

and equipment required to limit occupational exposure, impacts to building interiors/exterior, and environmental contamination with lead, cadmium, and chromium (VI) when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 [or 40 CFR 745] and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with these specifications, Navy and PNSY requirements, and federal, State, and local requirements.

3.3.2 Removal of Materials with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI)

Manual or power sanding or grinding of lead, cadmium, and chromium (VI) surfaces or materials is not permitted unless using wet methods and/or tools equipped with HEPA attachments. The dry sanding or grinding of surfaces that contain lead, cadmium, and chromium (VI) is prohibited. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Provide methodology for removing lead, cadmium, and chromium (VI) in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Select lead, cadmium, and chromium (VI) removal processes to prevent contamination of work areas outside the control area and/or areas that are required to be protected (i.e. HVAC systems, furniture, equipment, etc.), with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris or waste and to ensure that unprotected personnel are not exposed to concentrations of lead, cadmium, and chromium (VI) exceeding the applicable Action Levels. Describe this removal process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

While performing this work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3.2.1 LCC Work Categories

The purpose of this section is to provide work control practices for workers to minimize occupational exposure to LCC while working with Material with Lead, Cadmium and Chromium (VI) and to ensure that these practices are consistent with OSHA standards (29CFR§1926.62, §1926.1126, and §1926.1127). This applies to LCC operations conducted at Portsmouth Naval Shipyard (PNS) and other Areas of Responsibility (AOR) sites.

These categories are being established to define the general level of controls, monitoring, training, and other requirements associated with different work activities commonly performed on projects at PNSY and other AOR sites. These categorizations are intended to be general guidelines only and do not absolve the contractor from responsibility for performing an assessment of the materials, conditions, and activities and determining the appropriate worker safety requirements.

3.3.2.1.1 Category A LCC Work

This category is reserved for LCC operations expected to result in exposure below the Action Limit. Category A operations will require an initial industrial hygiene assessment and whenever the operation changes. Category A work typically requires awareness level training but does not require monitoring, use of engineering controls, or respiratory protection. The work shall be documented in an Activity Hazard Analysis form and a wash station shall be provided for workers to wash hands and face after completion of work. Dry sweeping and use of compressed air for cleanup is prohibited. After leaving the work area, workers must wash their hands and faces prior to eating, drinking, smoking or applying cosmetics. Eating, drinking, chewing, smoking, etc. is not permitted within the lead work area.

Typical operations that could be considered Category A tasks include work with materials where no heavy metals are detected, limited drilling or sawcutting activities of concrete coated with LCC paint where controls are already being employed to comply with the OSHA silica standard, whole component removal not subject to sawing, grinding, drilling or other aggressive methods, and application of chemical stripper in accordance with manufacturer's recommendations.

3.3.2.1.2 Category B LCC Operations

This category is reserved for LCC operations expected to result in exposure at or above the Action Limit but below the Permissible Exposure Limit. Category B operations will require an initial industrial hygiene assessment and whenever the operation changes. Category B work typically requires adequately trained personnel, periodic IH monitoring (initial and weekly thereafter), isolation/demarcation of regulated areas, may require use of wet methods and HEPA equipped vacuums, personal protective equipment, and suitable decontamination facilities. Dry sweeping and use of compressed air for cleanup is prohibited. After leaving the work area, workers must wash their hands and faces prior to eating, drinking, smoking or applying cosmetics. Eating, drinking, chewing, smoking, etc. is not permitted within the lead work area. This work shall be documented in a LCC compliance work plan in accordance with the appropriate section of this specification.

Typical operations that could be considered Category B tasks include drilling or sawcutting surfaces where LCC paint or dust is present without the use of wetting or HEPA equipped vacuum shrouds, removal of chemical strippers with LCC paint, surface cleaning where LCC materials are present, and exterior paint scraping operations.

3.3.2.1.3 Category C LCC Operations

This category is reserved for LCC operations expected to result in exposure above the permissible exposure limit (PEL). Category C operations will require an initial industrial hygiene assessment and whenever the operation changes. Category C work requires adequately trained personnel, daily IH monitoring, isolation/demarcation of regulated areas, negative pressure enclosures, use of wet methods and HEPA equipped vacuums, personal protective equipment, and 3-stage decontamination facilities with shower. Dry sweeping and use of compressed air for cleanup is prohibited. This work shall be documented in a LCC compliance work plan in accordance with the appropriate section of this specification.

Typical operations that could be considered Category C tasks include

interior scraping of LCC paint, dry hand scraping or sanding of LCC paint, abrasive blasting, grinding, burning or other hot work on LCC paint and other LCC materials.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, and chromium (VI) controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing before entering the contaminated change room.
- b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.
- c. Shower.
- d. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air, Wipe and Soil Sampling Before and During Work

Conduct sampling for lead, cadmium, and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and as specified herein. Air, wipe and soil sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing and/or performing the air, soil and wipe sampling and inspecting the project work to ensure that the requirements of the contract, local, state and federal statutes and regulations, Navy and PNSY requirements, and project specifications have been satisfied during all work performed under these specifications.
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, for each task or activity with a minimum of two rounds of testing.
- c. Conduct area air sampling as depicted above in areas adjacent to the lead, cadmium and chromium (VI) control area, at exhaust points for negative air machines, at entrances to the negative pressure enclosure (NPE), and at other locations determined appropriate for the project and site conditions. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above the applicable Action Levels. If the Action Level is reached or exceeded, stop work, correct the condition(s) causing the increased levels if known, and notify the Contracting Officer immediately. Determine if condition(s) require any further change in work methods. Resume LCC work only after the CP and the Contracting Officer give approval.
- d. Submit the analytical results for the personal and area air monitoring samples, signed by the CP, to the Contracting Officer within 72-hours

after the air samples are collected.

- e. Before any work begins, the CP must collect and analyze baseline wipe, air, and soil (if bare soil is present during exterior remediation operations) samples in accordance with methods defined by federal, State, and local standards inside and outside of the physical boundary to establish baseline conditions in the facility prior to performance of any and all work under these specifications. Provide Initial Sample Results to the Contracting Officer before work begins.
- f. Surface Wipe Samples- If at any time lead, cadmium and/or chromium (VI) concentrations in area air monitoring samples exceed Action Levels, stop work and collect surface wipe samples exterior to the work area. Samples must be collected at a sufficient frequency to properly evaluate area conditions. Surface wipe sample results must meet criteria in Section entitled CLEARANCE CERTIFICATION. If sample results exceed clearance criteria, then the area must be cleaned until the criteria is met. Work may recommence when the criteria are met with approval from the Contracting Officer.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

3.4.1.2 Wipe and Soil Sampling After Work

The clearance thresholds for required wipe sampling are as follows:

- (1)Lead- 40 ug/sq.ft. on floors; 250 ug/sq.ft. on interior window sills; 400 ug/sq. ft. on window troughs; or below concentrations detected in wipe samples collected before commencement of work- whichever is lower.
- (2)Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.
- (3)Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

For any and all work performed under these specifications, surface wipe sampling must be conducted and clearance determinations made according to the requirements detailed in Section entitled SAMPLING AND ANALYSIS herein and to the work practice standards presented in 40 CFR 745.227.

3.4.1.3 Testing of Material Containing Lead, Cadmium, and Chromium (VI) Residue

Test residue in accordance with 40 CFR 261 for hazardous waste.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, and chromium (VI) control area free of accumulations of dust and debris. Restrict the spread of dust and debris to other areas within the facility; keep waste from being distributed over & beyond the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the work performed under these specifications has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area in accordance with these specifications and the approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become impacted at any time during the work, clean, visually inspect, and then wipe sample all impacted areas. The CP must then certify in writing to the Contracting Officer that the area has been cleaned of lead, cadmium and chromium (VI) impacts before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing to the Contracting Officer and the PD&C HAZMAT POC that air samples collected outside the lead, cadmium, and chromium(VI) control area, including the Facility's HVAC Ductwork and Filters, during performance of any and all work under these specifications are less than the applicable Action Levels of 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; that there were no visible accumulations of material and dust containing lead, cadmium, and chromium (VI) left in the work site, and all applicable wipe and soil clearance sample analytical results are below applicable clearance thresholds. The Contractor is to clean all HVAC Ductwork if the facility's HVAC system is operational and cannot be isolated from the work control area or there is any evidence that the pre-filters were not effective. Do not remove the lead, cadmium, and chromium (VI) control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt and approval of the CP certification. Clearance sampling results must be submitted to the Contracting Officer within 72 hours after the samples are taken. The Contractor should allow fourteen (14) calendar days for the Government review period.

3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with local, state, and federal statutes and regulations, Navy and PNSY requirements, and project specifications. Ensure all waste is properly characterized in accordance with the Waste Characterization Sampling Plan developed under Section entitled WASTE CHARACTERIZATION SAMPLING PLAN here. The result of each waste characterization will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, and chromium (VI) contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, and chromium (VI) -contaminated clothing. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.

- c. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with 40 CFR 262 and corresponding state regulations.
- d. Manage lead, cadmium, and chromium (VI) or lead, cadmium, and chromium (VI) contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265.
- e. All lead, cadmium, and chromium (VI) waste generation, management, and disposal will be coordinated with the Shipyard environmental function (Code 106, B357) and the Contracting Officer.

All hazardous waste must be turned in to B357 not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor shall not bring hazardous wastes onto Government property.

Containers positioned within the work area boundaries shall have covers in place whenever containers are not in use.

3.5.2.1 Disposal Documentation

For non-hazardous waste, provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Non-Hazardous Waste

Payment for disposal of non-hazardous waste will not be made until a

signed copy of the manifest from the disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of non-hazardous waste delivered to the disposal facility.

-- End of Section --

SECTION 02 84 33

REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs)

02/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.145	Specifications for Accident Prevention Signs and Tags
29 CFR 1910.1000	Air Contaminants
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 174	Carriage by Rail
49 CFR 175	Carriage by Aircraft
49 CFR 176	Carriage by Vessel
49 CFR 177	Carriage by Public Highway
49 CFR 178	Specifications for Packagings
49 CFR 179	Specifications for Tank Cars

1.2 REQUIREMENTS

The work includes the removal and disposal of designated PCB containing material. Perform work in accordance with 40 CFR 761 and the requirements specified herein.

1.3 DEFINITIONS

1.3.1 Leak

Leak or leaking means any instance in which a PCB Article, PCB Container, or PCB Equipment has any PCBs on any portion of its external surface.

1.3.2 PCBs

PCBs as used in this specification mean the same as PCBs, PCB Article, PCB Article Container, PCB Container, PCB Equipment, PCB Item, PCB Transformer, PCB-Contaminated Electrical Equipment, as defined in 40 CFR 761, Section 3, Definitions.

1.3.3 Spill

Spill means both intentional and unintentional spills, leaks, and other uncontrolled discharges when the release results in any quantity of PCBs running off or about to run off the external surface of the equipment or other PCB source, as well as the contamination resulting from those releases.

1.4 QUALITY ASSURANCE

1.4.1 Training

Instruct employees on the dangers of PCB exposure, on respirator use, decontamination, and applicable OSHA and EPA regulations.

1.4.2 Certified Industrial Hygienist (CIH)

Obtain the services of an industrial hygienist certified by the American Board of Industrial Hygiene to certify training, review and approve the PCB removal plan, including determination of the need for personnel protective equipment (PPE) in performing PCB removal work.

1.4.3 Regulation Documents

Maintain at all times one copy each at the office and one copy each in view at the job site 29 CFR 1910.1000, 40 CFR 761, and Contractor work practices for removal, storage and disposal of PCBs.

1.4.4 Surveillance Personnel

Surveillance personnel may enter PCB control areas for brief periods of time provided they wear disposable polyethylene gloves and disposal polyethylene foot covers, as a minimum. Additional protective equipment may be required if respiratory hazard is involved or if skin contact with PCB is involved.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

PCB Removal Work Plan

SD-07 Certificates

Training certification

Qualifications of CIH

Notification

Certification of Decontamination for PCB Spill

Post Cleanup Sampling Data

1.6 EQUIPMENT

1.6.1 Special Clothing

Work clothes consist of PPE as required by OSHA regulations, including, but not limited to the following:

- a. Disposable coveralls
- b. Gloves (Disposable rubber gloves may be worn under these)
- c. Disposable foot covers (polyethylene)
- d. Chemical safety goggles
- e. Half mask cartridge respirator.

1.6.2 Special Clothing for Government Personnel

Provide PPE specified in paragraph SPECIAL CLOTHING to the Contracting Officer as required for inspection of the work.

1.6.3 PCB Spill Kit

Assemble a spill kit to include the following items:

<u>ITEM</u>	<u>MINIMUM QUANTITY</u>
1. Disposable gloves (polyethylene)	6 prs
2. Gloves with a high degree of impermeability to PCB	6 prs
3. Disposable coveralls with permeation resistance to PCB	4 ea
4. Chemical safety goggles	2 ea
5. Disposable foot covers (polyethylene)	6 prs

6. PCB Caution Sign: "PCB Spill--Authorized Personnel Only"	2 ea
7. Banner guard or equivalent banner material	100 feet
8. Absorbent material	5 bags
9. Blue polyethylene waste bags	5 ea
10. Cloth backed tape	1 roll
11. Area access logs, blank	10 ea
12. Brattice cloth, 6' x 6'	1 piece
13. Rags	20 ea
14. Ball point pens	20 ea
15. Herculite, 4' x 4' and 8' x 8'	2 ea and 1 ea
16. Blank metal signs and grease pencils	2 ea
17. Waste containers 55 gallon drum, may be used as container for kit	1 ea

1.7 QUALITY ASSURANCE

1.7.1 Training Certification

Submit certificates, prior to the start of work but after the main abatement submittals, signed and dated by the CIH and by each employee stating that the employee has received training. Organize certificates by individual worker, not grouped by type of certificates.

1.7.2 Qualifications of CIH

Submit the name, address, and telephone number of the Industrial Hygienist selected to perform the duties in paragraph CERTIFIED INDUSTRIAL HYGIENIST. Submit proper documentation that the Industrial Hygienist is certified, including certification number and date of certification/recertification.

1.7.3 PCB Removal Work Plan

Submit a detailed job-specific plan of the work procedures to be used in the removal of PCB-containing materials, not to be combined with other hazardous abatement plans. Provide a Table of Contents for each abatement submittal which follows the sequence of requirements in the contract. Include in the plan a sketch showing the location, size, and details of PCB control areas, location and details of decontamination rooms, change rooms, shower facilities, and mechanical ventilation system. Include in the plan, eating, drinking, smoking and restroom procedures, interface of trades, sequencing of PCB related work, PCB disposal plan, respirators, protective equipment, and a detailed description of the method of

containment of the operation to ensure that PCB contamination is not spread or carried outside of the control area. Include provisions to ensure that airborne PCB concentrations of 0.5 milligrams per cubic meter (mg/m³) of air are not exceeded outside of the PCB control area. Include air sampling, training and strategy, sampling methodology, frequency, duration of sampling, and qualifications of air monitoring personnel in the air sampling portion of the plan. Obtain approval of the plan prior to the start of PCB removal work.

1.7.4 PCB Disposal Plan

Submit a PCB Disposal Plan within 45 calendar days after award of contract for Contracting Officer's approval. Comply with applicable requirements of Federal, State, and local PCB waste regulations and address:

- a. Identification of PCB wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with PCB wastes.
- d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- e. Spill prevention, containment, and cleanup contingency measures to be implemented.
- f. Work plan and schedule for PCB waste containment and removal. Clean up and containerize wastes and bring to Building 357 daily by the end of shift in which it was generated.

1.7.5 Notification

Notify the Contracting Officer 20 days prior to the start of PCB removal work.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 PROTECTION

3.1.1 Decontamination Room, Clean Room and Shower Facilities

- a. Provide material and labor for construction of a decontamination room, a clean room, and shower facilities. Provide rooms with doors and attach to the exit ways of PCB work areas. Provide rooms of sufficient size to accommodate the Contractor's operation within. Provide portable toilet and shower facilities. Locate shower facilities between the clean room and decontamination room. Provide separate clothing lockers or containers in each room to prevent contamination of street and work clothes.

- b. Remove PCB-contaminated PPE in the decontamination room before proceeding to showers. Workers must shower before lunch and at the end of each day's work. Hot water, towels, soap, and hygienic conditions are the responsibility of the Contractor.

3.1.2 PCB Control Area

Isolate PCB control area by physical boundaries to prevent unauthorized entry of personnel. Food, drink and smoking materials are not permitted in areas where PCBs are handled or PCB items are stored.

3.1.3 Personnel Protection

Workers must wear and use PPE, as recommended by the Industrial Hygienist, upon entering a PCB control area. If PPE is not required per the CIH, specify in the PCB removal work plan.

3.1.4 Footwear

Work footwear remains inside work area until completion of the job.

3.1.5 Permissible Exposure Limits (PEL)

PEL for PCBs is 0.5 mg/m³ on an 8-hour time weighted average basis.

3.1.6 Special Hazards

- a. Do not expose PCBs to open flames or other high temperature sources since toxic decomposition by-products may be produced.
- b. Do not heat PCBs to temperatures of 135 degrees F or higher without Contracting Officer's concurrence.

3.1.7 PCB Caution Label

40 CFR 761, Subpart C. Affix labels to PCB waste containers and other PCB-contaminated items. Provide label with sufficient print size to be clearly legible, with bold print on a contrasting background, displaying the following: CAUTION: Contains PCBs (Polychlorinated Biphenyls).

3.1.8 PCB Caution Sign

29 CFR 1910.145. Provide signs at approaches to PCB control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area.

3.2 WORK PROCEDURE

Furnish labor, materials, services, and equipment necessary for the complete removal of PCBs located at the site as indicated or specified in accordance with local, State, or Federal regulations. Package and mark PCB as required by EPA and DOT regulations and dispose of off Government property in accordance with EPA, DOT, and local regulations at a permitted site.

3.2.1 No Smoking

Smoking is not permitted within 50 feet of the PCB control area. Provide "No Smoking" signs as directed by the Contracting Officer.

3.2.2 Work Operations

Ensure that work operations or processes involving PCB or PCB-contaminated materials are conducted in accordance with 40 CFR 761 and the applicable requirements of this section, including but not limited to:

- a. Obtaining advance approval of PCB storage sites.
- b. Notifying Contracting Officer prior to commencing the operation.
- c. Reporting leaks and spills to the Contracting Officer.
- d. Cleaning up spills.
- e. Maintaining an access log of employees working in a PCB control area and providing a copy to the Contracting Officer upon completion of the operation.
- f. Inspecting PCB and PCB-contaminated items and waste containers for leaks and forwarding copies of inspection reports to the Contracting Officer.
- g. Maintaining a spill kit as specified in paragraph PCB SPILL KIT.
- h. Maintaining inspection, inventory and spill records.

3.3 PCB TRANSFORMERS

3.3.1 Draining of Transformer Liquid

Perform work in accordance with 49 CFR 171, 49 CFR 172, 49 CFR 173, 49 CFR 174, 49 CFR 175, 49 CFR 176, 49 CFR 177, 49 CFR 178, and 49 CFR 179, Subchapter C and as specified herein. Drain the transformer, switches, and regulators of free flowing liquid prior to transportation. Place the drained liquids in DoT Spec drums. The drums must not contain more than 50 gallons of oil. If the equipment cannot be drained, then place it in DoT Spec drums. DoT drum specifications must meet UN 1A1/Y1.6 (steel drum) or UN 1H1/Y1.6 (plastic drum).

3.3.2 Markings

Provide drums and drained PCB-contaminated electrical equipment with caution label markings as specified in paragraph PCB CAUTION LABEL.

3.3.3 Laboratory Analysis

Provide all transformers with a laboratory analysis for turn-in. Building 357 prefers a gas chromatograph test. The only two exceptions to this rule are:

- a. The transformer is hermetically sealed (solder sealed or fusion sealed. No access ports or openings).
- b. The name plate states that the transformer contains Pyranol,

Interteen, etc.

Attach a copy of the lab analysis to both the DD 1348-1 and the transformer itself.

3.3.4 Markings

3.3.4.1 Transformers, Less Than 50 ppm

Add absorbent material to absorb residue oil remaining after draining. Write the date drained on the transformer. Turn in transformers to Building 357.

3.3.4.2 Transformers, 50-499 ppm

Same procedure as transformers in the less than 50 ppm range.

3.3.4.3 Transformers, Greater Than 500 ppm

Stencil date drained on the transformer. Turn in transformer to Building 357.

3.3.4.4 Drums

Drums will be obtained from Building 357 for all PCB waste disposal. Stencil on DOT-approved 55 gallon drums containing PCB liquid the following:

- a. ppm
- b. Date drum filled
- c. Serial number of transformer liquid came from
- d. National Stock Number
 - (1) "9999-00-OIL" for <50 ppm
 - (2) "9999-00-CONPCB" for 50-499 ppm
 - (3) "9999-00-PCBOIL" for >500 ppm

Do not mix different ppms in the same drum. Drums must have a 2 inch ullage space from the top of the drum.

3.4 PCB REMOVAL

Select PCB removal procedure to minimize contamination of work areas with PCB or other PCB-contaminated debris/waste. Handle PCBs such that no skin contact occurs. PCB removal process should be described in the work plan.

3.4.1 Confined Spaces

As feasible, do not carry out PCB handling operations in confined spaces. A confined space means a space having limited means of egress and inadequate cross ventilation.

3.4.2 Control Area

Establish a PCB control area around the PCB item as specified in paragraph PCB CONTROL AREA. Only personnel briefed on the elements in the paragraph TRAINING and on the handling precautions are allowed into the area.

3.4.3 Exhaust Ventilation

If used, discharge exhaust ventilation for PCB operations to the outside and away from personnel.

3.4.4 Temperatures

As feasible, handle PCBs at ambient temperatures and not at elevated temperatures.

3.4.5 Solvent Cleaning

Clean contaminated tools, containers, etc., after use by rinsing three times with an appropriate solvent or by wiping down three times with a solvent wetted rag. Suggested solvents are Stoddard solvent or hexane.

3.4.6 Drip Pans

Drip pans are required under portable PCB transformers and rectifiers in use or stored for use. Provide the pans with a containment volume of at least one and one-half times the internal volume of PCBs in the item.

3.4.7 Evacuation Procedures

Provide written procedures for evacuation of injured workers. Do not delay aid for a seriously injured worker for reasons of decontamination.

3.5 PCB SPILL CLEANUP REQUIREMENTS

3.5.1 PCB Spills

Immediately report to the Contracting Officer any PCB spills on the ground or in the water, PCB spills in drip pans, or PCB leaks.

3.5.2 PCB Spill Control Area

Rope off an area around the edges of a PCB leak or spill and post a "PCB Spill Authorized Personnel Only" caution sign. Immediately transfer leaking items to a drip pan or other container.

3.5.3 PCB Spill Cleanup

40 CFR 761, Subpart G. Initiate cleanup of spills as soon as possible, but no later than 48 hours of its discovery. To clean up spills, personnel must wear the PPE prescribed in paragraph SPECIAL CLOTHING of this section. If misting, elevated temperatures or open flames are present, or if the spill is situated in a confined space, notify the Contracting Officer. Mop up the liquid with rags or other conventional absorbent. Properly contain and dispose of spent absorbent as solid PCB waste.

3.5.4 Records and Certification

Document the cleanup with records of decontamination in accordance with 40 CFR 761, Section 125, Requirements for PCB Spill Cleanup. Provide certification of decontamination.

3.5.5 Sampling Requirements

Perform post cleanup sampling as required by 40 CFR 761, Section 130, Sampling Requirements. Do not remove boundaries of the PCB control area until site is determined satisfactorily clean by the Contracting Officer.

3.6 STORAGE FOR DISPOSAL

3.6.1 Storage Containers for PCBs

49 CFR 178. Store liquid PCBs in Department of Transportation (DOT) Specification 17E containers. Store nonliquid PCB mixtures, articles, or equipment in DOT Specification 5, 5B, or 17C containers with removable heads.

3.6.2 Waste Containers

Label with the following:

- a. "Solid (or Liquid) Waste Polychlorinated Biphenyls"
- b. The PCB Caution Label, paragraph PCB CAUTION LABEL
- c. The date the item was placed in storage and the name of the cognizant activity/building.

3.6.3 PCB Articles and PCB-Contaminated Items

Label with items b. through c. above.

3.6.4 Approval of Storage Site

Obtain in advance Contracting Officer approval using the following criteria without exception.

- a. Adequate roof and walls to prevent rainwater from reaching the stored PCBs.
- b. An adequate floor which has continuous curbing with a minimum 6 inch high curb. Such floor and curbing must provide a containment volume equal to at least two times the internal volume of the largest PCB article or PCB container stored therein or 25 percent of the total internal volume of all PCB equipment or containers stored therein, whichever is greater.
- c. No drain valves, floor drains, expansion joints, sewer lines, or other openings that would permit liquids to flow from the curbed area.
- d. Floors and curbing constructed of continuous smooth and impervious materials such as portland cement, concrete or steel to prevent or minimize penetrations of PCBs.

- e. Not located at a site which is below the 100-year flood water elevation.
- f. Post each storage site with the PCB Caution Sign, paragraph PCB CAUTION SIGN.

3.7 CLEANUP

Maintain surfaces of the PCB control area free of accumulations of PCBs. Restrict the spread of dust and debris; keep waste from being distributed over work area.

Do not remove the PCB control area and warning signs prior to the Contracting Officer's approval. Reclean areas showing residual PCBs.

3.8 DISPOSAL

3.8.1 Disposal by the Government

All PCB waste >50 ppm will be disposed of by the government. All waste will be brought to Building 357 by the end of the work shift in which it was generated.

-- End of Section --

SECTION 03 01 00

REHABILITATION OF CONCRETE

02/18

PART 1 GENERAL

1.1 SCOPE

This specification governs the rehabilitation of structural concrete.

1.2 DEFINITIONS

1.2.1 Bracing

Temporary supplemental members used to avoid local or global instability during construction, evaluation, or repair that are intended to be removed after completion of construction.

1.2.2 Delamination

A planar separation in a material that is roughly parallel to the surface of the material.

1.2.3 Rehabilitation

Repairing or modifying an existing structure to a desired useful condition.

1.2.4 Repair

The reconstruction or renewal of concrete parts of an existing structure for its maintenance or to correct deterioration, damage, or faulty construction of members or systems of a structure.

1.2.5 Shoring

Props or posts of timber or other material in compression used for the temporary support of excavations, formwork, or unsafe structures; the process of erecting shores.

1.2.6 Termination Joint

The interface where a placement of repair material meets existing concrete, the edge of an expansion joint, or other existing surfaces.

1.2.7 Unsound Concrete

Concrete that is fractured, delaminated, spalled, deteriorated, defective, contaminated or otherwise damaged.

1.3 REFERENCES

AMERICAN CONCRETE INSTITUTE (ACI)

- ACI 117 (2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
- ACI 440.5 (2008) Specification for Construction with Fiber-Reinforced Polymer Reinforcing Bars
- ACI 440.6 (2008) Specification for Carbon and Glass Fiber-Reinforced Polymer Bar Materials for Concrete Reinforcement
- ACI 440.8 (2013) Specification for Carbon and Glass Fiber-Reinforced Polymer (FRP) Materials Made by Wet Layup for External Strengthening of Concrete and Masonry Structures
- ACI 503.2-503.4 (2010, R 2003) Three Epoxy Specifications
- ACI 503.3 (2010) Specification for Producing a Skid-Resistant Surface on Concrete by the Use of Epoxy and Aggregate
- ACI 503.7 (2007) Specification for Crack Repair by Epoxy Injection
- ACI 548.12 (2012) Specification for Bonding Hardened Concrete and Steel to Hardened Concrete with an Epoxy Adhesive

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- ASCE/SEI 37 (2015) Design Loads on Structures During Construction

ASTM INTERNATIONAL (ASTM)

- ASTM C33/C33M (2018) Standard Specification for Concrete Aggregates
- ASTM C42/C42M (2020) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
- ASTM C387/C387M (2017) Standard Specification for Packaged, Dry, Combined Materials for Concrete and High Strength Mortar
- ASTM C496/C496M (2017) Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens

ASTM C881/C881M	(2020) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C882/C882M	(2020) Bond Strength of Epoxy-Resin Systems Used with Concrete by Slant Shear
ASTM C928/C928M	(2020) Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs
ASTM C1059/C1059M	(2013) Standard Specification for Latex Agents for Bonding Fresh to Hardened Concrete
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1438	(2013; R 2017) Standard Specification for Latex and Powder Polymer Modifiers for use in Hydraulic Cement Concrete and Mortar
ASTM C1583/C1583M	(2013) Standard Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method)
ASTM C1600/C1600M	(2017) Standard Specification for Rapid Hardening Hydraulic Cement
ASTM C1602/C1602M	(2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D93	(2019) Standard Test Methods for Flash-Point by Pensky-Martens Closed Cup Tester
ASTM D323	(2015a) Vapor Pressure of Petroleum Products (Reid Method)
ASTM D542	(2014) Index of Refraction of Transparent Organic Plastics
ASTM D1078	(2011) Standard Test Method for Distillation Range of Volatile Organic Liquids
ASTM D2103	(2015) Standard Specification for Polyethylene Film and Sheeting
ASTM D3418	(2015) Transition Temperatures of Polymers by Differential Scanning Calorimetry
ASTM D4016	(2014) Viscosity of Chemical Grouts by

Brook field Viscometer (Laboratory Method)

ASTM D4580/D4580M

(2012) Standard Practice for Measuring
Delaminations in Concrete Bridge Decks by
Sounding

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval . When used, a designation following the "G" designation identifies the office that will review the submittal for the government. Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

Work Plan; G

Quality Control Plan; G

SD-03 Product Data

Polymers, G

Miscellaneous Materials And Equipment

SD-04 Samples

Reinforcement And Reinforcement Supports

Polymers

Miscellaneous Materials And Equipment

SD-05 Design Data

Formwork And Shoring; G

Repair Procedures; G

Mixture Proportioning; G

SD-06 Test Reports

Mixture Proportioning, G

Quality Control, G

Reinforcement And Reinforcement Supports

Polymers

Miscellaneous Materials And Equipment

SD-07 Certificates

Qualifications

SD-08 Manufacturer's Instructions

Equipment For Concrete Preparation

Polymers

Miscellaneous Materials And Equipment

1.5 QUALITY ASSURANCE

1.5.1 General Requirements

- a. To protect personnel from overexposure to toxic materials, conform to the applicable manufacturer's Safety data sheets or local regulations. Submit manufacturer's Safety Data Sheets for all polymers as well as other potentially hazardous materials.
- b. Submit the [repair procedures](#) for executing the work as well as the test data and documentation on materials used for repair. Submittal must include component materials, mixture proportions, and supplier's quality control program.
- c. Inspection and testing of surface preparation as well as placement of reinforcing steel must be in accordance with provisions included herein and the Contract Document.
- d. Sampling and testing of materials, as well as inspection and testing of work, must be in accordance with established procedures, manufacturer's instructions, specific instructions from the Contracting Officer if given, or recommended practices as referenced herein and the Contract Documents.
- e. Trial batches and testing requirements for various repair materials specified are the responsibility of the Contractor.
- f. The testing agency must inspect, sample, and test repair materials and concrete production as required. When it appears that material furnished or work performed by Contractor fails to conform to Contract Documents the testing agency will immediately report such deficiency.

1.5.2 [Quality Control Plan](#)

Submit a quality control plan as specified in Sections [01 45 00.00 10](#)
QUALITY CONTROL .

1.5.3 Qualifications

The submittals must where applicable, identify agencies and individuals who will be working on this contract and their relevant experience. Do not make changes in approved agencies or personnel without prior approval of the Contracting Officer.

1.5.3.1 Testing Agencies

In addition to the requirements of Section 01 45 00.00 10 QUALITY CONTROL, agencies that test concrete materials must meet the requirements of ASTM C1077. Testing agencies that test or inspect placement of reinforcing steel must meet the requirement of ASTM E329. Submit data on qualifications of Contractor's proposed testing agency for acceptance.

1.5.3.2 Quality Control Personnel

Field tests of repair materials required must be made by an ICRI Concrete Surface Repair Technician Tier 2. Submit resumes, pertinent information, past experience, training and education of all operators of specialized demolition equipment if needed for this and the three paragraphs above.

1.5.3.3 Contractor Qualifications

The contractor performing the repair work must have been involved in a minimum of three concrete repair projects similar in size and scope to this project for at least five years. Submit information, including name, dollar value, date, and point-of-contact for similar projects which demonstrates the required experience and/or training.

1.5.3.4 Worker Qualifications

- a. Each worker engaged in the use of specialized removal or application equipment, including saw operators and epoxy injection , must have satisfactorily completed an instruction program and three years of experience in the operation of the equipment. The worker must have active experience with the equipment within five years of the project.
- b. Workers installing adhesive anchors must be ACI Adhesive Anchor Installer certified or equivalent.

1.5.3.5 Regulatory Requirements

Perform all work in accordance with applicable Federal, State, and local safety, health, and environmental requirements. The Contractor is responsible for obtaining all permits required by Federal, State, and local agencies for the performance of the work.

1.5.4 Pre-Construction Conference

Conduct a pre-construction conference to discuss repair materials performance requirements, control provisions, and roles and responsibilities for the Work to ensure that the Contractor's personnel understand all aspects of the repair material, its properties and application procedures. The conference must include the Contracting Officer or authorized representative, the Contractor's field superintendent and foreman, and a competent Technical Representative of the material

manufacturer, and other involved trades or supplier representatives. The Technical Representative must be fully qualified to perform the work.

1.5.5 Work Plan

Prepare a work plan describing the methods of concrete removal and repair, including methods, equipment and materials to be used for each feature. Submit the work plan for approval at least 30 days prior to the start of the work. The plan must include, but not be limited to, repair materials to be used with specific information on products and/or constituents, and requirements for handling, storage, etc., equipment to be used, surface preparation, and requirements for placement, finishing, curing and protection specific to the materials used. Include a description of field demonstrations in the work plan. Do not commence work until the work plan and field demonstration representative of the type of work are approved.

1.6 ACCEPTANCE OF REHABILITATION WORK

1.6.1 General Requirements

- a. Completed concrete rehabilitation work must conform to applicable requirements of Contract Document and this specification. The Contractor is responsible to bring Work into compliance with requirements of Contract Documents if the Concrete repair work fails to meet one or more requirements of Contract Documents.
- b. Correct rejected repair work by removing and replacing or by strengthening with additional construction acceptable to the Contracting Officer. Use repair methods that meet applicable requirements for function, durability, dimensional tolerances, and appearance.
- c. Submit proposed work plan, repair methods, materials, and modifications to the Work needed to correct rejected repair work to meet the requirements of Contract Documents.

1.6.2 Tolerances

- a. Construction tolerances for repairs must conform to ACI 117. Where existing conditions do not allow tolerances to conform to ACI 117, use the details and materials for such conditions as indicated in the Contract Documents. For conditions not shown or that are different than indicated in the Contract Documents, notify the Contracting Officer before proceeding with the work at those locations.
- b. Inaccurately formed concrete surfaces resulting in concrete members with dimensions that exceed ACI 117 tolerances are subject to rejection.

1.6.3 Appearance

Concrete surfaces not meeting the requirements of the Contract Documents must be brought into compliance.

1.7 PROTECTION OF COMPLETED REHABILITATION WORK

- a. Do not allow construction loads to exceed the loads that a structural member or structure is safely capable of supporting without damage.

Provide supplemental support if construction loads are expected to exceed safe load capacity.

- b. Protect repaired and adjacent areas from damage by construction traffic, equipment, and materials. During the curing period, protect repair materials from damage by mechanical disturbances, including load-induced stresses, shock, and vibration.
- c. Protect repair materials from environmental damage by weather events during the length of the curing period.

PART 2 PRODUCTS

Products or materials used must conform to the requirements included herein as well as the Contract Documents. The usage of other products or materials not covered by this requirement or specified in the Contract Documents are permitted upon approval by the Contracting Officer. Additional information and submittals for products and materials not included in this document including product data, samples, design data, test reports, certificates, manufacturer's instructions, and field reports must be submitted as requested by the Contracting Officer.

2.1 MATERIALS FOR SHORING AND BRACING

2.1.1 Shoring and Bracing Systems

Use commercially manufactured and engineered shoring and bracing systems and components, except where custom built assemblies of lumber or other suitable materials are permitted by the Contracting Officer.

2.1.2 Design Requirements

The design of the bracing and shoring must be based on [ASCE/SEI 37](#).

- a. Non-manufactured shoring and bracing systems must have calculations signed and sealed by a Licensed Design Professional.
- b. Members of non-manufactured shoring systems, must be designed in accordance with the provisions of the governing building code for the specific material of the member.
- c. Members of manufactured shoring systems, consisting of pre-engineered components designed and produced specifically for structural shoring, must be used in accordance with the manufacturer's recommendations.

2.2 [EQUIPMENT FOR CONCRETE PREPARATION](#)

Means and methods used for concrete removal and surface preparation must be selected and used such as to minimize damage to the structure and to the concrete substrate that remains.

2.2.1 Equipment for Concrete Removal

Removal equipment and techniques must be suitable to produce concrete surface profiles and level of cleanliness in designated areas as required by this specification and the contract Documents.

2.2.1.1 Cutting Equipment

- a. The following cutting equipment are permitted: Saw cutting, Diamond wire cutting, Mechanical shearing, Stitch drilling.
- b. Cutting, lifting, and transporting equipment must be adequate to cut, support, and transport concrete sections without incurring any damage to the existing structure.

2.2.1.2 Concrete Breakers

- a. Provide sharp tips on breaker equipment to minimize microcracking damage in partial depth removal.
- b. The use of the following impact equipment and methods is permitted: Hand-held breakers, Boom-mounted breakers, Scabblers, Needle scalers, Scarifiers.

2.2.2 Surface preparation and cleaning equipment

2.2.2.1 Abrasive Blasting

- a. Use dry or wet oil-free abrasive blasting capable of removing loose micro-fractured (bruised) or otherwise damaged or pulverized concrete surfaces, and rust from exposed steel reinforcement, and providing a surface profile in compliance with the Contract Documents.
- b. Use the following abrasive blasting methods: Sandblasting.

2.2.2.2 Low Pressure Water Cleaning

Use equipment capable of delivering 1000 psi to to 5000 psi at 2 gal/min to 10 gal/min for cleaning loose material from repair areas.

2.2.2.3 Other Cleaning Equipment

Use equipment that delivers oil free air capable of cleaning loose material and debris from repair areas. If necessary to dry the concrete surface, gas-fired torches or clean, dry, compressed air may be used. Also, use vacuums capable of removing loose material and debris.

2.3 MATERIALS FOR FORMWORK AND EMBEDDED ITEMS

2.3.1 Formwork Materials

a. Form-facing material in contact with concrete must be lumber, plywood, tempered concrete-form-grade hardboard, metal, plastic, or treated paper that creates specified appearance and texture of concrete surface. Submit product information on proposed form-facing materials if different from that specified herein.

- a. Design formwork, shores, reshores, and backshores to support loads transmitted to them and to comply with applicable building code requirements.
- b. Design formwork joints to inhibit leakage of mortar.

- c. Limit deflection of facing materials for concrete surfaces exposed to view to 1/240 of center-to-center spacing of facing supports.
- d. Submit shop drawings for formwork, shoring, reshoring, and backshoring. Shop drawings must be signed and sealed by a licensed design engineer.
- e. Submit design calculations for formwork, shoring, reshoring, and backshoring. Design calculations must be signed and sealed by a licensed design engineer.
- f. Submit manufacturer's product data on form liner proposed for use with each formed surface.

2.3.1.1 Wood Forms

Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects. Provide plywood that complies with NIST PS 1, B-B concrete form panels or better or AHA A135.4, hardboard for smooth form lining.

2.3.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood that conforms to NIST PS 1, B-B, concrete form, not less than 5/8-inch thick.

2.3.1.2 Plastic Forms

Provide plastic forms that contain a minimum of 50 percent post-consumer recycled content, or a minimum of 50 percent post-industrial recycled content.

2.3.1.3 Steel Forms

Provide steel form surfaces that do not contain irregularities, dents, or sags.

2.3.2 Formwork Accessories

- a. Use commercially manufactured formwork accessories, including ties and hangers.
- b. Form ties and accessories must not reduce the effective cover of the reinforcement.
- c. Install and remove formwork without damaging or staining the existing structure or repair material.
- d. Forms used for polymer concrete/mortars must be tight enough to hold the material that is used without leaking. All surfaces where bond is not desired, but which are exposed to the monomer or resin, must be treated with a form release agent.

2.4 REINFORCEMENT AND REINFORCEMENT SUPPORTS

2.4.1 Reinforcing Bars

- a. Reinforcing bars must be deformed, except spirals, load-transfer

dowels, and welded wire reinforcement, which may be plain.

- b. [ASTM A615/A615M](#) with the bars marked A, Grade 60; or [ASTM A996/A996M](#) with the bars marked R, Grade 60, or marked A, Grade 60.
- c. Reinforcing bars may contain post-consumer or post-industrial recycled content. Submit documentation indicating percentage of post-industrial and post-consumer recycled content per unit of product. Indicate relative dollar value of recycled content products to total dollar value of products included in project.
- d. Submit mill certificates for reinforcing bars.

2.4.2 Mechanical Reinforcing Bar Connectors

- a. Provide 125 percent minimum yield strength of the reinforcement bar.
- b. Mechanical splices for galvanized reinforcing bars must be galvanized or coated with dielectric material.
- c. Mechanical splices used with epoxy-coated or dual-coated reinforcing bars must be coated with dielectric material.
- d. Submit data on mechanical splices demonstrating compliance with this paragraph.

2.4.3 Welding

- a. Provide weldable reinforcing bars that conform to [ASTM A706/A706M](#) and [ASTM A615/A615M](#) and Supplement S1, Grade 60, except that the maximum carbon content must be 0.55 percent.
- b. Comply with [AWS D1.4/D1.4M](#) unless otherwise specified. Do not tack weld reinforcing bars.

2.4.4 Fiber-Reinforced Polymers

- a. Fiber-Reinforced Polymers (FRP) bars used as internal reinforcement in concrete and their supports must meet the product requirements of [ACI 440.5](#) and conform to [ACI 440.6](#).
- b. Submit test reports and certificates for FRP bars as required by [ACI 440.5](#) and the Contract Documents.
- c. Fiber-Reinforced Polymer (FRP) laminate materials externally bonded to concrete made by wet layup must meet the requirements of [ACI 440.8](#) and the Contract Documents. Submit product data sheets for materials used for FRP layup systems as described in [ACI 440.8](#).
- d. The use of externally bonded FRP systems other than wet layup systems are permitted upon approval by the Contracting Officer. Submit product and materials data, design data, test reports, certificates, manufacturer's instructions, and field reports for those systems as requested by the Contracting Officer and required by Contract Documents.

2.5 [POLYMERS](#)

- a. The requirements for the properties of polymers and aggregates used in polymers must meet the requirements specified in this paragraph as well as the properties specified in the referenced specifications and the Contract Documents.
- b. Polymers used must be compatible with other polymers and materials used on the project. Unless repair materials are specified in the contract documents, the Contractor is responsible for verifying material compatibilities.
- c. Submit product data, manufacturer's Safety Data Sheets, samples, design data, test reports, certificates, manufacturer's instructions, and field reports for materials as required by this document as well as the referenced specifications and the Contract Documents.

2.5.1 Epoxies

- a. Epoxy mortars and epoxy compounds must conform to [ASTM C881/C881M](#), Type III or IV; Class A, B, or C as applicable; Grade 1, 2, or 3.
- b. Epoxy mortars used for repairing defects in hardened portland cement concrete must meet the requirements of [ACI 503.2-503.4](#).
- c. Epoxy used for crack repair must meet the requirements of [ACI 503.7](#).
- d. Epoxy used to produce a skid-resistant surface on hardened concrete must meet the requirements of [ACI 503.3](#).
- e. Epoxy used for overlays must meet the requirements of [ACI 548.8](#)[ACI 548.9](#).
- f. Epoxy used for bonding freshly mixed concrete and hardened concrete must meet the requirements of [ASTM C881/C881M](#), Type V, Grade 2 or 3, Class A, B, C.
- g. Epoxy used for bonding hardened concrete and steel to hardened concrete must meet the requirements of [ACI 548.12](#).

2.5.2 Latexes

- a. Latex used in polymer modified portland cement concrete/mortar must meet the requirements of [ASTM C1438](#).
- b. Latex used in polymer modified portland cement concrete overlays must meet the requirements of [ACI 548.4](#).
- c. Latex used for bonding freshly mixed concrete and hardened concrete must meet the requirements of [ASTM C1059/C1059M](#), Type II.

2.5.3 [Repair Materials](#)

- a. [The use of urethanes or acrylics is permitted.](#)
- b. Submit product data, samples, design data, test reports, and manufacturer's instructions for acceptance by the Contracting Officer.

2.5.4 Aggregate

- a. Unless otherwise specified or recommended by the polymer material manufacturer, aggregate used with polymers must meet [ASTM C33/C33M](#) requirements.
- b. Aggregate properties and proportions used with polymers must meet the requirements of the polymer material manufacturer, the requirements of the referenced polymer standard, and the Contract Documents.
- c. Aggregate used with polymers must be dry and free of dirt, asphalt, and other organic materials. Aggregate moisture content must be less than 0.2 percent by weight.
- d. For patch repairs, the maximum-sized aggregate must not be greater than one third the depth of the patch area.

2.6 MISCELLANEOUS MATERIALS AND EQUIPMENT

2.6.1 Packaged and proprietary materials

The required properties for the materials listed in this paragraph must meet the properties specified in the Contract Documents. Submit Product data, samples, design data, test reports, manufacturer's instructions, and field reports as required by the Contracting Officer and the Contract Documents.

- a. Packaged, rapid hardening concrete repair materials must conform to [ASTM C928/C928M](#).
- b. Packaged, mortar and concrete must conform [ASTM C387/C387M](#).
- c. Rapid hardening cement must conform to [ASTM C1600/C1600M](#).

Water used with packaged and proprietary materials must meet [ASTM C1602/C1602M](#) requirements. Aggregates must meet the repair material manufacturer's requirements if available and [ASTM C33/C33M](#) if such requirements are not specified.

2.6.2 Bond Breakers

- a. Bond breaker materials must meet the requirements of [ASTM D2103](#), and must have a minimum thickness of [0.010 in.](#)
- b. Bond breaker materials used must not have detrimental effects on portland cement concrete and reinforcement.

2.6.3 Miscellaneous Equipment

- a. Equipment designed specifically for the application of repair materials must be used as required by the repair material manufacturer and the referenced specification.
- b. Equipment not listed in this specification but referenced or used for repairs must be clean and in good operating condition.

- c. All supplies and equipment must be available in sufficient quantities to allow continuity in the installation project and quality assurance.

2.7 MIXTURE PROPORTIONING

- a. Polymer concrete/mortar/resin/monomer proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements of the referenced material specifications and the repair material manufacturer's directions.
- b. Polymer-modified portland cement concrete proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements provided by the repair material manufacturer as well as ACI 548.4 when such materials are used for overlays.
- c. Proportioning and mixing materials not specified above must follow the requirements provided by the repair material manufacturer.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Examination

Locate area of unsound concrete or delamination using hammer sounding or chain drag sound methods in accordance to ASTM D4580/D4580M. Denote and mark perimeter boundaries and notify the Contracting Officer to approve the unsound concrete layout boundaries.

3.1.2 Protection

Protect pedestrians, motorized traffic, mechanical, electrical, and plumbing equipment, surrounding construction, project site, landscaping, and surrounding buildings from damage or injury resulting from concrete rehabilitation work.

- a. Construct dust and debris barriers surrounding repair work perimeter to control dust and to protect and control construction traffic.
- b. Dispose of runoff from wet demolition or surface preparation operations in accordance with all local ordinances. Disposal methods must avoid soil erosion, avoid undermining pavements and foundations, damage to landscaping and vegetation, and minimize water penetration through other parts of buildings.
- c. Collect and neutralize alkaline wastes and acid wastes and dispose in accordance with local, state, and federal regulations.
- d. Comply with local noise ordinances during demolition operations.
- e. Perform demolition work and surface preparation work in a manner that minimizes disturbances of operations. Coordinate work with the Contracting Officer.
- f. Submit a proposed protection plan for approval by owner representative

and Licensed Design Professional.

3.1.3 Formwork and Shoring

3.1.3.1 Formwork

- a. Construct forms to sizes, shapes, lines, and dimensions to match existing adjacent surfaces and textures. Provide forms that match openings, offsets, chamfers, anchorages, inserts and other features as described on Contract Documents. Construct forms to accommodate installation of products by other trades. Provide forms for easy removal to minimize damage to concrete surfaces and adjacent surfaces. Apply form release coating over formwork surfaces prior to each concrete placement. Form release agents must not be applied to or come in contact with the repair area concrete substrate or reinforcement.
- b. Do not damage repair material during removal of formwork for columns, walls, sides of beams, and other parts not supporting weight of concrete or repair material. Perform needed repair and treatment required on vertical surfaces at once and follow immediately with specified curing. Remove all formwork anchors embedded in existing concrete. Fill anchor holes and repair all damage to existing concrete at anchor holes.

3.1.3.2 Shoring

- a. Provide shoring as required for repairs. Shoring must be designed, documented, and stamped by a Licensed Design Professional. Shoring designs must be submitted to and approved by the Contracting Officer prior to work commencing.
- b. Leave formwork and shoring in place to support existing loads, construction loads and weight of repair material in beams, slabs, and other structural members until in-place strength of repair material determined in accordance with the Contract Documents.

3.1.4 Concrete preparation

- a. Remove concrete as needed per the removal requirements of this section. Limits on removal equipment are specified in the paragraph titled EQUIPMENT FOR CONCRETE PREPARATION.
- b. Remove foreign material, such as dirt, oil, grease, or other chemicals, from the cracks before injection using compressed air, low-pressure water, or vacuuming. Allow wet surfaces to dry at least 24 hours.
- c. Immediately before placing the repair material or installing formwork, make the repair area available for inspection by the Contracting Officer. Obtain acceptance by the Contracting Officer of surface preparation before proceeding with Work. If the Work is rejected, perform additional operations to the satisfaction of Contracting Officer.
- d. Perform tensile pull-off tests in accordance with ASTM C1583/C1583M and guidance at four locations specified by Contracting Officer. Pull-off strength must meet or exceed 250 psi.

3.1.5 Quality Control

3.1.5.1 Quality control of surface preparation

Evaluation of prepared substrate must be continuously monitored to assure that the prepared substrate surface meets project requirements.

3.1.5.2 Quality control of repair overlays

All components of overlay PPCC materials must be certified by the material manufacturer or aggregate supplier to meet all project testing requirements. During the PPCC overlay, take mixed samples and check that the materials are mixed properly. Confirm that the right PC overlay thickness was applied by recording the volume of PC overlay materials and the substrate surface area covered by the overlay.

3.1.6 Curing

- a. For polymer concrete/mortar Work, follow manufacturer's requirements for curing.
- b. For polymer modified portland cement concrete Work follow manufacturer's requirements for curing.

3.1.7 Clean up

- a. Clean and remove all spills and leaks of injection adhesive and stains caused by the injection adhesives.
- b. Dispose wastewater used for cutting and cleaning without staining or damaging the existing surfaces of the structure or the environment of the project area.

3.1.8 Safety

- a. Provide Safety Data Sheets (SDS) for products on site reviewing them before work begins.
- b. Provide safety guards, maintenance, and warnings for all machinery and equipment.
- c. Have personal protection equipment practice in place - eye protection and face guards.
- d. Have all workers in contact with wet cementitious material wear protective gloves and clothing.
- e. Provide eyewash facilities on-site with location signage.
- f. Provide dust masks for workers operating mixers.
- g. Have confined space procedures in place including adequate ventilation in closed spaces before operating equipment or using products that emit potentially dangerous or toxic exhaust, fumes, or dust.
- h. Provide secured storage available for all hazardous or flammable materials.

- i. Conduct safety meetings prior to beginning repair operations.

3.2 CRACK REPAIR

3.2.1 Preparation

3.2.1.1 General Requirements

- a. Clean all cracks in accordance with the paragraph titled Concrete Preparation.
- b. Do not repair cracks when the temperature of the concrete is below freezing and moisture conditions indicate the possibility of ice on the internal surfaces of the crack.
- c. Do not apply adhesive if the temperature of the concrete is not within the range of application temperatures recommended by the manufacturer of the adhesive.

3.2.1.2 Crack routing

Inspect surfaces adjacent to crack to receive repair material. If deteriorated, route a V-groove section at the crack face until sound concrete is reached.

3.2.1.3 Sealing

- a. For epoxy injection, apply a surface seal over all exterior faces of the crack that can be reached to contain the injection adhesive in the crack.
- b. For gravity fill repairs, apply a surface seal along the bottom surface of the element that can be reached to contain the repair material in the crack.

3.2.2 Application

3.2.2.1 Epoxy Injection

- a. Install the injection entry and venting ports using flush mounted or drilled fittings per proprietary manufacturer's instructions.
- b. Space the ports at 8 in.
- c. Inject the epoxy using material manufacturer's recommended equipment.
- d. Apply recommended manufacturer's injection pressure.
- e. For vertical or inclined cracks, apply injection by pumping epoxy into entry ports at the lowest elevation, cap, and move upward.
- f. For horizontal cracks, apply injection by proceeding from one end of the crack to the other until the crack is fully sealed.
- g. After 10 min., repeat injection procedure until all ports refuse injection.

- h. Remove ports and remove the surface seal by chipping, or grinding or other acceptable means after the injected epoxy has cured.

3.2.2.2 Gravity fill

- a. Mix resin or monomer per material manufacturer's instructions.
- b. Pre-fill cracks at least 0.125 in. wide with aggregate.
- c. Pour resin or monomer onto the surface, over the cracks and spread with brooms, rollers, or squeegees.
- d. Work material back and forth over the cracks to maximize fill in crack.
- e. Allow at least 20 minutes for material to penetrate cracks.
- f. Remove excess material once cracks have been filled to refusal.
- g. Allow material to cure per material manufacturer's recommendations.
- h. Remove sealant and grind smooth.

3.2.3 Quality Control

- a. Conduct quality [and control] tests for metering accuracy and mixing effectiveness of the continuous mixing pump in accordance with ACI 503.7.
- b. Qualify the test injection procedures in accordance with ACI 503.7.

3.2.4 Acceptance Criteria

3.2.4.1 Core Sampling

- a. Obtain core samples in accordance with ASTM C42/C42M.
- b. Allow 24 hours after injection before coring.
- c. Obtain cores in a manner that includes as much of the bond line of the repaired concrete as possible. Replace cores that do not intersect the crack for at least 75 percent of the length of the core.
- d. Obtain three diameter core from first 100 ft. and one core for each 100 ft. thereafter.
- e. If cores would sever reinforcing steel or other embedded items, do not core, and notify the Contracting Officer so that an alternative location can be chosen.
- f. Obtain cores at least 2 in. in diameter for visual inspections and at least 4 in. in diameter for the splitting tensile test. Perform a splitting tensile test on one core from the first 100 ft. and one core for each 250 ft. thereafter.
- g. Fill core holes with non-shrink grout.

3.2.4.2 Core Testing

- a. Test a portion of the core samples for the splitting tensile strength in accordance with [ASTM C496/C496M](#).
- b. Allow 72 hours after injection before beginning splitting tensile tests
- c. Prepare core sample per [ASTM C42/C42M](#).
- d. Align the core so that the crack is in a plane as close to vertical as possible.

3.2.4.3 Acceptance

Work is acceptable if at least 90 percent of the depth of the crack in each core is filled with adhesive.

3.3 CORROSION AND SURFACE REPAIR

3.3.1 Preparation

3.3.1.1 Identification of Extent of Concrete Removal

- a. Configure geometry of removal area to maximize the use of right-angle geometry, avoiding reentrant corners, and to obtain uniformity of depth. Determine the depth, location, and size of reinforcing bars prior to removal of concrete.
- b. Perform visual inspection and hammer tapping, chain drag sounding, or other methods acceptable by the Contracting Officer to identify cracked, delaminated, spalled, disintegrated, and otherwise unsound concrete for removal. Mark boundaries of repair area before concrete removal.
- c. Inspect the marked boundaries with the Contracting Officer prior to commencing with the concrete removal. Revise the repair area boundaries as instructed by the Contracting Officer.

3.3.1.2 Shoring and Formwork

- a. Provide shoring and formwork per the paragraph titled Formwork and Shoring.

3.3.1.3 Concrete Removal

- a. Remove concrete from repair areas to indicated depth and profile. Notify Contracting Officer if additional delaminated, fractured, or unsound concrete is present.
- b. Do not damage embedded reinforcing and adjacent concrete. The removal methods must produce minimal microcracking (bruising) of the prepared substrate surfaces. Avoid directly striking reinforcing steel with impact tools used for concrete removal.
- c. Provide perpendicular edges at perimeter of repair area. The perimeter of the repair areas must be saw cut to a depth of [0.50 to 0.75 in.](#) Do not cut or damage embedded reinforcement or other embedded items. If embedded reinforcing steel or other embedded items are too close to the

surface to provide the perpendicular edge cut, notify the Contracting Officer for direction before proceeding.

- d. Extend concrete removal along the corroded reinforcing steel to a point where there is no further delamination, concrete cracking, or reinforcing steel corrosion, and where the reinforcement is bonded to the surrounding concrete.
- e. Remove concrete around the exposed layer of reinforcement to a uniform depth beyond within the repair areas to provide a minimum clearance between exposed reinforcing steel and surrounding concrete of 0.75 in. or at least 0.25 in. larger than the maximum nominal size of the coarse aggregate in the repair material.
- f. Do not remove concrete behind vertical reinforcing bars in columns.

3.3.1.4 Preparation of Concrete Substrate Surface

- a. Confirm perpendicular edges at repair area perimeter, and reinstate if damaged by concrete removal process. Remove loosely bonded concrete, bruised or fractured concrete, and bond-inhibiting materials such as dirt, concrete slurry, or any other detrimental materials from the concrete substrate using approved methods. Where concrete has been removed by impact methods, abrasive blasting must be used to prepare the surface and remove bruised concrete.
- b. Provide substrate surface profiles as specified in the Contract Documents.
- c. Visually inspect and sound substrate surface to confirm that no further delaminations or otherwise unsound concrete remains. If encountered, notify the Contracting Officer.
- d. Clean the substrate per the paragraph titled Concrete preparation.

3.3.2 Application

3.3.2.1 Existing Reinforcement Preparation

- a. Clean existing reinforcement that will remain. Remove corrosion and/or other laitance and notify the Contracting Officer if section loss is greater than 20%.
- b. Permit evaluation of existing reinforcement and placement of new reinforcement by the Contracting Officer.

3.3.2.2 Placement of New Reinforcement

Place repair reinforcement in accordance with ACI 301.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

- a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support, and in accordance with ACI 301 and CRSI 10MSP. Do not use supports to support runways for concrete conveying equipment and similar

construction loads.

- b. Equip supports on ground and similar surfaces with sand-plates.
- c. Support welded wire reinforcement as required for reinforcing bars.
- d. Secure reinforcements to supports by means of tie wire. Wire must be black, soft iron wire, not less than 16 gage.
- e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.
- f. Bending of reinforcing bars partially embedded in concrete is permitted only as specified in the Contract Documents.

3.3.2.3 Placement of Other Repair Materials

- a. Equilibrate repair material(s) and substrate to the temperature, cleanliness of substrate and reinforcement, and moisture requirements of the repair material manufacturer's requirements.
- b. Comply with the repair material manufacturer's requirements for batching, mixing, placing and curing repair materials.
- c. Review consistency of the mixed repair material(s) relative to the parameters documented in the repair material manufacturer product data sheet. If non-conforming, adjust consistency in compliance with the repair material manufacturer's requirements.
- d. Apply or install repair material(s) within the application time frame (pot life) requirements of the repair material manufacturer's requirements, and place and consolidate to provide well-compacted repair.
- e. Finish and tool repair materials, finished in accordance with the repair material manufacturer's written instructions and as indicated in Contract Documents.
- f. Protect installed repair material(s) from damage, exposure to environmental conditions that are detrimental to the uncured or cured properties of the material. Cure in accordance with the requirements of the repair material manufacturer's requirements.

3.3.3 Welding

Welding must be in accordance with AWS D1.4/D1.4M.

3.3.4 Quality Control

- a. Protect concrete surfaces, beyond limits of surfaces receiving bonding agent adhesive, against spillage. Immediately remove any bonding agent adhesive that has spilled beyond desired area. Perform cleanup with

material designated by bonding agent adhesive manufacturer. Avoid contamination of work area.

- b. The bond strength between the existing concrete and the repair material must be a minimum of 250 psi per ASTM C1583/C1583M. Test a minimum of 3 specimens at locations no greater than 500 square yards of prepared surface.

-- End of Section --

SECTION 03 20 01.10

CONCRETE ANCHORS AND DOWELS- BID OPTION 1
04/06

PART 1 GENERAL

Work under this section covers reinforcement, dowels, anchors and miscellaneous appurtenances. Concrete anchors shall be tested for acceptance in accordance with this specification.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE INTERNATIONAL (ACI)

ACI 301 (2010; ERTA 2015) Specifications for Structural Concrete

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2009) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A276 (2013a) Standard Specification for Stainless Steel Bars and Shapes

ASTM A307 (2014) Standard Specification for Carbon Steel Bolts and Studs, 60 000 PSI Tensile Strength

ASTM A563 (2015) Standard Specification for Carbon and Alloy Steel Nuts

ASTM A615/A615M (2015a; E 2015) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM F844 (2007a; R 2013) Washers, Steel, Plain (Flat), Unhardened for General Use

ASTM E488/E488M (2015) Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements

State Of Maine, Department of Transportation (MaineDOT)

MaineDOT State of Maine, Department of Transportation, Standard Specifications

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Anchor Grout; G

Submit manufacturer's catalog cuts, manufacturer's instructions and product specifications.

Submit proposed materials, proportions, mixing and placing procedures, and curing times for cementitious grout.

Testing Equipment; G

SD-06 Test Reports

Testing Reports; G

SD-07 Certificates

Anchor Testing Personnel; G

1.4 DELIVERY AND STORAGE

Store materials delivered to the job-site in the original unopened containers within an appropriate facility capable of maintaining storage conditions consistent with the manufacturer's recommendations.

PART 2 PRODUCTS

2.1 ANCHORS

Reinforcement, dowels and anchors shall conform to the required specification sections 03 31 29 MARINE CONCRETE and 05 12 00 STRUCTURAL STEEL.

2.3 ANCHOR GROUT

Anchor grout shall be in accordance MaineDOT Qualified Products List of Anchoring Materials suitable for horizontal applications and approved by the Contracting Officer.

PART 3 EXECUTION

3.1 Drilled-In Anchors

Perform anchor installation in accordance with manufacturer instructions. Anchor embedment's shall be as shown on the drawings.

Drill holes with rotary impact hammer drills using carbide-tipped bits. Drill bits shall be of diameters as specified by the anchor manufacturer. All holes shall be drilled perpendicular to the concrete surface.

Clean all holes to remove loose material and drilling dust prior to

installation of grout. Holes shall be saturated surface dry. Pump or inject grout into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the grout. Sufficient grout shall be injected in the hole to ensure that the annular gap is filled to the surface. Install damp anchor into hole and remove excess grout from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load until the grout has cured for the specified time.

3.2 TESTING

Anchor testing shall be completed by an independent testing agency. Submit experience and qualification for the [Anchor Testing Personnel](#) and [Testing Equipment](#) prior to testing.

Test 10% of all concrete anchors. Test first reinforcing anchor installed. Anchor test locations shall be selected at random by the Contracting Officer. Anchor dowels shall be tested in tension utilizing the Incremental Load Application Test in accordance with [ASTM E488/E488M](#) Tension Testing. Proof load shall be maintained for a minimum of 10 minutes.

All tested anchors shall be tested to 70% of the anchor yield strength. The first five anchor dowels installed shall be tested to show appropriate materials and installation methods.

Anchors that fail testing shall be removed and replaced at no additional cost to the Government. Failure is considered one of the following:

1. Failure to reach Proof Load of 70% yield strength
2. A permanent set remains after load is released
3. Displacement greater than elastic stretch of the materials under required load.

After an anchor test failure, the next 5 anchors or more shall be tested as required by the Contracting Officer.

Contractor shall submit all [Testing Reports](#) to the Contracting Officer within 24 hours of test completion.

-- End of Section --

SECTION 03 31 29.10

MARINE CONCRETE

08/12

PART 1 GENERAL

This specification details the requirement for the design, production, placement, and testing of concrete required for this repair project. This is a specialty mix with an anticipated 50 year service life and requires development by a Concrete Specialist with experience in the development of specialty marine concrete mix designs. The mix design is required to have a low strength (f'c 4500-6500 psi) and high resistance to chloride ion penetration, high crack resistance and low shrinkage.

The Prime Contractor shall be responsible for ensuring that the concrete mix design shall be designed by a Concrete Specialist and that the concrete meets all testing requirements outlined in this specification.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 182	(2005; R 2012) Standard Specification for Burlap Cloth Made from Jute or Kenaf and Cotton Mats
AASHTO T 260	(1997; R 2011) Standard Method of Test for Sampling and Testing for Chloride Ion in Concrete and Concrete Raw Materials
AASHTO T 277	(2015) Standard Method of Test for Rapid Determination of the Chloride Permeability of Concrete

AMERICAN CONCRETE INSTITUTE INTERNATIONAL (ACI)

ACI 117	(2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 121R	(2008) Guide for Concrete Construction Quality Systems in Conformance with ISO 9001
ACI 201.2R	(2008) Guide to Durable Concrete
ACI 211.1	(1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete

ACI 214R	(2011) Evaluation of Strength Test Results of Concrete
ACI 301	(2010; ERTA 2015) Specifications for Structural Concrete
ACI 301M	(2010; ERTA 2015) Metric Specifications for Structural Concrete
ACI 304.2R	(1996; R 2008) Placing Concrete by Pumping Methods
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 305R	(2010) Guide to Hot Weather Concreting
ACI 306R	(2010) Guide to Cold Weather Concreting
ACI 308.1	(2011) Specification for Curing Concrete
ACI 309R	(2005) Guide for Consolidation of Concrete
ACI 318	(2014; Errata 1-2 2014; Errata 3-4 2015) Building Code Requirements for Structural Concrete and Commentary
ACI 318M	(2014) Building Code Requirements for Structural Concrete & Commentary
ACI 347	(2004; Errata 2008; Errata 2012) Guide to Formwork for Concrete
ACI 364.3R	(2009) Guide for Cementitious Repair Material Data Sheet
ACI SP-15	(2011) Field Reference Manual: Standard Specifications for Structural Concrete ACI 301-05 with Selected ACI References
ACI SP-2	(2007; Abstract: 10th Edition) ACI Manual of Concrete Inspection
ACI SP-66	(2004) ACI Detailing Manual

AMERICAN WELDING SOCIETY (AWS)

AWS D1.4/D1.4M	(2011) Structural Welding Code - Reinforcing Steel
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APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA PS 1	(2009) Structural Plywood (with Typical APA Trademarks)
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ASTM INTERNATIONAL (ASTM)

ASTM A496/A496M	(2007) Standard Specification for Steel Wire, Deformed, for Concrete Reinforcement
ASTM A497/A497M	(2007) Standard Specification for Steel Welded Wire Reinforcement, Deformed, for Concrete
ASTM A615/A615M	(2015a; E 2015) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A706/A706M	(2014) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement
ASTM A82/A82M	(2007) Standard Specification for Steel Wire, Plain, for Concrete Reinforcement
ASTM A934/A934M	(2013) Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars
ASTM C 1218/C 1218M	(1999; R 2008) Standard Specification for Water-Soluble Chloride in Mortar and Concrete
ASTM C1017/C1017M	(2013; E 2015) Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete
ASTM C1064/C1064M	(2011) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM C1074	(2011) Standard Practice for Estimating Concrete Strength by the Maturity Method
ASTM C1077	(2015) Standard Practice for Laboratories Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Laboratory Evaluation
ASTM C1107/C1107M	(2014a) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C1116/C1116M	(2010a; R 2015) Standard Specification for Fiber-Reinforced Concrete
ASTM C114	(2015) Standard Test Methods for Chemical Analysis of Hydraulic Cement
ASTM C1157/C1157M	(2011) Standard Specification for Hydraulic Cement
ASTM C1202	(2012) Standard Test Method for Electrical Indication of Concrete's Ability to Resist

Chloride Ion Penetration

ASTM C1218/C1218M	(1999; R 2008) Standard Specification for Water-Soluble Chloride in Mortar and Concrete
ASTM C1231/C1231M	(2014) Standard Practice for Use of Unbonded Caps in Determination of Compressive Strength of Hardened Concrete Cylinders
ASTM C1240	(2014) Standard Specification for Silica Fume Used in Cementitious Mixtures
ASTM C1260	(2014) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C138/C138M	(2014) Standard Test Method for Density ("Unit Weight"), Yield, and Air Content (Gravimetric) of Concrete
ASTM C143/C143M	(2012) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2015) Standard Specification for Portland Cement
ASTM C1567	(2013) Standard Test Method for Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
ASTM C157/C157M	(2008; R 2014; E 2014) Standard Test Method for Length Change of Hardened Hydraulic-Cement Mortar and Concrete
ASTM C1581/C1581M	(2009a) Standard Test Method for Determining Age at Cracking and Induced Tensile Stress Characteristics of Mortar and Concrete under Restrained Shrinkage
ASTM C1602/C1602M	(2012) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM C1611/C1611M	(2014) Standard Test Method for Slump Flow of Self-Consolidating Concrete
ASTM C171	(2007) Standard Specification for Sheet Materials for Curing Concrete
ASTM C172/C172M	(2014a) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2014) Standard Test Method for Air Content of Freshly Mixed Concrete by the

Volumetric Method

ASTM C192/C192M	(2015) Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C231/C231M	(2014) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2010a) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C294	(2012) Standard Descriptive Nomenclature for Constituents of Concrete Aggregates
ASTM C295/C295M	(2012) Petrographic Examination of Aggregates for Concrete
ASTM C309	(2011) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C31/C31M	(2012) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C311/C311M	(2013) Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete
ASTM C33/C33M	(2013) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2015a) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C42/C42M	(2013) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C494/C494M	(2013) Standard Specification for Chemical Admixtures for Concrete
ASTM C496/C496M	(2011) Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens
ASTM C595/C595M	(2015; E 2015) Standard Specification for Blended Hydraulic Cements
ASTM C618	(2012a) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C666/C666M	(2015) Resistance of Concrete to Rapid Freezing and Thawing

ASTM C78/C78M	(2015a) Standard Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading)
ASTM C805/C805M	(2013a) Rebound Number of Hardened Concrete
ASTM C856	(2014) Petrographic Examination of Hardened Concrete
ASTM C94/C94M	(2015) Standard Specification for Ready-Mixed Concrete
ASTM C989/C989M	(2014) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM D1179	(2010) Fluoride Ion in Water
ASTM D3867	(2009) Nitrite-Nitrate in Water
ASTM D512	(2012) Chloride Ion in Water
ASTM D516	(2011) Sulfate Ion in Water
ASTM D75/D75M	(2014) Standard Practice for Sampling Aggregates
ASTM E329	(2014a) Standard Specification for Agencies Engaged in the Testing and/or Inspection of Materials Used in Construction

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI 10MSP	(2009; 28th Ed) Manual of Standard Practice
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1.2 DEFINITIONS

- a. "Blending size" is an aggregate that complies with the quality requirements in ASTM C33/C33M and paragraph entitled "Aggregates" and as modified to produce a well graded combined grading.
- b. "Cementitious material" as used herein shall include portland cement and any pozzolanic material such as fly ash, natural pozzolans, ground granulated blast-furnace slag and silica fume.
- c. "Complementary cementing materials" (CCM) include coal fly ash, granulated blast-furnace slag, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in considerable improvement to sustainability, durability and in some cases a reduction in initial cost.
- d. "Concrete System" is the term describing a structural element comprised of concrete, reinforcing steel and concrete cover.
- e. "Design strength" (f'c) is the specified compressive strength of concrete at time(s) specified by Contracting Officer to meet structural

design criteria. Typical duration is 28 days; however, the Contracting Officer and Engineer of Record are encouraged to consider specifying strength at 56 or 90 days. For high-volume fly ash concrete mixtures, the duration shall be a minimum of 56 days.

- f. "Field test strength" (f_{cr}) is the required compressive strength of concrete to meet structural and durability criteria. Determine (f_{cr}) during mixture proportioning process.
- g. "High-volume fly ash concrete" has a minimum of 50 percent Class F fly ash as a partial replacement to portland cement.
- h. "Marine concrete" is all concrete that will be in contact with seawater or brackish water, tidal variations, splash, or spray from water in navigable waterways. All concrete shall be considered marine concrete unless otherwise noted.
- i. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, durability, and constructability requirements and of the project while minimizing the initial and life-cycle cost.
- j. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic yard or meter) of concrete.
- k. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.
- l. "Process control sampling" is sampling and testing conducted by the Contractor to monitor the quality of materials or processes. Process control sampling is intended to indicate the quality of materials at critical steps in production that allow intervention prior to using a material on the project.
- m. "Required compressive strength" (f'_{cr}) is the mean compressive strength of concrete required to meet structural criteria. The required strength is the mean concrete strength for tests of properly batched concrete at the age specified herein.
- n. "Test Section" is a slab or wall part of the main structure and constructed prior to main construction as an all inclusive demonstration of methods and materials. The adequacy of the Test Section must be approved by the Contracting Officer prior to construction of the project.

1.3 SUBMITTALS

The use of a "G" following a submittal indicates that a Government approval action is required. Submittals not having a "G" designation are for Contractor Quality Control Approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES, except as modified in this contract.

SD-01 Preconstruction Submittals

- Concrete Mixture Portions; G, CS
- Concrete Curing Plan; G
- Concrete Qualification Program; G
- Concrete Quality Control Program; G
- Concrete Placement and Compaction; G
- Concrete Pumping; G
- Curing concrete elements; G
- Form Removal Schedule; G
- Laboratory Qualifications; G
- Quality Control Personnel Qualifications; G
- Concrete Placement Plan; G

SD-02 Shop Drawings

- Formwork; G
- Reinforcing steel; G

Reproductions of contract drawings are unacceptable.

SD-03 Product Data

- Admixtures; G
- Air Entraining; G
- Aggregates; G
- Fiber Reinforcement; G
- Materials for curing concrete; G
- Material Safety Data Sheets; G
- Non-shrink grout; G
- Reinforcing Bars; G
- Crack Repair; G

SD-04 Samples

- Test Section; G

SD-05 Design Data

- Concrete Mixture Requirements; G
- Mixture designs; G

SD-06 Test Reports

- Air Entraining; G
- Aggregates; G
- Admixtures; G
- Cement; G
- Concrete Test Reports; G
- Complementary Cementing Materials; G
- Fresh Concrete Properties; G
- Hardened Concrete Properties; G
- Silica fume; G
- Reinforcing Bars; G
- Water; G

SD-07 Certificates

- Admixtures; G
- Cementitious Materials; G
- Cementitious material mill certificates; G
- Field testing technician and testing agency; G
- NRMCA Certificate Of Conformance or proof of NHDOT/MaineDOT approved Concrete Plant; G
- QC Inspection Checklist; G

SD-11 Closeout Submittals

- Aggregate Moisture Content; G
- Aggregate Sampling; G
- Concrete Test Reports; G
- Quality Control Charts; G
- Daily inspection reports; G
- Quality Team Meetings; G
- Sampling logs; G

1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with ACI publications except as modified herein. Consider the advisory or recommended provisions to be mandatory, as though the word "shall" had been substituted for the words "should" or "could" or "may," wherever they appear. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

1.5 DELIVERY, PLACING, STORAGE, AND HANDLING OF CONCRETE

Follow [ACI 301M](#) [ACI 301](#) and [ACI 304R](#) requirements and recommendations. Do not deliver concrete until forms, reinforcement, embedded items, and chamfer strips are in place, have been inspected, and are ready for concrete placement. Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground. Protect materials from contaminants such as grease, oil, and dirt. Ensure materials can be accurately identified after bundles are broken and tags removed.

1.6 CONCRETE QUALITY CONTROL

The objective of the [concrete quality control program](#) is to outline the procedures that will be used to construct a structure that will obtain the performance criteria outlined in this specification. Develop and submit for approval a concrete quality control program in accordance with the guidelines of [ACI 121R](#) and as specified herein. The plan shall include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. If concrete cylinders tested during production indicate inadequate strength, or inadequate mixing, then the Government may require concrete core samples from the hardened concrete for analysis at no additional expense to the Government to assure that the quality of the concrete as placed and cured will satisfy the defined service life.

Develop and submit for approval a concrete quality control program in accordance with the guidelines of [ACI 121R](#) and able to meet the defined service life using the methodology herein. Maintain a copy of [ACI SP-15](#) and [CRSI 10MSP](#) at the project site.

1.6.1 Quality Control Personnel

Submit for approval an organizational chart defining the quality control hierarchy, the responsibilities of the various positions, including the names and qualifications of the individuals in those positions.

1.6.1.1 Quality Manager Qualifications

The quality manager shall hold a current license as a professional engineer in a U.S. state or territory with experience on at least five (5) similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.6.1.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract shall be performed by an ACI Concrete Field Testing Technician Grade 1 qualified in accordance with [ACI SP-2](#) or equivalent. Equivalent certification programs shall include requirements for written and performance examinations as stipulated in [ACI SP-2](#).
- b. Testing agencies that perform testing services on [reinforcing steel](#) shall meet the requirements of [ASTM E329](#).
- c. Testing agencies that perform testing services on concrete materials shall meet the requirements of [ASTM C1077](#).

1.6.1.3 Additional Personnel

Also submit American Concrete Institute certification for the following:

- a. CQC personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians: ACI Concrete Field Testing Technician, Grade I.
- d. Laboratory Testing Technicians: ACI Concrete Strength Testing Technician and Laboratory Testing Technician, Grade I or II.
- e. Petrographer: Bachelor of Science degree in geology or petrography, trained in petrographic examination of concrete aggregate according to [ASTM C294](#) and [ASTM C295/C295M](#) and trained in identification of the specific deleterious processes and tests identified in this specification. Resume shall detail the education, training and experience related to the project-specific test methods and deleterious materials and shall be submitted at least 20 days before petrographic and deleterious materials examination is to commence.
- f. Concrete Batch Plant Operator: National Ready Mix Concrete Association (NRMCA) Plant Manager Certification at the Plant Manager level or Plant

Manager at NHDOT approved Concrete Plant. Submit [NRMCA Certificate of Conformance](#) or proof of NHDOT/MaineDOT approved Concrete Plant. The intent of the Plant Manager Certification is to ensure that the plant has a qualified person in charge of the concrete operations. In lieu of a Plant Manager Certification from NRMCA or DOT, a variance can be request during construction. The variance would need show substantial evidence (qualifications, training, and experience) that the plant manager is equivalently qualified. The final acceptance of the variance is up to the Contracting Officer.

1.6.2 [Laboratory Qualifications](#) for Concrete Qualification Testing

The concrete testing laboratory shall have the necessary equipment and experience to accomplish required testing. The laboratory shall meet the requirements of [ASTM C1077](#), be Cement and Concrete Reference Laboratory (CCRL) inspected.

1.6.3 Laboratory Accreditation

Laboratory and testing facilities shall be provided by and at the expense of the Contractor. The laboratories performing the tests shall be accredited in accordance with ASTM C1077, including [ASTM C78/C78M](#) and [ASTM C1260](#). The accreditation shall be current and shall include the required test methods, as specified.

- a. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by [ASTM C31/C31M](#).
- b. Contractor Quality Control: All sampling and testing shall be performed by an approved, onsite, independent, accredited laboratory.

1.6.3.1 [Fiber Reinforced Concrete](#)

Test to confirm in accordance with [ASTM C1116/C1116M](#).

1.7 CONCRETE DURABILITY

1.7.1 Concrete Mixture Proportions

The Prime Contractor shall submit preliminary and complete concrete mix designs within 45 days and 60 days of award, respectively. Concrete mix designs shall include concrete mixture proportions, ingredient material certificates and test data, and trial batch test data for each class of concrete proposed for use on the project. Submittal shall clearly indicate where each mixture will be used when more than one mixture design is submitted. The concrete mix design must be approved by Contracting Officer within 90 days of award.

1.7.2 Concrete Design Requirements

Proportion concrete mixtures to meet the requirements listed in Tables 1 and 2 in accordance with the procedures outlined in [ACI 201.2R](#) and [ACI 211.1](#).

Table 1 - Marine Concrete Mix Design Requirements		
Prescriptive Requirements	Minimum	Maximum
Water to cementitious material ratio		0.45
Nominal maximum aggregate size		3/4"
Air Content (1)	4.5%	7.5%
Cement Quality	505 lbs.	
Silica Fume Quantity		25 lbs.
Fly Ash Quantity (% of total cementitious material by weight)		30

(1) - Concrete that arrives onsite with an air content +/- 1.0% of the target will be accepted, and modifications will be made at the batch plant for subsequent concrete deliveries. Concrete with an air content outside the acceptable tolerance of +/- 1.0% will be rejected.

Table 2 - Marine Concrete Performance Requirements		
Property	Test Method	Criteria
Compressive Strength (f'c) @ 56 days	ASTM C39/C39M	4500 psi (minimum), 6500 maximum
Slitting Tensile Strength	ASTM C496/C496M	300 psi (minimum) @ 28 days
Drying Shrinkage	ASTM C157/C157M modified by ACI 364.3R	0.05% (maximum) @ 28 days
Restrained shrinkage cracking	ASTM C1581/C1581M	No cracks within 14 days
Resistance to the penetration of chloride ions, coulombs @ 56 days	ASTM C1202 OR AASHTO T 277	1500 coulombs (maximum)
Bond Strength	ICRI Guideline No.	75 psi (minimum) @ 28 days

1.7.3 Concrete Mixture Qualifications

1.7.3.1 Concrete Mixtures

- a. Concrete Mixture shall be designed to be low strength and minimize shrinkage. Factors that serve to minimize shrinkage in a concrete mixture are (but are not limited to): minimum cement content, maximum practical aggregate size, minimum sand content, minimum water content,

and minimum concrete mixture temperature.

- b. Submit complete ingredient material test data, including applicable reference specifications. Submit additional data regarding concrete aggregates if the source of aggregate changes.
- c. Submit copies of test reports by independent test lab conforming to [ASTM C1077](#) showing that the mixture has been successfully tested to produce concrete with the properties specified and that mixture will be suitable for the job conditions as described. Test reports shall be submitted along with the concrete mixture proportions. Obtain approval before concrete placement.
- d. Test trial batch of production concrete to establish the tolerance limits. If batching facilities are located such that the haul-time will exceed 30 minutes, a simulated haul time shall be included in the trial batch. Concrete for test batches shall be produced at the batch plant.
 - (1) Test and report fresh concrete property tests of each trial batch as follows:
 - (a) Slump in accordance with [ASTM C143/C143M](#).
 - (b) Air content in accordance with [ASTM C231/C231M](#) or [ASTM C173/C173M](#).
 - (c) Unit weight in accordance with [ASTM C138/C138M](#).
 - (d) Temperature in accordance with [ASTM C1064/C1064M](#).
 - (2) Cast specimens, test, and report hardened concrete property tests of each trial batch as follows:
 - (a) Compressive strength at 3, 7, 28 and 56 days in accordance with [ASTM C39/C39M](#). Use of unbonded caps in accordance with [ASTM C1231/C1231M](#) is permitted.
 - (b) Drying shrinkage may be determined from one batch.
 - (c) Tensile strength may be determined using specimens cast from one batch.
 - (d) Freeze-thaw durability factor shall be determined as the mean of six test specimens comprised of at least two specimens cast from each batch.
 - (e) Initial chloride ion content may be determined from one batch.
 - (3) Moist cure concrete intended for cast-in-place applications in accordance with the standard moist curing conditions described in [ASTM C192/C192M](#) unless otherwise specified.

1.7.3.2 Previously Approved Concrete Mixtures

For identical concrete mixtures previously approved for use within the past 12 months, the previous mixture qualification submittal may be re-submitted

without further trial batch testing if accompanied by:

- (1) A copy of prior approvals indicating project name, project number, and location.
- (2) Ingredient material test data conducted within 6 months of the submittal date.
- (3) Copies of previously approved trial batch test data.
- (4) A log containing at least fifteen sequential test results with the calculated mean and standard deviation of the production concrete for air content, unit weight, and compressive strength.

If the Contractor changes material, type, class, sources, or suppliers; chemical composition; and/or mix proportions, the Contractor shall provide a written opinion of the significance of the change(s). The change(s) may require additional testing at the discretion of the Contracting Officer.

1.7.4 Concrete Qualification Program

1.7.4.1 Fresh Concrete Properties

- a. Air Content: Concrete that is air entrained shall conform to the air limits specified in [ACI 301M ACI 301](#) for exposure and the aggregate size used and tested in accordance with [ASTM C231/C231M](#). Variations outside the limits specified shall not be reason to reject the concrete in locations not subject to freeze-thaw conditions.
- b. Slump: The concrete mixture shall be proportioned to have, at the point of deposit, a slump of [100 mm 4 inches](#) as determined by [ASTM C143/C143M](#) when admixtures that affect slump are not used. Where an [ASTM C494/C494M](#), Type F or G admixture is used, the slump after the addition of the admixture shall be [7 inches](#). Slump tolerances shall comply with the requirements of [ACI 117](#).

1.7.4.2 Hardened Concrete Properties

- a. Compressive Strength: The minimum compressive strength results at 56 days. Determine compressive strength (f'_{cr}) for qualification of concrete mixtures and for quality acceptance testing. A compressive strength test result is defined as the mean of three properly conducted tests on [100 by 200 mm 4 by 8 inch](#) cylinders in accordance with [ASTM C39/C39M](#). Alternatively and for concrete mixtures containing a maximum size aggregate greater than [25.4 mm 1 inch](#), a strength test result shall be defined as the mean of two properly conducted 56-day tests on [150 by 300 mm 6 by 12 inch](#) cylindrical specimens in accordance with [ASTM C39/C39M](#). In addition:
 - (1) Specified Compressive Strength: For structural concrete elements exposed in a marine environment, the minimum specified 56 day design strength is denoted as (f'_c).
 - (2) Required Average Strength: The concrete shall be proportioned such that the minimum required average compressive strength (f'_{cr}) exceeds the specified design strength (f'_c) as per [ACI 301M ACI 301](#).

- (3) If the average compressive strength is above or below the specified strength at the same age by more than 20 percent, the quality control process shall be reviewed by the Contracting Officer, Prime Contractor and concrete supplier to identify and alleviate issues and to ensure subsequent concrete deliveries are within the project specifications.
- (4) Strength of any individual concrete placement shall be considered satisfactory if the following requirement is met:
 - (a) The arithmetic mean of any three consecutive lot strength tests is between 0.9 and 1.2f'c, and;
- (5) In the event that a placement is represented by single sampling lot, strength shall be considered satisfactory if either:
 - (a) The mean of the initial test is between 1.0 and 1.2 f'c, or;
 - (b) The mean of the initial test and retest is between 1.0 and 1.2 f'c, and neither strength test result is less than 0.90 f'c.
- b. Drying Shrinkage: Determine drying shrinkage for qualification of concrete mixtures prior to the fabrication of the Test Section and from samples made during the fabrication of the Test Section (see the subpart entitled "Test Section"). All test results shall not exceed 0.05 percent. A drying shrinkage test result shall be the mean value from three or more individual specimens constituting a test set. If an individual specimen's measurements deviate from the mean value by more than 0.009 percent length change the specimen's measurements shall be discarded and a new average established. Casting more than three specimens for each set is permitted. Test procedures and test specimens shall conform to the following:

Drying shrinkage specimens, typically 75 by 75 by 285.8 mm 3 by 3 by 11.25 inch prisms for 25.4 mm 1 inch maximum size aggregate or smaller, shall be fabricated, cured, dried, and measured at 28 days in the manner delineated in ASTM C157/C157M except as modified by ACI 364.3R.
- c. Tensile strength: Determine splitting-tensile strength of concrete only for qualification of concrete mixtures. Determine and report the splitting-tensile strength result of each class of concrete in accordance with ASTM C496/C496M as the mean of three properly conducted tests at the age specified for f'c and again at 90-days age for information only.
- d. Freeze-thaw durability: Determine the freeze-thaw durability factor of concrete for qualification of concrete mixtures, if required by environmental conditions. Determine and report the freeze-thaw durability factor of each class of concrete in accordance with ASTM C666/C666M Method-A. Start testing after 28 days of moist curing. The minimum acceptable durability factor after 300 cycles of rapid freezing and thawing is 90 percent.
- e. Water Soluble Chloride Ion Content: Determine the chloride ion content only for qualification of concrete mixtures. Determine water soluble chloride ion content in accordance with ASTM C1218/C1218M, ASTM C114, or AASHTO T 260. The limits for allowable water-soluble chloride ion concentrations in hardened concrete are listed in Table 3.

Table 3 - Maximum water soluble Chloride Ion Content for Corrosion Protection	
Type of Member	Maximum water soluble chloride ion in concrete, percent by weight of cementitious material
Reinforced concrete exposed to chloride in service	0.08 percent

1.8 CONCRETE

1.8.1 Drawings

Fabrication Drawings for concrete formwork, reinforcement materials, precast elements, wall forms, and bulkhead forms must indicate concrete pressure calculations with both live and dead loads, along with material types. Provide design calculations by a registered Civil or Structural Engineer for the formwork.

1.8.1.1 Formwork

Prior to commencing work submit drawings for approval showing details of formwork including, but not limited to: joints, supports, studding and shoring, and sequence of form and shoring removal. Reproductions of contract drawings are unacceptable.

Design, fabricate, erect, support, brace, and maintain formwork so that it is capable of supporting without failure all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork.

ACI 347. Include design calculations indicating arrangement of forms, sizes, species, and grades of supports (lumber), panels, and related components. Indicate placement schedule, construction, and location and method of forming control joints. Include locations of inserts, pipe work, conduit, sleeves, and other embedded items. Furnish drawings and descriptions of shoring and reshoring methods.

1.8.1.2 Reinforcing Steel

ACI SP-66. Provide bending and cutting diagrams, assembly diagrams, splicing placement and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Only complete drawings will be accepted.

Provide coating manufacturer's and coating applicator's test data sheets certifying that applied coating meets the requirements of **ASTM A934/A934M**. Provide mill certificates indicating conformance with **ASTM A615/A615M** or **ASTM A706/A706M**.

1.8.1.3 Joints

Submit a plan indicating the type and location of each construction joint. Final joint locations are subject to Government approval.

1.8.2 Pre-Construction Submittals

1.8.2.1 Curing Concrete Elements

Submit proposed materials and methods for curing concrete elements.

1.8.2.2 Concrete Curing Plan

Submit proposed materials, methods, and duration for curing and cooling concrete elements in accordance with ACI 308.1.

Minimum moist curing duration shall be seven days, immediately followed by a 14-day chemical cure period.

Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

1.8.2.3 Form Removal Schedule

Submit schedule for form removal indicating element and minimum length of time for form removal. Submit technical literature of forming material or liner, form release agent, form ties, and gasketing to prevent leakage at form and construction joints. Provide a full description of materials and methods to be used to patch form-tie holes.

1.8.2.4 Concrete Placement and Compaction

- a. Submit technical literature for equipment and methods proposed for use in placing concrete. Include concrete pumping or conveying equipment including type, size and material for pipe, valve characteristics, and the maximum length and height concrete will be pumped. No adjustments shall be made to the mixture design to facilitate pumping.
- b. Submit technical literature for equipment and methods proposed for vibrating and compacting concrete. Submittal shall include technical literature describing the equipment including vibrator diameter, length, frequency, amplitude, centrifugal force, and manufacturer's description of the radius of influence under load. Where flat work is to be cast, provide similar information relative to the proposed compacting screed or other method to ensure dense placement.
- c. Submit Concrete Placement Plan detailing the dimensions, layout, and sequence of placements. Floor placements shall not have a width that exceeds 20 feet, unless otherwise noted or approved. Wall placements shall not have a height that exceeds 20 feet, unless otherwise noted or approved. Concrete demolition shall not exceed 2000 sqft at any time.

1.8.2.5 Concrete Report

Provide a Report inclusive of materials and methods used, test results, and the field test strength (fcr) for concrete that shows compliance with the structural and performance requirements.

1.8.2.6 Preconstruction Testing of Materials

All sampling and testing shall be performed by, and at the expense of, the Contractor. Use an approved commercial laboratory or, for cementitious materials and chemical admixtures, a laboratory maintained by the manufacturer of the material. No material shall be used until notice of acceptance has been given. The Contractor will not be entitled to any additional payment or extension of time due to failure of any material to meet project requirements, or for any additional sampling or testing required. Additional tests may be performed by the Government at the discretion of the Contracting Officer; such Government testing will not relieve the Contractor of any testing responsibilities.

1.8.2.7 Material Safety Data Sheets

Submit Material Safety Data Sheets (MSDS) for all materials that are regulated for hazardous health effects. Prominently post the MSDS at the construction site.

1.8.2.8 Mixture Designs

Provide a detailed report of materials and methods used, test results, and the field test strength (fcr) for marine concrete required to meet structural and durability requirements.

1.8.3 Sampling

Concrete production process control sampling and testing shall be conducted in compliance with this specification.

1.8.3.1 Ingredient Material Sampling

- a. [Cementitious material mill certificates](#) and test reports shall be provided for each shipment. Record the date delivered and quantity of material represented by the certificate.
- b. Conduct and log [aggregate moisture content](#) at a minimum frequency of twice daily for each day's production. Use of moisture sensors in storage bins is recommended practice, but does not satisfy this requirement.
- c. [Aggregate sampling](#) for gradation and dry-rodded unit weight shall be conducted for each 100 tons delivered for use on the project, or portion thereof.

1.8.4 Reporting

1.8.4.1 Daily Inspection Reports

Prepare daily inspection reports for all inspection activities such as base preparation, formwork preparation, reinforcement installation, concrete placement log, and temperature control activities. Submit sample forms and describe the procedure used to organize, archive, and retrieve inspection records in the Quality Program submittal.

1.8.4.2 Sampling Logs

Maintain a concrete placement log as an electronic spreadsheet or database identifying each placement date, placement location, volume of concrete, batch ticket numbers, lot identification code, fresh concrete properties, compressive strength results, transport properties, inspection comments, and acceptance status. Provide/transmit the concrete testing log to the Contracting Officer weekly. Provide copies of supporting documents for any placement requested by the Contracting Officer immediately upon request.

1.8.4.3 Quality Control Data

Prepare, maintain, and report separate [quality control charts](#) illustrating the slump, temperature, air content, and compressive strength test results for each lot of each concrete mixture used on the project.

1.8.4.4 Quality Team Meetings

Conduct regular quality control team meetings to review plans for future placements, review test results, and discuss dispensation of non-conforming materials. The quality team shall include the Contractor's quality manager, the project manager, the project superintendent, the Contracting Officer, and representatives of the testing agency and concrete producer, or approved substitutes.

Prepare quality control team meeting minutes for each meeting. The minutes shall include the date of each meeting, attendees, key discussion points, findings, recommendations, assigned tasks, assigned personnel, task completion dates and status of each task.

1.8.4.5 Non-conforming materials

The exact location of non-conforming concrete as placed shall be identified and the Contracting Officer shall be notified immediately. There are numerous possible indicators that the as-placed concrete is non-conforming including (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, and concrete delivery records that indicate excessive time between mixing and placement and/or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing to quantify the concrete properties. If justified, cores may be extracted for testing, and an investigation into the cause for non-conformance shall be conducted.

Investigations into non-conforming materials shall be conducted at the Contractor's expense. The Contractor shall be responsible for the investigation and shall make written recommendations to adequately mitigate or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

1.8.5 Test Reports

[Concrete Test Reports](#) shall be identified by a sequential report identification code. Each report shall identify the placement date, placement location, weather, name of testing technician, time of sampling, batch ticket number, fresh concrete test results, and hardened concrete test results.

1.8.5.1 Concrete Mixture Requirements

- a. Submit copies of test reports conforming to ASTM C1077 showing that the mixture has been successfully tested to produce concrete with the properties specified and that mixture will be suitable for the job conditions. Test reports shall be submitted along with the concrete mixture proportions. Obtain approval before concrete placement.
- b. Fully describe the processes and methodology whereby mixture proportions were developed and tested and how proportions will be adjusted during progress of the work to achieve, as closely as possible, the designated levels of relevant properties.

1.8.5.2 Complementary Cementing Materials

Submit test results in accordance with ASTM C618 and the physical and chemical analysis in accordance with applicable ASTM standards such as ASTM C311/C311M for fly ash. Submit test results performed within 6 months of submittal date. Update this report during construction as necessary to assure that the complementary cementing materials used on the projects meets the ASTM criteria and the report on file is never older than 6 months.

1.8.5.2.1 Ground Granulated Blast-Furnace Slag

Submit test results in accordance with ASTM C989/C989M for ground granulated blast-furnace slag. Submit test results performed within 6 months of submittal date. Update this report during construction as necessary to assure that the report on file is never older than 6 months.

1.8.5.2.2 Ultra Fine Fly Ash or Pozzolan

Submit test results in accordance with ASTM C618 as a Class F fly ash or Class N pozzolan with the following additional requirements:

- a. The strength activity index at 28 days shall be at least 95 percent of the control.
- b. The average particle size shall not exceed 6 microns.
- c. The sum SiO_2 plus Al_2O_3 plus Fe_2O_3 shall be greater than 77 percent.

Submit test results performed within 6 months of submittal date. Update this report during construction as necessary to assure that the report on file is never older than 6 months.

1.8.5.3 Silica Fume

Submit test results in accordance with ASTM C1240 for silica fume. Data shall be based upon tests performed within 6 months of submittal. Update this report during construction as necessary to assure that the report on file is never older than 6 months.

1.8.5.4 Aggregates

Aggregate samples shall be obtained in accordance with ASTM D75/D75M and shall be representative of the materials to be used for the project. Submit test results for aggregate quality in accordance with ASTM C33/C33M,

and the combined gradation curve proposed for use in the work and used in the mixture qualification, and [ASTM C295/C295M](#) for results of petrographic examination. Confirm that the potential for alkali-silica reaction are within allowable limits by conducting tests in accordance with [ASTM C1260](#). Maximum allowable expansion is 0.08 percent at 28 days. Submit results of all tests during progress of the work in tabular and graphical form as noted above, describing the cumulative combined aggregate grading and the percent of the combined aggregate retained on each sieve.

1.8.5.5 [Admixtures](#)

Submit test results in accordance with [ASTM C494/C494M](#) and [ASTM C1017/C1017M](#) for concrete admixtures, [ASTM C260/C260M](#) for air-entraining admixture, and manufacturer's literature and test reports for corrosion inhibitors and anti-washout admixture. Submitted data shall be based upon tests performed within 6 months of submittal. Submit certified copies of test results for the specific lots or batches to be used on the project. Test results shall be not more than 6 months old prior to use in the work. Chemical admixtures that have been in storage at the project site for longer than 6 months or that has been subjected to freezing will be retested at the expense of the Contractor.

1.8.5.6 [Portland Cement](#)

Portland cement, ground granulated blast furnace (GGBF) slag, and pozzolan will be accepted on the basis of manufacturer's certification of compliance, accompanied by mill test reports showing that the material in each shipment meets the requirements of the specification under which it is furnished. Mill test reports shall be no more than 1 month old, prior to use in the work. No cementitious material shall be used until notice of acceptance has been given by the Contracting Officer. Cementitious material may be subjected to check testing by the Government from samples obtained at the mill, at transfer points, or at the project site. If tests prove that a cementitious material that has been delivered is unsatisfactory, it shall be promptly removed at Contractor's expense from the site of the work. Cementitious material that has not been used within 6 months after testing shall be retested at the Contractor's expense and shall be rejected if test results are not satisfactory. Submit test results in accordance with [ASTM C150/C150M](#) portland cement and/or [ASTM C595/C595M](#) and [ASTM C1157/C1157M](#) for blended cement.

1.8.5.7 [Testing During Construction](#)

During construction, the Contractor is responsible for sampling and testing aggregates, cementitious materials, and concrete as specified herein. The Government will sample and test concrete and ingredient materials as considered appropriate. Provide facilities and labor as may be necessary for procurement of representative test samples. Testing by the Government will in no way relieve the Contractor of the specified testing requirements.

1.8.5.8 [Test Section](#)

- a. **Vertical Placements:** Prior to Full-scale demolition activities construct a Test Section, at same location as the demolition Test Section, that represents the vertical placements at the construction site. Test Section shall meet all specification requirements and being acceptable to the Contracting Officer in all respects including, but not limited to, delivery time, placement, consolidation, curing and

surface texture. Use the Test Section to develop and demonstrate to the satisfaction of the Contracting Officer the proposed techniques of mixing, hauling, placing, consolidating, finishing, curing, initial saw cutting, start-up procedures, testing methods, plant operations, and the preparation of the construction joints. The mixing plant shall be operated and equipment calibrated prior to start of placing the Test Section. Use the same equipment, materials, and construction techniques on the Test Section as will be used in all subsequent work. Concrete production, placing, consolidating, curing, construction of joints, and all testing shall be in accordance with applicable provisions of this specification. At a minimum of three days after completion of the Test Section, extract a sufficient number of concrete cores to evaluate homogeneity, consolidation, segregation and transport properties. If any of the Test Results are unacceptable the Contracting Officer may require that a new Test Section be accomplished at no additional cost to the Government. Once accepted by the Contracting Officer, this Test Section will become part of the repaired area.

1.8.5.9 Acceptability of Work

The materials and the structure itself will be accepted on the basis of tests made by the Contractor and shall be in compliance with the criteria herein. The Government may make check tests at its expense to validate the results of the Contractor's testing. Testing performed by the Government will in no way relieve the Contractor from the specified testing requirements.

PART 2 PRODUCTS

2.1 CEMENTITIOUS MATERIALS

Cementitious materials shall be portland cement or cement blended with complementary cementing materials. New submittals are required when the cementitious materials change sources or types.

Provide cementitious materials meeting the requirements of the applicable specification, and as modified herein. Provide mill certificates and test results conducted within six-months of the submittal date as part of the concrete mixture qualification submittal.

Provide a single manufacturer of cementitious material for each type of cement and complementary cementing materials supplied to the project.

2.1.1 Portland Cement

Provide portland cement conforming to **ASTM C150/C150M**, Type I/II low-alkali with tri-calcium aluminates (C3A) content between 6 and 8 percent and a maximum cement-alkali content of 0.60 percent Na₂O_e (sodium oxide) equivalent.

ASTM C150/C150M cements shall be combined with complementary cementing materials in the concrete mixture.

2.1.2 Blended Cements

Blended cement shall conform to **ASTM C595/C595M**, Type IP or IS, including the optional requirement for mortar expansion and consist of a mixture of

ASTM C150/C150M Type I, or Type II cement and a complementary cementing material. The slag added to the Type IS blend shall be ASTM C989/C989M ground granulated blast-furnace slag. The pozzolan added to the Type IP blend shall be ASTM C618 Class F and shall be interground with the cement clinker. The manufacturer shall state in writing that the amount of pozzolan in the finished cement will not vary more than plus or minus 5 mass percent of the finished cement from lot-to-lot or within a lot. The percentage and type of mineral admixture used in the blend shall not change from that submitted for the aggregate evaluation and mixture proportioning.

2.1.3 Pozzolan

2.1.3.1 Fly Ash

Fly ash shall conform to ASTM C618, Class F, including the optional requirements for uniformity and effectiveness in controlling Alkali-Silica reaction and shall have a loss on ignition not exceeding 6 percent. Class F fly ash for use in mitigating Alkali-Silica Reactivity shall have a Calcium Oxide (CaO) content of less than 8 percent and a total equivalent alkali content less than 1.5 percent. Add with cement.

2.1.3.2 Raw or Calcined Natural Pozzolan

Natural pozzolan shall be raw or calcined and conform to ASTM C618, Class N, including the optional requirements for uniformity and effectiveness in controlling Alkali-Silica reaction and shall have an on ignition loss not exceeding 3 percent. Class N pozzolan for use in mitigating Alkali-Silica Reactivity shall have a Calcium Oxide (CaO) content of less than 13 percent and total equivalent alkali content less than 3 percent.

2.1.3.3 Ultra Fine Fly Ash and Ultra Fine Pozzolan

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) shall conform to ASTM C618, Class F or N, and the following additional requirements:

- a. The strength activity index at 28 days of age shall be at least 95 percent of the control specimens.
- b. The average particle size shall not exceed 6 microns.
- c. The sum of $\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$ shall be greater than 77 percent.

2.1.4 Ground Granulated Blast-Furnace (GGBF) Slag

Ground Granulated Blast-Furnace Slag shall conform to ASTM C989/C989M, Grade 100 or Grade 120. Add with cement.

2.1.5 Silica Fume

Silica fume shall conform to ASTM C1240, including the optional limits on reactivity with cement alkalis. Silica fume may be furnished as a dry, densified material or as slurry. Proper mixing is essential to accomplish proper distribution of the silica fume and avoid agglomerated silica fume which can react with the alkali in the cement resulting in premature and extensive concrete damage. Supervision at the batch plant, finishing, and curing is essential. Provide at the Contractor's expense the services of a manufacturer's technical representative, experienced in mixing,

proportioning, placement procedures, and curing of concrete containing silica fume. This representative must be present on the project prior to and during at least the first 4 days of concrete production and placement using silica fume. A High Range Water Reducer (HRWR) shall be used with silica fume. Finishing may be more difficult. Proper curing is essential because there is a tendency for plastic shrinkage cracking.

2.1.6 Complementary Cementing Materials (AKA Supplementary Cementitious Materials (SCM) Content (SCM))

The concrete mix shall always contain complementary cementing materials whether or not the aggregates are found to be reactive in accordance with the subpart entitled "Aggregates". Concrete mixtures shall be designed and proportioned to meet the requirements for strength, constructability, shrinkage, etc.

2.2 AGGREGATES

Comply with [ASTM C33/C33M](#) Class 4S, except as modified herein.

The quantities to be retained on each sieve may be adjusted only where available aggregates are elongated or slivered and cause interference with mix mobility, or available aggregate gradations do not comply with the 18-8 requirement. When necessary to satisfy local conditions and when permitted, the combined aggregate percentages may be changed to not more than 22 percent nor less than 6 percent retained on any individual sieve. The combined aggregates in the mixture (coarse, intermediate, and fine) shall be well graded with no more than 18 percent nor less than 8 percent of the combined aggregate retained on any individual sieve, unless satisfactory performance can be demonstrated. The [300 micrometers No. 50](#) sieve may have less than 8 percent retained; sieves finer than [300 micrometers No. 50](#) shall have less than 8 percent retained, and the coarsest sieve may have less than 8 percent retained. Use intermediate sizes for blending where necessary, to provide a well graded combined aggregate.

- a. Provide gradation of individual aggregate sizes using standard concrete aggregate sieves including [37.5 mm 1-1/2 inches](#), [25 mm one inch](#), [19 mm 3/4 inch](#), [12.5 mm 1/2 inch](#), [9.5 mm 3/8 inch](#), [4.75 mm No. 4](#), [2.36 mm No. 8](#), [1.18 mm No. 16](#), [600 micrometers No. 30](#), [300 micrometers No. 50](#), and [150 micrometers No. 100](#).
- b. Provide aggregates for exposed concrete from one source. Aggregate reactivity shall be limited per the subpart entitled "Aggregates". Provide aggregate containing no deleterious material properties as identified by [ASTM C295/C295M](#).
- c. Where a size designation is indicated, that designation indicates the nominal maximum size of the coarse aggregate.
- d. Aggregate tests shall be conducted within 6 months from the date of concrete mixture submittal.
- e. Provide [ASTM C1260](#) test results conducted within 6 months of the submittal date showing the proposed coarse and fine aggregates are either: innocuous to alkali silica reaction; or that reactivity has been mitigated by the proposed cementitious materials as modified herein. Conduct [ASTM C1260](#) tests on each aggregate source separately.

Fine and coarse aggregates to be used in all concrete shall be evaluated and tested for alkali-aggregate reactivity. Both coarse aggregate size groups shall be tested. Test results of the individual aggregates shall have a measured expansion equal to or less than 0.08 percent after 28 days of immersion in a 1N 1M NaOH solution at 176 degrees.

- f. Should the test data indicate an expansion of greater than 0.08 percent, the aggregate(s) shall be rejected or additional testing shall be performed as follows: utilize the Contractor's proposed low alkali portland cement, blended cement, and/or complementary cementing materials, in combination with each individual aggregate. If SCMs are being evaluated, the testing shall be in accordance with ASTM C1567. Determine the quantity that will meet all the requirements of these specifications and that will lower the expansion equal to or less than 0.08 percent after 28 days of immersion in a 1N NaOH solution. Mixture proportioning shall be based on the highest percentage of SCM required to mitigate ASR-reactivity.
- g. Perform ASTM C1260 and ASTM C1567 tests simultaneously in order to expedite test results, submittal review and mix design approval.

2.3 WATER

Water shall be potable and free from injurious amounts of oils, acids, alkalis, salts, and organic materials. Where non-potable water or water from reprocessed concrete is proposed for use in the work, submit results of tests in accordance with ASTM C1602/C1602M. Submit test results in accordance with ASTM D512 and ASTM D516.

2.4 ADMIXTURES

- a. Provide certifications that chemical admixtures comply with the requirements shown in Table 4 and are compatible with each other. Use admixtures in accordance with manufacturer's recommendations, as appropriate for the climatic conditions and construction needs.
- b. Do not use calcium chloride or admixtures containing chloride ion content in more than trace amounts from impurities in admixture ingredients or potable water. Provide maximum concentrations of corrosion-inducing chemicals as shown in Table 4.

Table 4 - Limits on Corrosion-Inducing Chemicals		
Chemical*	Limits, Percent**	Test Method
Chlorides	0.10	ASTM D512
Fluorides	0.10	ASTM D1179
Nitrates	0.17	ASTM D3867
* Limits refer to water-soluble chemicals		

Table 4 - Limits on Corrosion-Inducing Chemicals		
Chemical*	Limits, Percent**	Test Method
** Limits are expressed as a percentage of the mass of the total cementitious materials.		

- c. The total alkali contribution of chemical admixtures shall not increase the total sodium-oxide equivalent content of the concrete mixture by more than 0.3 kg/m³ 0.5 lb/yd³.

2.4.1 Air Entraining

Provide air entraining admixtures conforming to ASTM C260/C260M.

2.4.2 Accelerating

ASTM C494/C494M, Type C.

2.4.3 Retarding

ASTM C494/C494M, Type B, D, or G.

2.4.4 Water Reducing

ASTM C494/C494M, Type A, E, or F.

High Range Water Reducer (HRWR) shall be ASTM C494/C494M, Type F and ASTM C1017/C1017M.

2.4.5 Fiber Reinforcement

Provide fiber reinforced concrete in accordance with ASTM C1116/C1116M, Type III, synthetic fiber-reinforced concrete, and as follows. Synthetic reinforcing fibers shall be 100 percent virgin polypropylene fibrillated fibers. Fibers shall have a specific gravity of 0.9, a minimum tensile strength of 70 ksi, graded per manufacturer, and specifically manufactured to an optimum gradation for use as concrete secondary reinforcement. A minimum of 1.5 pounds of fibers per cubic yard of concrete shall be used. Fibers shall be added at the batch plant. Provide the services of a qualified technical representative to instruct the concrete supplier in proper batching and mixing of materials to be provided. Submit product data for fiber reinforcement.

2.5 NON-SHRINK GROUT

ASTM C1107/C1107M.

2.6 MATERIALS FOR FORMS

Provide wood, plywood, or steel. Use plywood or steel forms where a smooth form finish is required. Lumber shall be square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects.

Plywood: APA PS 1, B-B concrete form panels or better. Steel form

surfaces shall not contain irregularities, dents, or sags.

2.6.1 Form Ties and Form-Facing Material

- a. Provide a form tie system that does not leave mild steel after break-off or removal any closer than 50 mm 3 inches from the exposed surface. Do not use wire alone. Form ties and accessories shall not reduce the effective cover of the reinforcement.
- b. Form-facing material shall be structural plywood or other material that can absorb air and some of the high water-cementitious materials ratio surface paste that may be trapped in pockets between the form and the concrete. Maximum reuse is three times. Provide forms with a form treatment to prevent bond of the concrete to the forms. Use a controlled permeability form liner in strict accordance with the manufacturer's recommendations.

2.7 REINFORCEMENT

2.7.1 Reinforcing Bars (Epoxy Coated Purple)

ACI 301MACI 301 unless otherwise specified. Bars shall be ASTM A615/A615M or ASTM A706/A706M and epoxy coated. Epoxy coating shall be in accordance with ASTM A934/A934M (purple).

2.8 ACCESSORY MATERIALS

2.8.1 Materials for Curing Concrete

2.8.1.1 Impervious Sheeting

ASTM C171; waterproof paper, clear or white polyethylene sheeting, or polyethylene-coated burlap.

2.8.1.2 Pervious Sheeting

AASHTO M 182 or carpet covering the free surface and kept continuously wet throughout the curing period.

2.8.1.3 Liquid Membrane-Forming Compound

Comply with ASTM C309, white-pigmented, Type 2, Class B.

2.9 Urethane Grout Injection

Urethane grout for use in sealing cracks. Product shall be capable of stopping water flow and suitable on both horizontal and vertical surfaces.

PART 3 EXECUTION

3.1 FORMS

- a. Provide formwork with clean-out openings to permit inspection and removal of debris. Formwork shall be gasketed or otherwise rendered sufficiently tight to prevent leakage of paste or grout under heavy, high-frequency vibration. Use a release agent that does not cause surface dusting. Inspect formwork between uses and replace frequently

to prevent irregular surfaces of the finished concrete. Reuse may be further limited by the Contracting Officer if it is found that the pores of the plywood are clogged with paste so that the wood does not absorb air and some of the high water-cementitious materials ratio paste that may be trapped in pockets between the form and the concrete.

- b. Patch form tie holes with a non shrink patching material in accordance with the manufacturer's recommendations and subject to approval.

3.1.1 Coating

Before concrete placement, coat the contact surfaces of forms with a non staining mineral oil, non staining form coating compound, or two coats of nitrocellulose lacquer.

3.1.2 Removal of Forms and Supports

After placement, concrete shall be moist cured for a minimum of 7 days. Formwork shall remain in place during the 7 day moist cure time period. Prevent concrete damage during form removal. Allow 14 days between adjacent concrete placements.

3.1.3 Reshoring

Do not allow construction loads to exceed the superimposed load which the structural member, with necessary supplemental support, is capable of carrying safely and without damage. Reshore concrete elements where forms are removed prior to the specified time period. Do not permit elements to deflect or accept loads during form stripping or reshoring. Perform reshoring operations to prevent subjecting concrete members to overloads, eccentric loading, or reverse bending. Reshoring elements shall have the same load-carry capabilities as original shoring and shall be spaced similar to original shoring. Firmly secure and brace reshoring elements to provide solid bearing and support.

3.2 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

ACI 301 **MACI 301**. Remove rust, scale, oil, grease, clay, or foreign substances from reinforcing. Do not tack weld. Inspect and verify proper reinforcement grade, quantity, spacing, and clearance requirements prior to concrete placement.

3.2.1 Epoxy Coated Reinforcing and Accessories

Shall meet the requirements of **ASTM A934/A934M** including Appendix X2, "Guidelines for Job Site Practices" except as otherwise specified herein.

3.2.1.1 Epoxy Coated Reinforcing Steel Delivery, Handling, and Storage

Record coating lot on each shipping notice and carefully identify and re-tag bar bundles from bending plant. Provide systems for handling coated bars which have padded contact areas, nylon slings, etc., all free of dirt and grit. Lift bundled coated bars with strong back, multiple supports, or platform bridge to prevent sagging and abrasion. Bundling bands shall be padded where in contact with bars. Do not drop or drag bars or bundles. Store coated bars both in shop and in field, aboveground, on wooden or padded cribbing. Space the dunnage close enough to prevent excessive

sags. Stack large quantities of straight bars with adequate protective blocking between layers. Schedule deliveries of epoxy coated bars to the job site to avoid the need for long term storage. Protect from direct sunlight and weather. Bars to be stored longer than 12 hours at the job site shall be covered with opaque polyethylene sheeting or other suitable equivalent protective material.

3.2.1.2 Epoxy coated Steel Reinforcing Steel Placement and Coating Repair

Carefully handle and install bars to minimize job site patching. Use the same precautions as described above for delivery, handling, and storage when placing coated reinforcement. Do not drag bars over other bars or over abrasive surfaces. Keep bar free of dirt and grit. When possible, assemble reinforcement as tied cages prior to final placement into the forms. Support assembled cages on padded supports. It is not expected that coated bars, when in final position ready for concrete placement, will be completely free of damaged areas; however, excessive nicks and scrapes which expose steel will be cause for rejection. Inspect for defects and provide required repairs prior to assembly. After assembly, reinspect and provide final repairs.

- a. Immediately prior to application of the patching material, any rust and debonded coating shall be manually removed from the reinforcement by suitable techniques employing devices such as wire brushes and emery paper. Care shall be exercised during this surface preparation so that the damaged areas are not enlarged more than necessary to accomplish the repair. Damaged areas shall be clean of dirt, debris, oil, and similar materials prior to application of the patching material.
- b. Repair and patching shall be done in accordance with the patching material manufacturer's recommendations. These recommendations, including cure times, shall be available at the job site at all times.
- c. Allow adequate time for the patching materials to cure in accordance with the manufacturer's recommendation prior to concrete placement.
- d. Rinse placed reinforcing bars with fresh water to remove chloride contamination prior to placing concrete.

All epoxy coated bars or accessories placed adjacent to existing uncoated steel shall have a minimum separation of 2 inches to provide electrical isolation.

3.2.2 Reinforcing Supports

Place reinforcement and secure with nonmetallic chairs, spacers, or epoxy coated metal hangers.

ASTM A934/A934M. Epoxy-coated reinforcing bars supported from formwork shall rest on coated wire bar supports. Reinforcing bars used as support bars shall be epoxy coated. Spreader bars, where used, shall be epoxy coated. Proprietary combination bar clips and spreaders used in construction with epoxy-coated reinforcing bars shall be epoxy coated. Epoxy-coated bars shall be tied with plastic-coated tie wire; or other materials acceptable to the Contracting Officer.

3.2.3 Splicing

As indicated. For splices not indicated, comply with **ACI 301M ACI 301**. Do not splice at points of maximum stress. Welded splices shall comply with **AWS D1.4/D1.4M** and be approved prior to use.

3.2.4 Future Bonding

Plug exposed, threaded, mechanical reinforcement bar connectors with a greased bolt. Bolt threads shall match the connector. Countersink the connector in the concrete. Caulk the depression after the bolt is installed.

3.2.5 Cover

Concrete cover over steel reinforcement shall be 3 inches unless otherwise noted. Use **ACI 117** to determine allowable tolerances for the placement of the steel.

3.2.6 Setting Miscellaneous Material

Place and secure anchors, bolts, pipe sleeves, conduits, and other such items in position before concrete placement. Plumb anchor bolts and check location and elevation. Temporarily fill voids in sleeves with readily removable material to prevent the entry of concrete. Electrically isolate exposed steel work and its anchor systems from the primary steel reinforcement with at least **50 mm 2 inches** of concrete. Take particular care to ensure against corrosion on edges and horizontal surfaces.

3.2.7 Construction Joints

Locate joints to least impair strength. Continue reinforcement across joints unless otherwise indicated. Final joint locations are subject to Government approval or substantiating calculations from the Contractor.

3.3 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

ASTM C94/C94M, ACI 301M ACI 301, and **ACI 304R**, except as modified herein. Batching equipment shall be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch tickets imprinted with mix identification, batch size, batch design and measured weights, moisture in the aggregates, and time batched for each load of ready mix concrete. When a pozzolan is batched cumulatively with the cement, it shall be batched after the cement has entered the weight hopper.

3.3.1 Measuring

Make measurements at intervals as specified in subparts entitled "Sampling" and "Field Quality Control."

Adjust batch proportions to replicate the mixture design using methods provided in the approved quality assurance plan. Base the adjustments on results of tests of materials at the batch plant for use in the work. Maintain a full record of adjustments and the basis for each.

3.3.2 Mixing

Comply with **ASTM C94/C94M** and **ACI 301M ACI 301**. If time of discharge exceeds time required by **ASTM C94/C94M**, submit a request along with description of precautions to be taken.

3.3.3 Transporting

Comply with **ACI 304R**.

3.4 PLACING CONCRETE

Comply with **ACI 304R** and **ACI 304.2R**. Place concrete as soon as practicable after the forms and the reinforcement have been inspected and approved. Do not place concrete when weather conditions prevent proper placement and consolidation; in uncovered areas during periods of precipitation; or in standing water. Prior to placing concrete, remove dirt, construction debris, water, snow, and ice from within the forms. Deposit concrete as close as practicable to the final position in the forms. Do not exceed a free vertical drop of **one m 3 feet** from the point of discharge. Place concrete in one continuous operation from one end of the structure towards the other or lifts for vertical construction.

3.4.1 Vibration

Comply with the requirements of **ACI 309R** using vibrators with a minimum frequency of 9000 vibrations per minute (VPM). Use only high cycle or high frequency vibrators. Motor-in-head 60 cycle vibrators shall not be used. For walls use a minimum of two vibrators with the first to melt down the mixture and the second to thoroughly consolidate the mass. Provide a spare vibrator at the casting site whenever concrete is placed. Place concrete in **500 mm 18 inch** maximum vertical lifts. Insert and withdraw vibrators approximately **500 mm 18 inches** apart. Penetrate at least **200 mm 8 inches** into the previously placed lift with the vibrator when more than one lift is required. Extract the vibrator using a series of up and down motions to drive the trapped air out of the concrete and from between the concrete and the forms.

For slab construction use vibrating screeds designed to consolidate the full depth of the concrete. Vibrators shall be equipped with rubber vibrator heads.

3.4.2 Cold Weather

Comply with **ACI 306R**. Do not allow concrete temperature to decrease below **10 degrees C 50 degrees F**. Obtain approval prior to placing concrete when ambient temperature is below **4 degrees C 40 degrees F** or when concrete is likely to be subjected to freezing temperatures within 24 hours. Placement of concrete shall be halted whenever the ambient temperature drops below **5 degrees C 40 degrees F**. When the ambient temperature is less than **10 degrees C 50 degrees F** the temperature of the concrete when placed shall be not less than **10 degrees C 50 degrees F** or more than **25 degrees C 75 degrees F**. Heating of the mixing water or aggregates may be necessary to regulate the concrete placing temperature. An accelerating admixture may be used when the ambient temperature is below **10 degrees C 50 degrees F**. Covering and other means shall be provided for maintaining the concrete at a temperature of at least **10 degrees C 50 degrees F** for not less than 7

days after placing, and at a temperature above freezing for the remainder of the curing period..

3.4.3 Hot Weather

Comply with ACI 305R. Maintain required concrete temperature using Figure 2.1.5, "Effect of Concrete Temperatures, Relative Humidity, and Wind Velocity on the Rate of Evaporation of Surface Moisture From Concrete" in ACI 305R to prevent the evaporation rate from exceeding one kg per square meter 0.2 pound of water per square foot of exposed concrete per hour. If necessary, cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. If the evaporation rate exceeds 0.5 kg per square meter 0.1 pound of water per square foot per hour, fog spray the exposed concrete surfaces until active moist curing is applied. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

3.4.4 Prevention of Plastic Shrinkage Cracking

During weather with low humidity, and particularly with high temperature and appreciable wind, develop and institute measures to prevent plastic shrinkage cracks from developing. If plastic shrinkage cracking occurs, halt further placement of concrete until protective measures are in place to prevent further cracking. Periods of high potential for plastic shrinkage cracking can be anticipated by use of Figure 2.1.5 of ACI 305R. In addition to the protective measures concrete placement shall be further protected by erecting shades and windbreaks and by applying fog sprays of water, the addition of monomolecular films, or wet covering. When such water treatment is stopped, curing procedures shall be immediately commenced. The methods and materials to remove or repair areas affected by plastic shrinkage cracks shall be suggested by the Contractor, reviewed by the Navy's Subject Matter Expert in Concrete Materials, and approved by the Contracting Officer. Cracks shall never be troweled over or filled with cement slurry.

3.5 SURFACE FINISHES EXCEPT HORIZONTAL CONCRETE SURFACES

3.5.1 Defects

Repair formed surfaces by removing minor honeycombs, pits greater than 600 square mm one square inch surface area or 6 mm 0.25 inch maximum depth, or otherwise defective areas. Provide edges perpendicular to the surface and patch with non-shrink grout. Patch tie holes and defects when the forms are removed. Concrete with extensive honeycomb including exposed steel reinforcement, cold joints, entrapped debris, separated aggregate, or other defects which affect the serviceability or structural strength will be rejected, unless correction of defects is approved. Obtain approval of corrective action prior to repair. The surface of the concrete shall not vary more than the allowable tolerances of ACI 347. Exposed surfaces shall be uniform in appearance and finished to a smooth form finish unless

otherwise indicated.

3.5.2 Formed Surfaces

3.5.2.1 Tolerances

Comply with **ACI 117** and as indicated on drawings. Note specific tolerances required for inner and outer seats and sill.

3.5.2.2 As-Cast Form

Provide form facing material producing a smooth, hard, uniform texture on the concrete. Arrange facing material in an orderly and symmetrical manner and keep seams to a practical minimum. Support forms as necessary to meet required tolerances. Material with raised grain, torn surfaces, worn edges, patches, dents, or other defects which will impair the texture of the concrete surface shall not be used. Patch tie holes and defects and completely remove fins.

3.6 FINISHES FOR HORIZONTAL CONCRETE SURFACES

3.6.1 Finish

Comply with **ACI 301M ACI 301**. Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater.

3.6.1.1 Broomed

Perform a floated finish, then draw a broom or burlap belt across the surface to produce a coarse scored texture. Whether floating with a wood, magnesium, or composite hand float, with a bladed power trowel equipped with float shoes, or with a powered disc, float shall begin when the surface has stiffened sufficiently to permit the operation. Permit surface to harden sufficiently to retain the scoring or ridges. Broom transverse to traffic or at right angles to the slope of the slab.

3.7 CURING AND PROTECTION

Comply with **ACI 301M ACI 301** and **ACI 308.1** unless otherwise specified. Prevent concrete from drying by misting surface of concrete. Begin curing immediately following final set. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, by rain or running water, adverse weather conditions, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period.

3.7.1 Wet Curing

Wet cure marine concrete using ASTM C1602/C1602M compliant water for a minimum of 7 days. Do not allow construction loads to exceed the superimposed load which the structural member, with necessary supplemental support, is capable of carrying in current condition safely and without damage.

Leaving the forms in place for seven days is a suitable alternative to wet curing.

3.7.1.1 Fog Spraying or Sprinkling

Apply water uniformly and continuously throughout the curing period. For temperatures between 4 and 10 degrees C 40 and 50 degrees F, increase the curing period by 50 percent.

3.7.1.2 Pervious Sheeting

Completely cover surface and edges of the concrete with two thicknesses of wet sheeting. Overlap sheeting 150 mm 6 inches over adjacent sheeting. Sheeting shall be at least as long as the width of the surface to be cured. During application, do not drag the sheeting over the finished concrete or over sheeting already placed. Wet sheeting thoroughly and keep continuously wet throughout the curing period.

3.7.1.3 Impervious Sheeting

Wet the entire exposed surface of the concrete thoroughly with a fine spray of water and cover with impervious sheeting throughout the curing period. Lay sheeting directly on the concrete surface and overlap edges 300 mm 12 inches minimum. Provide sheeting not less than 450 mm 18 inches wider than the concrete surface to be cured. Secure edges and transverse laps to form closed joints. Repair torn or damaged sheeting or provide new sheeting. Cover or wrap columns, walls, and other vertical structural elements from the top down with impervious sheeting; overlap and continuously tape sheeting joints; and introduce sufficient water to soak the entire surface prior to completely enclosing.

3.7.2 Liquid Membrane-Forming Curing Compound

After 7 day cure apply a curing compound to all surfaces. Maintain compound for minimum of 14 days. Seal or cover joint openings prior to application of curing compound. Prevent curing compound from entering the joint. Apply in accordance with the recommendations of the manufacturer immediately after any water sheen which may develop after finishing has disappeared from the concrete surface. Provide and maintain compound on the concrete surface throughout the curing period. Do not use this method of curing where the use of Figure 2 .1.5, "effect of Concrete Temperatures, Relative Humidity, and Wind Velocity on the Rate of Evaporation of Surface Moisture From Concrete" in ACI 305R indicates that hot weather conditions will cause an evaporation rate exceeding one kg pf water per square meter per hour 0.2 pound of water per square foot per hour.

3.7.2.1 Application

Mechanically agitate curing compound thoroughly during use. Use approved power-spraying equipment to uniformly apply two coats of compound in a continuous operation. The total coverage for the two coats shall be 5

square meters maximum per L 200 square feet maximum per gallon of undiluted compound unless otherwise recommended by the manufacturer's written instructions. The compound shall form a uniform, continuous, coherent film that will not check, crack, or peel. Immediately apply an additional coat of compound to areas where the film is defective. Respray concrete surfaces subjected to rainfall within 3 hours after the curing compound application.

3.7.2.2 Protection of Treated Surfaces

Prohibit pedestrian and vehicular traffic and other sources of abrasion at least 72 hours after compound application. Maintain continuity of the coating for the entire curing period and immediately repair any damage.

3.7.3 Curing Periods

Moist cure concrete using ASTM C1602/C1602M compliant water for a minimum of 7 days. Continue chemical curing for a total period of 21 days. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing shall be subject to approval by the Contracting Officer.

3.8 FIELD QUALITY CONTROL

3.8.1 Fresh Concrete Properties

For each concrete mixture, take samples in accordance with ASTM C172/C172M, test and record the slump, and temperature. If the slump deviates from the previous batch by more than 25.4 mm 1 inch, air content shall also be determined. Adjustment of air content and/or slump with chemical admixture is permitted provided the water to cementitious material ratio is not exceeded.

3.8.1.1 Slump Tests

ASTM C143/C143M. Test concrete samples from the truck upon its arrival and at mid placement from the end of the pump hose.

3.8.1.2 Temperature Tests

- a. Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions below 10 degrees C and above 27 degrees C below 50 degrees F and above 80 degrees F for each batch (minimum) or every 40 cubic meters 50 cubic yards (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.
- b. Determine temperature of each concrete sample in accordance with ASTM C1064/C1064M. Temperatures must comply with the Concrete Temperature Control Plans.

3.8.1.3 Air Content Tests

ASTM C231/C231M or ASTM C173/C173M. Test concrete samples from the truck

upon its arrival and at mid placement from the end of the pump hose.

3.8.1.4 Unit Weight Test

ASTM C138/C138M. Test concrete samples from the truck upon its arrival and at mid placement from the end of the pump hose.

3.8.2 Hardened Concrete Properties

Sample and test at a minimum one truck from each placement.

Cast and cure specimens in accordance with **ASTM C172/C172M**, **ASTM C31/C31M**, and applicable requirements of **ACI 305R** and **ACI 306R**.

For each lot, record the date and time sampled, the batch ticket code, cylinder ID code the location of placement, total volume of concrete represented by the sample, and fresh concrete properties; **ASTM C143/C143M** for slump or **ASTM C1611/C1611M** for slump flow and visual stability index (VSI), **ASTM C231/C231M** for air content, **ASTM C1064/C1064M** for temperature, and **ASTM C138/C138M** for unit weight.

For each lot sample, cast twelve 150 by 300 mm, 6 by 12 inch cylinder specimens for strength and seven 100 by 200 mm, 4 by 8 inch cylinder specimens for transport property testing. Special handling will be necessary for shipments of transport property specimens. These cylinders shall be wrapped completely with slightly dampened paper towels with spring water only. The wrapped cylinders shall be placed in either a vacuum package or double layers of sealed plastic bags. Package cylinders to prevent damage and ship priority mail to the approved testing laboratory.

In the event the results of cylinder tests fail to satisfy transport properties, then an additional pair of specimens shall be tested. In the event quality acceptance test results and retest results fail to meet the quality acceptance criteria, the entire lot shall be considered non-conforming material, refer to the subpart entitled "REPAIR, REHABILITATION and REMOVAL".

For every [382.3 cubic meters] 500 cubic yards placed perform a petrographic examination in accordance with **ASTM C856**.

Provide additional cylinders for radcon testing as required by the Contracting Officer.

3.8.2.1 Compressive Strength Tests

ACI 214R tests for strength – conduct strength tests of concrete during construction in accordance with the following procedures:

- a. Test cylinders in accordance with **ASTM C39/C39M**. Test three cylinders at 3 days, three cylinders at 7 days, and three cylinders at the age when the compressive strength requirement was specified. Hold the remaining three cylinders in storage. If one specimen in a test shows evidence of improper sampling, molding or testing, discard the specimen and consider the strength of the remaining cylinder to be the test result. If more than one specimen shows excess defects, the Contracting Officer may allow the entire test to be discarded. Test results shall not exceed the specified compressive strength by more

than 20 percent for the age specified.

- b. If the average strength test results are less than the specified strength (f'_c), extract three core samples from the structure in accordance with [ASTM C42/C42M](#) from the area that correlates to the low test results. These extracted cores shall not contain steel reinforcing. Repair core holes with non-shrink grout. Match color and finish of adjacent concrete. For concrete not meeting strength criteria, prepare a remediation strategy for the review by the Contracting Officer.
- c. Strength test reports shall be provided within 7 days of test completion.

3.8.2.2 Chloride Ion Concentration

Comply with [ACI 318M ACI 318](#). Determine water soluble chloride ion concentration. Perform test once for each mix design. The limits for average chloride ion content are provided in Table 3.

3.8.2.3 Non-Destructive Tests

Use of a rebound hammer to obtain data on the strength of the concrete surface shall be in accordance with [ASTM C805/C805M](#). Test results from the rebound hammer and other non-destructive testing may be helpful in selecting areas to extract concrete cores for destructive testing.

3.8.3 Core Samples and Compressive Strength Testing

Obtain and test cores in accordance with [ASTM C42/C42M](#).

If concrete in the structure is dry under service conditions, air dry cores (temperature [16 to 27 degrees C](#) [60 to 80 degrees F](#), relative humidity less than 60 percent) for 7 days before testing and test dry. Otherwise, test the cores, after moisture conditioning, in accordance with [ASTM C42/C42M](#).

Acceptance criteria for cylinder compressive strength are provided in subpart entitled "Acceptance of Concrete Strength".

Take at least three representative cores from each member or area of concrete in place that is considered potentially strength deficient. Impair the strength of the structure as little as possible. If, before testing, extracted cores show evidence of having been damaged subsequent to or during removal from the structure, take replacement cores.

Fill core holes with low slump concrete or mortar of a strength equal to or greater than the original concrete.

The Contracting Officer will evaluate and validate core tests in accordance with the specified procedures.

3.8.4 Acceptance of Concrete Strength

3.8.4.1 Standard Molded and Cured Strength Specimens

The acceptance of concrete strengths shall be based on averages of results from three consecutive compressive strength tests. When the averages of

all sets of three consecutive compressive strength test results are between 1.0 and 1.2 times the field test strength (fcr), and no individual strength test falls below fcr by more than 3.45 MPa 500 psi, the strength of the concrete is satisfactory. These criteria also apply when accelerated strength testing is specified unless another basis for acceptance is specified.

3.8.4.2 Non-Destructive Tests

Non-destructive tests may be used when permitted to evaluate concrete where standard molded and cured cylinders have yielded results not meeting the criteria.

3.8.4.3 Extracted Core Tests

When the average compressive strengths of the representative cores are between 0.85 fcr and 1.2 fcr and if no single core is less than 0.75 fcr,

the strength of concrete is satisfactory.

3.8.5 Preparations for Placing

The licensed Professional Engineer (PE) of record must inspect prepared surfaces, forms, and embedded items in sufficient time prior to each placement of repair material to certify that the surfaces are ready to receive the repair material. The professional engineer of record shall fill out QC Inspection Checklist and submit prior to placing repair material. Blank [QC Inspection Checklist](#) included. [The PE shall be the QC manager specified in paragraph 1.6.1.1.](#)

3.9 FINAL INSPECTION AND ACCEPTANCE TESTING

Following completion of the work, inspect surfaces for damage, staining, and other distresses. Inspect repairs for cracking, crazing, delamination, unsoundness, staining and other defects. Inspect the finish and surface tolerances of the repairs to verify that all requirements have been met. Repair all surfaces exhibiting defects as directed at no additional cost to the Government when defects are due to Contractor workmanship or procedures. Submit procedures to repair defective work prior to completing repairs.

3.10 REPAIR, REHABILITATION AND REMOVAL

Before the Government accepts the structure and final payment is made inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the quality of the structure. A report documenting these defects shall be prepared which includes recommendations for repair, removal and/or remediation which, will be reviewed by the Navy's Subject Matter Expert in Concrete Materials and submitted to the Contracting Officer for approval before any corrective work is accomplished.

3.10.1 [Crack Repair](#)

Crack inspections shall be completed 120 days after concrete placement. Crack inspections completed earlier will not be accepted. Repair all cracks 0.006 inch and wider with urethane grout injection. The Contracting Officer may require the Contractor to repair cracks smaller than 0.006 inch with urethane injection. The proposed method and materials to repair the cracks shall be submitted to the Contracting Officer for approval.

3.10.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than [6 mm 1/4 inch](#) thick shall be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than [6 mm 1/4 inch](#) thick shall be removed and new material provided or mitigated in a manner acceptable to the Contracting Officer.

3.10.3 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor to restore the quality shall be reviewed by the Navy's Subject Matter Expert in Concrete Materials and approved by the Contracting Officer prior to proceeding.

<u>QC INSPECTION CHECKLIST</u>			
PROJECT:			CONTRACTOR:
PROJECT LOCATION:			
DEFECT ID:			
DEFECT LOCATION:			
<u>INSPECTION TASKS</u>			
TASK:	DATE COMPLETED:	QC INSPECTOR SIGNATURE:	
LAYOUT			
CHIPPED			
3/4" BEHIND REBAR			
SURFACE PROFILE			
REINFORCING CLEANED			
FORM PLACED			
FORM SEALED			
FORM STRIPPED			
DEFECT INSPECTION			
ANCHOR BOLT HOLES PATCHED			
<u>REPAIR QUANTITIES</u>			
TASK:	QUANTITY:	DATE INSTALLED:	
REINFORCING STEEL (LBS)			
CONCRETE (SF)			
CRACK INSPECTION (LF)			
CRACK REPAIRS (LF)			

-- End of Section --

SECTION 05 12 00

STRUCTURAL STEEL

08/18, CHG 2: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO LRFD (8th Edition; 2017) Bridge Design
Specifications

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 207 (2016; R 2017) Certification Standard for
Steel Fabrication and Erection, and
Manufacturing of Metal Components

AISC 303 (2016) Code of Standard Practice for Steel
Buildings and Bridges

AISC 325 (2017) Steel Construction Manual

AISC 326 (2009) Detailing for Steel Construction

AISC 341 (2016) Seismic Provisions for Structural
Steel Buildings

AISC 360 (2016) Specification for Structural Steel
Buildings

AISC 420 (2010) Certification Standard for Shop
Application of Complex Protective Coating
Systems

AISC DESIGN GUIDE 10 (1997) Erection Bracing of Low-Rise
Structural Steel Buildings

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189 (2020) ASNT Standard for Qualification and
Certification of Nondestructive Testing
Personnel

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 (2020) Surface Texture, Surface Roughness,
Waviness and Lay

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	(2012) Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS D1.1/D1.1M	(2020) Structural Welding Code - Steel
AWS D1.8/D1.8M	(2016) Structural Welding Code—Seismic Supplement
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M	(2017a) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling
ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A193/A193M	(2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A276/A276M	(2017) Standard Specification for Stainless Steel Bars and Shapes

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2019) Structural Engineering
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR Part 1926, Subpart R	Steel Erection
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erection and Erection Bracing Drawings; G

SD-02 Shop Drawings

Fabrication Drawings Including Details of Connections; G

SD-03 Product Data

Shop Primer

Welding Electrodes and Rods

Non-Shrink Grout

Recycled Content for Structural Steel; S

Recycled Content for Structural Steel Tubing; S

Recycled Content for Steel Pipe; S

SD-05 Design Data

Shoring and Temporary Bracing; G [If necessary]

SD-06 Test Reports

Bolts, Nuts, and Washers

Weld Inspection Reports

SD-07 Certificates

Steel

AISC Structural Steel Fabricator Quality Certification

Welding Procedures and Qualifications

Welding Electrodes and Rods

Certified Welding Inspector

NDT Technician

Welding Procedure Specifications (WPS)

1.3 AISC QUALITY CERTIFICATION

Work must be fabricated by an AISC Certified Structural Steel Fabricator, in accordance with AISC 207, Category BU. Submit AISC Structural Steel Fabricator quality certification.

1.4 QUALITY ASSURANCE

1.4.1 Preconstruction Submittals

1.4.1.1 Erection and Erection Bracing Drawings [IF NECESSARY]

Submit for record purposes. Indicate the sequence of erection, temporary shoring and bracing. The erection drawings must conform to AISC 303.

1.4.2 Fabrication Drawing Requirements

Submit [fabrication drawings](#) for approval prior to fabrication. Prepare in accordance with [AISC 303](#), [AISC 326](#) and [AISC 325](#). Fabrication drawings must not be reproductions of contract drawings.[Sign and seal fabrication drawings by a registered professional engineer.] Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use [AWS A2.4](#) standard welding symbols. [Shoring and temporary bracing](#) must be designed and sealed by a registered professional engineer and submitted for record purposes, with calculations, as part of the drawings. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

1.4.3 Certifications

1.4.3.1 Welding Procedures and Qualifications

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests.[If the qualification date of the welder or welding operator is more than 6 months old, the welding operator's qualification certificate must be accompanied by a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.]

Conform to all requirements specified in [AWS D1.1/D1.1M](#).

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide the structural steel system, including [shop primer] [galvanizing], complete and ready for use. Provide structural steel systems including design, materials, installation, workmanship, fabrication, assembly, erection, inspection, quality control, and testing in accordance with [AISC 303](#), [AISC 360](#), and [UFC 3-301-01](#) except as modified in this contract.

2.2 STEEL

2.2.1 Structural Steel

Wide flange and WT shapes, [ASTM A276 S316 Stainless Steel](#). Angles, Channels and Plates, [ASTM A36/A36M](#).

2.2.2 Steel Pipe

[ASTM A53/A53M](#), Type E or S, Grade B, weight class STD (Standard) or as indicated. Provide steel pipe containing a minimum of 50 percent recycled content. Submit data identifying percentage of [recycled content for steel pipe](#).

2.3 BOLTS, NUTS, AND WASHERS

Submit the certified manufacturer's mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied fasteners.

2.3.1 Common Grade Bolts

2.3.1.1 Bolts

ASTM A307, Grade A, plain finish [hot dipped zinc coating][mechanically deposited zinc coating]. The bolt heads and the nuts of the supplied fasteners must be marked with the manufacturer's identification mark, the strength grade and type specified by ASTM specifications.

2.3.1.2 Nuts

ASTM A563, Grade A, heavy hex style.

2.3.1.3 Washers

ASTM F844.

2.4 STRUCTURAL STEEL ACCESSORIES

2.4.1 Welding Electrodes and Rods

AWS D1.1/D1.1M . Submit product data for welding electrodes and rods.

2.4.2 Non-Shrink Grout

ASTM C1107/C1107M, with no ASTM C827/C827M shrinkage. Grout must be nonmetallic. Submit product data for non-shrink grout.

2.5 FABRICATION

Fabrication must be in accordance with the applicable provisions of AISC 325. Fabrication and assembly must be done in the shop to the greatest extent possible. Punch, subpunch and ream, or drill bolt [and pin] holes perpendicular to the surface of the member.

Compression joints depending on contact bearing must have a surface roughness not in excess of 500 micro inch as determined by ASME B46.1, and ends must be square within the tolerances for milled ends specified in ASTM A6/A6M.

Shop splices of members between field splices will be permitted only where indicated on the Contract Drawings. Splices not indicated require the approval of the Contracting Officer.

2.5.1 Markings

Prior to erection, identify members by a painted erection mark. Connecting parts assembled in the shop for reaming holes in field connections must be match marked with scratch and notch marks. Do not locate erection markings on areas to be welded. Do not locate match markings in areas that will decrease member strength or cause stress concentrations.

2.5.2 Shop Primer (For ASTM A36 AND A53 steel)

SSPC Paint 20 or SSPC Paint 29, (zinc rich primer). Shop prime structural steel, except as modified herein, in accordance with SSPC PA 1. Do not prime steel surfaces embedded in concrete, galvanized surfaces, [surfaces to receive sprayed-on fireproofing,] [surfaces to receive epoxy coatings,] [surfaces designed as part of a composite steel concrete section,] or surfaces within 0.5 inch of the toe of the welds prior to welding (except surfaces on which metal decking and shear studs are to be welded). If flash rusting occurs, re-clean the surface prior to application of primer. Apply primer [in accordance with endorsement "SPE-P1" of AISC 420 or approved equal NACE or SSPC certification to a minimum dry film thickness of 2.0 mil. Submit shop primer product data.

Prior to assembly, prime surfaces which will be concealed or inaccessible after assembly. Do not apply primer in foggy or rainy weather; when the ambient temperature is below 45 degrees F or over 95 degrees F; or when the primer may be exposed to temperatures below 40 degrees F within 48 hours after application, unless approved otherwise by the Contracting Officer. Repair damaged primed surfaces with an additional coat of primer.

2.5.2.1 Cleaning

SSPC SP 6/NACE No.3, except steel exposed in spaces above ceilings, attic spaces, furred spaces, and chases that will be hidden to view in finished construction may be cleaned to SSPC SP 3 when recommended by the shop primer manufacturer. Maintain steel surfaces free from rust, dirt, oil, grease, and other contaminants through final assembly.

2.5.3 Surface Finishes

ASME B46.1 maximum surface roughness of 125 for pin, pinholes, and sliding bearings, unless indicated otherwise.

PART 3 EXECUTION

3.1 ERECTION

- a. Erection of structural steel, except as indicated in item b. below, must be in accordance with the applicable provisions of AISC 325, AISC 303 and 29 CFR Part 1926, Subpart R.
- b. For low-rise structural steel buildings (60 feet tall or less and a maximum of 2 stories), erect the structure in accordance with AISC DESIGN GUIDE 10.

After final positioning of steel members, provide full bearing under base plates and bearing plates using nonshrink grout. Place nonshrink grout in accordance with the manufacturer's instructions.

3.1.1 STORAGE

Store the material out of contact with the ground in such manner and location as to minimize deterioration.

3.2 CONNECTIONS

Except as modified in this section, design connections indicated in accordance with AISC 360. Build connections into existing work. Do not

tighten anchor bolts set in concrete with impact torque wrenches. Holes must not be cut or enlarged by burning. Bolts, nuts, and washers must be clean of dirt and rust, and lubricated immediately prior to installation.

3.2.1 Common Grade Bolts

Tighten [ASTM A307](#) bolts to a "snug tight" fit. "Snug tight" is the tightness that exists when plies in a joint are in firm contact. If firm contact of joint plies cannot be obtained with a few impacts of an impact wrench, or the full effort of a man using a spud wrench, contact the Contracting Officer for further instructions.

3.3 GAS CUTTING

Use of gas-cutting torch in the field for correcting fabrication errors is not permitted on any major member in the structural framing. Use of a gas cutting torch will be permitted on minor members not under stress only after approval has been obtained from the Contracting Officer.

3.4 WELDING

Welding must be in accordance with [AWS D1.1/D1.1M](#). Grind exposed welds smooth as indicated. Provide [AWS D1.1/D1.1M](#) qualified welders, welding operators, and tackers.

Develop and submit the [Welding Procedure Specifications \(WPS\)](#) for all welding, including welding done using prequalified procedures. Submit for approval all WPS, whether prequalified or qualified by testing.

3.5 SHOP PRIMER REPAIR

Repair shop primer in accordance with the paint manufacturer's recommendation for surfaces damaged by handling, transporting, cutting, welding, or bolting.

3.5.1 Field Priming

Field prime steel exposed to the weather, or located in building areas without HVAC for control of relative humidity. After erection, the field bolt heads and nuts, field welds, and any abrasions in the shop coat must be cleaned and primed with paint of the same quality as that used for the shop coat.

3.6 GALVANIZING REPAIR

Repair damage to galvanized coatings using [ASTM A780/A780M](#) zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied.

3.7 FIELD QUALITY CONTROL

Perform field tests, and provide labor, equipment, and incidentals required for testing[, except that electric power for field tests will be furnished as set forth in Division 1]. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

3.7.1 Welds

3.7.1.1 Visual Inspection

AWS D1.1/D1.1M. Furnish the services of AWS-certified welding inspectors for fabrication and erection inspection and testing and verification inspections. A **Certified Welding Inspector** must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Submit certificates indicating that certified welding inspectors meet the requirements of **AWS QC1**.

Inspect proper preparation, size, gaging location, and acceptability of all welds; identification marking; operation and current characteristics of welding sets in use.

3.7.1.2 Nondestructive Testing

Nondestructive testing must be in accordance with **AWS D1.1/D1.1M**. Ultrasonic testing must be performed in accordance with Table 6.2 of **AWS D1.1/D1.1M**. Test locations must be as indicated. All personnel performing NDT must be certified in accordance with **ANSI/ASNT CP-189** in the method of testing being performed. Submit certificates showing compliance with **ANSI/ASNT CP-189** for all **NDT technicians**. If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing, and all fillet welds made by that welder must be inspected by magnetic particle testing (MT) or dye penetrant testing (PT) as approved by the Contracting Officer. When groove welds made by an individual welder are required to be tested, magnetic particle or dye penetrant testing may be used only in areas inaccessible to ultrasonic testing. Retest all repaired areas. Submit **weld inspection reports**.

Testing frequency: Provide the following types and number of tests:

Test Type	Number of Tests
Ultrasonic	50 percent of CJP Welds
Magnetic Particle	50 percent of PJP and Fillet Welds
Dye Penetrant	50 percent of PJP and Fillet Welds

-- End of Section --